

— Fifth Edition —

Hegde's PocketGuide to **TREATMENT** in Speech-Language Pathology

M. N. Hegde

Autism

Stuttering

Voice
Disorders

Dysarthria

AAC

Treatment

Literacy

TBI

Right Hemisphere
Disorder

Apraxia of Speech

Language Disorders

Aphasia

Voice Disorders

Speech Sound
Disorders

Dementia



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PocketGuide to
Treatment in
Speech-Language
Pathology**

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M. N. Hegde, PhD





9177 Aero Drive, Suite B
San Diego, CA 92123

email: information@pluralpublishing.com
website: <https://www.pluralpublishing.com>

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Preface to the Fifth Edition

The fifth edition of this *PocketGuide to Treatment in Speech-Language Pathology* has been updated to include newer research-based treatment procedures. All existing entries have been revised in light of recent developments in treatment approaches and techniques. Guidelines on treating all disorders of communication have been revised in accordance with evidence-based practice principles. Unfounded treatment procedures have been eliminated. Available controlled or other kinds of evidence for described major treatment procedures are briefly noted. To further assist critical evaluations and treatment procedure selections, comprehensive levels of evidence are described. Most entries have been revised to make them more succinct to allow inclusion of newer information. Current references to major entries have been added and older ones eliminated.

A companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, has been simultaneously revised to update and expand information on assessment approaches and procedures. The two guides provide comprehensive information to advance evidence-based clinical practice. The guides serve a dual function: First, they are detailed enough for the student clinicians as well as the more established practicing clinicians. Second, the guides are succinct enough to provide an overview of the entire range of knowledge related to assessment and treatment in speech-language pathology.

This fifth edition of the *PocketGuide to treatment procedures* is designed for clinical practitioners and students in communication disorders. The *PocketGuide* combines the most desirable features of a specialized dictionary of terms, a clinical resource book, and textbooks on the treatment of communicative disorders. It works as a quick reference book like a dictionary because the entries are alphabetized, but it offers more than a dictionary because it specifies treatment

procedures the clinicians can apply step-by-step in treating any disorder of communication in any age group. The PocketGuide is like a resource book in that its main objective is to describe practical treatment procedures while minimizing theories that are often unnecessary to implement the procedures. However, the guide offers more than a resource book because it clearly specifies all the steps involved in treating clients with various communication disorders.

The PocketGuide is unlike any other standard textbook or resource book on treatment because it is uniquely comprehensive in addressing treatment procedures for all disorders of communication in both children and adults. The technical treatment actions the clinicians take are highlighted to show that the same set of techniques are applicable in treating disorders from A through V—aphasia to voice disorders and all disorders in between. Easy accessibility for a specific disorder's treatment is the main reason why the guide is written as an alphabetized dictionary.

How the PocketGuide Is Organized

All main entries for the treatment of communication disorders are printed in bold typeface and **blue** color. Each cross-referenced entry is underlined. Each main disorder of communication is entered in alphabetical order. Subcategories or types of a given disorder are described under the main entry (e.g., Broca's Aphasia under Aphasia, Ataxic Dysarthria under Dysarthria).

A major concern or confusion of clinicians is the multiplicity of advocated treatment procedures for every communication disorder. Selection of a treatment procedure may be baffling, especially to student clinicians. This guide simplifies the treatment selection process. For each of the major communication disorders, the clinician will first find a description of a comprehensive and complete treatment procedure that is self-sufficient to treat that disorder. Clinicians may use this procedure because it integrates all necessary and research-supported procedures to treat that disorder. The procedure is comprehensive in that it begins with establishing baserates, moves through steps in treatment, and ends with maintenance strategies and caregiver training. The comprehensive procedure also describes treatment at progressively more complex levels, beginning with the sound-syllable-word level to conversational speech. Most specialized and named treatment programs do not offer a comprehensive picture of treating a disorder; most are limited to a single skill and limited skill levels.

Organization of entries varies somewhat for different disorders, but they follow a general format that begins with a brief description of the disorder, general guidelines on treatment, ethnocultural considerations in treatment, and comprehensive treatment procedures in many cases, ending with specific treatment programs or procedures when available (e.g., social approaches to treating aphasia, enhanced milieu teaching, or syllable prolongation in stuttering). Major and current

references are given at the end of each main entry and at the end of most specific treatment techniques or programs.

Many specific techniques that are a part of the comprehensive treatment program and are applicable to treating multiple disorders have their independent entries (e.g., Baselines, Discrete Trials, Modeling, Biofeedback, or Reinforcement Schedules, Response Cost, Pause-and-Talk). Disorders for which a particular technique is especially appropriate are also described.

How to Use This PocketGuide

There are two methods for the clinician to use this guide. In the first method, the clinician may look up treatment procedures by disorders in their alphabetical order. *Major Entries* (in the table of contents) will quickly refer the reader to specific communication disorders described in the guide. Clinicians can rapidly gain access to treatment procedures, from aphasia to voice disorders, described in their alphabetical order. Under each of the main entries for major disorders, the clinician may look up subentries or specific types of disorders. For example, under Dysarthrias, the clinician will find the various types of dysarthria, also entered alphabetically, from ataxic dysarthria to unilateral upper motor neuron dysarthria. Similarly, following the main entry for Aphasia, the different varieties of aphasia are described in their alphabetical order.

In the second method, the clinician may look up a treatment procedure by its name. For example, the clinician can look up such specific treatment techniques as the following in their alphabetical order: activity-based language intervention, airflow management in stuttering, conversational repair strategies, delayed auditory feedback, functional equivalence training, mand-model, melodic intonation therapy, and so forth. In many cases, the reader who finds a specific treatment technique in the alphabetized order is referred to the specific disorder for which the technique is applicable.

A Caveat

Conscientious attempts have been made to include most treatment techniques described in the literature. However, the author is aware that not all techniques have been included. Some have been excluded because of their transparent lack of logic, appropriateness, or even an expectation of desirable effects. Others have been omitted because of negative evidence. A few popular methods are described briefly with the cautionary note that the method may be useful but more

evidence is needed to support its practice. Most importantly, in any task such as this that requires an encyclopedic review of the entire treatment literature in speech-language pathology, the omission of a procedure that deserves inclusion is an acknowledged limitation and an unintended consequence. The reader more often may be correct in assuming that a technique was omitted inadvertently than to assume that it was considered and rejected. After careful consideration, rejecting or accepting a technique, included or excluded in this guide, is still the clinician's prerogative as well as responsibility; the author hopes that this guide will facilitate that consideration. Finally, the author will be grateful to clinicians and researchers who might suggest procedures to be included in future editions.

About the Author

M. N. Hegde, PhD, is Professor Emeritus of Speech-Language Pathology in the Department of Communicative Disorders at California State University, Fresno. A highly regarded and proficient author in speech-language pathology, his books include leading texts for academic courses and valuable resources for practicing clinicians. Dr. Hegde enjoys world renown as a researcher, presenter, contributor of original articles to leading national and international journals, and is also the critically acclaimed author of more than two dozen highly regarded books in speech-language pathology. He has edited more than 25 books for different publishers and is on the editorial board of several scientific journals and has been a guest editor of international journals. He also has served on the editorial boards of scientific and professional journals and continues to serve as an editorial consultant to the *Journal of Fluency Disorders* and the *American Journal of Speech-Language Pathology*.

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A

Abstract and Figurative Language. These language skills, also described as figurative (non-literal) language skills, include production and comprehension of proverbs, metaphors, similes, and idioms; recognizing absurd statements; distinguishing logical and illogical utterances; making correct inferences from indirect expressions (understanding implied meanings); understanding humor and jokes; and so forth.

Abstract and Figurative Language Intervention/General Guidelines

- Abstract and figurative language skills may be treatment targets in several kinds of communication disorders, including those based on neurological impairments (e.g., traumatic brain injury, right hemisphere disorder, dementia), autism spectrum disorder, significant language disorders in older students and adolescents, and so forth
- Such skills may be targets for individuals with any type of communication disorder if assessments reveal deficits
- Controlled experimental research on practical treatment methods are extremely limited; use behavioral intervention procedures that are known to be effective in teaching various language skills
- Target these advanced features after establishing the basic language skills, including words, sentences, narratives, discourse, and other [Social Communication Skills](#)
- Select treatment targets that are client-specific; consider the level of education and occupation for all individuals; in addition, consider the individual's bilingual status and ethnocultural variables because of their effects on proverbs, common sayings, idioms, and metaphors; these tend to be culture-specific
- Find common proverbs, metaphors, and similes online
- See the cited Nippold book for developmental data on abstract language and multiple examples of proverbs, metaphors, simile, idiom, riddles, jokes, and so forth

Treat Impaired Inference With Multiple Strategies

- **Tell brief stories or describe short episodes**
- Ask questions to have the client describe information not explicitly stated but implied in the narratives; for example:
 - Tell the client that *a man in a business suit, tie, and a tablet in his hands walks into a restaurant and asks the hostess, “Do you have a special room that is quiet? Need a table for five.”* Then ask the client, “Why do you think the man wants a quiet room to eat?” Reinforce the correct response, which may be something like, “Perhaps he is going to have a business luncheon meeting.”
 - Describe and teach inferences from multiple scenarios
- **Use various kinds of pictures or illustrations**
- Find such stimuli for inferences in books and magazines; for example:
 - Show a picture in which a man is packing a duffel bag and a pair of hiking boots are placed next to it; and a woman is putting a fancy dress into a suitcase and a pair of high-heeled shoes are placed next to it; ask the client “Where are they going?”
 - Reinforce correct responses (“The man is going on a hiking trip; the woman is going on a business trip”); prompt if no responses; correct wrong responses by modeling the right inference
 - Show a picture that prompts multiple inferences (e.g., show the picture of a woman buying a large bouquet of flowers in a flower shop and ask, “Why is she buying flowers?”); multiple answers may be good inferences (e.g., “She is buying flowers for her husband’s birthday”; “She is buying flowers for her friend’s wedding”; and so forth)
 - Use similar strategies with other kinds of pictures (e.g., such as those by Normal Rockwell) and ask the client to infer what is going on and what had just gone on

- Show such line drawings as that of a dog on a beach, shaking its body, and ask what the dog had been doing
- Show pictures that show sad and happy faces and ask the client why they might be so
- Train multiple exemplars and test generalized inferencing skills by presenting new and untrained stimuli

Treat Impaired Recognition of Absurdities

- **Use pictures that depict logical and absurd events**
 - Show the picture of a cat chasing a rat and another of a rat chasing a cat
 - Show the picture of an elephant eating leaves and another eating a dead cow
 - Ask the client to point out the one that is logical (correct)
 - Ask the client to point out the one that does not make sense (absurd)
 - Use other pairs of absurd and logical pictures
- **Present verbal or written statements that are logical or absurd**
 - Present printed absurd and sensible statements and ask the client to separate them (e.g., “Yesterday, I was so hungry when coming home from work that I went to a park and sat there” versus “Yesterday, I was so hungry when coming home from work that I stopped by a restaurant to eat before reaching home”)
 - Present other absurd and logical statements
 - Present a series of absurd statements without their logical counterparts and ask the person to explain why a statement is absurd
 - Test for generalized recognition of absurd statements by presenting untrained statements

Treat Impaired Comprehension of Proverbs

- **Teach client-specific proverbs**
 - Proverbs mean something other than what the words directly say

- For the selected proverbs, write literal interpretations and the abstract meanings they imply; for instance, for the proverb *Birds of a feather flock together*, write (a) birds with similar feathers stay together (literal) and (b) similar people associate with each other (implied meaning)
- Ask the client to select statements that give literal meanings
- Ask the client to select statements that give abstract meanings
- Prompt correct responses
- Reinforce and give corrective feedback
- Tell a few proverbs and ask the person to state the abstract meanings
- Ask the person to state proverbs and explain their abstract meanings
- Start with more common proverbs (e.g., *Honesty is the best policy*) and move on to less common proverbs (e.g., *Don't put too many irons in the fire*)
- Teach multiple exemplars and test for generalization with untrained proverbs
- **Teach common metaphors and similes**
 - Metaphors (figurative speech) equate two things, implying some kind of similarity; similes point out similarities between two things without equating them; *this is that* is a metaphor (e.g., *Summer in the desert is the oven at 400 degrees*; *She is a sun in the day, a moon in the night*), but *this is like that* is a simile (e.g., *Driving through fog in Fresno is like swimming in mushroom soup*; *the hail storm was like a pebble storm*)
 - Write both literal and figurative explanations of metaphors or similes you present
 - Ask the person to explain their meanings
 - Prompt and reinforce correct interpretations; verbally correct literal interpretations
 - Teach multiple exemplars and test for generalization of correct interpretations by presenting untrained metaphors and similes

Treat Impaired Comprehension of Idioms

- Idioms are also figurative expressions that mean something other than what the words say; while most figuratives like metaphors and similes are found in literary writings (stories, novels, and poems), idioms are everyday expressions that do not compare anything (metaphors and similes do); while proverbs may convey a lesson learned from life or a common belief in the community, idioms indirectly say something or refer to an action or event but not what is literally meant.
- Select a few idioms that are appropriate for the client's educational and cultural background
- Write down the literal and implied meanings of each target idiom
- Present an idiom in writing or verbally (e.g., *read between the lines, kick the bucket, a piece of cake, spill the beans, saving up for a rainy day*)
- Ask the person to either point to the correct written statement or to verbally state the correct meaning
- Teach multiple idiomatic exemplars and prone for generalization with untrained idioms

Treat Impaired Comprehension of Humor

- **Use cartoons to teach correct comprehension of humor**
 - Show cartoons and their captions separately
 - Ask the client to study the cartoons and then match them with their captions
 - Ask the client to say what is humorous about the cartoon
 - Ask the client to read a few jokes whose body of the joke and the punchlines are separated
 - Ask the client to match the joke with its punchline
 - Ask the client why a joke is funny
 - Teach multiple joke exemplars and test generalization with untrained jokes

Lundgren, K., & Brownell, H. (2016). Figurative language deficits associated with right hemisphere disorder. *Perspectives, SIG 2, 1*(2), 1-C15.

Nippold, M. A. (2016). *Later language development* (4th ed.). Pro-Ed.

Agaphia. To treat lost or impaired writing skills associated with cerebral pathology or injury that may also be associated with reading problems ([Alexia](#)), see [Treatment of Aphasia: Writing Problems](#).

Aided AAC Modeling. A procedure that combines a clinician's oral modeling of target responses with augmentative communication methods to children and adults with severely limited oral communication skills (e.g., children with autism spectrum disorder, intellectual disabilities, severe early child language disorders). To implement this procedure:

- Select a system of [Augmentative Communication—Aided](#)
- Combine it with oral modeling, missing in some typical augmentative methods
- When the child points to a picture, symbol, word, or phrase on a nonelectronic (e.g., communication board) or an electronic speech display system (e.g., iPad), model the target expression immediately and do what the pointing suggests (e.g., model “bathroom” when the child touches the picture of a toilet and then take the child to the bathroom; model “I want ice cream” if the child touches the icons for *I want* and *ice cream* and then give ice cream)

Binger, C., & Light, J. (2007). The effect of aided AAC modeling on the expression of multi-symbol messages by preschoolers who use AAC. *Augmentative and Alternative Communication*, 23(1), 30–43.

Airflow Management. A stuttering treatment target in fluency shaping procedure; includes inhalation of air, slight exhalation before initiating phonation, and sustained airflow throughout an utterance; for details, see [Stuttering; Treatment of Stuttering: Specific Techniques or Programs](#).

Alaryngeal Speech. Speech without a biological larynx. For intervention procedures, see [Laryngectomy](#).

Alerting Stimuli. Various means of drawing clients' impaired attention to the imminent treatment stimuli; include such statements as “Get ready! Here comes the

picture!” or “Look at me, I am about to show you how,” or such nonverbal cues as touching the individual’s hand just before presenting a stimulus; may be needed in treating autism spectrum, aphasia, dementia, right hemisphere disorder, and children with attention-deficit disorders.

Alexia. Reading problems in adults who have neurological impairments (e.g., strokes, neurodegenerative diseases); for treatment of alexia, see [Treatment of Aphasia: Reading Problems](#). To address reading problems of children who have not mastered grade-appropriate reading skills, see [Literacy and Literacy Skills in Children](#).

Alzheimer’s Disease. A major cause of dementia in older individuals; intervention may slow down deterioration and help manage communication skills as long as possible; see [Dementia](#) for management details.

American Indian Hand Talk (Plains Indian Sign Language, PISL). A manual interlanguage of nonverbal signs Native Americans use to communicate with members of other tribes with different languages; gestures may be produced in series to express more complex ideas, called agglutination; many signs are one-handed; may be used in teaching [Augmentative Communication, Gestural \(Unaided\)](#).

American Sign Language (ASL or AMESLAN). A highly developed manual (gestural) language used mostly by Deaf persons in the United States but may also be a means of communication for nonverbal or minimally verbal hearing persons; each sign or gesture may represent a letter of the English alphabet, a word, or a phrase; signs provide phonemic, morphologic, and syntactic information; see [Augmentative Communication, Gestural \(Unaided\)](#).

Amyotrophic Lateral Sclerosis (ALS). To treat motor speech disorders associated with this progressive neurological disease in which the upper and lower motor neurons degenerate, see [Dysarthrias](#).

Anomia. Treatment of naming difficulties (anomia) may be needed for individuals with aphasia, traumatic brain injury, dementia, and other neurologically based communication disorders; see [Aphasia](#) for treatment strategies.

Antecedents. Stimuli or events that clinicians present to evoke target responses in treatment; to make treatment stimuli effective:

- Select stimuli that are ethnoculturally appropriate for the client
- Select common objects or objects from the individual's home environment (e.g., a child's favorite toy or book)
- Select pictures that are colorful, unambiguous, and attractive
- Re-create or enact events to show actions and scripts
- Use instructions, demonstrations, modeling, prompting, manual guidance, and other special stimuli

Aphasia. Treatment of aphasia, a language disorder typically caused by strokes, targets multiple skills that tend to be impaired; initially, a team of professionals in a hospital manage persons with stroke and aphasia; eventually, most individuals may receive communication treatment as outlined; see the sources cited at the end of this entry to learn more about aphasia and its assessment.

Treatment of Aphasia: General Guidelines

- Note that both controlled and uncontrolled evidence support communication treatment for aphasia
- Conduct a detailed assessment; see the cited sources and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*
- Give a realistic prognosis to the individual and the family members
- Develop a variety of task- and individual-specific treatment procedures
- Let the client and the family members help select functional communication targets rather than grammatical accuracy
- Sequence target behaviors; move from simpler to more complex tasks

- Offer intensive treatment; the greater the frequency of weekly sessions, the higher the progress
- Use such extra stimuli as instructions, demonstrations, prompts, modeling, pictures, and objects in initial stages and fade them
- Use natural social communication stimuli in later states (e.g., asking a question instead of modeling or prompting)
- Use natural reinforcers for functional communication targets (e.g., smile and approval to reinforce verbal expressions; real objects to reinforce requests for objects); verbally correct incorrect responses
- Teach Self-Control (Self-Monitoring) skills to reduce errors and sustain treatment gains in the natural environment
- Offer group treatment sessions to promote generalization of verbal skills to social communication and social integration
- Consider telerehabilitation to continue treatment after discharge and to promote maintenance
- Offer Augmentative and Alternative Communication (AAC) strategies to reduce the residual deficits
- Treat a coexisting Apraxia of Speech; those who have Broca's aphasia are more likely to need it
- Counsel family members to help them cope with the residual deficits
- Train family members to evoke, prompt, reinforce, and maintain communicative behaviors
- Judge when it is not useful or ethical to continue the treatment

Treatment of Aphasia:

Ethnocultural Guidelines

- See Ethnocultural Variables in Treatment for details
- Consider the ethnocultural, language, and economic background of the individual in planning treatment
- Evaluate the individual's dialect (e.g., African American English or Spanish-influenced English), premorbid literacy level, and the current need for literacy skills

- Assess communication needs of a bilingual individual in both languages or the dominant language
- Select treatment targets that the client and the family members approve; select treatment stimuli that are available in the individual's home and, if appropriate, work environment
- Carefully describe the treatment procedures and note the effects they produce or fail to produce; modify the treatment procedure in light of the individual's performance and ethnocultural background

Treatment of Aphasia: Auditory Comprehension.

There is no strong and replicated evidence to support auditory comprehension treatment in persons with aphasia; having clients point to objects or printed words on repeated trials may not promote comprehension of conversational speech; moreover, improved comprehension (if that is demonstrated) does not improve production; dealing with evident comprehension problems in the context of teaching functional and social communication skill production may be the best strategy; evidence suggests that when production skills improve, comprehension skills also improve with no additional and direct treatment for comprehension; critically evaluate the need or usefulness of auditory comprehension treatment for a given individual; if you choose to teach this nonverbal skill, consider Treatment for Wernicke's Aphasia, described under Treatment of Aphasia: Specific Techniques or Programs; also consider the following to address auditory comprehension deficits during production training:

Promote Auditory Comprehension During Speech Production Training

- Select picturable verbs and other words that give a clue to auditory comprehension
- Select unambiguous treatment stimuli
- Speak to the client in direct, shorter, simpler, and active sentences
- Build personally relevant information into treatment tasks

- Speak at a slower rate, pause frequently, and give additional stress on key terms
- Conduct treatment in quieter environment with little or no distraction
- Give redundant messages and instructions; repeat them
- Speak in connected speech that gives context, rather than isolated words or sentences that do not
- Give the individual limited response choices; do not confuse by demanding multiple responses at the same time
- Pair auditory treatment stimuli with appropriate visual stimuli; use objects whenever possible; if not, use realistic, colorful, unambiguous pictures
- Make your face visible to the individual as you speak
- Draw the individual's attention before presenting treatment trials if necessary (give alerting stimuli; e.g., "Look at my face," "Here comes the picture," "Listen! I am going to ask you to do something")
- Target comprehension during conversation and narrative discourse training to combine production as well as comprehension; ask questions to assess comprehension during typical conversation and reinforce correct answers; reiterate misunderstood statements; tell brief stories and ask the individual to retell them; read aloud a brief story and ask the individual to retell it; ask questions about the details and sequence of the story to assess and positively reinforce comprehension; prompt details and sequences to reinforce narrative skills

Treatment of Aphasia: Naming. Most clinicians tend to begin aphasia intervention with naming skills; confrontation naming (naming objects, pictures, or photographs of people when asked, "What is this?" or "Who is this?") is a common treatment target; teaching confrontation naming may be justified if (a) the words taught generalize to social communication and if (b) training some nouns (names) results in generalized production of other, untrained, nouns; there is no evidence that

either of these outcomes follows confrontation naming intervention; moreover, confrontation naming is not a functional skill because very few persons with or without impaired naming are repeatedly asked, “What is this?” in their environment; if naming skills are taught, it is critical to advance from single-word naming to more complex verbal expressions, including Social Communication Skills.

Treatment of Naming: General Considerations

- Select words that the individual needs to utter frequently (e.g., nouns related to the individual’s hobbies, interests, and occupation; names of family members, friends, colleagues, and pets)
- Select the names of manipulable objects the individual may have used (e.g., a hammer for a carpenter, a painting brush for a painter, a pen for a writer)
- Use realistic drawings or colorful photographs rather than line or abstract drawings to teach the naming skills
- Let the individual regulate the rate of stimulus presentation (e.g., slower rate the individual may prefer)
- Give extra time to respond
- Display the stimuli for a longer duration if necessary
- Present simultaneously visual and auditory stimuli

Confrontation Naming: Treatment Techniques

- Start with more familiar objects and move on to less familiar objects; generate the list based on the individual’s own experiences
- Use cueing hierarchies as described next; note that most but not all *cues* are *prompts* in treating other communication disorders
- Use a stronger cue only when weaker cues do not evoke the response
- Start with a single cue and add additional cues only when necessary
- Use different types of cues
- Reinforce the correct response; verbally correct incorrect responses

- Teach the selected names on discrete trials
- Place a picture or an object in front of the individual
- Ask, “What is this?” (or “Who is this?” to evoke proper names)
- **Ask a question and give a cue**
 - Note that cues are the same as prompts
 - Sentence completion as a cue (“You write with a . . .” to evoke “pen”; “He is your . . .” to evoke “husband”; “This is a spoo . . . to evoke “spoon”)
 - Phonemic (“The name starts with a b . . .” to evoke “ball” or “Bob”)
 - Silent articulatory gestures for the initial sound (press the lips together to evoke “puppy”)
 - Personalized (“Once you hit your finger with it” to evoke “hammer”)
 - Functional description (“This is a round object you roll or kick” to evoke “ball”; “you write with it” to evoke “pen”)
 - Client’s description of function (“Tell me what you use this for and then tell me its name” to evoke “I use it to write, it’s a pen”)
 - Client’s description of use or demonstration (“Show me how you use this and then tell me its name,” to evoke the demonstration of drinking and the name “cup”)
 - An associated sound (“It goes *urf urf*” to evoke “dog”)
 - A synonym (“What is the same as a dwelling?” to evoke “house”)
 - An antonym (“What is the opposite of woman?” to evoke “man”)
 - An associated word (“What goes with the word chair?” to evoke “table”)
 - A superordinate (“It is a food item” to evoke “cake”)
 - A rhyming word (“It rhymes with *hog*” to evoke “dog”)
- **Fade the special cues**
 - Gradually reduce the amount and extent of cues to have the individual respond to more natural stimuli

- Reinstate previously successful cues when necessary and fade them again
- Fade cues in various ways (e.g., fading a verbal cue into a mere gestural cue; fading the vocal intensity of any verbal cue until it is no longer heard and then completely withdrawn)
- **Consider a nonconfrontational method of teaching naming**
 - Have the person read the target word, repeat it, write it to dictation, and then ask to name the target stimulus item
 - Model multiple target words several times, and each time, have the person repeat after you, and then ask for a stimulus picture description that may contain those words
 - Have the person name different exemplars of the same target objects to be named (e.g., show three different kinds of chairs, have the person name them)
- **Teach self-cueing**
 - Select a naming strategy that has worked for the individual (e.g., production of an antonym, reading the printed word or repeating it multiple times, or description-demonstration of the use of an object to be named); teach the individual to self-generate effective cues
 - Prompt the individual to use the strategy; fade prompting

Treatment of Aphasia: Expansion of Verbal Expressions. Expand words into phrases, phrases into sentences, and sentences into narratives, discourse, and conversational speech. For expansion, select verbal expressions that are:

- Most useful to the individual, the family members, and the caregivers; seek their input on selecting target expressions
- Helpful in expressing personal experiences, basic needs, emotions, and thoughts

- Most meaningful in social contexts, help sustain conversation, and promote social participation
- Request descriptions of pictured scenes and model simple sentences to be imitated
- Fade the model and ask a question to evoke the previously imitated sentence
- Ask questions about the individual's daily activities
- Request descriptions of pictured actions
- Supply functional words and ask the individual to make sentences with those words
- Tell a story and request a retell
- Tell a story and ask questions about the details
- Show sequenced pictures and ask the individual to construct a story
- Give such cues as "say more" or "elaborate on that" to prompt expanded expressions
- Ask for descriptions of familiar tasks (e.g., making an omelet or a favorite sandwich, planting flowers, mowing the lawn, changing flat tires or light bulbs)
- Target more naturalistic conversation (e.g., ask about what the person did that day, what activities are planned for the weekend)
- Have family members engage the individual in conversation in and out of the clinic
- Reinforce the individual for correct or functionally appropriate productions
- Correct incorrect responses ("That is not correct," "That was a wrong word," "How about this _____?")
- Repeat successful trials several times to strengthen the responses

Treatment of Aphasia: Functional Communication and Maintenance. Functional communication skills are effective in social situations and are final targets of aphasia treatment; use the following basic treatment procedure and select one of several special programs described later under Treatment of Aphasia: Specific Techniques or Programs; integrate compatible elements of different procedures.

- **Target effective communication, not necessarily grammatical accuracy**
 - Do not correct grammatical errors if the verbal expressions communicate effectively
 - Accept any mode of communication (words, phrases, gestures)
- **Select socially effective words, phrases, and sentences**
 - Let the client, family members, and health care staff suggest functional targets
 - Include expressions of personal experiences, bodily needs, emotions, and thoughts
 - Design individual-specific treatment programs; shape progressively longer utterances
 - Start with what the individual can say, perhaps a few words or even syllables
 - Add other syllables to create words or words to create phrases
 - Add more words to create sentences
- **Evoke a variety of sentence structures**
 - Noun and verb combinations
 - Active declarative sentences
 - Requests, commands, demands
 - Wh-questions
 - Sentences with adjectives and comparatives
 - Yes/no questions
 - Sentences with prepositions, pronouns, present progressives, and so forth
 - Keep the emphasis on functional (effective) communication
 - Use pictures, modeling, prompting, and such other special stimuli only when necessary
- **Fade out the special stimuli and fade in the naturalistic stimuli**
 - Stop modeling the full word (such as, “Say, *pen*”), but provide only the first letter of the word; then show only the articulatory posture for the initial sound of the word; eventually just ask a question, such as “What is this?” or “Tell me something about it?”

- Withdraw pictures the individual has been describing; ask the individual to describe a personal experience
- Use natural reinforcers (e.g., smile, nod, agreement, positive comments)
- **Move to conversational speech**
 - Engage the individual in meaningful, functional conversation
 - Ask the individual to describe personal or professional experiences, hobbies, family-related events, favorite foods, entertainment, books read, vacations taken, and so forth
 - Narrate a story and ask the individual to retell it
 - Role-play turn taking
 - Reinforce the individual for staying on a topic; extend the duration of Topic Maintenance, described under Social Communication Skills
- **Implement a maintenance program**
 - Teach self-monitoring skills; when errors occur, stop and ask what went wrong; teach the person to pause and correct the errors
 - Implement treatment in naturalistic settings
 - Use natural response consequences (e.g., smile, nod, agreement)
 - Conduct group sessions in which the participants learn to monitor and reinforce each other's verbal or nonverbal expressions and to strengthen social communication; use the following two ways of structuring the group sessions
 - More tightly structure the group format to directly teach the skills or to reinforce the previously taught skills
 - More loosely structure the group format to incidentally and indirectly reinforce skills that the clients exhibit in interactions with each other
 - Teach the group members to reinforce each other in a positive and supportive manner
 - Train health care professionals to support and socially reinforce the communicative behaviors

- **Train the caregivers**

- Target family members as well as professional caregivers
- Train them to evoke and reinforce speech; teach them to prompt, occasionally model, and provide cues that have worked
- Teach them ways of reducing communication demands when necessary (e.g., teach the family members to ask one question at a time)
- Support the individual's strengths (e.g., if the individual is very good with gestures, reinforce them)
- Express emotional support for the individual
- Include the individual in social activities

Treatment of Aphasia: Reading Problems.

Treatment of reading problems may or may not be needed for all individuals. When it is, use the following guidelines:

- Assess whether reading skills are important for the individual
- Consider the level of premorbid reading skills and the current need to read
- Teach first the functional reading skills to persons who have mild or moderate aphasia
- Target comprehension of silently read material rather than oral reading
- Target newspaper- and book-reading skills only when functional and basic reading skills are intact or have been regained
- **Teach comprehension of printed words in the beginning**
 - Have the individual read aloud selected printed words
 - Model and prompt the responses
 - Repeat successful trials for each word
 - Have the individual read the words silently and state their meaning
 - Provide positive reinforcement and verbal correction
- **Construct phrases and sentences with comprehended words**

- Have the client read the words aloud with the help of modeling and prompting
- Ask the client to read the words silently and state their meanings
- Target progressively complex material and assess comprehension at each level of complexity

Treatment of Aphasia: Writing Problems. Use the following guidelines to reinstate writing skills:

- Assess the level of premorbid writing skills and the current need to write
- Target functional writing skills for persons with mild to moderate aphasia
- Consider the preferred hand and whether it is free from neuromuscular disorders
- Consult with the individual, family members, and other caregivers to select words, phrases, and sentences that are important to the individual and are useful in daily living (e.g., names of family members, address and phone numbers, grocery lists, short letters, filling-out forms, writing down appointments)
- **Target correct spelling and grammatical accuracy in writing**
 - Trace printed letters and words
 - Copy letters and words
 - Write letters and words to dictation
 - Copy sentences
 - Spontaneously write sentences
 - Write paragraphs, short letters, lists, and so forth
 - Complete writing homework you assign
 - Train family members to help sustain the writing skills at home

Treatment for Aphasia: Telerehabilitation. Telepractice is useful to (1) monitor a client's status and progress at home and to (2) offer treatment when face-to-face contact is unfeasible; needs efficacy data; there are two main varieties of telepractice intervention; in both the varieties, periodic face-to-face contact should be arranged; regular telephone or online video contact

should help monitor progress, correct mistakes in home practice, and offer counseling and support:

- **Synchronous telepractice**
 - Meet your clients in two-way videoconferencing sessions
 - Select the target behaviors in consultation with the client
 - Provide stimuli, model target responses, and reinforce imitations and correct wrong responses
 - Fade modeling and teach progressively more complex skills as done in face-to-face sessions
- **Asynchronous telepractice**
 - There is no visual contact between the clinician and the client
 - Create treatment videos and have the client download them
 - Let the client practice treatment at home and video-record the sessions
 - Monitor the videorecordings and suggest improvements to the client
 - Modify the treatment sessions as needed

Aphasia: Quality of Life After Treatment. Quality of Life (QoL) is a matter of post-treatment assessment, not a treatment procedure per se; nonetheless, improving the QoL of individuals with aphasia is a treatment goal; improved QoL should be considered in planning a treatment program for persons with aphasia.

- See Quality of Life After Treatment for procedures that applies to all disorders of communication
- Administer ASHA's *Quality of Communication Life Scale (ASH-QCL)* (get it at asha.org)
- Administer *Stroke and Aphasia Quality of Life Scale-39 (SAQOL-39)*; available in Spanish and several other languages; find it online
- Consider other tools reviewed in the cited Babael & Yadegari
- Consider the following in planning treatment for persons with aphasia:

- The most common reason for lower QoL after treatment is inadequately treated functional communication skills; individuals who cannot communicate their basic needs will be dissatisfied with their QoL; stabilize functional communication skills in natural environments before discharge
- Other reasons include depression and chronic fatigue; if you suspect these, refer the person to specialists who can treat them
- Follow up the individual to assess maintenance of communication skills; offer booster treatment when warranted to support better QoL
- See the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for details on QoL assessment

Babael, Z., & Yadegari, F. (2024). Tools for assessing quality of life in people with stroke-induced aphasia: A literature review. *Folia Phoniatrica*, September 24, Online ahead of print, 1-19.

Paul et al. (n.d.). *Quality of Communication Life Scale (ASH-QCL)*. American Speech-Language-Hearing Association.

Cassiante, L. et al. (2021). Telerehabilitation for people with aphasia: A systematic review and meta-analysis. *Journal of Communication Disorders*, 92, 106111.

Elman, R. J. (Ed.). (2006). *Group treatment for neurogenic communication disorders: The expert clinician's approach* (2nd ed.). Plural Publishing.

Hegde, M. N. (2022). *A coursebook on aphasia and other neurogenic language disorders* (5th ed.). Plural Publishing.

Hegde, M. N., & Freed, D. (2020). *Assessment of communication disorders in adults: Resources and protocols* (3rd ed.). Plural Publishing.

Helm-Estabrooks, N. et al. (2014). *Manual of aphasia therapy* (3rd ed.). Pro-Ed.

Treatment of Aphasia: Specific Types. Experimentally validated treatment procedures that are specific to different types of aphasia are limited. In individuals with any type of aphasia, behavioral procedures may be used in teaching target skills. To treat a specific type of aphasia, use the procedures described under the main entry, Aphasia. The following highlights a few unique features of different types of aphasia.

Broca's Aphasia. In treating clients with this type of nonfluent, effortful, agrammatic speech, use the procedures described under [Aphasia](#) and consider the following:

- Increase length of utterances in gradual steps; model progressively longer utterances and ask the individual to imitate; reinforce imitative productions
- Increase complexity of responses in gradual steps
- Reinforce the correct production of grammatical morphemes in sentences and conversational speech
- Treat naming difficulties
- Verbally correct stereotypic utterances
- Reinforce pointing, gestures, drawing, writing, and reading to improve functional communication
- Combine gestures with verbal expressions, as this combination is known to facilitate naming and other verbal expressions
- Incorporate into training compensatory strategies an individual uses (e.g., singing, gesturing, writing key words to effectively communicate) to reinforce their use in and out of the clinic
- Expand and model telegraphic productions into more complete utterances
- Reinforce the individual's imitation of more complete utterances
- Select one of the special programs described under [Aphasia](#); [Treatment of Aphasia: Special Programs: the Helm Elicited Language Program for Syntax Stimulation](#), [Promoting Aphasics' Communicative Effectiveness](#), and [Response Elaboration Training](#)

Global Aphasia. To treat this type of severe aphasia with impaired comprehension and expression, use the procedures described under [Aphasia](#) and consider the following:

- **Train the caregivers to:**
 - Be alert to the individual's communicative efforts of any kind, including eye contact, head nodding, facial expressions, postures, simple gestures, and so forth

- Eliminate distractions while talking to the individual
- Face the individual while talking; draw the individual's attention before saying something, and speak slowly
- Pause at syntactic junctures and between stimulus presentations
- Use short, simple sentences, use appropriate stress and intonation, and pause between sentences
- Supplement oral communication with nonverbal cues
- Allow extra time for the person to respond
- Be unhurried
- State verbally the guessed response from the individual to get confirmed
- Verbally state that you do not understand (e.g., "I am sorry, I do not understand")
- **Implement a more formal treatment program**
 - Establish realistic goals for the individual
 - Select basic, simple, functional words and phrases for initial treatment
 - Teach *yes/no* responses to basic questions
 - Teach a few unequivocal gestures, simple line drawing, and words and phrases that express basic needs
 - Accept verbal, gestural, or signed responses
 - Provide both auditory and visual stimulation; combine verbal responses with gestures
 - Provide multiple stimuli (modeling, pictures, written stimuli, objects, gestures)
 - Model target responses and require immediate imitation; fade modeling
 - Shape the response to achieve more complex forms if found appropriate
 - Provide manual guidance in shaping gestures, head nodding, and object pointing
 - Give prompt, natural, and social reinforcement
 - Teach responses to simple questions
 - Teach simple requests and descriptions

- Move to basic conversational skills training if judged appropriate
- Teach an organized gestural system and consider techniques described under [Augmentative Communication](#)
- Consider one of the special programs ([Aphasia; Treatment of Aphasia: Special Programs](#), including [Visual Action Therapy](#))
- Counsel the family about the effects of stroke, the communication problems and prospects of treatment, home strategies to enhance communication, and so forth

Transcortical Motor Aphasia. To treat this type of nonfluent, agrammatic, paraphasic, and telegraphic speech with intact repetition, use the procedures described under [Aphasia](#) and consider the following:

- Use imitation and naming to improve speech
- Select pictures as stimuli
- Prompt nouns and verbs that the pictures suggest
- Model sentences that include nouns and verbs the person produces
- Ask the individual to form longer sentences with other words
- Reinforce all attempts in the right direction

Transcortical Sensory Aphasia. Treat this type of fluent aphasia as you would [Wernicke's Aphasia](#), and address the following special features:

- Target naming skills; see [Aphasia](#), main entry
- Reduce echolalia; use such procedures as nonexclusion time-out and extinction; do not pay attention or reinforce them otherwise
- Reduce neologism and paraphasia with extinction and by teaching specific words in conversational speech
- Improve listening skills and reduce the rush of speech by giving prompts to stop or slow down

Wernicke's Aphasia. To treat this type of fluent aphasia with good or even excessively fluent but meaningless

speech, use the procedures described under [Aphasia](#) and consider the following:

- Structure the treatment sessions and reduce distracting stimuli
- Ask the individual to listen
- Use gestures and manual guidance to stop the individual who rambles (touch your lips with your index finger to suggest “be quiet,” give a gesture to stop talking)
- Ask yes/no questions and accept only such answers, not elaborate utterances
- Expand utterances gradually
- Train the individual to listen carefully instead of rushing to speak
- Reduce the rate of speech directed to the individual to improve comprehension
- Train the individual to self-monitor and reduce the rate of speech
- Treat auditory comprehension deficits as described under [Treatment of Aphasia; Auditory Comprehension](#)
- Use one of the special programs described under [Aphasia; Treatment of Aphasia: Special Techniques or Programs](#), including [Treatment for Wernicke’s Aphasia \(TWA\)](#)

Treatment of Aphasia: Specific Techniques or Programs

Communicative Drawing Program. Teaches drawing to persons with aphasia with limited oral language skills to express their basic needs; may be considered an augmentative and alternative form of communication, but some individuals who learned to draw pictures of objects with their preferred or nonpreferred hand also began to verbally name them; consult the cited source for details; essentially, the program consists of multiple steps in which the client is trained to:

- Classify objects into superordinate categories (e.g., placing the picture of a *hammer* into the *tool* category)

- Color selected line drawings with their appropriate color (e.g., red for watermelon)
- Outline pictures of objects on paper without touching the lines of the picture
- Copy geometric shapes without touching the exemplars
- Complete drawings with missing elements, draw objects from memory, draw objects when a printed word is shown (e.g., present the printed word *eyeglasses* and ask the individual to draw one), draw objects in a superordinate category when presented a printed word (e.g., present a printed word *vegetables* and ask the individual to draw a vegetable), and draw as many types of animals and modes of transportation as possible on blank paper
- Draw one-, two-, and three-paneled cartooned stories first by showing such cartoons for study and then asking the individual to draw each from memory

Helm-Estabrooks, N. et al. (2014). *Manual of aphasia therapy* (3rd ed.). Pro-Ed.

Constraint-Induced Language Therapy. This approach limits (constrains) to verbal responses only and removes reinforcement for nonverbal responses; only verbal responses, not gestures, drawings, and any other forms of communication are reinforced; an intensive group treatment program, with up to 4 hours of daily treatment for 2 to 3 weeks; a more comprehensive enhanced protocol (CAIT II) targets speech repetition, activities of daily living phrase repetition, language card game, picture description, role-play with caregivers, and activities to promote generalization; another modified constraint approach, called the *Intensive Language Action Therapy*, accepts and reinforces nonverbal behaviors that evoke verbal behaviors; to implement the basic program:

- Have two to three clients play a language card game
- Distribute to the participants duplicate (paired) picture cards showing action, colors, number, phone-

mically similar words, and nouns of both high and low frequency

- Raise cardboard barriers between the participants to make the cards invisible to each other
- Have a client request a card (e.g., “Mr. Jones, do you have a card with a blue mug?”) and ask the client who has a duplicate of that to give it to the requester (saying, “Yes, Mrs. Smith, I do have it”)
- Have the client who doesn’t have the card respond verbally (“No, Mrs. Smith, I don’t have it”)
- Do not allow any type of nonverbal responses or clues
- Prompt and shape more complex, more polite, and expanded utterances

Johnson, M. L. et al. (2014). An enhanced protocol for constraint-induced aphasia therapy II: A case series. *American Journal of Speech-Language Pathology*, 23, 60–72.

Kurland, J. et al. (2016). Intensive language action therapy in chronic aphasia: A randomized clinical trial examining guidance by constraint. *American Journal of Speech-Language Pathology*, 25, S798–S812.

Melodic Intonation Therapy (MIT). An intervention program for alert and motivated individuals with severe nonfluent aphasia with good auditory comprehension; uses musical intonation, continuous voicing, and rhythmic tapping to teach verbal expression; appropriate for individuals with Broca’s aphasia who produce some words accurately in a familiar song they sing; contraindicated for individuals with Wernicke’s, transcortical motor or sensory, and global aphasia; the program’s scoring procedure is not summarized here; the clinician should consult the cited source to use this method.

General Procedures of MIT

- Select a variety of high-probability words (minimally two syllables), phrases, and sentences
- Use pictures or environmental cues for each target utterance
- Intone each word, phrase, or sentence slowly and with constant voicing; intoning involves singing

the words, phrases, and sentences with melodic patterns

- Maintain pitch and stress variations of normal speech
- Hold the individual's left hand with your right hand, and raise it to tap on the table top once for each intoned syllable
- Signal with your left hand when to listen and when to intone
- Move to an earlier step when the individual fails at any step
- **Level I**
 - Humming: Show a picture, hum the target item, and tap; no response is required
 - Unison singing: Intone in unison with the individual and tap
 - Unison with fading: Intone, tap, and fade half-way through the phrase
 - Immediate repetition: Ask the individual to listen to you as you intone the phrase and tap; ask for imitation
 - Response to a probe question: Following a correct imitation, intone a probe question (e.g., "What did you say?")
- **Level II**
 - Introduction of item: Intone the phrase twice and tap; no response is required
 - Unison with fading: Intone, tap, and fade half-way through the phrase
 - Delayed repetition: Intone and tap, and after 6 seconds of delay, let the individual tap with assistance; ask the individual to intone without help
 - Response to a probe question: Six seconds following the individual's response, intone the probe question; do not hand tap; let the individual intone the phrase
- **Level III**
 - Delayed repetition: Tap and intone and let the individual intone the phrase after 6 seconds and give tapping assistance

- Introducing *sprechgesang* (speech song) with fading: Present the target phrase twice slowly, without singing, much like in choral speaking, but with exaggerated rhythm and stress; tap the hand but require no responses
- *Sprechgesang* with fading: Begin the target phrase intoning along with the individual, but fade your voice to let the individual continue alone and complete the phrase
- Delayed spoken repetition: Present the phrase in normal prosody, without hand tapping, and let the individual imitate after 6 seconds in normal prosody
- Response to a probe question: Ask a probe question with normal prosody after a 6-second delay; let the individual respond with normal prosody

Helm-Estabrooks, N. et al. (2014). *Manual of aphasia therapy* (3rd ed.). Pro-Ed.

Promoting Aphasics' Communicative Effectiveness (PACE).

A program designed to promote face-to-face conversation; emphasizes exchange of new information, functional communication with turn taking, communication in any modality, and natural feedback.

- Use a large number of stimulus cards that contain pictured objects, actions, and stories; stack the cards face down on the table
- Take turns drawing cards from the stack; communicate information about the stimulus
- Encourage any mode of expression (words, gestures, drawings, writing, pointing, or a combination of these)
- Add new stimulus cards to promote the exchange of new information
- Provide natural consequences (e.g., What did you say? Do you mean_____? I am not sure . . .)
- Acknowledge the individual's message while suggesting the correct word or words (e.g., "I understand. You mean *book*, right?")

- Make variations and adaptations
- Exchange the roles of speaker and listener with the individual

Davis, G. A. (2005). PACE revisited. *Aphasiology*, 19(1), 21–38.

Response Elaboration Training. A program designed to expand utterances of individuals with aphasia; emphasizes shaping and chaining utterances the client initiates; allows a wide variety of responses as against a predetermined correct response; may be used in group therapy; may be useful with clients who have both aphasia and apraxia.

- Select line drawings to stimulate speech
- Show a stimulus card and evoke an initial response, any response (e.g., the individual may say “Man . . . sweeping” to a line drawing of a man with a broom)
- Reinforce the individual; also, shape and model the individual’s response (e.g., say, “Great. The man is sweeping”)
- Ask a Wh-question to evoke an elaboration of the initial utterance (e.g., ask, “Why is he sweeping?”)
- Reinforce the individual’s elaboration and shape and model the initial response combined with the subsequent elaboration (e.g., if the response is “wife . . . mad,” you may say, “Way to go! The man is sweeping the floor because his wife is mad”)
- Model the longer response a second time and ask the individual, “Try and say the whole thing after me. Say . . .”
- Ask the individual to imitate after a delay if successful at the previous step
- Continue until the individual fails to elaborate any more
- Introduce another picture for a similar sequence or initiate a different initial response for the same picture

Wambaugh, J. L. et al. (2012). Modified response elaboration training: A systematic extension with replications. *Aphasiology*, 26(12), 1407–1439.

Script Training. The method uses scripts the client and the clinician collaboratively write to promote conversational skills; scripts may be monologues or dialogues; in using this approach:

- Ask the client to suggest personally meaningful topics for monologues or dialogues (e.g., personal stories or experiences); suggest topics that might be of interest to the person
- Help the client generate a script on the selected topic
- Read the script aloud with the person; then have the client read it aloud and read it at home
- Have the client practice the script with conversational partners

Cherney, L. R. (2012). Aphasia treatment intensity, dose parameters, and script training. *International Journal of Speech-Language Pathology*, 14(5), 424–431.

Semantic Feature Analysis (SFA). A method to treat naming problems; in this evidence-based method, words that share certain semantic features are grouped (e.g., semantic features of objects such as *balls* share features like *round, red, small, big, rough, smooth*); they may be acted upon in various ways (e.g., *kick, bounce, throw*); such features are used to evoke target names; the procedure is more detailed than what is outlined here with a single, specific stimulus item; for details, consult the cited sources:

- **To teach a naming response such as “fork”:**
 - Place a fork on the table; above the fork, on three cards, write a single word on each (Group, Description, Function) and place them above the fork; below the target word, place three cards on which single words that suggest additional features may be placed (e.g., Context, Special Features, Other/Personal)
 - Ask a question (e.g., “What is this?”); if no response, then
 - Ask additional questions in succession, each referring to a semantic feature until the client names the item (e.g., “What category (group) does it

belong to?” “Where can you find it?” “What is it used for?”)

- Reinforce all correct responses with verbal praise

Boyle, M., & Coelho, C. (1995). Application of semantic feature analysis as a treatment for aphasic dysnomia. *American Journal of Speech-Language Pathology*, 4(4), 94–98.

Efstratiadou, E. A. et al. (2018). A systematic review of semantic feature analysis therapy studies for aphasia. *Journal of Speech, Language, and Hearing Research*, 61(5), 1261–1278.

Sentence Production Program for Aphasia. Based on the earlier *Language Program for Syntax Stimulation*, this revised program is designed to increase the production of varied sentence types in individuals with aphasia; especially appropriate for clients with agrammatic speech (limited verbal expressions); the method helps teach the following eight sentence types, each with 15 target sentences:

1. Imperative Intransitive (“Lie down”)
 2. Imperative Transitive (“Wash the dishes”)
 3. Wh-interrogative (“What” and “Who” questions)
 4. Wh-interrogative (“Where” and “When” questions)
 5. Declarative Transitive (“She cleans teeth”)
 6. Declarative Intransitive (“She skates”)
 7. Comparative (“They’re funnier”)
 8. Yes/No Questions (“Did you buy the paper?”)
- Level A targets sentences in an imitative form within a story context
 - Level B targets correct answers to questions about a story the client is told
 - Consult the program manual for procedural details
- Helm-Estabrooks, N. et al. (2014). *Sentence production program for aphasia* (3rd ed.). Pro-Ed.

Social Approaches to Aphasia Rehabilitation. These approaches emphasize social communication, social integration of the individual with aphasia, and training family and friends to become conversational partners of the individual; they deemphasize grammatical

correctness or nonfunctional speech; traditional language treatment and functional communication skill training are still important; group format is typical; treatment methods of social approaches need more methodological clarity and precision; the approach is social, but the treatment and training procedures are behavioral.

- Target conversational skills in social situations as the ultimate goal of aphasia treatment
- Let the caregivers (family members and other professionals) take active roles; consider the treatment targets the individual and the family members prefer
- Have the individual and the caregivers evaluate improvement in social communication skills and satisfaction with the treatment
- Address social isolation of people with aphasia; in addition to language treatment, target social integration; get the individual involved in community activities, to volunteer in hospitals, participate in social clubs and organizations, speak at support groups, talk to students in academic classes on aphasia, and so forth
- Help rejuvenate such premorbid social activities as playing cards with friends, participating in cook-outs and picnics, watching sports or games, and so forth
- Help create new social skills such as painting with friends or playing musical instruments, going to movies with spouses or friends, tending a garden, and so forth
- Try to move treatment sessions out of the clinic room and into the more natural settings where the skills are prompted or reinforced in subtle ways (e.g., conduct an informal conversational skill training session at the individual's home or in a restaurant)
- Train persons who regularly interact with the client in effective social communication
- Train health care workers and volunteers in nursing homes to get your client involved in social activities and to support any effective form of communication

- Train the caregivers in adaptive strategies, as the deficits may not be completely eliminated even with effective treatment (e.g., talking slowly to the individual, accepting gestures from the individual, minimizing demands while not ignoring the individual)

Elman, R. J. (2007). *Group treatment of neurogenic communication disorders: The expert clinician's approach*. Butterworth-Heinemann.

Simmons-Mackie, N. et al. (Eds.). (2013). *Supporting communication for adults with acute and chronic aphasia*. Paul H. Brookes.

Treatment for Wernicke's Aphasia (TWA). A method to treat auditory comprehension and verbal expression problems; appropriate for individuals with severe Wernicke's aphasia who can read and understand single picturable words; reinforced responses to written stimuli help strengthen verbal comprehension:

- Select a corpus of words printed in lowercase that the individual can read aloud and point to pictured stimuli
- Present one target printed word and five foils and ask the individual to select a picture that goes with the word
- Ask the individual to read the word aloud
- Turn over the card so the printed word is invisible, say the target word, and ask the individual to repeat after you
- Say the target word and ask the individual to select the correct picture from among an array of six pictures
- Introduce new words as performance improves
- Reevaluate the procedure if new words cannot be introduced by about the fifth session; select another procedure
- Chart correct and incorrect responses on a recording sheet

Helm-Estabrooks, N. et al. (2014). *Manual of aphasia therapy* (3rd ed.). Pro-Ed.

Visual Action Therapy (VAT). A nonvocal, visual/gestural communication approach for individuals with

limited oral or written language skills due to global or mixed nonfluent aphasia, associated with some degree of limb and oral apraxia; should be able to spontaneously produce a few everyday gestures (e.g., pointing, waving goodbye); neither the clinician nor the individual talk during treatment; an individual who cannot match an object with the tracing of that object is not a good candidate for VAT; more treatment efficacy data are needed; because of multiple levels and training steps, consult the cited source to use this procedure.

- Select 15 each of real objects, shaded line drawings of these objects, and action pictures involving the objects
- Select some contextual props (e.g., a screw in a block of wood to use a screwdriver)
- **Level I.** Use real objects, pictures of objects, and action pictures (e.g., someone sawing a piece of wood, a hand holding a cell phone); steps include placing objects on pictures, placing pictures on objects, pointing to objects, pointing to pictures, object use training, action picture demonstration, following action picture commands, pantomimed gesture demonstration and recognition, representation of hidden object demonstration, and production of gestures for hidden objects
- **Level II.** Implement action picture demonstration, following action picture commands, pantomimed gesture demonstration and recognition, pantomimed gesture production, representation of hidden action pictures demonstration, and production of gestures for action pictures
- **Level III.** Implement object picture demonstration, following picture commands, pantomimed gesture demonstration and recognition, pantomimed gesture production, pantomimed hidden object pictures demonstration, and production of gestures for hidden object pictures

Helm-Estabrooks, N., & Albert, M. L. (2014). *Manual of aphasia therapy* (2nd ed.). Pro-Ed.

Aphonia. Loss of voice; see [Voice Disorders](#) and [Psychiatric Problems Associated With Communication Disorders](#) for functional aphonia.

Apraxia of Speech (AOS) in Adults. A speech disorder with documented neuropathology in the left cerebral hemisphere with no muscle weakness, paralysis, or muscle fatigue; impaired speech movement patterns are presumed to be a consequence of central speech planning and programming deficits; treated primarily as a speech articulation and prosodic disorder; see also [Childhood Apraxia of Speech](#); may be associated with [Aphasia](#) and [Dysarthria](#); consult the cited resources for details on AOS.

Treatment of Apraxia of Speech: General Considerations

- Make a thorough assessment of apraxia of speech; for details, see the cited sources and [Hegde's PocketGuide to Assessment in Speech Language Pathology](#)
- Behavioral treatment of AOS is highly structured, intensive, and targets speech production on repetitive trials
- Treatment efficacy research on AOS lags behind descriptive research; many treatment procedures have not been experimentally evaluated for their effectiveness; nonetheless, there is clinical evidence that treatment produces improvement, although effectiveness cannot be claimed from such evidence
- Articulatory accuracy is a target of treatment, but effective and functional communication may be more practical and useful; all communication styles and skills, including gestures and facial expressions, may be reinforced to reduce the impairments
- Carefully selected prosthetic devices (e.g., palatal left prosthesis, nasopharyngeal obturator to reduce hypernasality) may enhance functional communication
- Individuals with severe apraxia of speech with limited oral communication may benefit from suitable [Augmentative and Alternative Communication](#) techniques

- Oral sensation and form recognition exercise are unlikely to result in improved speech production
- Some individual may not prefer certain techniques (e.g., melodic speech or clinician's manipulation of articulators)
- Treatment may be deferred for individuals with a coexisting aphasia, for those scheduled for medical or surgical treatment (e.g., removal of a brain tumor), and for those who are depressed or seriously cognitively impaired until their problems are expertly addressed; family members should be counseled and the client reassessed for speech treatment candidacy at a later date
- Management may be started as early as possible, but not when the individual is just coming out of an acute phase of the disease (e.g., soon after a stroke), that makes it difficult to concentrate on treatment tasks
- All treatment decisions, including the target behaviors, should be decided in consultation with the individual and significant caregivers
- Frequent treatment sessions produce better results, but the frequency depends on the individual's physical condition and concentration during treatment
- Moving from easy to difficult treatment tasks may be helpful
- Sessions may be ended with success; if errors persist on a certain treatment target, a few practice trials on an earlier learned target might be helpful
- Articulatory accuracy may be targeted in the context of effective and functional communication

Treatment of Apraxia of Speech: Common Procedures

Counsel the Family Members, the Individual, and Other Caregiver

- Give a reasonable statement of prognosis
- Discuss the severity of accompanying aphasia and need for additional treatment
- Ask the family members and health care workers to speak slowly, use shorter sentences, reduce back-

ground noise, talk only when the individual is attentive, and use Total Communication

- Teach family members and health care staff to give various prompts (e.g., the first letter or the first syllable of the word), use the Cloze Procedure, paraphrase what the individual may have said to evoke *yes* or *no* responses, and so forth.
- Finalize the treatment targets only after consulting with the individual, family members, and professional caregivers
- Ask the family to observe treatment to learn to reinforce speech or other types of communication attempts at home
- Inform the individual what you have trained the family members to do
- Educate the other members of the team about the individual's communication problems, strengths, and the treatment program
- Let the client and the family suggest treatment targets; incorporate their targets in treating all individuals with apraxia and at all levels of severity

Establish Reliable Baselines

- Baselines are quantitative, reliable, and detailed measures of communication skills, measured prior to treatment to document improvement under treatment
- Baselines may include a rating of speech intelligibility, although rating scales are less objective than frequency (e.g., frequency and percentage of individual speech sound errors)

Use the Behavioral Treatment Procedures

- Give instructions on how to produce the target skills and demonstrate the actions
- Frequently model the target skills
- Require immediate imitation of the modeled productions
- Reinforce correct responses verbally and naturally; explain why the response is correct

- Correct the errors verbally; experiment with instrumental or biofeedback to judge their effectiveness
- Use Discrete Trials to repeatedly evoke the same target responses
- Try to improve physiologic support for speech production before starting direct treatment for speech (e.g., consider modifying poor posture or improving respiratory support for speech), but do not waste too much time trying to improve respiratory support that is unrelated or unnecessary for speech
- Reduce the speech rate initially to improve articulatory accuracy and to convince the individual of the desirable effect of slower rate on intelligibility; increase the speech rate as articulatory accuracy improves and stabilizes
- Use Phonetic Placement Method and Phonetic Derivation with almost all clients
- Use a variety of sounds and sound combinations to promote generalization and maintenance
- Have the individual practice sound productions with meaningful material; avoid nonsense syllables or words that are not functional to the individual
- Target speech tasks during treatment, not unrelated nonspeech tasks (e.g., nonspeech oral-motor tasks)
- Use Contrastive Stress Drills to improve prosodic features in all clients
- Use the Key Word technique; use words in which an otherwise misarticulated sound is correctly produced to teach the correct production of the sound in other words
- Use cueing to fade modeling (e.g., show a tongue tip placement or bilabial lip posture for a phoneme production without oral modeling)
- Use automatic speech tasks (e.g., counting, reciting the days of the week) to initially evoke speech and quickly move to more spontaneous word, phrase, and sentence productions
- Push on abdomen to achieve vocal fold closure and phonation for the speechless individual

- Use the shaping method to promote natural prosody; a gradual increase in speech rate while still maintaining intelligibility, varied stress patterns, and variations in intonation will help achieve a more natural-sounding speech
- Emphasize total communication (combined use of verbal expressions, gestures, writing, augmentative devices)
- Teach Self-Control (Self-Monitoring) skills to all clients through the treatment and especially in the final stages
- Assess and treat a coexisting Aphasia
- Use the techniques of teaching Speech Sound Disorders as found appropriate
- Use a specific technique or program or elements of it described under Treatment of Apraxia of Speech: Specific Techniques or Programs

Sequence the Treatment Targets

- Automatic speech before spontaneous speech
- Frequently occurring sounds before less frequently occurring sounds
- Stimulable sounds before nonstimulable sounds
- Sounds in word-initial positions before those in other positions
- Visible before nonvisible sounds
- Oral-nasal distinctions before voicing distinctions
- Voicing distinctions before manner distinctions
- Manner distinctions before place distinctions
- Bilabial and lingua-alveolar sounds before others
- Singletons before clusters
- High-frequency words before low-frequency words
- Single-syllable words before multisyllable words
- Single words before phrases or sentences
- Self-monitoring skills in advanced treatment stage

Treatment of Mild Apraxia of Speech. Persons with mild AOS (a) are good candidates for massed-trial treatment of articulatory accuracy, (b) may learn to communicate well in social and occupational situations, and (c) are unlikely to exhibit severe forms of aphasia

- Target articulatory accuracy, good prosody, and appropriate rate of speech
- Teach sound productions in meaningful and personally relevant words
- Model sound productions in words initially
- Fade modeling in gradual steps
- Move to modeled and evoked productions of phrases and sentences in graduated steps
- Use visible and simple utterances in the beginning
- Give visual feedback of the movement of articulators (e.g., “See how I produce the sound and do the same”)
- Encourage the clients to listen to their own sound productions to judge adequacy
- Require immediate imitation of modeled productions because it is easier than delayed imitation
- Reduce the individual’s rate of speech
- Extend treatment to utterances that are more complex and sound productions that are less visible in carefully graded steps
- Ask open-ended questions
- Have the individual ask questions to practice normal rhythm
- Encourage the individual to read aloud and self-correct mistakes
- Encourage self-correction
- Increase speed of response (reduced reaction time)

Treatment of Moderate Apraxia of Speech.

Moderate AOS may be associated with (a) hemiparesis or hemiplegia, (b) mild or moderate aphasia, and (c) non-verbal apraxia (e.g., limb apraxia, buccofacial apraxia)

- Encourage the individual to make decisions about the future (returning to work, changing assignments at work, driving, and so forth)
- **Model selected speech sound contrasts**
 - Provide individuals with auditory and visual cues
 - Ask the individual to imitate your model
 - Place a single target in varied linguistic contexts (e.g., for the target /t/, a typical list of stimuli might be *tea*, *tie*, *toe*, and *two*)

- Ask the individual to contrast the target with other sounds
- Replace single-syllable words with polysyllabic ones
- Construct phrases and sentences out of practiced words for more practice
- Make speech sound contrasts harder
- Use slow rate initially with difficult targets
- Use varying rhythm and stress (e.g., begin with equal and even stress and progress toward normal)
- Use a question-and-answer dialogue

Use Reading in Treatment

- Ask the individual to read aloud
- Fade the printed stimuli by having the individual (1) look at the text and talk about it and (2) look at the text and wait before talking about it
- Teach the individual to self-monitor rate, rhythm, stress, and errors

Treatment of Severe Apraxia of Speech. Severe AOS may be associated with (a) extremely limited spontaneous speech, (b) limb and buccofacial apraxia, (c) moderate aphasia, and (d) hemiparesis or hemiplegia.

- Begin direct treatment with modeling-imitation-verbal praise sequence
- Use the Phonetic Placement Method to help improve articulatory accuracy
- **Teach manner-place-voicing distinctions**
 - Use manner distinctions (especially plosive and fricative)
 - Use simultaneous manner and place distinctions
 - Make voicing distinctions (evoke any kind of sound, including humming or grunting, and then shape it)
 - Make oral-nasal distinctions
- Combine modeling, phonetic placement, and shaping (phonetic derivation) techniques
- Use rhythm to evoke speech sounds, syllables, and words; use aspects of Melodic Intonation Therapy described under Aphasia; Treatment of Aphasia: Specific Techniques or Programs

- Consider teaching augmentative or alternative communication skills to persons with the most severe apraxia

Apraxia of Speech: Quality of Life After Treatment. Although a matter of assessment, not a treatment procedure per se, build treatment programs to enhance the Quality of Life (QoL).

- Administer ASHA's *Quality of Communication Life Scale (ASH-QCL)*; get it from asha.org
- Give adequate and effective communication treatment; establish functional communication skills
- Assess generalization and maintenance as these two are the key factors in enhancing QoL of individuals with treated AOS
- Follow up the individual to assess maintenance of communication skills
- Offer booster treatment to remediate residual deficits to support better QoL
- See [Quality of Life After Treatment](#) for additional suggestions that are relevant for all disorders of communication

Ballard, K. J. et al. (2015). Treatment for acquired apraxia of speech: A systematic review of intervention research between 2004 and 2012. *American Journal of Speech-Language Pathology*, 24(2), 316–337.

Wambaugh, J. L. et al. (2013). Treatment for acquired apraxia of speech: Examination of treatment intensity and practice schedule. *American Journal of Speech-Language Pathology*, 22, 84–102.

Treatment of Apraxia of Speech: Specific Techniques or Programs

Combined Aphasia and Apraxia of Speech Treatment (CAAST). A method to simultaneously treat language production and speech sound production in the same individuals; combines [Response Elaboration Training](#) to enhance language (described under [Aphasia](#)) and [Sound Production Treatment](#) for articulation, described in this section. To implement:

- Evoke narrative discourse with pictorial stimuli for language; evoke short sentences for measuring articulatory accuracy

- Present an action picture and ask questions (e.g., “Tell me anything about the picture; What does it remind you of? What’s happening?”)
- If no response, prompt (e.g., “You could say something like man spills”)
- Reinforce any correct or approximate response (e.g., “Spill”)
- Ask another question (e.g., “What does the man spill?”)
- Reinforce the response and combine the two responses (e.g., “Good. Spill milk”)
- Model the combined response (e.g., “Repeat after me . . . spill milk”)
- Verbally correct the inaccurate sound production (e.g., if *spill* is produced as *pill*, say, “Not quite correct; try again, spill milk”)
- Ask the person to repeat the correct production (e.g., spill milk) three more times
- Remove the picture and after 5-second delay, and say, “Say it again.”
- Verbally correct the wrong response and ask the person to watch and say along with you (e.g., spill milk)

Wambaugh, J. L. et al. (2014). Combined aphasia and apraxia of speech treatment (CAAST): Effects of a novel therapy. *Journal of Speech, Language, and Hearing Research*, 57, 2191–2207.

Eight-Step Continuum Treatment for AOS. A procedure for shaping word productions with the use of cues (prompts) that are subsequently faded out.

- **Select the sounds and words for training**
 - Select words for initial training that contain easier sounds in them (vowels, nasals, and stops as against fricatives and affricates)
 - Select words that contain sounds produced with closer articulatory contacts for initial training (e.g., words with only bilabial sounds) and increase the distance between the articulatory contact gradually (e.g., words with bilabial and lingua-alveolars moving on to velar sounds)

- Select short, simple, and frequently used words for initial training and increase the length of words gradually
- **Apply a sequenced treatment**
 - Say “Watch me” and “Listen to me” and then model a target word
 - Ask the individual to say the word along with you; this is unison production
 - Say “Watch me” and “Listen to me” and then say the word; ask the individual to produce the word aloud as you silently mouth the word
 - Say “Watch me” and “Listen to me” and then say the word; ask the individual to produce the word without any cues
 - Say “Watch me” and “Listen to me” and then say the word; ask the individual to produce the word several times without any cues
 - Present the target word printed on a card and ask the individual to produce it
 - Show the printed target word briefly, remove it, and then ask the individual to produce it
 - Ask a question to evoke the target word and let the individual respond with it
 - Arrange role-playing conversational situations in which the individual has opportunities to produce target words; get family members and others involved in this role-playing

Rosenbek, J. C., & Associates. (1973). A treatment for apraxia of speech in adults. *Journal of Speech and Hearing Disorders*, 38, 462–472.

Phonomotor Treatment (PMT) for Apraxia of Speech.

Described as an articulatory kinematic approach, the method may be used to simultaneously treat aphasia and apraxia of speech; the method involves targeting speech sounds in error and involves motor description of target sounds, speech production, modeling, verbal requests, and intensive practice (frequent trials); for all steps and details of a modified version of PMT, see the cited source.

- Select single sounds or a sound combination produced with less than 60% accuracy
- Train the target sounds in nonwords and conduct repeated probes for generalization in nonwords and real words
- Proceed from simpler consonant-vowel (CV; e.g., VC, CV, CVC) to more complex target productions (e.g., VCC, CCV, CCVC, CVCC, CCVCC); form more complex targets by incorporating sounds produced correctly
- In Stage 1, train sounds in isolation; in Stage 2, train more complex productions by combining sounds
- Initially, teach only one target at a time with repeated trials
- Randomize the stimulus presentation as the client's production accuracy improves
- Have the client repeat consecutively a correctly produced target five times
- Use verbal correction in the beginning
- **Use Socratic questioning as feedback**
 - Ask question about a speech sound's auditory, visual, articulatory, and tactile/kinesthetic properties
 - Show a mouth picture of a sound production; model its production and ask the client to imitate it; then ask Socratic questions (e.g., "What do you feel when you make that sound?")
 - Describe how a sound is produced (e.g., /p/), model the sound, and after the client imitates it, ask Socratic questions (e.g., "Did your lips blow apart or stay together?")
 - Have the client produce the target sound; if correct, ask for five repetitions; if incorrect, ask Socratic questions (e.g., "Make the sound where your lips come together and blow apart")
- Reduce feedback as the client's production of target sounds improves
- Proceed to Stage 2 to train sounds in syllables; use the same treatment procedure

Bislick, L. (2020). A phonomotor approach to apraxia of speech treatment. *American Journal of Speech-Language Pathology*, 29(4), 2109–2130.

Prompts for Restructuring Oral Muscular Targets (PROMPTS). This procedure emphasizes the use of tactile-kinesthetic cues in teaching correct production of phonemes; uses manual guidance to position articulators or suggest their appropriate movement; for details, see the cited sources.

- Note that kinesthetic and tactile cues may be effective only in the context of speech production; by themselves, such cues may be ineffective
- Learn the cues that are described in the cited source; note that cues are given for articulatory positions, voicing, extent of jaw opening, syllable timing, manner of articulation, coarticulation, and so forth
- Learn the many contact positions on the face, neck, jaw, and so forth
- Provide cues by touching the individual's face, neck, jaw, and other structures
- Manually guide the articulators to correct positions
- Model target words for the individual to imitate
- Give cues if the imitative production was incorrect
- Manually guide the articulators to correct positions
- Ask the individual to produce the sound as the articulators are being moved to correct positions
- Give other cues to suggest articulatory movements that cannot be manually guided

Bose, A. et al. (2001). Effects of PROMPT therapy on speech motor function in a person with aphasia and apraxia of speech. *Aphasiology*, 15(8), 767–785.

Sound Production Treatment. A treatment procedure that has been supported by evidence, although generalization and maintenance need to be better researched; to use this procedure:

- Select the sounds that are consistently difficult for the individual with AOS
- Develop monosyllabic and multisyllabic words, phrases, and sentences that contain those sounds

- Use the Minimal Pair Contrast method to select stimulus words
- Model the target words, phrases, or sentences
- Ask the client to repeat correct productions five times in succession and proceed to the next stimulus item
- Give feedback for incorrect responses and model a minimal pair word
- Show the letter representing the target sound produced correctly in the minimal pair and ask the participant to produce the word
- Request for five repetitions of the correct production and move to the next item
- Provide articulatory placement cues and integral stimulation (“watch me, listen to me, and say it with me”) if the response is incorrect; if the client fails, move to the next stimulus item
- Consider using Biofeedback via Electropalatography to improve articulatory accuracy

Bailey, D. et al. (2015). Sound Production Treatment: Synthesis and quantification of outcomes. *American Journal of Speech-Language Pathology*, 24, S798–S814.

Wambaugh, J. L., & Mauszycki, S. C. (2010). Sound production treatment: Application with severe apraxia of speech. *Aphasiology*, 24(6–8), 814–825.

Artificial Larynx. Mechanical larynges for individuals with laryngectomy; generates a mechanical sound that is articulated into speech; the same as Electronic Device for Alaryngeal Speech or Pneumatic Device for Alaryngeal Speech.

Assessment. Procedures that include (a) description of an individual’s existing and nonexistent communicative behaviors, background variables, and associated factors to evaluate or diagnose a communicative problem and (b) clinical measurement of a person’s communicative behaviors.

- Obtain case history
- Interview individual, the caregivers, family members, and others concerned

- Conduct an orofacial examination
- Make individual-specific judgments on use of standardized or nonstandardized measures
- Use measures appropriate to the individual and their ethnic, cultural, and linguistic background
- Screen hearing
- Obtain a speech-language sample
- Administer the selected tests
- Analyze results
- Draw conclusions; make a diagnosis; recommend treatment; counsel the individual and the family members
- Consult the cited sources and the *PocketGuide to Assessment in Speech-Language Pathology*

Hegde, M. N., & Freed, D. (2020). *Assessment of communication disorders in adults: Resources and protocols* (3rd ed.). Plural Publishing.

Hegde, M. N., & Pomaville, F. (2021). *Assessment of communication disorders in children: Resources and protocols* (4th ed.). Plural Publishing.

Assistive Listening Devices. See Hearing Assistive Technology Systems.

Ataxic Dysarthria. To treat this type of motor speech disorder resulting from damage to the cerebellum, see Dysarthria: Specific Types under Dysarthria.

Attention Disorders. Intervention for attention deficits may be integrated with communication training for individuals with aphasia, apraxia of speech, dementia, traumatic brain injury, right hemisphere disorder, intellectual disabilities, and autism spectrum disorder; treating attention as an isolated skill in noncommunicative contexts may not be useful; for a review of different approaches, see the cited resource.

Treating Attention Disorders: General Guidelines

- During assessment and early treatment sessions, take note of slow reaction, lack of sustained attention to assessment materials and tasks, frequent shift in

attention from one stimulus to the other, losing track of topics of conversation, irrelevant responses due to inattention to the speaker, and so forth

- Integrate attentional training activities with functional communication training
- Work directly on functional attentional tasks (e.g., paying attention to speech or instructions, environmental cues, treatment stimuli) instead of meaningless and nonfunctional attentional tasks (e.g., crossing out circles intermixed with squares on paper or backward counting) that do not generalize to communication contexts
- Consult with the family and health care staff in identifying attentional deficits of concern and functional treatment activities and goals they prefer
- Prompt the individual to pay attention to spoken speech, conversational topics, and speaking for longer durations in narrative and discourse training
- Ask family members and health care staff to use the strategies that have worked in your formal sessions
- Select treatment stimuli that are relevant to the individual's interests and premorbid hobbies; prepare colorful and attention-getting stimuli for treatment

Attention Disorders: Treatment Strategies

- Begin treatment in a quiet, simple, clutter-free place with minimal stimuli; gradually extend the treatment to more natural settings in which you reinforce attentional behaviors
- Sit in front of the individual in treatment sessions; make sure there is enough lighting on your face and on the treatment stimuli and activities
- Give all treatment instructions in simple, brief, and direct language; repeat, if necessary; break down the instructions into smaller steps; talk slowly; ask questions to assess comprehension of instructions
- Give alerting signals to draw the individual's attention before presenting all treatment stimuli; include the person's name (e.g., "Watch for this now, Mr. Triumph," "Here it comes, Mrs. Robinson," "I am going to

show you a picture now, Ms. Lopez,” “Mr. Rodriguez, look at me now,” “Mr. Woo, please pay close attention”); make sure the individual has established eye contact with you or with the stimulus before you take the next step

- Verbally reinforce the individual for maintaining eye contact, looking at the treatment stimuli, concentrating on treatment tasks, performing accurately on tasks at hand, and paying attention for longer durations; inform the client what it is that you are reinforcing (e.g., “I am glad you are looking at the picture!”; “Good job of maintaining eye contact with me”)
- Offer verbal correction; stop the individual when you see their attention wandering; reinstruct, reorient, draw attention, and continue with treatment trials
- Experiment with the loudness of your voice; some individuals may need a louder and more intoned voice; others may react better to soft voice and speech
- Limit choices offered in treatment tasks (e.g., put only a few stimulus items in front of the individual)
- Touch the individual to draw attention; use manual guidance to orient the individual’s face toward the stimuli (e.g., for correct pointing, guide the individual’s hand toward the stimulus picture placed in an array)
- Give necessary intertrial interval time; do not rush stimulus presentations, but do not wait too long between stimulus presentations, as this would give room for competing responses
- Judge the individual’s physical endurance; hold brief sessions if the individual fatigues easily; take short breaks during treatment; hold sessions at times when the individual is well rested
- Shape attending behaviors; increase the duration of required attention gradually (e.g., initially set a realistic goal of attending to a task for only a few seconds and gradually increase the duration); reinforce for all on-task behaviors
- Teach the individual to make requests that help concentration (e.g., requesting others to turn down the

music; close a door or a window to reduce outside noise; speak louder, softer, or slower)

- Use high-probability behaviors to reinforce attending behaviors (e.g., good attending behavior in treatment may be reinforced by a brief period of rest or a break to have a sip of water)
- Teach the individual to self-assess concentrated work during treatment to promote self-correction; reinforce realistic self-assessments
- Measure duration of attending behaviors and the frequency of inattentive behaviors to document changes and improvements; give informational feedback to the individual (e.g., “Yesterday you could concentrate for only 2 minutes; today you did for 3 minutes”)
- Teach the individual to self-cue (e.g., by repeating instructions and reading a list of steps to be taken in completing a task before beginning the task)
- Teach family members to prompt and reinforce attending behaviors, draw attention before speaking, repeat instructions, minimize distractions, and so forth

Peach, R. K. et al. (2019). Clinical outcome following language-specific attention treatment versus direct attention training for aphasia: A comparative effectiveness study. *American Journal of Speech, Language, and Hearing Research*, 62, 2785–2811.

Auditory Discrimination Training. Teaching individuals to distinguish clinician-produced correct and incorrect speech sound production; based on a questionable assumption that individuals who misarticulate speech sounds do not hear differences between different speech sounds and auditory discrimination training is a precursor to speech sound production training; evidence suggests that production training will induce discrimination but discrimination training may not improve production; if needed, discrimination may be trained after production training; the same as Perceptual Training; to implement:

- Teach production of misarticulated sounds first
- Train discrimination if the client still gives confused responses to words in minimal pairs, an unlikely event
- Describe the target sound and how it is produced

- Produce words and phrases that contain the target sound as well as nontarget sounds
- Ask the individual to respond in some way to the production of the target sound (by raising the right hand or by pointing to the drawing of a smiling face)
- Produce a word with the correct sound and a word with a sound the individual substitutes for a correct sound (e.g., *radio* and *wadio*)
- Ask the individual to respond to the correct production (/r/ in *radio*) in one manner (pointing to the printed letter r) and to the incorrect production (/w/ in *wadio*) in a different manner (pointing to the drawing of a frowning face)
- Produce many words and phrases that are loaded with the target sounds to provide intense auditory stimulation
- Move on to production training, as there is little or no evidence that auditory discrimination training will result in correct production of speech sounds

Augmentative and Alternative Communication (AAC). Augmentative communication enhances and expands limited oral means of communication by multiple methods; alternative communication replaces oral communication by nonoral modes of expression; the two methods may be integrated into rehabilitation; both include various means, some of which are more technologically oriented than others; appropriate for persons with severe forms of aphasia, apraxia of speech, autism spectrum disorder, cerebral palsy and other neurological disorders, severe hearing loss, dementia, dysarthria, glossectomy, intubation, laryngectomy, intellectual disabilities, tracheostomy, and traumatic brain injury; aided and unaided are the two kinds of AAC systems.

Basic Principles of Selecting an Augmentative Communication Mode or System

- **Use the participation model to assess:**
 - The opportunities and access barriers the individual faces
 - Existing speech as well as nonspeech communication skills

- Strengths and limitations
- Sensory and motor capabilities and limitations
- Spoken language comprehension
- Communicative demands the individual faces
- **Select a mode or system that:**
 - Gives the maximum advantage to the individual
 - Is affordable
 - Is acceptable to the individual and the family members
 - Meets the communicative demands the individual faces
 - Is easier to train the client and the family members in its use

Augmentative Communication—Aided. Methods of communication in which an individual (a) selects or scans messages displayed on a nonmechanical device (e.g., a communication board), (b) displays messages on a mechanical speech-generating device (e.g., a computer monitor, a smartphone, a tablet screen), or (c) uses a dedicated speech synthesizer that “speaks” on the client’s behalf; specialized AAC software programs are available for electronic devices.

Functional Communication Through Pictures, Drawings, and Symbols

- Have the person use regular or miniaturized objects to communicate basic needs (e.g., the individual points to a fork to indicate the need to eat)
- Teach how to express a particular message through a picture (e.g., to suggest tiredness or need to sleep, the individual points to or looks at a picture of a person relaxing or sleeping)
- Train the person to use line drawing to express bodily states (e.g., the person may point to or look at a specific body part on the drawing to suggest pain)
- Train the individual to sequentially touch the letters of the alphabet to spell a word
- Teach the individual to touch numbers printed on a sheet of paper to communicate the quantity of something

- Teach the Picture Exchange Communication System (PECS) to pick up a picture and hand it to a communication partner to express needs or wants
- Train the person to use the Picsyms, which is a set of graphic symbols that represent nouns, verbs, and prepositions
- Teach the person to use Pic Symbols (Pictogram Ideogram Communication), which are white drawings on a black background
- Teach the person to communicate through Rebus (pictures that represent objects or events along with words, grammatical morphemes, or both)
- Teach the individual to add grammatic morphemes to a picture or a word (e.g., adding s to the picture of a book to suggest *books*)

Functional Communication Through Nonelectronic Communication Boards

- Design a board of paper, cardboard, fabric, wood, or cork; prepare a book of symbols and written messages; select a board that all conversational partners can see simultaneously; portable, if necessary; attractive to look at; big enough to contain critical elements of the system; not overwhelmingly big or complex
- Write symbols (alphabets, orthographic messages, pictures, various kinds of symbols) on separate cards that can be mounted on the board
- Teach the individual who cannot point because of motoric limitations to scan the message; offer selections and have the individual indicate *Yes* to the right selection (e.g., You point to the word “food,” a symbol for it, or a picture of a food item; the individual gestures *Yes* or *No*)
- Teach the individual to encode a message by pointing to a number printed on a separate, smaller, portable selection chart; have the messages on a larger communication board numbered; let the individual point to a number on the selection chart; decode the number into the message on the board (e.g., if the

individual points to #5, it may mean “I am hungry” as per the communication board)

- Teach the individual to directly select the message on the board, instead of a number that stands for a message; teach individuals to select by means of pointing and other hand gestures, finger movements, eye gestures, gaze, headpointers, or headsticks
- Combine spoken model of target words or messages with a symbol or display system (e.g., as you touch a picture, a word, a symbol, or a message on a communication board or an electronic communication display system, model the communication target); see Aided AAC Model

Functional Communication Through Drawing and Writing

- Teach the individual to draw symbols, pictures, or simple line drawings to communicate; let the individual use paper, magic slate, or any other convenient surface; drawing gives more flexibility than a fixed communication board for those who can draw
- Teach the individual to communicate by writing (Traditional Orthography); teach conventional writing to nonverbal children who can master it; teach them initially to write simple, functional messages; gradually increase response complexity

Functional Communication Through Electronic Systems

- Select an electronic system that is a good fit for the individual; consider the cost, ease of use, and efficiency of the system
- Select a practical message-accessing mechanism that the individual can easily learn to use and generate signals for the electronic device (such as those that are specially constructed or modified or regular microcomputers with specially designed keyboards); eye gaze or eye-tracking and touch electronic screens may be the easiest to use
- Select a display system to show messages the individual wishes to communicate by touch or switch

activation; these may be electronic screens of a computer, tablet, smartphone, or a dedicated AAC unit

- Select a control electronic unit (a computer with special applications, a dedicated AAC unit, or a hand-held electronic device such as a tablet or smartphone with special software applications)
- Teach the individual to use the device; start with simpler messages; give plenty of practice in using the method of message generation; increase the complexity of messages in gradual steps; train the individual's typical communicative partners
- Combine spoken model of target words or messages with a symbol or display system

Augmentative Communication—Unaided. Methods of communication that use patterned muscle movements (gestures) to enhance oral communication but do not use instruments or external aids; appropriate for all persons with severely limited oral expressions.

- **Teach gestures for Yes or No**
 - Target *Yes* or *No* for individuals with severely impaired motor performance but relatively intact language comprehension
 - Teach the individual to gesture *Yes* or *No* in response to a series of common questions with the carrier phrase “Do you want _____?”
 - Shape a clear gesture that all communication partners can understand
 - Model the gesture if necessary
 - Reinforce consistently discriminated responding (individual always gives the gesture that is meant)
 - Consider the following gestures: head movements (side to side for *No* and up and down for *Yes*); eye movements (looking up for *Yes* and down for *No*); blinking once for *Yes* and twice for *No*; blinking the right eye for *Yes* and the left eye for *No*); hand movements (thumbs up for *Yes* and thumbs down for *No*) and feet movements (right foot movement for *Yes* and left foot movement for *No*)

- **Teach a pattern of eye blinks**
 - Blink three times to say *I am hungry*
 - Blink four times to say *I am thirsty*
 - Blink five times to say *I need to go to the bathroom*
- **Teach pointing to objects needed**
 - Teach finger pointing
 - Teach pointing by directing gaze
- **Teach the left-hand manual alphabet**
 - Consider teaching the left-hand manual alphabet for individuals whose right hand is paralyzed
 - Consider teaching the manual shorthand, which combines gestures with letters from the left-hand manual alphabet (*talking hand system*)
- **Teach pantomime**
 - Teach pantomime along with speech
 - Teach basic needs communication with mimes
 - Use progressively a greater number of mimes
 - Stop miming when the individual regains or improves oral speech
- **Teach American Indian Hand Talk (AMER-IND)**
 - Teach signs that express mands (basic needs, requests)
 - Teach the use of functionally intact hand when the other is paralyzed
 - Teach the expression of concrete ideas before abstract ideas
- **Teach American Sign Language (ASL or AME-SLAN)**
 - Teach the use of signs that express mands (basic needs, requests)
 - Teach the use of signs for concrete ideas before those for abstract ideas

Quality of Life (QoL) of Persons Who Use AAC

- Conduct an Ethnographic Interview to assess the quality of life and satisfaction with AAC
- QoL is better if the device and the method of communication the persons use is suitable to their needs; receive adequate training; enjoy continued support from the caregivers who accept the device and the method of

communication; and have good accessibility in various social, educational, and occupational contexts

- Assess reasons for unimproved QoL in persons who have been trained to use AAC
- Address the reasons in continued intervention and improve the AAC effectiveness for the individuals
- See Quality of Life After Treatment for additional strategies to enhance QoL of individuals who use AAC

Beukelman, D. R., & Light, J. C. (2020). *Augmentative and alternative communication: Supporting children and adults with complex communication needs* (5th ed.). Paul H. Brookes.

Hall, N. et al. (2022). *Fundamentals of AAC: A case-based approach to enhancing communication*. Plural Publishing.

Hegde, M. N., & Pomaville, F. (2021). *Assessment of communication disorders in children: Resources and protocols* (4th ed.). Plural Publishing.

Aural Rehabilitation. An educational and clinical program implemented, for the most part, by audiologists; includes the assessment of hearing loss in adults and children; counseling; selection and fitting of hearing aids and auditory training; use of group amplification systems in educational and communication training sessions; an audiologist may lead a team of specialists, including otologists, special education specialists, psychologists, and speech-language pathologists; speech-language pathologists are mainly involved in treating oral communication disorders in persons with hearing problems; see Hearing Loss for details.

Montano, J. J., & Spitzer, J. B. (2020). *Adult audiologic rehabilitation* (3rd ed.). Plural Publishing.

Autism Spectrum Disorder (ASD). Treatment of ASD requires a team of specialists, including speech-language pathologists, applied behavior analysts, and special educators; this entry is concerned with speech-language services that are typically coordinated with other services; severe behavioral impairments may make it difficult to teach communication skills until those impairments are addressed; Asperger's disorder, childhood disintegrative

disorder, and Rett disorder are no longer separate disorders; the treatment procedure described here is comprehensive and integrative of specific evidence-based approaches (e.g., Milieu Therapy or Discrete Trials); when their effective components are analyzed, many treatment programs described as unique or special contain core behavioral procedures described under this entry.

Communication Treatment of Autism Spectrum Disorders: General Guidelines

- Use experimentally supported behavioral methods of teaching speech and language skills
- Avoid such popular but unsupported procedures as sensory integration therapy, auditory integration training, and facilitated communication; be wary of many “new and revolutionary” methods without research backing periodically advocated and sometimes hailed in public media
- Work with behavior analysts, special educators, and other professionals to integrate communication training with other skill training
- Select a set of target behaviors that all members of an interdisciplinary team will support
- Give a list of your communication targets to other professionals; request them to incorporate them into their treatment (e.g., give the list of basic words you are teaching to other professionals so that they can prompt and reinforce those word productions during their sessions)
- Understand what other professionals are doing; integrate their goals into your work (e.g., if the educator is teaching the child to answer questions appropriately, teach correct responses to different types of questions in language therapy)
- Make language therapy relevant to the child’s curriculum; select language therapy goals in consultation with the regular and special education teachers
- Teach observable and measurable skills so the progress or lack of it can be evident

- Consider the three types of evidence-based procedures: direct teaching, activity-based teaching, and incidental teaching
- Use direct teaching to establish initial communication skills (e.g., words and phrases; grammatic morphemes; articulation of speech sounds); use the massed but briefly separated discrete trials to teach the target skills; use instructions, demonstration, modeling, prompting, shaping, fading, and similar behavioral techniques; positively reinforce correct responses and verbally correct incorrect responses
- Teach more advanced language skills in a play-oriented setup and as you engage the child in conversation (e.g., the production of plural or other grammatic morphemes in sentences)
- Use the Incidental Teaching Method to promote naturalistic language production; reinforce a child's attempt to communicate in naturalistic contexts (e.g., when a child points to *banana* on *top of the refrigerator*, say the word *banana* several times and give it to the child); see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques
- Expect resistance, interfering behaviors, aggressive behaviors, self-stimulation, inattention, stereotypic behaviors—all disrupting your communication treatment; acquire the basic skills in behavior management techniques that help control such problems (e.g., extinction, differential reinforcement of desirable behaviors that replace problem behaviors, prevention of escape, simplifying tasks, teaching requests that will reduce uncooperative behaviors, and so forth; see Direct Methods of Response Reduction, Indirect Methods of Response Reduction, and Differential Reinforcement)
- Keep the child motivated and on-task, hold short sessions, positively reinforce small improvements in behaviors, reinforce quiet sitting for as short a duration as 1 minute, allow frequent and short breaks for the child to play, and demand what the child is capable of doing

- Increase gradually the task complexity, the demand level, and session durations; decrease gradually the frequency and duration of breaks
- Sit in front of the child, with the child's legs between yours; gradually, move the chairs closer to the table and begin working off the tabletop
- Select effective reinforcers to motivate children with ASD; use tokens and have a variety of backup reinforcers available; select activities the child chooses and have them as backup reinforcer for tokens

Teaching Language and Communication Skills

- **Use a variety of treatment stimuli to promote generalization**
 - Use objects whenever possible
 - Use colorful and clear pictures
 - Ask parents to bring the child's toys, books, hobby materials, and so forth to be used in treatment activities
- **Vary the language contexts**
 - Teach grammatic morphemes in several phrases and sentences with varied verbal contexts
 - Teach the same sentence structure (e.g., simple active declarative form) with different words in them
- **Teach multiple exemplars of the same target skill**
 - Use Discrete Trials
 - Select at least 20 exemplars for each language target (e.g., 20 phrases or sentences that contain the present progressive *ing* or a regular plural morpheme)
 - Teach multiple exemplars
 - Probe for generalization with untrained stimuli
 - Teach additional exemplars if the probe data do not meet the generalization criterion (e.g., 80% accuracy)
- **Teach in a variety of environments**
 - Use the Milieu Teaching approach (see under Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs)
 - Train parents to conduct home treatment sessions

- Teach initially in a quiet room, but shift training to less formal situations and eventually to the child's typical environments
- Take the child to the cafeteria or library or for a walk on the campus to informally and naturally reinforce the target communication skills (e.g., a nod, a smile, subtle verbal praise)
- Train teachers to praise the child for appropriate communication skills in the classroom
- **Reduce echolalia**
 - Find out if echolalia is functional; hand the object to the child or ask if the child wants what the echolalic response included or implied; if the child agrees or accepts what is offered, perhaps echolalia is functional
 - Teach verbal requests to replace functional echolalic responses
 - Ignore nonfunctional echolalia completely; it may have been maintained by attention
 - Give corrective feedback whenever the child echoes
 - Teach alternative communicative skills
- **Give direct, intensive training**
 - Target useful skills and teach them directly
 - Use the method of Discrete Trials, known for its effectiveness
 - Repeat trials on the same target skill until the behavior is reliably produced
 - Provide frequent weekly training, which is known to be effective
- **Reinforce all attempts at effective communication**
 - Accept all appropriate communication attempts, even inadequate ones, in the initial stage
 - Teach words, phrases, gestures, pointing, and any other form of effective communication
 - Gradually shift reinforcement to more precise, socially appropriate, and complex verbal responses
- **Reduce autistic leading**
 - Reduce the typical behavior of grasping an adult's hand and leading to a desired object

- First, teach the child to point to objects desired; give the object pointed to
- Next, model the names of common objects and teach the child to request them
- Withdraw reinforcement for nonverbal responses if verbal requests may be maintained with sustained reinforcement
- Shape longer durations of eye contact
- **Reduce pronoun reversal**
 - Teach the correct use of *I* and *you*
 - To teach the pronoun *I*, prompt the child to perform an action such as clapping hands
 - Ask, “What are you doing?” or “Who is clapping?”; model the correct response: “*I* am clapping” and praise the child for the correct response
 - To teach the pronoun *you*, perform a similar action and ask, “What am I doing?”; model the response, “*You* are clapping”
 - Use verbal praise or tokens backed up by reinforcers; verbally correct incorrect pronoun usage
 - Fade modeling
- **Teach verbal requests**
 - Use the evidence-based Mand-Model method
 - Teach parents to observe the child approaching them or looking at a desired object placed out of reach
 - Teach parents to immediately prompt a verbal request by saying, “Tell me what you want”
 - Teach parents to model the mand if necessary
 - Teach parents to expand the child’s response if necessary (e.g., the parents might expand the child’s single-word response, “ball,” into a sentence by saying, “Tell me, *I want the ball*”)
 - Use the same procedure in treatment sessions
- **Teach appropriate social behaviors**
 - Use the evidence-based video modeling technique
 - Show videotaped social interactions among children
 - Show multiple and varied patterns of appropriate interactions
 - Instruct, model, and reinforce attentive watching of the videotaped interactions

- Measure the effect of video watching on the child's subsequent social interactions
- **Teach spontaneous requests or other kinds of verbal behaviors**
 - Use the evidence-based time delay procedure
 - Present a picture to the child that is expected to evoke a response
 - Do not immediately prompt the response; instead, delay your prompt a few seconds to see if the child would respond
 - Reinforce a comment, a description, or any kind of response to the stimulus; however, prompt if the child did not respond within a few seconds
 - Delay prompts once again; progressively, increase the delay duration as the child begins to respond spontaneously to stabilize such requests, comments, or other kinds of verbal behaviors
- **Regulate the flow of behavior**
 - Use the visual schedule method, a pictorial representation of scheduled activities at certain times
 - Teach parents to prepare a visual display of daily activities (breakfast time, dinner time, homework time, bedtime); have them pictorially represent the different actions involved in that activity (e.g., setting the table, sitting on the chair, eating during breakfast time); train parents to draw the child's attention to the schedule and use it to regulate the child's behavior
 - Have the teacher prepare a visual display of daily classroom activities (to be used in school); train the teacher in teaching the child to use the schedule
 - Prepare a schedule of treatment session activity into which communication treatment is integrated (e.g., painting, working with puzzles, listening to stories); teach the child to consult the schedule and move from one activity to the other in an orderly fashion
- **Teach the child to request objects by exchanging pictures**
 - Use the evidence-based Picture Exchange Communication System (PECS)

- Display several pictures of objects in front of the child; the displayed objects must be available to dispense
- Extend your hand to receive the picture the child picks up
- Give the object depicted on the picture to immediately reinforce the child
- Model the word the picture represents, but initially give the picture even if the child did not say the word; subsequently, reinforce the child only for verbal requests
- **Reduce socially inappropriate verbal expressions**
 - Try extinction; completely ignore inappropriate verbal expressions (avoid eye contact, refrain from verbal corrections, and sit still)
 - Do not inadvertently reinforce unacceptable verbal responses because they are funny or “cute”
 - Reinforce appropriate expressions to reduce inappropriate expressions
 - Verbally correct all inappropriate expressions when not using extinction
 - Use Differential Reinforcement techniques to teach alternative, desirable expressions
- **Teach social communication skills**
 - Teach eye contact in all treatment sessions; frequently instruct and prompt; reinforce eye contact
 - Teach topic initiation, topic maintenance, turn taking, and conversational repair strategies during conversations
 - See Social Communication Skills for the procedures to teach each of those skills
- **Teach augmentative and alternative communication (AAC)**
 - Use the procedures described under Augmentative Communication—Aided and Augmentative Communication—Unaided
 - Combine oral modeling of target responses with an augmentative or alternative mode of expression (e.g., when the child points to the picture of a glass of

- water* on a communication board, touch the picture and say, “water,” before giving it to the child)
 - Teach other forms of nonverbal communication (e.g., American Sign Language); preferably combine oral speech with signs as in Total Communication (e.g., sign a word and say the word as you sign)
- **Implement generalization and maintenance strategies**
 - Work with caregivers to promote generalization and maintenance of clinically established skills
 - Conduct a few treatment sessions at home
 - Train parents and other caregivers to prompt and reinforce desirable behaviors at home; train them to conduct brief treatment sessions at home
 - Review the home intervention sessions online (telepractice) or recorded videos to give feedback to the caregivers
 - Periodically assess the child and offer booster treatment
 - Work closely with other specialists
 - Review the progress of other professionals (e.g., teachers, behavior analysts); consult or give feedback as necessary
 - Supply periodic progress data to other professionals and consult with them about the future course of intervention

Quality of Life (QoL) After Communication Treatment for Autism

- QoL in persons with autism may be lower for a variety of reasons, not all related to communication skills; aggressive, self-injurious, and emotional behavioral problems inadequately treated by other professionals may be responsible to varying extents
- Administer ASHA’s *Quality of Communication Life Scale (QCL)*; the scale is not specific to autism, but may still be useful; get it from asha.org
- Administer the *Pediatric Quality of Life Inventory (PEDsQL)*; visit <https://www.pedsql.org>

- Administer *Autism Spectrum Quality of Life (ASQoL)* —a questionnaire specific to autism; see the McConachie citation
- Conduct a client-specific Ethnographic Interview of the client and the caregivers to understand their ratings of QoL and satisfaction with the treatment
- Analyze reasons for dissatisfaction and lower QoL that may be related to communication treatment
- Address those reasons by offering additional communication treatment; assess generalization of communication skills to the client's home, school, or work situations
- See Quality of Life After Treatment for several treatment strategies to improve QoL of individuals with autism
- Assess maintenance of communication skills and offer booster treatment to help sustain skills

Binger, C., & Light, J. (2007). The effect of aided AAC modeling on the expression of multi-symbol messages by preschoolers who use AAC. *Augmentative and Alternative Communication*, 23(1), 30–43.

Fuller, E. A., & Kaiser, A. P. (2020). The effects of early intervention on social communication outcomes for children with autism spectrum disorder: A meta-analysis. *Journal of Autism and Developmental Disorders*, 50, 1683–1700.

Makrygianni, M. K. et al. (2018). The effectiveness of applied behavior analytic interventions for children with autism spectrum disorder: A meta-analytic study. *Research in Autism Spectrum Disorders*, 51, 18–31.

McConachie et al. (2018). Enhancing the validity of a quality of life measure for autistic people. *Journal of Autism and Developmental Disorder*, 48, 1596–1611

Automatic Reinforcers. Sensory consequences of responses that reinforce those responses (e.g., the sensation a child with autism derives from head-banging or persistent clapping); because the consequences are automatic, reinforcement is swift; to reduce behaviors that are automatically reinforced:

- Prevent automatic sensory consequences of undesirable behaviors during treatment

- Make the child wear a helmet that eliminates sensory stimulation from head-banging
- Cover the table with padded material to prevent reinforcing auditory stimulation by hitting the table
- Provide stimulating and reinforcing activities to reduce undesirable behaviors maintained by automatic reinforcers

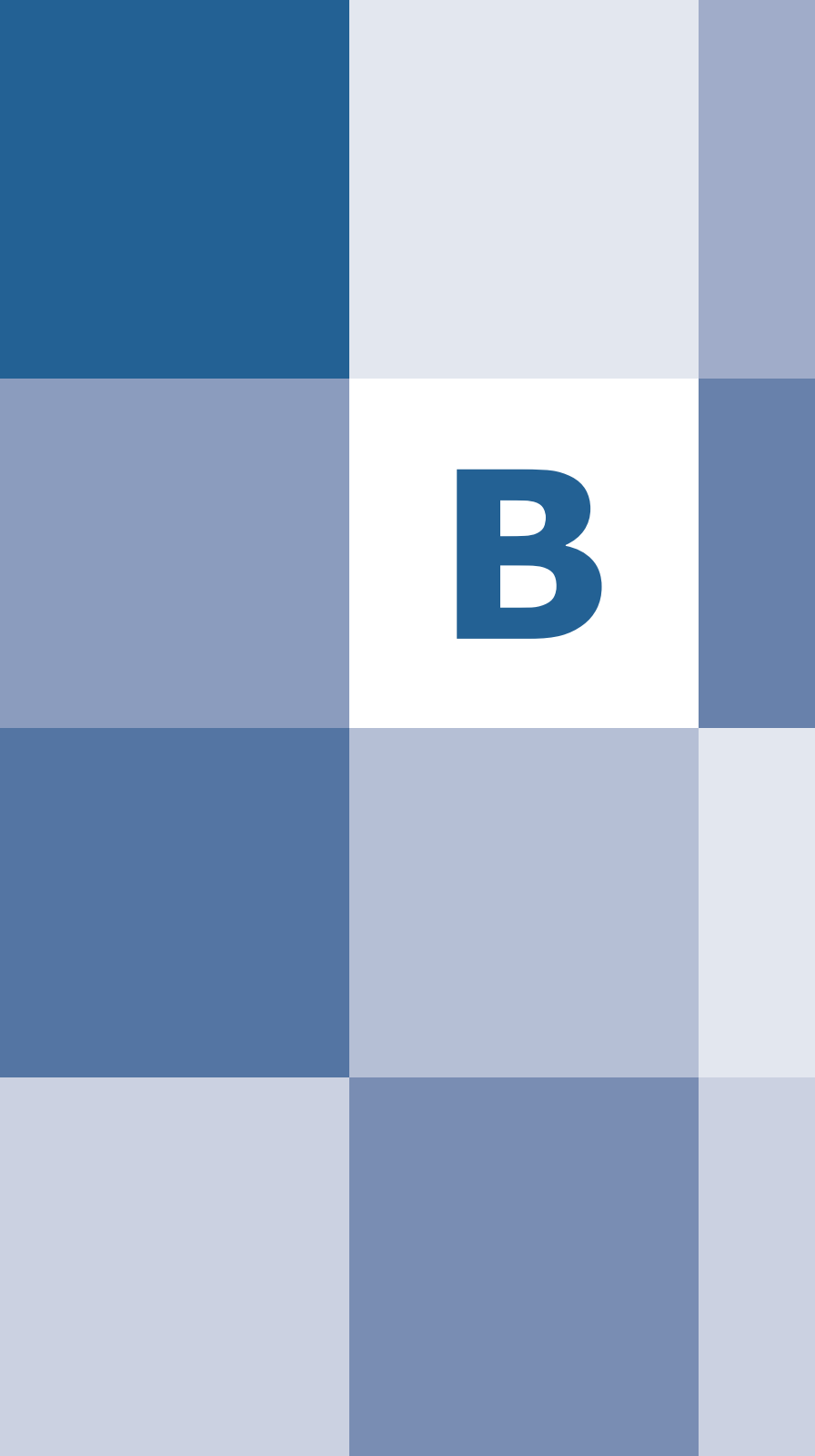
Aversive Stimuli. Events that people work hard to avoid or move away from; reduction in aversive stimulation provides negative reinforcement; a behavior that reduces negative experiences tends to increase in frequency; in treatment, positive reinforcement is preferable to negative reinforcement; however, negative reinforcement derived from avoiding aversive treatment may maintain many kinds of undesirable behaviors children exhibit during treatment; to control them:

- Find out what maintains undesirable behaviors (e.g., if the child leaves the seat during treatment—a negatively reinforced behavior—it may be due to aversive treatment trials from which the child seeks escape)
- Reduce treatment aversiveness; present more powerful reinforcers; simplify treatment tasks
- Alternatively, physically prevent escape

Avoidance. A behavior that prevents the occurrence of an aversive event and hence gets *negatively* reinforced; target is to reduce avoidance if judged undesirable, a typical target in persons who stutter; the individual is made to face previously avoided situations with appropriate clinical support; for instance, to reduce avoidance of certain speaking situations by persons who stutter:

- Build a hierarchy of most-to-least frequently avoided speaking situations or tasks
- As the individual becomes more fluent during treatment, introduce the individual to least frequently avoided situations first and move up the hierarchy
- **Offer training in situations the individual avoids**
 - Take the client to a restaurant and have the person order food (an avoided response); monitor fluency and provide subtle reinforcement and verbal correction

- Have the individual make phone calls as you monitor fluency (e.g., prompt the person to slow down)
- Arrange group situations in which the individual will speak or make brief presentations
- Introduce strangers to the treatment setting and let the individual practice fluency skills in front of them
- Generally, reinforce the individual for facing previously avoided situations and tasks



B

Backup Reinforcers. Events, objects, and opportunities for selected actions that become available to individuals who exchange their earned tokens in treatment sessions.

- Have a collection of backup reinforcers
- Have the child select a backup reinforcer for that particular session (e.g., a child might select a toy car, a chance to draw)
- Give tokens to reinforce target responses (e.g., for correct articulations, language responses, fluent productions)
- Exchange tokens for selected events, objects, or opportunities for actions at the end of the session (e.g., the child gets the toy or a chance to draw for 3 minutes at the end of the session)

Backward Chaining. A behavioral procedure to teach a complex skill by breaking it down into components and teach them in their reversed order; instead of beginning with the first component of a target response and moving to the final as in *Shaping*, backward chaining begins with the last component of a target response and moves to first or the final component. See also *Chaining*. Speech sounds and language structures are best suited for backward chaining; as an example, in teaching a child to produce the word *walking*:

- Model *ing* and reinforce correct imitation on at least five trials
- Model *king* and reinforce correct imitation on at least five trials
- Model *walking* and reinforce correct imitation
- Stop modeling and evoke *walking* on discrete trials
- Continue training until the stated objectives are met

Baselines. Measured frequency of responses before treatment is initiated; essential to show improvement in routine treatment and effectiveness in treatment research; reliable (repeated) measures help establish the need for treatment and demonstrate improvement during treatment; in experimental treatment research, baselines, along with other controls, help rule out extraneous variables.

Baselines of Conversational Speech

- Establish a measure of target behaviors produced in conversational speech in the absence of treatment before starting treatment
- Do not reinforce or offer verbal corrections
- Record a naturalistic conversational speech
- Have toys, pictures, books, and other materials to evoke speech from children; engage the child in conversational speech in a play-like setup
- Hold a conversation with adults on their favorite topics; use the clinical interview to establish baselines of target behaviors (e.g., language characteristics, stuttering, vocal quality)
- Measure the correct and incorrect productions of the target behaviors in the sample
- Calculate the percent correct baseline response frequency to be compared with the same under treatment

Baselines With Discrete Trials:***General Procedure***

- Establish baselines in discrete trials that are easily transitioned into treatment trials; most language and speech sound treatment need discrete trials
- Do not reinforce or verbally correct
- Select target behaviors (phoneme productions, grammatic morphemes, sentence structures, pragmatic skills, fluent productions, naming skills, etc.)

Specify Target Behaviors in***Measurable Terms***

- Production of /s/ in word-initial positions
- Production of present progressive *ing* in sentences
- Naming pictures or photographs
- Reduced rate of speech
- Elimination of hard glottal attacks

Prepare the Physical and Verbal***Stimulus Materials***

- Prepare 20 stimulus items for each speech and language target response
- Twenty pictures that help evoke 20 words with /s/ in the initial position

- Twenty sentences with the present progressive *ing* in them (e.g., The boy is walking)
- Plan the response evoking and modeling procedures
- Write down such questions as, “What are these?” “Who is this?” “What is she doing?” and so forth
- Write down the exact way of modeling the response (e.g., “Say, *these are books*”; “Say, *this is my wife*”; “Say, *she is walking*”)
- Prepare a recording sheet
- Administer the two types of trials: Baseline Evoked Trials and Baseline Modeled Trials

Baseline Evoked Trials

- Do not model but evoke the target responses with natural kinds of stimuli (e.g., such questions as “What is this?”); administer evoked trials before modeled trials
- Measure each attempt to produce a target behavior separately
- Do not reinforce or verbally correct
- Place a stimulus item in front of the individual (e.g., a picture of a *ball*) or demonstrate an action (e.g., moving a toy car)
- Ask the relevant predetermined question (e.g., “What is this?” “What am I doing?” or “What is happening?”)
- Wait a few seconds for the individual to respond
- Record the individual’s response on the recording sheet
- Remove the stimulus item (move it toward you, away from the individual)
- Wait 2–3 seconds to signify the end of a trial
- Begin the next trial with a different item

Baseline Modeled Trials

- Model the correct response for the individual to imitate
- Measure the client’s attempts at target behavior imitation separately
- Do not reinforce or verbally correct
- Place a stimulus item in front of the individual or demonstrate an action
- Ask the predetermined question (e.g., “What is this?”)

- Immediately model the correct response (e.g., “Johnny, say *ball*”)
- Wait a few seconds for the individual to respond
- Record the individual’s response on the recording sheet
- Remove the stimulus item (move it toward you, away from the individual)
- Wait 2–3 seconds to signify the end of a trial
- Begin the next trial with a different stimulus item

Analyze the Baseline Data

- Analyze data to calculate the percentage of correct responses (e.g., 50% correct production of the /s/ in word-initial positions; 75% correct production of *ing* in sentences)
- Calculate the percentages of correct responses separately for modeled and evoked trials
- Repeat measures; compare the discrete trial and conversational speech measures
- When measures are stable, begin treatment

Behavioral Contingency. In behavioral analysis and treatment, a dependent relationship between antecedents, responses, and consequences; to manage this contingency:

- Provide antecedents (physical stimuli, modeling, instruction, demonstration, prompts, etc.)
- Require a specified response
- Provide immediate consequences (positive reinforcers or verbal corrections)

Behavioral Momentum. Clinician rapidly and repeatedly evokes a high-probability response and then immediately commands a low-probability response; often used to reduce noncompliance and increase the frequency of desirable responses of low probability:

- Find a response the individual readily performs (e.g., hand clapping, a high probability response)
- Model and have the child imitate that high-probability response repeatedly and in rapid succession

- While the child is still performing the high-probability action, quickly interject a request to perform a low-probability target response (e.g., ask the child to open the mouth to say *hi*, a low-probability response)
- Reinforce the occurrence of the low-probability response, which is likely; continue to teach it

Binswanger Disease. To treat communication disorders associated with vascular pathology resulting in dementia, see [Dementia](#) and [Vascular Dementia](#).

Biofeedback. A method used to reduce incorrect responses or shape and increase desirable responses in treatment; includes mechanical feedback given to the individual on vocal pitch and intensity, respiration, electropalatography, and muscle action potential level.

Bite Block. A custom-made small block of rigid plastic or rubber placed on the right and left back molar teeth; individual bites down on them while speaking; may reduce dystonia, stabilize the jaw during speech, and improve speech intelligibility; used in treating some individuals with [Dysarthrias](#).

Booster Treatment. Treatment offered any time after a client was dismissed from the original treatment; a follow-up may suggest a need for booster treatment because of deterioration in clinically established skills; part of response maintenance strategy.

- Conduct periodic follow-ups
- If the follow-up measures show a decline in response rate, give booster treatment
- Use the original or newer, more effective procedures

Botulinum Toxin Injection. A medical treatment procedure for neurogenic or idiopathic adductor spasmodic dysphonia that may be combined with behavioral voice treatments; botulinum toxin is injected into the thyroarytenoid muscle unilaterally or bilaterally; paralyzing effect of the drug diminishes the spasmodic bursts and increases the airflow rate; effects last about 3 months.

Bound Morphemes. Significant treatment targets for children with language disorders; teaching grammatic morphemes that are inflected with words (hence the name, *bound*) is essential to remediate childhood language disorders; intervention targets include such bound morphemes as the present progressive *ing*, the allomorphic variations of the regular plural and regular past, various prefixes (e.g., *pre-*, *post-*, *anti-*) and various suffixes (which include the regular plural and past inflections); to teach bound morphemes:

- Select treatment target morphemes that are missing in reliable language samples
- Develop stimulus materials (words, phrases, and sentences) and pictures or objects
- Present a stimulus item and ask a relevant question (e.g., present the picture of two books and ask the question, “What are these?”) and immediately model the correct response (e.g., “Say, *these are books*”)
- Reinforce with verbal praise if the imitated response is correct, and verbally correct if incorrect
- Fade modeling, train multiple exemplars, and probe to see if the child would produce the trained morpheme in untrained contexts (words, phrases, or sentences)
- See [Language Disorders in Children](#) for additional information

Breathiness. A voice quality that results from excessive air leakage during phonation because of inadequate approximation of the vocal folds, caused by various factors; treatment varies by cause; see [Voice Disorders](#).

Broca’s Aphasia. To treat this type of nonfluent and agrammatic aphasia with effortful speech production, see [Aphasia: Specific Types](#).

Buccofacial Apraxia. Impaired buccofacial movements when the clinician requests them; a treatment target for persons with [Apraxia of Speech](#).

C

C

Carryover. The same as Generalization; an important goal of treatment, it is the generalized production of clinically established behaviors in natural and untreated settings, in responses to untreated stimuli.

Cerebral Palsy. Treatment of cerebral palsy (CP), a congenital, nonprogressive neurological disorder that affects motor control, generally follows the procedure of treating Dysarthria, as the symptoms are similar; injury to the cerebral levels during the prenatal or perinatal period results in CP; not being a degenerative disorder, the symptoms tend to improve with age; treatment targets include speech sound production, normal voice and resonance features, velopharyngeal adequacy, and potential language and literacy skills.

Treatment of Cerebral Palsy:
General Guidelines

- Make a thorough assessment of communication disorders and the associated neuromuscular problems; see the cited sources
- Work closely with the team of specialists serving children with cerebral palsy; the work of physical therapists is essential in improving muscle strength and controlling interfering reflexes
- Counsel parents about the effects of cerebral palsy on communication and their role in stimulating language at home
- Work closely with parents throughout the treatment duration
- Design treatment to suit the individual child's problems, needs, and strengths
- Work closely with educators to align communication treatment with educational targets to have the child meet current or future academic demands
- Borrow techniques from other communicative disorders in children (e.g., Language Disorders in Children, Speech Sound Disorders in Children, Dysarthria, and Voice Disorders) as cerebral palsy is not the name of a unique speech or language disorder; modify the standard techniques to suit the individual child's specific symptom complex

Treatment Procedures for CP

- **To use or not to use nonspeech oral-motor exercises**
 - Note that some experts advocate the use of feeding and oral-motor exercises to improve on-speech neuromuscular control, while others reject this because nonspeech oral movements are mediated at a lower level in the central nervous system than the speech movements that are mediated at the cortical level
 - There is no controlled evidence to show that oral-motor exercises are effective in promoting better speech production
 - It is generally more efficient to improve speech-related oral movements through speech training instead of nonspeech movement training
- **Treatment of language disorders**
 - Train parents to stimulate language at home; see [Language Disorders in Infants, Toddlers, and Preschoolers](#); [Language Stimulation by Parents](#)
 - Assess the child's language development periodically to determine the need for formal clinical treatment
 - Implement formal language treatment if necessary; use the treatment procedures described under [Language Disorders in Children](#) and [Language Disorders in Infants and Toddlers](#); modify the procedures to suit the child with cerebral palsy
 - Consider teaching literacy skills integrated with communication training; see [Literacy](#) and [Literacy Skills in Children](#) for procedures
- **Treatment of voice disorders**
 - Diagnose the specific voice disorder; when appropriate, use one or more treatment techniques described under [Voice Disorders](#) with suitable modifications; be aware that voice disorders may be due to respiratory problems associated with cerebral palsy
 - Treat associated respiratory problems; prescribe exercises to improve breath support for speech; use techniques described under [Dysarthria](#)

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- Treat velopharyngeal incompetence only if there is enough tissue mass and behavioral training is likely to be effective; see [Velopharyngeal Dysfunction](#)
- **Treatment of speech sound disorders**
 - Assess the child's specific sound errors and error patterns
 - Assess the compensatory articulatory postures the child uses
 - Modify or eliminate inappropriate and ineffective compensatory postures
 - Teach the specific phonemes or classes of phonemes based on phonological patterns
 - Use the treatment techniques described under [Speech Sound Disorders in Children](#) with appropriate modifications; see also [Dysarthria](#) for additional treatment suggestions
- **Manage the associated attentional problems**
 - Treat the attention deficits the child may exhibit during communication treatment sessions
 - Positively reinforce progressively longer periods of attention to treatment stimuli; consider using a token system to reinforce attending behaviors
 - Draw attention to treatment stimuli as soon as the child shows signs of distraction; praise the child for paying attention
 - See [Attention Disorders](#) for details and additional procedures
- **Manage the associated sensory problems**
 - Refer the child to an audiologist for needed auditory diagnosis
 - Check the child's hearing aid to ensure that it is working properly
 - Use FM and other auditory trainers if needed
 - Have the child wear prescription glasses during treatment
 - Make needed accommodations for any auditory or visual limitations (e.g., conducting treatment in well-lit rooms, making sure that the light is on your face when you talk)

Hegde, M. N. (2026). *Hegde's pocketguide to assessment in speech-language pathology* (5th ed.). Plural Publishing.

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Chaining. A behavioral technique of linking elements of a complex skill; similar to shaping; useful in teaching both nonverbal and verbal skills, although most frequently employed in teaching self-help skills; includes Shaping (which uses forward chaining) and Backward Chaining.

- Make a task analysis and identify the steps involved in achieving a complex task; use either shaping or backward chaining
- **Forward chaining**
 - Use shaping, which is the same as forward chaining; analyze the component skills of the final target response; teach skills in their typical order: for instance, to teach the word *soccer* to a child:
 - Identify such individual skills as (1) the production of *sa*, (2) the production of *sock*, and finally, (3) the production of *soccer*
 - Model the first component *sa* for the child to imitate; reinforce or verbally correct and repeat the trials until the child produces at least five correctly imitated responses
 - Model the second component *sock*; reinforce or verbally correct; train until five correctly imitated responses
 - Model the third and final target word *soccer*, reinforce or verbally correct; fade modeling, and continue to teach on evoked trials until a training criterion is met
- To use Backward Chaining to teach the same target word *soccer*, teach first the syllable *er*, then *sock*, and finally the target word *soccer*. Use the modeling, imitation, reinforcement, and verbal correction procedures.

Changing Criterion Treatment Procedure. A method of shaping desirable behaviors by using performance criteria that change every time the individual meets

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a prior criterion; the criterion may change in either direction (lower or higher) depending on the target behavior; in reducing the speech rate of individuals (e.g., in dysarthria or stuttering), the criteria are progressively lower; in shaping longer utterances in language intervention, the criteria are progressively higher.

- **Use changing criteria to reduce undesirable behaviors**

- To reduce misarticulation of a phoneme, set a target for the child (a criterion such as “Try *not* to make more than five errors in this session; in the last session, you made eight errors”) and state the criterion at the beginning of each session
- Decrease the number of errors allowed by setting progressively lower criteria of performance (e.g., “Try *not* to make more than three errors today; last time you made five errors”; “Try *not* to make more than one error today; last time you made three errors”) as the previous criterion was met and sustained
- Use similar criteria to reduce or eliminate the frequency of asking irrelevant questions, leaving the chair, dysfluencies, naming errors, and so forth
- Use progressively lower temporal criteria to reduce such problem behaviors as high-pitched voice (e.g., “Try *not* to produce a high-pitched voice for more than 5 minutes; last time your pitch was too high for 7 minutes”; “Try *not* to produce a high-pitched voice for more than 3 minutes; last time your pitch was too high for 5 minutes”); use similar criteria to reduce other behaviors measured on the basis of duration (including hoarse or harsh voice, resonance disorders, and such undesirable behaviors as looking away from the stimulus material or fidgeting)

- **Use changing criteria to increase desirable target skills**

- To increase the frequency of correct production of phonemes, grammatic morphemes, naming skills, and so forth, require progressively higher criteria of performance; set a target performance criterion at the beginning of each session and state it (e.g., “Let us see if you

can increase correct responses to 80% today; you were 76% correct last time”)

- Set progressively higher criteria of performance as the previous one is met and sustained
- Use similarly progressive temporal criteria to increase behaviors measured for their duration (e.g., “Let us see if you can sit quietly and look at the pictures for 2 minutes today; you did it for one minute last time”; “Try to talk fluently for 5 minutes today; you did it for 3 minutes last time”)

Chant-Talk. To use this voice therapy technique to reduce problems associated with hyperfunctional voice, see [Voice Disorders: Specific Normal Voice Facilitation Techniques](#).

Chewing Method. To use this voice therapy technique, see [Voice Disorder: Specific Normal Voice Facilitation Techniques](#).

Child-Centered Approach. A general intervention philosophy that assigns a more active role to the child; treatment sessions may be play oriented and the treatment may be indirect; the clinician takes the child’s lead in targeting language structures for intervention; see [Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs](#).

Childhood Apraxia of Speech (CAS). Less often called *developmental apraxia of speech*, CAS is a speech disorder in children that shares some common characteristics with [Apraxia of Speech \(AOS\) in Adults](#), but without documented neuropathology which has led to some controversy; treatment of CAS primarily targets articulatory accuracy to promote appropriate positioning of articulators and sequential and smooth speech muscle movements during speech production; additional targets include associated prosodic problems, coexisting language disorders, reading problems, and writing deficiencies; controlled treatment efficacy data are limited but evolving.

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Treatment of CAS: General Guidelines

- Speech sound production training procedures similar to those used in children with typical speech sound disorders are part of CAS treatment
- A strictly phonological approach that emphasizes patterns and rules and auditory discrimination drills may be unnecessary, whereas imitation of articulatory postures, correct placement and movement of articulators, and repeated practice of speech sound production are useful; emphasis on sequencing the target sounds may be more productive than drills on individual sounds
- There is a clinical opinion that apraxia of speech in adults and children is more difficult to treat than regular speech sound disorders; in severe cases, prognosis may be guarded, although intensive and sustained treatment can still make a difference in functional communication skills
- The need for oral-motor exercises and oral sensory training that are unrelated to speech production is unclear; it may be more efficient to teach and stabilizing movements that are involved in speech production
- Teach correct articulatory placement and movement to improve speech production
- Avoid auditory discrimination training, as the main difficulty is sequenced production of speech sounds
- Provide intensive treatment; two or three weekly sessions conducted over several months may be needed to show improvement in speech production and intelligibility
- Effective behavioral treatment to address the dominant symptoms that seem to affect intelligibility the most is a pragmatic approach to treating CAS

Treatment Approaches for CAS

- **Teach through a treatment hierarchy**
 - Move hierarchically from easy to difficult tasks; shape progressively more complex patterns of speech movements; alternatively, use the Concurrent Treatment Approach in which you randomly train both simple and complex targets in the same

session; see Speech Sound Disorders in Children: Specific Treatment Techniques or Programs

- Determine the level at which the child will respond (phonemes, syllables, words); start with that level and move through more complex skill levels (shaping)
- Use meaningful speech stimuli in treatment instead of nonsense syllables or nonspeech movements
- Teach first the vowels and consonants that children normally produce early; alternatively, teach first the more advanced or more difficult sounds the mastery of which may speed up the learning of earlier-acquired sounds
- Teach consonants that are visible before you teach those that are not visible
- Teach phonemes that occur often or the phonemes the child misarticulates most consistently
- Teach voiceless consonants before voiced consonants
- Model the productions
- Cue the production of the sound or sound sequences before the child's attempted production
- Teach progressively more complex utterances
- Fade modeling and cueing in gradual steps
- Reduce the speech rate to improve speech intelligibility; gradually shape the more typical rate
- Reinforce the correct responses with effective reinforcers
- Use multiple repetitions of speech productions
- Use intensive discrete trials; stress sequence of movements involved in speech production
- Use selected aspects of treating Speech Sound Disorders
- Teach Self-Control (Self-Monitoring) Skills
- **Provide a single or a combination of cues**
 - Provide visual, auditory, kinesthetic, or tactile cues or a combination that works best with a child
 - Give touch, pressure, kinesthetic, and proprioceptive cues as in some specific programs described in the next section

- Give visual cues that may be hand shapes and gestures or may involve biofeedback through computerized programs and electropalatography
- **Address associated problems**
 - Treat such resonance problems as hypernasality or nasal emission; see [Cleft Palate](#) and [Voice Disorders](#) for details
 - Treat prosodic problems by teaching variable intonation; teach normal stress patterns with selected sentences in which the stress may be varied; consider using [Contrastive Stress Drills](#)
 - Treat language problems; see [Language Disorders in Children](#) for specific procedures for teaching semantic, grammatic, and pragmatic aspects of language production
 - Treat fluency disorders if they are clinically significant; see [Stuttering](#) for treatment options
- **Consider the needs of a bilingual child with CAS**
 - Determine the dominant language of the child
 - Do not expect generalized production across the two languages of treated sounds that are unique to one or the other language
 - Treat the child in both languages if you can
 - Teach sounds that are common to the child's two languages
 - Use stimuli that are familiar to the child's bilingual environment
 - Teach errors unique to a given language, but do not expect generalization to the other language; for instance, teaching word-final consonants taught in English may not generalize to Spanish if those consonants are not in that position in Spanish
 - See [Language Disorders in Ethnoculturally Diverse Groups](#) for additional suggestions
- **Consider augmentative and alternative communication (AAC)**
 - Children with extremely severe forms of CAS may be candidates for AAC either as a temporary form of communication at the beginning of verbal speech training or as a potentially permanent form

- See Augmentative and Alternative Communication (AAC)

Maas, E. (2024). Treatment for childhood apraxia of speech: Past, present, and future. *Journal of Speech, Language, and Hearing Research*, 67(9S), 3495–3520.

Treatment of CAS: Specific Techniques or Programs. Most successful programs are thought to be based on motor-based learning principles, which essentially include modeling-imitation sequence, shaping, various kinds of cueing, massed treatment trials, positive reinforcement, and verbal correction—essentially the same as the comprehensive program described in the previous section; some named treatment packages contain several elements whose independent effects are not clear and may target nonsense syllables; it may be more efficient to directly train speech sounds in meaningful words; because of expanding research, clinicians need to regularly check newer publications in scientific journals.

Dynamic Temporal and Tactile Cueing (DTTC).

Known earlier as *integral stimulation*, the method uses a modeling and imitation sequence; provides auditory and visual cues for speech production; targets sound production in real words, not nonsense syllables; tactile cues are added as needed.

- Slow the speech rate initially and gradually shape the more typical rate
- Give visual cues; ask the child to watch your face as you model; provide gestural cues (e.g., pointing to your retracted lips); give tactile cues (e.g., manually positioning the jaw and lips for target sound productions); note that tactile manipulations are technically not *cues* but methods of Manual Guidance or Phonetic Placement; it is physical shaping of articulators
- Reinforce and verbally correct imitative and spontaneous responses
- Introduce a temporal hierarchy in which you and the child produce the sound simultaneously; model the sound and ask the child to imitate immediately;

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model and ask the child to imitate after a delay; ask questions to evoke more spontaneous productions

- Fade cues as quickly as possible
- Maximize the treatment trials in each session
- Require 10 to 15 accurate productions at each level before moving to the next level of training
- Begin prosodic training after the sound productions are stabilized; model the target responses quieter or louder and with different emotions (e.g., sad, happy, excited); ask the child to “Say it just like me”; fade modeling and move to spontaneous production training

Strand, E. (2020). Dynamic temporal and tactile cueing: A treatment strategy for childhood apraxia of speech. *American Journal of Speech-Language Pathology*, 29, 30–38.

Prompts for Restructuring Oral Muscular Phonetic Targets (PROMPT).

Targets speech sound production in meaningful, functional words; the procedure is not exclusive to CAS treatment as it has been used to treat adults with apraxia of speech and those with autism spectrum disorder; though described as such, most of the techniques are not *prompts*; they are manual (physical) manipulations of a child’s articulators for specific speech sound productions; special training is needed to fully implement the procedure.

- Use tactile-kinesthetic-proprioceptive (TKP) cues to shape speech production; manually guide the articulator (e.g., lip rounding or jaw opening) for correct positions for target sound productions
- Use modeling, require imitation, offer feedback, fade the amount of guidance, and train with repeated trials as done in almost all other approaches
- Move on to the next stimulus item when the response is correct; use TKPs if the response is incorrect
- See [Apraxia of Speech \(AOS\) in Adults](#); [Specific Techniques or Programs](#) for additional information

Dale, P. S., & Hayden, D. A. (2013). Treating speech subsystems in childhood apraxia of speech with tactual input:

The PROMPT approach. *American Journal of Speech-Language Pathology*, 22(4), 644–661.

Cleft. An opening in an anatomic structure that is normally closed (e.g., clefts of the upper lip or the palate).

Cleft Lip. To treat communication disorders associated with it, see [Cleft Palate](#).

Cleft Palate. Treatment of children with cleft palate with or without cleft lip is a team effort, with speech-language pathologists attending to the needs of communication; see the sources cited at the end of this main entry for epidemiology, etiology, symptomatology, and communication impairments associated with clefts of the palate with or without the clefts of the lips; speech-language pathologists need to have a basic understanding of surgical management of the clefts because this management has speech-language consequences.

Surgical Management of the Clefts

- *Lip Surgery.* Surgical methods to close unilateral or bilateral clefts; usually done when the baby is about 3 months old or weighs about 10 pounds
- *Palatal Surgery.* Surgical procedures performed to close the cleft or clefts of the palate; done when the baby is between 9 and 24 months, with many between 9 and 12 months; the earlier the closure of the cleft, the better the speech development
- *Variations in Surgical Methods.* Lip and palatal surgical methods vary across surgeons; find out the child-specific information from the surgeons
- *Delayed Hard Palate Closure.* A surgical sequence in which the cleft of the soft palate is closed first and the hard palatal cleft is closed later
- *Palatal Lengthening.* Surgical procedures to increase the length of the soft palate to improve velopharyngeal function
- *Primary Surgery for the Clefts.* The initial surgery in which the clefts are closed

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- *Posterior Pharyngeal Flaps.* A secondary palatal surgical procedure designed to improve the velopharyngeal functioning for speech; a muscular flap is cut from the posterior pharyngeal wall, raised, and attached to the velum; the flap is open on either side to allow for nasal breathing, nasal drainage, and production of nasal speech sounds; helps close the velopharyngeal port and thus reduce hypernasality
- *Pharyngoplasty.* A surgical procedure designed to improve velopharyngeal incompetence; such substances as Teflon, silicone, Dacron wool/silicone gel bag, and cartilage may be implanted or injected into the posterior pharyngeal wall to make it bulge and thus help close the velopharyngeal port
- *Presurgical Management.* Procedures that include taping of the lip with surgical tapes, use of various intraoral appliances (e.g., a nasopalveolar molding), and lip adhesion to reduce the cleft width before surgery
- *Secondary Surgeries for Clefts.* Surgical procedures done after the primary surgery to improve functioning and facial appearance

Treatment of Communication Disorders in Children With Cleft Palate/General Principles

- Establish a good working relationship with all the members of the cleft palate team; in public school setting, stay in touch with the child's physician, the plastic surgeon, the otolaryngologist, the audiologist, and educational specialists that serve the child
- Make a thorough assessment of speech sound production skills and velopharyngeal function before starting treatment; see the cited sources for details
- See Language Disorders in Infants and Toddlers to train parents of young children with clefts to stimulate language development at home; refrain from intensive direct treatment when structural deficiencies are still there
- Begin direct treatment when the medical-surgical interventions justify

- Do not use nonspeech oral-motor exercises because they are ineffective
- Target improved intelligibility in conversational speech as the final goal of treatment
- Do not treat speech sound errors that are a direct result of structural deficiencies (*obligatory errors*); target them only after surgical interventions have eliminated those deficiencies
- Treat resonance disorders, especially hypernasality if the child has at least a marginal velopharyngeal competence
- Conduct trial therapy to judge intervention feasibility and prognosis
- Use such behavioral procedures as instructions, modeling, phonetic placement, manual guidance, prompting, shaping, reinforcement, and verbal correction in teaching speech sound productions; these procedures are often described as *phonetic* and *motor program based*; see [Speech Sound Disorders in Children](#) for details
- Note that some experts advise against using a phonological approach; however, some evidence suggests that the approach might produce positive results; this is not surprising because in all phonological approaches, the clinician teaches sound productions
- Combine speech sound and language interventions to develop a comprehensive program
- Work with teachers and special education specialists serving the child to meet the communication as well as educational needs of the child; select speech-language targets that support the child's academic success in the classroom
- Integrate literacy goals (e.g., reading and writing skills) with communication treatment
- Take note of middle ear infections and potential hearing loss; make referral to otologists and audiologists
- Collaborate with an audiologist to develop a speech-language and auditory rehabilitation plan for a child with significant hearing loss; see also [Hearing Loss](#)

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Treating Speech Sound Disorders Associated With Cleft Palate

- Use the procedures of treating Speech Sound Disorders
- Educate the parents about the speech mechanism, the effects of the clefts, and speech and resonance problems that accompany the clefts
- Ask parents to withhold reinforcement for undesirable compensatory behaviors, when medical or speech treatment has eliminated the need for them
- Target compensatory speech sound errors, including glottal stops, nasal, velar, and pharyngeal fricatives, pharyngeal stops, mid-dorsum palatal stops, and phoneme-specific nasal emission; these are likely mislabeled and can be expected to show improvement

Teaching Speech Sound Productions: Special Considerations

- Teach the more visible sounds before the less visible except for the linguadentals
- Teach stops and fricatives before other classes of sounds
- Avoid or postpone training on /k/ and /g/ if the velopharyngeal functioning is inadequate
- Teach fricatives, affricates, or both after teaching stops
- Teach linguapalatal sounds, lingua-alveolars, and linguadentals in that order
- Progress from syllables to words, phrases, and sentences
- Give auditory and visual cues; model the correct productions and have the child imitate
- Provide systematic practice and reinforce correct productions; verbally correct incorrect productions
- Introduce compensatory articulatory positioning where appropriate
- Teach the individual to direct the breath stream orally; let the child feel the airstream on hand or see the movement of a piece of tissue
- Teach the child to avoid posterior articulatory placements

- Teach the child to articulate with less effort and facial grimacing
- Give tactile cues and instruction to improve tongue positioning
- Work on generalization and maintenance; train parents to reinforce correct articulation at home
- Use the Enhanced Milieu Teaching with an emphasis on phonological (speech sound production) skills

Treatment of Language Disorders in Children With Cleft Palate

- **Train parents to stimulate language**
 - Conduct a few language treatment sessions in the clinic to demonstrate the procedure to the parents; train them in conducting naturalistic treatment sessions at home
 - Ask parents to encourage free verbal expression in their child
 - Ask parents to model and reinforce correct articulation and language features at the same time
 - Integrate information about all aspects of rehabilitation in your discussion with the parents
 - Ask parents to socially reinforce the child's spontaneous verbal productions
 - Teach parents to reduce negative feedback, and make more positive statements about the child's communicative attempts
 - Meet with parents regularly to review progress and modify their home language stimulation program
- **Offer formal language treatment in the clinic**
 - Use the procedures of treating Language Disorders in Children
 - Assess the child's language skills at regular intervals to evaluate the need and level of language intervention
 - Consider the imminent educational demands and plan language intervention to help meet them

Cleft Lip and Palate: Treatment of Phonatory Disorders

- Evaluate if the phonatory problems are due to compensatory behaviors or velopharyngeal insufficiency

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- Use the techniques described under Voice Disorders if the problems are independent of velopharyngeal insufficiency (possibly due to vocal abuse)
- Discontinue phonatory treatment if there is no improvement or other speech symptoms worsen
- Do not offer phonatory treatment to children with a clear diagnosis of velopharyngeal incompetence
- **Treat hyperfunctional voice disorder**
 - Describe how voice is produced to the child and the parents
 - Reduce vocally abusive behaviors in the child and in other members of the family
 - Counsel the family about good vocal behaviors (e.g., talking less in noisy environments, practicing soft speech, good conversational turn taking, clapping instead of shouting or yelling)
 - Contrast the child's voice with those of children without vocal nodules or other vocal pathology
 - Use such biofeedback instruments as the VisiPitch in training
- **Train healthy voice production**
 - Reduce vocal loudness
 - Eliminate hard glottal attacks and teach soft initiation of phonation
 - Initiate words that start with vowels
 - Self-monitor voice
- Follow up treated children because some improve, some deteriorate, and some stay the same

Cleft Lip and Palate: Treatment for Resonance Disorders

- Do not treat Hypernasality and nasal emission that are due to velopharyngeal inadequacy (insufficient muscle mass)
- Treat hypernasality and nasal resonance only if they are functional, following surgical or prosthetic intervention, or only as trial therapy to assess the suitability of continued treatment

- See Velopharyngeal Dysfunction for additional suggestions and guidelines on treating hypernasality and audible nasal emission
- Use techniques described under Voice Disorders to increase vocal loudness, promote discrimination between oral and nasal resonance, lower the pitch, and increase oral opening
- Use respiratory training to improve loudness
- Attempt articulation with the nares occluded
- Decrease intraoral breath pressure on stop consonants and fricatives while simultaneously using loose articulatory contacts
- Consider using such biofeedback instruments as a nasometer to reduce hypernasality and increase oral resonance, although studies on their effectiveness are limited

Hegde, M. N. (2026) *Hegde's pocketguide to assessment in speech-language pathology* (5th ed.). Plural Publishing.

Kotlarek, K. J., & Krueger, B. I. (2023). Treatment of speech sound disorders in cleft palate: A tutorial for speech-language pathology assistants. *Language, Speech, and Hearing Services in Schools*, 54, 171–188.

Zajac, D. J., & Vallino, L. D. (2024). *Evaluation and management of cleft lip and palate* (2nd ed.). Plural Publishing.

Client-Specific Strategy. A method of selecting relevant, useful, and functional target behaviors for individuals to promote effective communication in social contexts; select client-specific communication targets in treating all speech-language disorders.

- Observe the individual's environment for clues to functional targets; interview family members, colleagues, teachers, peers or friends, professional caregivers, and others who interact with the individual regularly to select treatment targets
- Select targets that are useful and relevant and help meet the educational, occupational, and social demands made on the individual

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- Select targets that will immediately enhance the individual's communication in natural settings; increase speech intelligibility and teach social communication skills
- Select targets that have potential for generalized productions; these tend to be everyday communication skills (functional)
- Select targets that may be expanded into more complex communication skills (words to phrases, phrases to sentences, and sentences to conversational speech)

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.). Plural Publishing.

Closed-Head Injury. To treat communication disorders and related deficits associated with closed-head injury, also known as *nonpenetrating head injury*, see [Traumatic Brain Injury](#).

Cloze Procedure. Use this treatment procedure in which you model parts of an utterance and pause for the child to produce words and phrases to complete the utterance; it is essentially a sentence completion method; the same as [Partial Modeling and Completion](#).

- Model only the initial portion of a target response (e.g., say “The boy is . . .” and wait for the response)
- Let the child complete the partial model (e.g., the child says, “walking”)
- Reinforce the child's response

Cluttering. Primary treatment target in this fluency disorder is an abnormally fast and irregular speech rate that reduces articulatory accuracy and speech intelligibility; other potential targets include language disorders, disorganized thought, reading and writing problems, impaired prosody (monotonous speech), lack of concern about one's own speech problem, and, in some individuals, an associated stuttering.

- **Increase self-awareness of cluttering**
 - Show the individual's videorecorded conversational speech sample and discuss the problems that are evident; note that most people who clutter are not convinced of their speech difficulty

- Discuss with the individual the erratic rate, excessively fast rate, and lack of intelligibility of speech, impulsive talking, incomplete sentences, syllable telescoping, deletion of sounds and syllables
- Discuss any language problems or poor organization of thoughts that may be evident on the taped language sample
- **Prioritize treatment targets**
 - Reduce the speech rate and dysfluencies
 - Add additional treatment targets (e.g., language, reading, and writing difficulties)
 - Have the individual practice multiple target skills in an integrated manner in later sessions
- **Teach a slower and even rate of speech to improve speech intelligibility**
 - Prefer instructions, modeling, and positive verbal reinforcement to delayed auditory feedback devices (DAFs) or metronomes to teach a slower rate; use devices only if instructions and modeling are ineffective
 - Have the individual slightly prolong syllables and reduce the overall rate of speech
 - Target not only a slower rate of articulation but also distinct, slightly exaggerated articulation with emphasis on syllables and words, especially the first words
 - Increase the speech rate beyond baseline for a brief duration and then slow the rate down to encourage discrimination
- **Modify prosodic features**
 - Teach the client to pause at natural junctures in speech (e.g., between clauses and sentences) and at punctuation marks in oral reading
 - Model and reinforce production of syllables with deliberate stress, especially the final and unstressed syllables of words
 - Teach varying the stress on different words in the same sentence
 - Model and reinforce varied intonation patterns
- **Treat an independent speech sound disorder**
 - Target either individual phonemes or a pattern of errors

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- Use the methods described under Speech Sound Disorders in Children
- **Teach self-monitoring skills**
 - Give prompt, contingent verbal corrections on cluttered speech to increase awareness
 - Reinforce appropriate and even speech rate, normal rhythm, prosody, and well-formed verbal expressions
 - Teach the individual to pay attention to listeners who signal lack of comprehension (e.g., listener's subtle facial expressions that betray a lack of comprehension); show such expressions on your face and teach the individual to pause, slow down, or change the speech in other ways
- **Teach appropriate responses to listeners' request for clarification**
 - Make frequent requests for clarification (e.g., "I am not sure what you just said," "I didn't understand you")
 - If the individual did not modify their way of speaking, model one, and ask the individual to use the strategy
 - Teach a variety of ways in which one might modify one's own production to help the listener's understanding
- **Treat language and thought impairments in conversational speech**
 - Treat word-finding difficulties by having the individual name rapidly and learn words in semantically varied categories
 - Target specific syntactic errors and verbally correct them
 - Ask the individual to describe objects, events, or activities
 - Ask the individual to list synonyms and antonyms
 - Ask the individual to describe how two objects are similar or different
 - If necessary and found appropriate, use the techniques described under Language Disorders in Children
- **Teach social communication skills**
 - Teach conversational turn taking by stopping the client who interrupts you; ask the individual to speak only when told "It's your turn"
 - Reinforce well-organized expressions

- Reinforce coherent talking; stop when speech becomes incoherent
- Teach the individual to clearly and slowly greet others and self-introduce
- **Assign a topic for discussion during subsequent sessions**
 - Ask the individual to come prepared for the next session with a systematic, well-planned, and well-organized presentation
 - Instruct the individual not to hurry, pause to collect thoughts, and give a systematic speech on the topic
 - Reinforce organized expression of ideas on the topic
 - Control the rate if the individual goes too fast or uses an erratic rate
- **Teach self-control (self-monitoring) skills**
 - Ask individuals to rate their own intelligibility; compare the self-rating with your rating
 - When you gesture lack of comprehension, pause to see if the individual would modify expressions; if not, draw attention to the missed sign
 - Ask the individual to chart errors along with you; help improve accuracy of self-charting
 - Use procedures described under Self-Control (Self-Monitoring) Skills
- **Treat a coexisting stuttering**
 - Target stuttering if it is a coexisting disorder, which is likely in many cases
 - Evaluate whether a slower speech rate reduces dysfluency frequency
 - If a high dysfluency rate (5% or more of words spoken) persists, use one of the stuttering treatment procedures
 - Consider the Contemporary, Comprehensive Treatment Procedure for Stuttering in Older Children and Adults described under Stuttering
 - Consider the Direct Stuttering Reduction Strategy: Pause-and-Talk (Time-Out) for an older child or an adult
 - Consider combining Pause-and-Talk with a slight prolongation of only the initial syllable of phrases and sentences for an older child or adult

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- Use the Direct Stuttering Reduction Strategy: Response Cost for a younger child (perhaps under age 10, but experiment informally)
- **Implement a maintenance program**
 - Train family members to prompt and praise a slower speech rate at home
 - Work with teachers to have them prompt and reinforce appropriate speech and language skills
 - Follow up with the individual and give periodic booster treatment if necessary

Completion. Use this sentence completion method in treating language disorders in adults and children; it is the same as Cloze Procedure and Partial Modeling; for instance, presenting two pictures, one showing a small house and the other a big house, you might say, “This is a small house. But this one is a . . .” and look expectantly; the individual is expected to complete the sentence by saying “big house.”

Complex Communication Needs. Individuals with complex communication needs tend to have a severe single disability (e.g., severe cerebral palsy, autism spectrum disorder, intellectual disability) or a combination of multiple disabilities (e.g., autism and intellectual disability); intervention for and rehabilitation of persons with complex communication needs are equally complex; often verbal speech-language training needs to be combined with Augmentative and Alternative Communication (AAC).

Conditioned Generalized Reinforcers. Use these powerful reinforcers to teach various skills, especially to children; they include Tokens, money, and opportunities to indulge in favorite activities; all are effective reinforcers in a wide range of conditions; token reinforcement is effective in teaching speech sound disorders and stuttering in young children; they are Secondary Reinforcers that have a generalized effect; use them to:

- Promote generalized productions of target behaviors
- Enhance the effectiveness of the reinforcers used in treatment (e.g., combine tokens with verbal praise)

- Exchange tokens for a prize the child will have selected at the beginning of the treatment session; see Tokens; see also Direct Stuttering Reduction Strategy under Stuttering for an example of its application; see Response Cost to understand the use of token reinforcement in teaching various target skills

Conditioned Reinforcers. Use these kinds of reinforcers when primary reinforcers (e.g., food and drink) are not appropriate or not needed; conditioned reinforcers are events that reinforce behaviors because of past learning experiences (e.g., verbal praise or tokens); the same as Secondary Reinforcers; they are effective except for certain kinds of mands (e.g., request for food or drink cannot be reinforced by such conditioned reinforcers as verbal praise; they can be reinforced only by primary reinforcers—the requested food and drink); see Unconditioned Reinforcers.

Conditioning and Learning. Use this behavioral method of treatment to teach virtually any type of communication skill; the method involves selecting and strengthening behaviors in individuals; technique of teaching new behaviors; techniques to decrease or increase the probability of existing behaviors by arranging different consequences for behaviors:

- **Avoidance conditioning:** Naturally learned behaviors that help avoid aversive conditions, stimuli, and persons; once learned, hard to extinguish; often found in persons who stutter (e.g., avoiding speaking on a telephone); modifying such avoidance behaviors is a treatment target
- **Classical conditioning:** Also known as Pavlovian conditioning, classical conditioning involves systematic pairing of two stimuli—one, an unconditioned stimulus (UCS) and the other, a conditioned stimulus (CS) so that the CS begins to elicit the response typically elicited by the UCS; most fears and anxieties associated with speech or speaking situations are learned through classical conditioning; pairing relaxation with feared speech situations in persons who stutter is an example of the use of the classical conditioning principle in treatment

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- **Operant conditioning:** Creating, shaping, selecting, strengthening, or weakening behaviors of an individual by arranging different consequences for those behaviors; the most researched and effective techniques known to teach new skills, including communication skills; includes positive reinforcement, negative reinforcement, differential reinforcement, shaping, prompting, modeling, verbal corrections, discrimination training, generalization, and so forth

Conductive Hearing Loss. To treat communication deficits associated with inefficient conductance of sound to the middle or inner ear due to the abnormalities of the external auditory canal, the eardrum, or the ossicular chain of the middle ear, see [Hearing Loss](#).

Conduction Aphasia. To treat this type of fluent aphasia, see [Aphasia](#) and [Aphasia: Specific Types](#).

Confrontation Naming. Treat confrontation naming deficits in individuals with [Aphasia](#), [Dementia](#), and [Traumatic Brain Injury](#) who typically have this skill impaired; it is naming a stimulus when asked to do so; to teach this skill:

- Select functional and individual-specific names to teach; consult with the individual and the family members for useful words (names)
- Show objects, pictures, or photographs
- Ask the common question, “What is this?”
- Model the correct response and have the individual imitate it
- Fade modeling by giving a cue (e.g., such a phonetic cue as “The word starts with a *p*” to evoke the word *pen*; such a personalized cue as “It was your last birthday gift . . .” to evoke “book”)
- Reinforce the correct responses and verbally correct the incorrect ones
- Give multiple trials on the same word
- See [Aphasia](#) for details

Congenital Palatopharyngeal Incompetence.

Velopharyngeal mechanism that cannot close the velopharyngeal port for the production of nonnasal speech sounds; to treat communication disorders associated with it, see Velopharyngeal Dysfunction; see also Cleft Palate and Voice Disorders.

Consequences. See Contingent Consequences.

Constituent Definitions. Dictionary definitions of terms with no reference to how the defined skills are measured (e.g., *the goal of treatment is better communicative competence*) contrasted with Operational Definitions (e.g., *the goal of treatment is to increase the production of selected words; to reduce specified dysfluencies to below 3% of the words spoken*).

Contact Ulcers (Contact Granuloma). Benign lesions on the posterior third of the glottal margin; possibly due to trauma, reflux, or vocally abusive behaviors; voice symptoms include low pitch, effortful phonation, and vocal fatigue.

- Do not recommend complete vocal rest or surgical treatment
- Do not recommend forced whispering
- Ask the individual to talk less
- Reduce Vocally Abusive Behaviors
- Teach the individual to speak with less effort and force
- Teach relaxed phonation and speaking
- Teach the individual to speak more softly
- Eliminate glottal attacks

Contingency. Essence of behavioral procedures for treating communication disorders; to use procedures effectively, efficiently manage the treatment contingencies; it is an interdependent relation between events or factors; it is a dependent relation between antecedents (stimuli), the individual's responses, and the clinician-provided response-contingent consequences (e.g., verbal praise or

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verbal correction) to the individual; see [Contingent Consequences](#) and [Environmental Contingency](#).

Contingent Consequences. Consequences are *contingent* when the clinician's verbal or nonverbal responses occur immediately after a client's response; during treatment, consequences (e.g., verbal praise or verbal correction) are what the clinician says or does immediately following a client's response; in the beginning stages of teaching, behaviors change only when the consequences closely follow them; delayed consequences are not response-contingent and may be ineffective, although after a few initially successful trials, delays between responses and consequences may be shaped to strengthen responses.

Contingent Queries. Questions the clinician asks immediately following a client's unclear or truncated statement during language therapy; a technique to encourage more specific or elaborate responses.

- Ask a question immediately following an unclear or telegraphic response from the child (e.g., the child says, "Kick ball"; you ask, "Who is kicking the ball?")
- Vary the type of question you ask ("Do you want to kick the ball?" or "What is happening to the ball?") to encourage different kinds of elaborations or differently clarified productions from the child

Continuous Airflow. Use continuous airflow as a stuttering treatment target while using either the regulated breathing or a comprehensive fluency shaping in which it is an element; require the individual to maintain uninterrupted and even airflow throughout an utterance; for procedures, see [Stuttering, Treatment of Stuttering: Specific Techniques or Programs](#).

Continuous Reinforcement. To establish a target behavior in the beginning stage of treatment, reinforce every occurrence of a response; effective in establishing new skills, but inefficient in maintaining already established skills; therefore:

- Use this schedule only in the initial stage of treatment
- Gradually shift from continuous to Intermittent Reinforcement

Contrastive Stress Drills. A treatment method used to promote both articulatory proficiency and natural prosodic features, especially the stress and rhythm aspects of spoken language; used in treating Apraxia of Speech (AOS) in Adults; using different phrases and sentences, the clinician teaches the individual to stress different words; stressed words may be used to promote articulatory proficiency or simply to vary prosodic features of speech.

- **Improve articulatory proficiency**
 - Construct phrases and sentences preferably with a single target sound in them (e.g., “My name is Peter” for /p/; “Sam did it” for /s/)
 - Ask a series of questions such that the individual will respond with the target phrase placing extra stress on the target word (e.g., “Is your name Tom?”; individual will respond “No, my name is *Peter*”; the individual is likely to stress the word *Peter*, especially the initial sound, and thus improve the articulatory precision of /p/; similarly, ask, “Tom did it?”; the individual will respond “*Sam* did it”)
 - Reinforce articulatory proficiency
- **Teach typical prosodic features**
 - Create a series of phrases and sentences (e.g., “Tom does not read mystery novels”)
 - Ask questions that will force stress on different words in target phrases and sentences (e.g., “Does Tom read *romance* novels?” may evoke “No, Tom reads *mystery* novels.” “Does Tom *never* read mystery novels?” may evoke “Tom *reads* them all the time.”)
 - Reinforce varying stress on different words

Control Group. The group that does not receive treatment in group treatment research study and hence shows no change in the target disorder or disease; part of the group design strategy that includes an experimental group to evaluate treatment effects.

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- Select subjects randomly (see [Random Selection](#))
- Assign subjects into control and experimental groups randomly
- Alternatively, match subjects in the experimental and control groups (see [Matching](#))
- Assess the control group
- Withhold treatment to the control group while the experimental group receives treatment
- Demonstrate that the control group did not change (improve) while the experimental group did

Controlled Evidence. In selecting treatment procedures, look for controlled evidence, which constitutes data that show that a particular treatment, not some other factor, was responsible for the positive changes observed in the participant's behavior; evidence gathered through controlled experimentation with either group or single-subject design strategy; data that show that treatment is significantly better than no treatment; needed for [Evidence-Based Clinical Practice](#); one of several [Treatment Selection Criteria](#).

Conversational Probes. Use conversational probes to assess generalized production of clinically established behaviors in conversational speech and language; conversational probes help assess whether the individual has met the final goal of treatment: production of the target skills in naturalistic conversational speech.

- Take a naturalistic conversational speech sample
- Do not direct the conversation or do so only minimally to adequately sample speech or language behaviors under probe
- Count the number of opportunities (obligatory contexts) for producing the skill under probe and the number of correct production within those contexts
- Calculate the percent correct production of probed behaviors
- Give additional training at the conversational level if the adopted probe criterion is not met (e.g., 90% accuracy)

- End intervention only when the individual meets the criterion

Conversational Turn Taking. Teach this important social communication skill to children and adults with language disorders; turn taking is a social language skill and treatment target; involves appropriate exchange of speaker and listener roles during conversation; to teach appropriate turn taking; See Social Communication Skills for procedural details.

Corrective Feedback. A contingent response to a client's incorrect productions; see Verbal Correction, Non-verbal Corrective Feedback, and Mechanical Corrective Feedback.

Craniocerebral Trauma. Brain injury due to external forces; to treat communication disorders associated with it, see Traumatic Brain Injury.

Criteria for Making Clinical Decisions. Rules to make various clinical judgments during treatment; includes such decision-making rules as when to model, when to stop modeling, when to probe for generalization, and when a behavior is trained.

- Model most target behaviors for most individuals, especially in the initial stages
- Discontinue modeling when the individual gives five consecutively correct, imitated responses
- Reinstate modeling if errors persist
- Consider an exemplar of a target behavior (e.g., a target speech sound in a word, a grammatical morpheme in a sentence) trained when the individual gives 10 consecutively correct nonimitated responses
- Consider a behavior clinically trained when the individual gives 90% correct responses on untrained exemplars on Probes
- Consider a behavior functionally (socially) trained when the individual gives 90% or better correct responses in

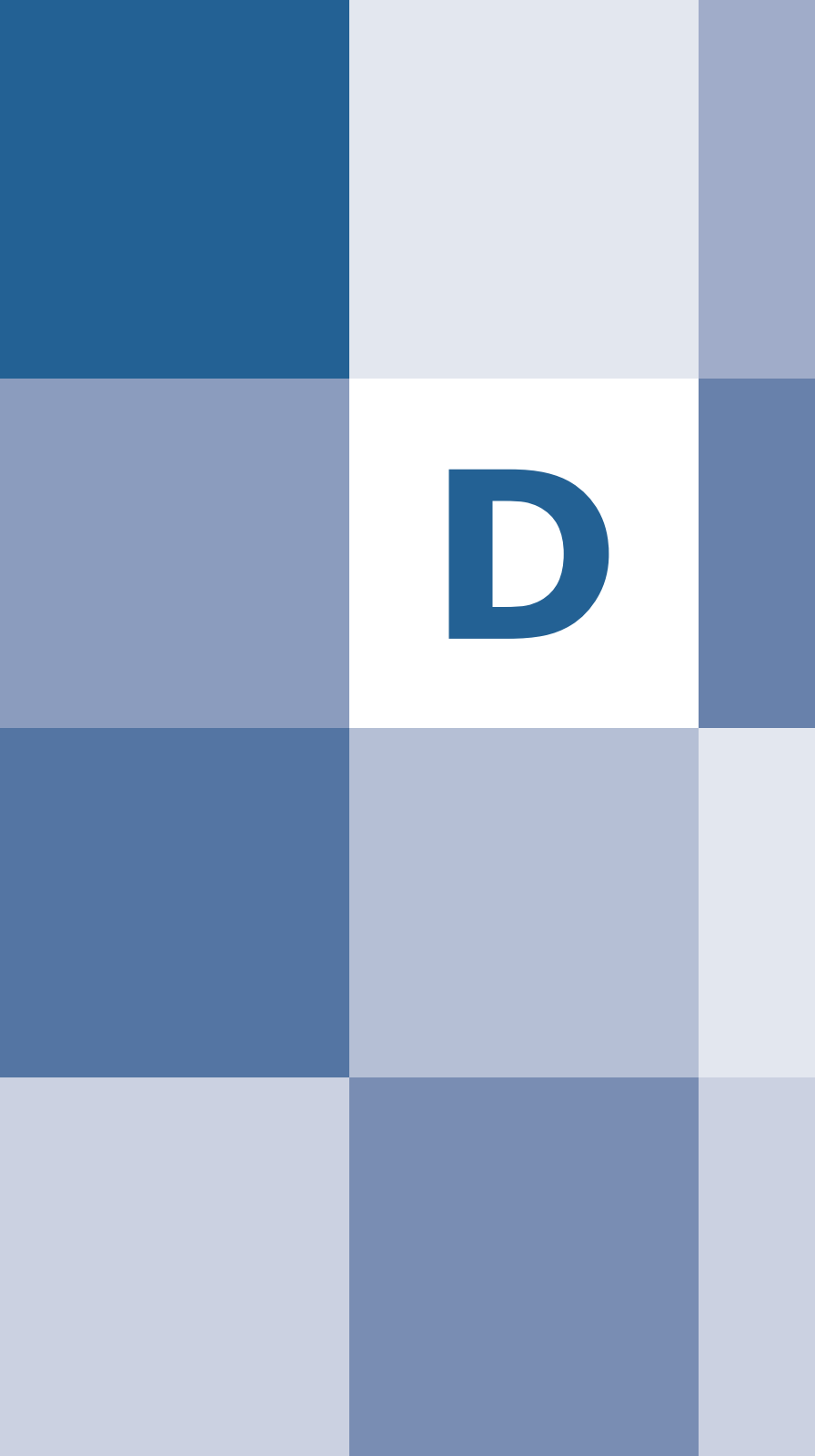
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Conversational Probes produced in clinical as well as extraclinical situations

Cued Speech. Teach this system of nonverbal communication as a supplement to speech reading; consists of eight hand shapes that represent categories of consonants and four positions about the face that represent categories of vowels; these hand shapes and positions suggest speech sounds in running speech; known to promote better reading skills in children who are deaf.

Cultural Diversity and Treatment Procedures. In designing treatment programs, consider factors related to ethnic background, culture, and language status that may influence treatment outcome; select individual- and family-specific treatment targets that are consistent with the family's ethnocultural and language background; use unbiased treatment materials; select treatment targets that are consistent with the patterns found in a bilingual child's first language; see Ethnocultural Variables in Treatment.

Cysts. Acquired or congenital, fluid-filled lesions of the vocal folds; vocal abuse; often removed surgically.



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Delayed Auditory Feedback (DAF). A procedure to slow down the speech rate in persons who stutter or clutter and those who have dysarthria; speaker wears headphones attached to a DAF machine and speaks; the machine feeds the speech back to the speaker after a delay, causing a slower and monotonous-sounding speech; see [Cluttering](#), [Stuttering](#), and [Dysarthria](#).

- Select one of the several DAF machines available on the market
- Experiment with different durations of delay that induce slower speech that is free from stuttering or cluttering or improves intelligibility in speakers with dysarthria
- Train and stabilize the target speech skills with the selected delay
- Fade DAF and reinforce more typical rate and normal prosodic features

Dementia. Continued research supports vigorous efforts to treat the behavioral and communication deficits in people with dementia to improve their quality of life and to offer sustained support to the family members; a combination of behavioral intervention, physical exercise, and the latest and effective medical treatment holds the greatest promise; sustained clinical efforts and family intervention may slow down the rate of regression to the extent possible; in all cases, both the individual and their family need counseling, support, and direct intervention; before starting treatment, make a thorough assessment of dementia and the family communication patterns and needs; see the cited sources.

Counsel and Educate the Family Members.

Dementia in an individual produces emotional, physical, and behavioral consequences for the family members and other caretakers; therefore, counsel and educate them about dementia.

- **Educate the family members about dementia and its intervention**
 - As a first step, ascertain their knowledge about dementia, ways of handling the current difficulties, and their plans for future problems

- Give information on the causes, symptoms, and course of dementia and the person's specific type of dementia; discuss, do not lecture
- Give an overview of general medical, rehabilitative, and behavioral management procedures; intertwine your information with that of theirs
- Give family members a realistic picture of what is ahead; discuss with them, in light of what they know, the need to cope with a difficult, prolonged, and expensive problem
- **Prepare the family members for difficulties ahead:**
 - Frustration, worry, anxiety, and withdrawal, especially in the initial stages
 - Memory deficits
 - Abusive, aggressive, and violent behaviors in later stages
 - Potential emotional outbursts and angry exchanges over trivial matters; catastrophic reactions
 - Delusions and hallucinations
 - Suspicious behaviors and baseless accusations
 - Uncooperativeness at bathing and at mealtimes
 - Communication problems
 - Unreasonable demands
 - Unsafe driving, cooking
 - Hiding things
 - Daytime wandering and night walking
 - Point out the need for vigilant supervision or institutionalization of the individual in the later stages of dementia
 - Let them appreciate the eventual need to feed, clothe, and take total care of the person; subsequent incontinence that will increase the burden of care; extended period of home care (from 1 to 2 years to 10 to 15 years); and the need for emotional, social, and financial help for themselves (the caregivers)
 - Alert the family members to potential abuse of individuals, especially older women, with dementia
- **Let them appreciate the worthwhileness of rehabilitation**

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- Inform the family members that rehabilitative efforts are still worthwhile to slow down deterioration in the physical, social, and intellectual status of the person with dementia
- Let the family members and caregivers know that a person with dementia may be capable of making decisions in the early and middle stages of the disease
- Let them understand that some individuals with Alzheimer's disease may have a slow progression with several years of relatively stable behavior patterns, suggesting both a need and the usefulness of prolonged rehabilitation efforts
- **Help them find community support**
 - Discuss the family members' financial capabilities to meet the cost of long-term care of the individual with dementia; help them find and contact local, state, and federal agencies and private organizations that offer financial support to families affected by dementia
 - Offer them help in finding appropriate nursing home placement when the time comes
 - Give them consumer-oriented printed information on dementia and management of communication problems
 - Give them names and addresses of local professional and social organizations concerned with aging and dementia
 - Introduce them to local and online support groups
 - Inform them about accessible specialty clinics, research centers, and universities where unique intervention programs or newer medical treatment evaluations are being conducted
 - Guide them to access information on websites that offer suggestions, newsletters, and information on new developments in medically and behaviorally managing persons with dementia
 - Refer them to other specialists as needed (e.g., neurologists, psychiatrists, psychologists, social workers)

Clinical Management of Individuals With Dementia. As a care team member, design a program to help the person with dementia sustain skills and behaviors to the extent possible and cope with progressive deterioration in skills and behaviors; teach compensatory strategies.

Minimize the Effects of Memory Problems

- To help persons with failing memory, design various kinds of memory aids, including calendars, cue cards, diaries, written to-do lists, weekly planners, written daily reminders, watch alarms, timers, vibrating signals, prominently displayed display information, orientation boards and signs, and so forth; see also Memory Impairments
- Design **memory wallets**, a set of index cards on which various kinds of personal and useful information may be written; teach the client to frequently consult the memory wallets
- Use the **spaced retrieval** procedure in which you teach the person to recall specific information with progressively longer intervals between recalls; for example, ask the client, “What do you do when you want to know what time is your doctor’s appointment?” The client says, “I look into my list of daily activities.” Reinforce the response, and then talk on some other topic (for a specific duration, such as 1 minute, 2 minutes, and so forth); then ask the question again; increase the duration between the repeated questioning
- Direct individuals who repeatedly ask questions seeking information of various sorts to memory books, index cards, and bulletin boards where such information is made available
- Teach the client to write down instructions and needed information; teach the individual to ask others to write down information for their use
- Give **verbal and nonverbal prompts** that remind the individual of specific activity (e.g., a social gathering, a game, a party), which may be more

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effective than other kinds of reminders (e.g., a bright poster printed in large letters)

- **Reinforce the individual** for following through because just making materials and opportunities available may not be effective
- Draw attention of institutionalized individuals to reminders, signs, posters, announcements, and scheduled activities; reinforce their attending behaviors
- Train individuals to use, and prompt them to consult, a diary that they keep about appointments and schedules of events; prompt them to consult their diary and praise them when they comply
- Have caregivers in an institution wear name tags printed in larger letters and train the caregivers to draw attention to their name tags, and have individuals read them or caregivers read them aloud; train the caregivers to reinforce the individuals for saying the name or reading it aloud

Help Manage the Daily Activities

- Teach individuals to make a written list of *what to do* every day; train the individual to use the list by frequent prompts and verbal praise; train caregivers to prompt the individual to consult the list at appropriate times
- Teach the individual to use portable alarms that remind them of scheduled activities; monitor the use of the devices and reinforce the individual for maintaining this skill
- Give **written instructions** on daily living chores (closing the windows, locking the doors, turning the stove and the lights off); teach the individual to follow the instructions and reinforce them for doing so
- Train staff members in health care facilities to give frequent and systematic reminders to the individuals and to reinforce the individual for performing daily activities
- Teach the individual to **rehearse information** (e.g., just before entering and leaving a party,

rehearsing the names of people at a party or class reunion; visualizing the faces of people)

- Teach the individual to **self-monitor**; reinforce the skills and evaluate their maintenance
- Create a **naming wallet** containing pictures of family members, their names, and sentences and phrases about them, and train the individual to carry it and use it
- Teach the individual to keep personal belongings (keys, clothing items, eyeglasses, pens) in a specific, invariable place; frequently monitor and reinforce this skill
- Teach the individual to keep **related objects together** (e.g., paper and pencil; socks and shoes; coffee and sugar); monitor and reinforce this skill
- Train the individual to **carry an identification card** that contains the name, address, and telephone numbers of a family member and a health care professional; frequently check the individual to see if this is maintained
- Teach the individual to wear a bracelet that contains personal identification; check its continued use
- Instruct the individual to exploit their strengths to compensate for weaknesses (e.g., writing down everything when memory tends to fail)
- Teach individuals to ignore relatively minor problems (e.g., forgetting certain words)
- Encourage the individual to get enough rest and sleep to reduce the effect of fatigue on memory

Control Disorientation and Confusion

- Place a large calendar in the client's bedroom and cross off the current date every night; draw the individual's attention to this activity and tell the individual what you are doing
- Make the individual wear a **digital calendar** watch with a large display of AM and PM, date, and day; frequently draw attention to it; have the individual use it in realistic situations (e.g., instructing the individual to look at the watch and saying that it is now 12 noon and time to have lunch)

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- Ask questions about time and place (orientation), model the correct responses, and verbally reinforce correctly imitated responses; gradually fade the modeling to stabilize responses to questions
- Provide **maps of frequently visited places** (e.g., homes of relatives or friends, doctor's office, shops, favorite restaurants); before visiting the places, prompt the individual to consult the maps

Work With Other Professionals

- Work with physical therapists to design and promote strength and fitness training
- Work with social workers and volunteers to promote social interactions at social organizations
- Work with recreational therapists to offer supervised recreational activities

Offer Direct Intervention for Communication and Related Problems

- Do not prejudice the usefulness of treatment; offer direct intervention and measure progress
- Provide cues to evoke words and then teach self-cueing techniques, especially in the early stages of dementia; see [Aphasia](#) for details
- Give phonemic cues for words (e.g., say, "The word starts with an s" to evoke *spoon*)
- Give such semantic cues as a generic class (e.g., say, "woman" to evoke the word "wife"), a synonym (e.g., say, "dwelling" to evoke "house"), an antonym (e.g., say, "good" to evoke "bad"), a category name (e.g., say, "it is a fruit" to evoke "apple"), or an associated word (e.g., say, "cup" to evoke "plate")
- Train the individual to describe an object and then naming it
- Teach self-cueing to produce the intended word
- Provide positive reinforcement and verbal correction, known to be effective in modifying communication in individuals with dementia

Teach Social Communication Skills

- Target such skills especially in the earlier stages of dementia; see [Social Communication Skills](#) for details

- Reinforce turn taking, eye contact, and topic initiation, and topic maintenance in conversation
- Use Script and Script Therapy; have the individual write a few sentences about their personal experiences (hobby, travel, work, movie, grandchildren), read them repeatedly, and then orally narrate the script; fade the scrip to maintain more spontaneous storytelling
- Ignore inappropriate, irrelevant, vulgar, delusional, and any other type of unacceptable or bizarre verbal behaviors and reinforce any appropriate verbal responses (differential reinforcement)
- Reinforce the individuals continuously (reinforcement for all desirable responses); if needed, reinforce the individuals with tangible reinforcers
- Teach gestures as a means of communication even before the oral skills begin to fade so that nonverbal means of communication are established
- Teach Augmentative and Alternative Communication; train the person to use such no-technology devices as a communication board, a technology device such as a smartphone or a tablet to achieve or maintain basic communication skills

Conduct Group Therapy Sessions

- Include family members or other individuals with similar problems to sustain social and communication skills as long as possible
- Direct the group sessions and manage the behavioral contingencies to promote the selected goals of the group session; provide refreshments during these sessions
- Have the individual tell stories, maintain topics of conversation, take conversational turns, and express their own feelings and thoughts about their difficulties
- Arrange for generous verbal reinforcers for all targeted expressions and any other appropriate expressions
- Ignore inappropriate responses and teach the group members to do the same

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- Reinforce continued use of such memory aids as memory wallets

Sustain Skills to the Extent Possible

- Progressively simplify the task, depending on the individual's performance
- Encourage the individual to continue to cook but with support
- Arrange for cooking simpler foods
- Sustain reading skills with routine materials (e.g., TV listings, newspapers, labels on packaged foods, restaurant menus)
- Encourage participation in social and family group activities as long as possible
- Get the individual to continue the previous social activities as long as possible (e.g., volunteering in social clubs, seeing movies with friends)
- Continue to reinforce previously established physical exercise programs (resulting in improved mood and a sense of physical well-being)
- Reinforce the individual's effort to sustain skills and social behaviors

Management Strategies for the Family Members and Other Caregivers

Help Create a Supportive Home Environment

- Be observant about changes in the individual's behavior that might signal a significant change in health status
- Help sustain the individual's physical health by making sure that the individual takes the prescribed medications, eats properly, exercises regularly, and gets enough sleep
- Reduce distractions
- Design and maintain a consistent routine for the individual
- Reassign household chores the individual cannot perform anymore, but do not expropriate the individual's responsibilities too soon
- Design and maintain a constant and simplified environment

- Let the individual have a say in arranging or rearranging their personal space (room design, color, furniture placements, decorations)
- Encourage the individual to be responsible for maintaining their personal space for as long as possible
- Allow the individual some privacy (do not relentlessly supervise when it is unnecessary)
- Use contrasting colors to enhance orientation to handrails, hallways, communication boards, table settings, and room decorations; train the individual to make use of them

Support Communication and Social Participation

- Create and maintain communication opportunities for the individual
- Incorporate as much relevant speech as possible with daily routines
- Do not restrict communication because of the fear of irrelevant speech
- Remove unduly restrictive rules against talking and social interactions in institutional settings
- Find regular conversational partners for the individual at home and in institutional settings; recruit friends, grandchildren, family members who do not see the individual regularly, neighbors, former colleagues, and volunteers (especially in an institution)
- Help the individual maintain contact with grandchildren and other younger acquaintances; work with both the younger persons and the individual to foster this relationship; if possible, include them in informal therapy sessions
- Encourage individuals to observe social activities even if they do not participate in them; gently guide them to participate
- Encourage individuals in the early stage of dementia to help with daily activities at home and institutions (e.g., helping spouses in cooking, cleaning, or shopping; helping new individuals with institutional routines)
- Arrange regular small group activities at home; facilitate and reinforce the individual's participation

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- Find similarly bilingual conversational partners for the bilingual person with dementia
- Ask family members to regularly visit their relative in an institution
- Have family members participate in social events at the institution
- Include family members in social activities arranged at the institution

Use Helpful Communication Strategies

- Approach the individual slowly, with calm and inviting expressions, and within their visual field (do not surprise the individual)
- Establish eye contact before speaking
- Let the individual see your face clearly while talking (stoop down if the individual is in a wheelchair)
- Supplement speech with gestures, smiles, and posture
- Speak clearly and directly
- Speak in simple terms
- Use requests instead of commands (e.g., “Should we do this together?” and “Would you give me a hand and help?”)
- Specify referents for speech (e.g., “Let’s go to the dining hall” instead of “Let’s go over there”); use proper names (e.g., your wife Jane, physical therapist Tom) instead of pronouns (e.g., *he, she, they*)
- Have only one or two people engage the individual in conversation at any one time
- Do not argue with the individual; instead, rearrange the environment (e.g., do not try to convince a suspicious person that no one has taken their possessions; instead, make sure that the possessions are always kept in one predictable, easily accessible place)
- Avoid asking open-ended questions; instead, ask yes/no questions, either/or questions, or short and simple questions
- Be specific and direct in your communication with the individual

- Avoid vague references, sarcasm, humor, indefinite referents, analogies, similes, proverbs, and statements with double or abstract meanings
- Say only a little at a time and make sure the individual understands before saying more
- Repeat questions if necessary
- Be redundant, repeat, restate, and paraphrase your descriptions and explanations
- Talk about familiar and concrete topics and directly observable objects
- Use photographs and drawings to improve understanding
- Touch the individual while talking with them; draw the individual's attention and improve communication with gentle and affectionate touch
- Praise the individual frequently for appropriate behaviors
- Say goodbye, wave, or give other departing signals
- Use the same phrase to suggest daily routines (e.g., "Let's eat," "Let's go to bed now")

Attend to the Sensory Needs of the Individual

- Have the individual's vision checked and provide new glasses if needed
- Provide magnifying glasses and printed material with larger print to sustain reading skills
- Place all visual information at the individual's eye level
- Keep the individual's living environment visually attractive
- Have the individual examined by an otologist and take necessary steps to maintain the individual's otological health
- Have the individual's hearing checked by an audiologist and provide hearing aids as recommended
- Monitor the use of hearing aids on a daily basis; offer assistance in inserting and removing the aids and in volume adjustments

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- Discuss with an audiologist the need for Assistive Listening Devices described under Aural Rehabilitation and follow the recommendations
- Reduce or eliminate noise in communicative situations (e.g., turn off the TV or radio while talking to the individual; monitor such mechanical noises as those of air conditioners and fans)
- Evaluate whether the individual whose visual and auditory acuity is diminished reacts better to improved tactile and olfactory cues

Reduce Conditions That Trigger Problem Behaviors

- Observe and record situations that lead to problem behaviors
- Look for early warning signs of emotional or aggressive outbursts (e.g., body rigidity, a certain look, crying)
- Reduce physical pain and discomfort
- Minimize the side effects of medicines
- Teach functional communication, including Augmentative and Alternative Communication skills (deficient communication skills may trigger aggressive behaviors)
- Engage the individual in a distracting and productive activity
- Reduce difficult demands; do not insist on remembering useless facts
- Limit choices about food and clothing to minimize confusion
- Take over the individual's actions when necessary but perform them discretely (e.g., pay the bills or manage finances when the individual is not in sight)

Support Nutrition and Eating

- Feed all meals in a constant place with no distractions
- Place food in clear view of the individual
- Place the eating utensils in the individual's hand
- Get the individual's attention and model eating
- Manually guide the act of eating (use touch and hand guidance)

- Teach the individual to pace eating (same time between bites)
- Offer the preferred beverage
- Offer finger foods as often as practical and nutritionally appropriate
- Offer plenty of social reinforcers for eating behaviors

Take Safety Precautions

- Install sensors under the rug in the house to monitor inappropriate roaming
- Install complicated locks on doors to prevent the individual from leaving the house and getting lost
- Install stair gates
- Ensure adequate lighting throughout the house, including hallways and bathrooms, and day and night
- Install refrigerator locks and automatic shutoff and locks on the cooking stove
- Secure knives, scissors, blenders, grinders, and weapons in the house
- Secure hand or power tools (saws, pruners, hammers) in the garage and shed

Support Adequate Sleep and Rest

- Reduce the frequency and duration of daytime naps
- Put the individual to bed at the same hour every night
- Require and encourage the individual to exercise every day for about 30 minutes
- Feed the individual a light snack an hour before the regular bedtime
- Make the individual use sleepwear that does not disturb sleep by twisting or binding around the body
- Close the windows and doors to keep the bedroom quiet
- Maintain a nightlight to avoid confusion or anxiety in the night when the individual wakes up

Suggest Self-Sustaining Measures to the Family Members

- Join or form a support group of families that have individuals with dementia

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- Take breaks from caregiving to renew themselves (respite care)
- Maintain their hobbies and recreational activities at a practical level
- Recognize their own need for professional help and seek it (e.g., counseling)
- Seek financial support to meet the cost of caring for the individual at home and to be able to take a break from their demanding caregiver duties

Bayles, K. A. et al. (2018). *Cognitive-communication disorders of MCI and dementia: Definition, assessment, and clinical management* (3rd ed.). Plural Publishing.

Hegde, M. N. (2022). *A coursebook on aphasia and other neurogenic language disorders* (5th ed.). Plural Publishing.

Hegde, M. N. (2026). *Hegde's pocketguide to assessment in speech-language pathology* (5th ed.). Plural Publishing.

Swan, K. et al. (2018). Speech-language pathologist interventions for communication in moderate-severe dementia: A systematic review. *American Journal of Speech-Language Pathology*, 27, 836–852.

Demonstration. Showing how a target response is performed after giving instructions.

- Describe the target behavior for the individual
- Model the response
- Show how the response is produced (e.g., the tongue position for the production of /t/)
- Give maximum feedback (use a mirror if necessary)
- Reinforce the correct response or an approximation of it

Denasality. The same as Hyponasality.

Deteriorating Baselines. Typically, baselines of target behaviors should be stable before treatment; if variable, baselines should be repeated until they are stable; a single exception to this rule is *deteriorating baselines* that show the problem is getting worse; desirable behaviors (e.g., fluency) may be lower each time they are measured; in such cases, introduce treatment immediately.

Developmental Apraxia of Speech (DAS). The same as Childhood Apraxia of Speech.

Dialect. A spoken form of a language with its own sound patterns, word usage, grammar, and social communication patterns; carefully consider the need and appropriateness of intervention to change a speaker's dialect because dialects are natural and culturally rooted variations of a language; even when a particular variation (e.g., the general American English dialect) is considered standard, all variations are acceptable forms of communication; dialects of the secondary language in bilingual speakers may be influenced by their primary language; African American English dialect is the product of a cultural and verbal heritage; such dialectal variations are not a basis for diagnosing a communication disorder; see Ethnocultural Variables in Treatment for more detailed guidelines on treating children and adults with varied dialects.

Differential Reinforcement. Treatment procedures that help establish discriminated responding by reinforcing a response in the presence of one stimulus and not reinforcing the same response in the presence of another stimulus; it is also a collection of indirect methods of reducing undesirable behaviors by increasing another, desirable behavior; all are especially effective in reducing undesirable behaviors in nonaversive ways because the methods reinforce desirable behaviors that replace undesirable behaviors; use the following evidence-based techniques to increase desirable behaviors while decreasing undesirable behaviors simultaneously.

Differential Reinforcement of Alternative Behaviors (DRA). An undesirable behavior is reduced by reinforcing a specified desirable behavior that serves the same function as the one to be reduced; also known as Functional Equivalence Training.

- Find out what function (purpose) the undesirable behavior to be reduced seems to serve (e.g., fussing in treatment sessions may serve the purpose of requesting help when the task for the child is too difficult and cannot verbally request help)

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- Select a behavior that is a desirable alternative to the behavior to be reduced (e.g., the response “help me” may serve the same function as fussing)
- Reinforce the production of the alternative, desirable response (e.g., teach the child to say “help me” instead of fussing)

Differential Reinforcement of Incompatible Behaviors (DRI).

An undesirable behavior is reduced by reinforcing a behavior that is incompatible with the one targeted for reduction.

- Specify the behavior to be reduced (e.g., leaving the chair and walking in the therapy room)
- Specify a behavior that is incompatible (e.g., sitting quietly and looking at the stimulus items presented)
- Systematically reinforce the child for exhibiting the desirable behavior (for sitting quietly and looking at the stimulus items) that is incompatible with the undesirable
- Suspend training on the target communicative skill for a while if necessary and until the prerequisite desirable behaviors (e.g., quiet sitting) are stabilized

Differential Reinforcement of Low Rates of Responding (DRL).

An undesirable behavior is reduced by reinforcing its progressively lower frequency of occurrence; the method shapes down an undesirable behavior.

- Specify the undesirable behavior to be reduced (e.g., interrupting treatment by irrelevant questions)
- Specify an acceptable level of the undesirable behavior (e.g., two questions in a 10-minute period)
- Reinforce the individual for not exceeding the set level (“Good! You asked only two questions during the last 10 minutes!”)
- Specify a new, more stringent criterion in successive stages until the behavior is eliminated or kept to a minimum

Differential Reinforcement of Other Behaviors (DRO).

An undesirable behavior is reduced by rein-

forcing any one of many unspecified behaviors; the behavior that will not receive reinforcement is clearly stated.

- Specify the undesirable behavior to be reduced (e.g., leaving the chair and walking around)
- Tell the individual that there will be no reinforcers for that behavior; also tell the person that you will give a reinforcer as long as the undesirable behavior is not exhibited
- Periodically reinforce the child's desirable behavior and for not exhibiting the undesirable behavior (perhaps for sitting quietly, reading, coloring, working on other assignments, but none specified as the response to be reinforced)

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.). Plural Publishing.

Digital Manipulation. Physical manipulation of the larynx during voice therapy to produce changes in vocal quality; for the procedure, see [Voice Disorders: Specific Normal Voice Facilitating Techniques](#).

Direct Language Treatment Approaches. Clinician-implemented language intervention with specified target behaviors, initially tightly structured and subsequently flexible treatment sessions, modeling-imitation-reinforcement-verbal correction sequence, and more spontaneously produced language targets; see [Language Disorders in Children](#).

Direct Methods of Response Reduction. Procedures to reduce undesirable behaviors by directly placing a contingency on them; contrasted with [Indirect Methods of Response Reduction](#).

- Specify the undesirable behavior to be reduced
- Place one of the following contingencies on it:
 - [Verbal Correction](#) (e.g., say “No”)
 - [Time-Out](#) (say “Stop,” turn your face away for 5 seconds and then reestablish eye contact and resume conversation)

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- Response-Cost (give a token contingent on every correct response, and take back a token contingent on every incorrect response)
- Extinction (ignore the response)
- Imposition of Work (ask a child who disrupts your stimulus materials to reorganize them for you)

Direct Stuttering Reduction Strategies. Use these treatment techniques to reduce stuttering by placing behavioral contingencies directly on stuttering itself; effective procedures include Pause-and-Talk and Response Cost; for procedural details, see Stuttering.

Discrete Trials. These are temporally separated opportunities for target response production; each trial ends with a brief pause, during which the clinician records the client's response and initiates the next trial soon after; used to baserate, treat, and probe for generalization; efficient in establishing responses and to measure their frequency to calculate the percent correct responses, but they are not good at promoting generalization and maintenance; see Baseline Evoked Trials and Baseline Modeled Trials, both described under Baselines; see Treatment Modeled Trials and Treatment Evoked Trials, both described under Treatment Discrete Trials; see also Probes.

Discrimination. Teach discrimination between speech or language elements that are confused but need to be distinguished; a behavioral method of establishing different (and appropriate) responses to different stimuli; opposite of generalization in which similar responses are given to different (though similar) stimuli; needed to teach such discriminated responding as plural words to plural stimuli and singular words to singular stimuli; nasal resonance on nasal sounds, but oral resonance on oral sounds; and so forth.

Dysgraphia. To treat writing problems associated with recent neurological injury and unrelated to poor instruction or lack of schooling, see Agraphia and Aphasia: Treatment of Writing Problems.

Dysarthrias. A motor speech disorder in which the peripheral or central nervous system damage impairs muscular control of the speech mechanism; respiratory control for speech, articulation, resonance, and prosodic features are all potential treatment targets; this main entry gives the general guidelines, treatment targets, and procedures that apply to dysarthrias in general; a subsequent section, Treatment of Dysarthria: Specific Types, lists unique treatment targets for each type; see the sources cited at the end for details on dysarthria and its diagnosis; conduct a thorough assessment of dysarthria and its type before beginning treatment.

Treatment of Dysarthria: General Guidelines

Counsel the Family and Select Treatment Targets and Procedures

- Counsel the client and family members about the course of the disorder, prognosis, and treatment plan for the individual as well as the family members
- Evaluate the increasing evidence for treatment of dysarthria and select experimentally evaluated procedures; most treatments are packages of different elements whose individual effects may be masked
- Finalize the treatment plan only after a thorough discussion with the client and family members; consider their goals and preferences for treatment
- Prioritize functional communication, speech intelligibility, improved prosodic features, and social participation to enhance the quality of life and reduce social isolation
- Increase the client's motivation for treatment by contrasting improved intelligibility at a slower speech rate and reduced intelligibility at a faster rate; videotape the client's speech and have them watch to appreciate the difference
- Be fully knowledgeable about medical, surgical, pharmacological, and prosthetic management, their limitations, and how they affect communication training
- Consider the complicating neurodegenerative disease associated with dysarthria, associated health

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problems, and their prognosis in planning treatment timing and intensity

Structure the Treatment

- Start management early, but wait until the end of the acute phase and the individual is more tolerant of treatment efforts
- Provide treatment frequently; three times a week is preferable
- Organize sessions to move from easy to difficult tasks

Use Effective Behavioral Procedures

- Use instruction, demonstration, modeling (followed by imitation), shaping, prompting, fading, differential reinforcement, verbal correction, and other proven behavioral management procedures
- Give intensive, systematic, and repetitive treatment trials, but consider the client's tolerance levels and fatigue in treatment sessions
- Use Phonetic Placement and its variations to teach better articulation skills
- Teach Self-Control (Self-Monitoring) skills
- Provide immediate, specific, and social and natural reinforcement and verbal correction
- Use instrumental feedback or biofeedback when necessary
- Give repeated trials on the same target skill in the initial stages; subsequently, give trials on different speech tasks
- Emphasize accuracy of speech sound production initially at a slower rate of speech, and as accuracy improves, target increased speech rate while still maintaining intelligibility
- Restore lost function to the extent possible; teach compensatory behaviors to replace lost or reduced functions
- Avoid nonspeech exercises designed to increase oral muscle strength because of lack of evidence; target muscle strength in the context of speech production

- Teach Augmentative and Alternative Communication (AAC), if necessary
- Design a generalization and maintenance plan by including formal or informal treatment sessions involving other communication partners; consider holding treatment sessions in more naturalistic settings; teach communication partners to support the individual's newly established communication skills

Communication Strategies for the Client and Family Members

- Teach the client to tell others how to communicate with them (e.g., showing others how to use of a communication board or a tablet)
- Train the individual to set the context and topic before beginning a conversation
- Train the individual to communicate with any and all means (short verbal utterances, gestures, augmented and alternative means)
- Teach the individual to monitor listener comprehension
- Teach significant others to modify the physical environment, be active listeners, and maximize their own hearing and visual acuity
- Teach the individual and significant others to maintain eye contact, establish effective communication strategies, and determine methods of feedback

General Treatment Targets and Procedures.

Treatment targets for persons with dysarthria include modification of problems in (a) respiratory support for speech, (b) phonation, (c) resonance, (d) articulation, and (e) prosody: most treatment procedures are behavioral; because of variable symptoms and neuropathology across subtypes, some techniques produce temporary effects; others are contraindicated for certain individuals; many techniques are suggested based on clinical experience without controlled experimental evidence; continue to use a technique only when it produces a clear and positive effect on the individual's speech; abandon ineffective procedures and modify those that seem to hold promise.

Modification of Respiratory Flow and Force

- Target improved respiratory flow and force for speech production
- Train consistent generation of subglottal air pressure; use manometer or air pressure transducer
- Train maximum vowel prolongation; reinforce sufficient loudness as the person learns to sustain the vowel production for longer durations
- Shape production of longer phrases and sentences produced on a single breath
- Shape a greater number of words and phrases said on a single breath (called *optimal breath groups*)
- Prompt the individual to inhale before speaking and reinforce deeper inhalations and more forceful exhalations to produce speech; push manually the abdomen to induce a more forceful expiration
- Teach controlled exhalation to stabilize even flow of speech
- Teach sustained exhalation throughout utterances to maintain loudness and other characteristics of speech
- Teach prompt initiation of phonation when exhalation begins to make the air supply for speech last longer and terminate speech earlier during exhalation to prevent voice fading
- Teach pausing and breathing at appropriate junctures in speech
- Teach the individual to push, pull, or bear down during speech production that will increase the breath support for speech
- Modify postures to promote respiratory support (e.g., upright sitting position, some may need a prosthetic device to keep that position)
- Use adjustable beds and wheelchairs to make postural adjustments
- Let the individual use neck and trunk braces if found to be helpful
- Use girdles and wraps around the abdominal area to increase muscle strength for respiration; do not use

this for extended periods of time, as such devices limit inhalation and pose the risk of pneumonia

- Use an Expiratory Board to stabilize the abdominal muscles for respiration
- Teach the individual to exhale slowly and in a controlled fashion after deep inhalations
- Train the individual to begin phonation as soon as the exhalation begins

Modification of Phonation

- Discuss with medical staff the need, effects, and effectiveness of medical treatments, including Pharyngeal Flap Operation, Laryngoplasty, Recurrent Laryngeal Nerve Resection, Botulinum Toxin Injection, and pharmacological measures; consider them in the total management of the individual and in treating communication disorders
- Increase vocal intensity (loudness of speech) and production of relatively longer utterances per breath
- Use biofeedback devices to give the individual immediate feedback on vocal intensity to effect changes in excessive or too little loudness; always also provide verbal reinforcement along with biofeedback; see Voice Disorders
- Shape vocal intensity to desirable levels with modeling, reinforcement, and verbal corrections; this avoids the use of biofeedback units that need to be faded out
- Train the individual with an excessively soft voice in using a portable amplification system if behavioral methods to increase vocal intensity have not been successful or as a temporary measure until the behavioral methods produce desirable results
- Train aphonic individuals in the use of an Artificial Larynx
- Ask individuals with aberrant neck movements or neck muscle weakness to wear Neck Braces
- Teach Effortful Closure Techniques for individuals with vocal cord paralysis (e.g., pulling or pushing while phonating)

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- Teach the individual to initiate phonation at the beginning of exhalation
- Teach the individual to turn the head toward the weak side during speech; try digital manipulation of the thyroid cartilage to increase loudness; be aware of temporary effects of these that need to be shaped and stabilized
- Try relaxation exercises and laryngeal massage to increase loudness
- Teach the individual to tilt the head back, initiate speech after a deep inhalation, and increase pitch to reduce strained voice quality
- Teach the individual with vocal fold hyperadduction to initiate phonation with breathy onset or a sigh

Modification of Resonance

- Discuss with medical staff the need, effects, and effectiveness of medical treatments, including pharyngeal flap surgery, botulinum toxin injection into the posterior pharyngeal wall, and palatal lift prosthesis to treat velopharyngeal incompetence
- Provide feedback on nasal airflow and hypernasality by using a mirror, nasal flow transducer, or a nasendoscope
- Train the individual to open the mouth wider to increase oral resonance and vocal intensity
- Teach the individual to articulate more precisely to increase oral resonance as well as to improve intelligibility
- Use a nasal obturator or nose clip; have the individual speak in the supine position; be aware of temporary improvement
- Avoid such nonspeech exercises as sucking, blowing, whistling and icing, stroking, or brushing the structures to improve velopharyngeal closure and to reduce hypernasality, as such exercises are ineffective

Modification of Articulation

- Discuss with medical staff the need, effects, and effectiveness of medical treatments, including Neu-

ral Anastomosis and pharmacological treatment in relation to communication training

- Analyze the error patterns and their potential reasons before developing a treatment program
- Encourage the individual to assume the best posture for good articulation
- Use the Bite Block to improve jaw control in individuals whose jaw control is severely impaired; it is a custom-fitted putty or acrylic material the individual holds between the upper and lower teeth
- Provide instructions and demonstrations
- Simplify the task; use shaping
- Model frequently for the individual to imitate
- Use phonetic placement techniques
- Reduce speech rate to improve intelligibility
- Ask the individual to exaggerate the production of medial and final consonants
- Reinforce immediately and offer verbal corrections
- Use minimal contrast pairs (e.g., *peet-beet*; *stop-top*)
- Teach self-monitoring skills
- Modify techniques in light of data
- Move from a simpler level of training to more complex levels
- Use electromyographic biofeedback to reduce hyper-tonicity and spasm of speech muscles
- Teach compensatory articulatory movements (e.g., use of tongue blade to make sounds normally made with the tongue tip)
- Use Intelligibility Drills
- Ask the individual to read texts or describe pictures you are not familiar with
- Retell what you hear
- Reinforce improved articulation to promote better listener understanding

Speech Rate Control

- Target reduced rate of speech to improve speech intelligibility of dysarthric speakers who have a more rapid or variable rate than normal (e.g., in hypokinetic dysarthria)

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- Do not teach a slower rate if the individual's speech intelligibility is acceptable
- Teach the individual to slightly increase pause durations between words, phrases, and sentences to affect improved intelligibility
- Use such prosthetic devices as Delayed Auditory Feedback (DAF), a Pacing Board, an Alphabet Board, or a metronome to induce Metronome-Paced Speech; fade their use to eliminate monotonous speech these procedures induce
- Prefer to teach rate reduction without external mechanical help; model a slower rate, let the individual determine a rate that is slower but sounds less unnatural, and reinforce that rate
- Use hand or finger tapping to reduce rate, but fade this, as tapping to syllables also induces an unnatural speech rhythm
- Provide visual feedback from a computer or storage oscilloscope on the desired rate of speech
- Fade all external cues, mechanical devices, finger tapping, and cueing in gradual steps while the individual still maintains a slower rate and improved intelligibility
- Teach individuals to self-monitor their speech rate because slow speech in the absence of external cues is difficult to sustain; have the individuals monitor their own rate; reinforce for slowing down without your help; ask the individual to frequently self-evaluate the rate of speech

Modification of Prosody

- Reduce the rate of speech and increase the frequency and duration of pauses at appropriate junctures to achieve improved prosodic features
- Modify pitch with the help of instruction, modeling, differential feedback, or such instruments as Visi-Pitch; be aware that direct work on pitch modification may not be needed in many cases because of successful modification of rate, intonation, and stress
- Shape louder speech through modeling, shaping, and differential reinforcement of greater inhalation,

increased laryngeal adduction, and wider mouth opening

- Teach the individual to chunk utterances into natural syntactic units to promote more natural-sounding speech
- Increase breath control to extend breath groups
- Use Contrastive Stress Drills (sentences with the same words that change meaning when different words are stressed)
- Teach the individual to signal stress by using other means (e.g., prolongation of syllables or pausing before a stressed word)
- Teach the individual to vary the number of words per breath group
- Begin treatment with structured tasks and make transition to conversational speech
- Teach self-monitoring skills

Modification of Interactions Between the Individual and Communication Partners

- Go beyond direct speech modifications and work on how people with dysarthria and their communication partners interact with each other in natural settings
- Promote two kinds of strategies: those that the individuals with dysarthria use to help their listeners better understand their speech and those that the listeners use to better comprehend the speech of dysarthric speakers
- **Teach individuals with dysarthria**
 - Alert listeners by saying “I am going to say something to you” or “Excuse me, I have something to tell you” before actually saying something
 - Establish firm eye contact with the listener to signal readiness to speak
 - Make a hand gesture to command attention from listeners or conversational partners
 - Request the conversational partners to speak slowly, write the key words, repeat what they say, and so forth

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- Introduce the topic by name or tell when topics are about to be shifted (e.g., “I am going to tell you about . . .,” “My topic of discussion is . . .,” “I am now going to change the topic to . . .”)
- Speak in the most effective and efficient manner (e.g., if a single-word response or a shorter response will do, just say that, instead of unnecessary and possibly difficult-to-understand elaborations)
- Use gestures and facial expressions to supplement oral communication
- Learn to understand the cues listeners give when they don’t understand the message; get sensitized to facial gestures, general postures, signs of boredom, and so forth to gauge the listener’s difficulty
- Modify statements when listeners fail to understand; say it differently, slowly, loudly, and so forth
- Request that distractions (e.g., TV, radio) and extraneous noise be minimized or eliminated before speaking, and make sure that the room is well lit
- Request or carry a personal amplification system when speaking to a group of listeners
- **Teach strategies to the individual’s conversational partners**
 - Ask caretakers and other family members to pay close attention and learn the the speech patterns of the dysarthric speaker
 - Maintain eye contact and watch the face of the speaker to pick up facial expressions that may help understand the speech
 - Ask questions and request clarifications when communication breaks down
 - Minimize or eliminate extraneous noise and distractions; get closer to the speaker and hold conversations in well-lit places

Complete a Skill Maintenance Program

- Train Self-Control (Self-Monitoring) skills during all phases of treatment

- Probe for generalization and reinforce generalized responses in nonclinical settings
- Train conversational partners to prompt and reinforce any kind of successful attempts at communication
- Train then to reinforce communication attempts
- Follow-up and give booster treatment when needed
- See Maintenance Strategy for additional procedures

Quality of Life After Treatment for Dysarthria

- Administer the *Quality of Communication Life Scale (ASHA QCL)*; check asha.org
- Administer the *Dysarthria Impact Profile* (See the Walshe et al. citation)
- Analyze the results to identify potential skill limitations that negatively affect the QoL of the treated individuals
- Remediate the skill deficits with additional training if judged feasible
- See Quality of Life After Treatment for additional suggestions and strategies

Walshe, M. et al. (2009). Dysarthria Impact Profile: Development of a scale to measure psychosocial effects. *International Journal of Language and Communication Disorders*, 44(5), 693–715.

Treatment of Dysarthria: Specific Types. Assess the type of dysarthria and tailor the treatment to the unique communication deficits of the individual; combine the relevant procedures that apply to dysarthria in general with those that are specific to the particular type under treatment; design a client-specific intervention plan; use the modeling-imitation-reinforcement and fading techniques in teaching all targets across all types; verbally correct all errors; implement a Maintenance Strategy to train family members and other caregivers who will help sustain treatment gains.

Ataxic Dysarthria. Target imprecise consonants, excess and equal stress, and irregular articulatory breakdowns; use modeling, prompting, reinforcement, and verbal correction.

- **Modify respiratory problems associated with speech production**
 - Teach the individual to inhale more deeply and exhale in a slow and controlled manner to sustain speech
 - Measure the duration of exhalation and reinforce progressively longer (more controlled) exhalation
 - Teach the individual to start speaking as soon as the exhalation begins to avoid wasted airflow; reinforce prompt phonation upon initiation of exhalation
 - Teach the individual to end an utterance well before running out of air; stop the individual when signs of airflow dissipation are evident, and ask the individual to inhale again
 - Shape longer utterances per breath
 - Teach the individual to stop and inhale at natural junctures in a sentence (e.g., at the beginning of a grammatical clause)
- **Modify prosodic problems**
 - Control irregular, slow, or even rapid rate of speech by slowing the rate of speech with frequent modeling-imitation-reinforcement sequence
 - Instruct the person to pace syllables to metronome beats; fade the metronome use
 - Use finger or hand tapping to generate an even and appropriate rate of speech; fade the tapping
 - Provide such consecutive cues as pointing to printed words to generate an even oral reading rate; have the person point to each word before reading; fade the cues
 - Teach appropriate stress on words in sentences; use [Contrastive Stress Drills](#)
 - Teach variations in pitch by using printed sentences on which you mark intonation cues: draw lines under words to suggest flat intonation, above words to prompt higher intonation, and under words to prompt lower intonation
 - Model and reinforce normal intonation patterns in conversational speech

- Control excess loudness variation by modeling normal variations
- **Modify speech articulation problems**
 - Ask the individual to produce words on a list while you judge their intelligibility without looking at the list or the individual's face
 - Verbally correct errors to encourage appropriate production of sounds in words you do not understand
 - Teach correct production of sounds by using the Phonetic Placement Method
 - Teach slightly overarticulated production of medial and final consonants
- Use the Minimal Contrast Method to improve intelligibility of words
- Model and reinforce more natural-sounding conversational speech

Flaccid Dysarthria. Target breathy and harsh voice quality, hypernasality, nasal emission, imprecise production of consonants, audible inspiration, monopitch and loudness, and short phrases; use modeling, prompting, reinforcement, and verbal correction:

- Avoid exercises to strengthen muscles of the velum, pushing and pulling techniques and whistle or bubble-blowing activities, because they do not improve velar function; increase muscle strength while speaking; ask the individual to increase effort while speaking
- **Modify respiratory problems**
 - Ask the individual to sit straight in the chair to increase breath support for speech
 - Place a padded lap tray on the wheelchair, positioned next to the abdomen, and ask the individual to lean against it to increase breath support
 - Teach the individual to phonate at the beginning of exhalation to conserve air supply for speech
 - Teach the individual to inhale deeply and exhale in a controlled manner
 - Increase the number of words per breath group

- **Modify phonatory problems**
 - Model louder speech and reinforce stepwise increases in the client's vocal loudness; use computer programs to give feedback on loudness
 - Demonstrate how to push on the arm of the chair while phonating to promote effortful vocal fold closure
 - Ask the individual to inhale deeply and hold the breath to promote firmer approximation of folds
 - Teach phonation initiation with hard glottal attacks
 - Ask the individual with a unilateral vocal fold weakness or paralysis to turn head toward the *affected* side or manually push the larynx toward the *unaffected* side to help achieve better closure
- **Modify resonance problems**
 - Target hypernasality, which is the main resonance problem caused by damage to the pharyngeal branch of the vagus nerve; the soft palate may be weak or paralyzed
 - Reduce the speech rate, which may reduce hypernasality
 - Model and reinforce greater mouth opening while speaking
 - Prompt louder speech, which may reduce hypernasality
 - Discuss the suitability of the Pharyngeal Flap Operation with a surgeon to improve velopharyngeal closure; note that the results have been inconsistent
 - Discuss the suitability of Pharyngoplasty with a surgeon to improve velopharyngeal closure; note that the results have been inconsistent
 - Recommend Palatal Lift Prosthesis, as this has been more effective in reducing hypernasality than pharyngeal flap or pharyngoplasty; recommend this device to individuals who (a) are severely hypernasal and have not improved with behavioral treatment; (b) do not have a spastic velopharyngeal mechanism, which may dislodge

the device; (c) have teeth for anchoring the device; (d) have good articulation and phonation, without which the device will not improve speech intelligibility; (e) do not have a hyperactive gag reflex or swallowing problems; and (f) are motivated to wear it and care for it

- Use modeling, shaping, positive reinforcement, and verbal correction to progressively decrease hypernasal voice
- **Modify speech articulation problems**
 - Reduce the rate of speech to improve speech intelligibility (and, to some extent, hypernasality); use finger tapping to cue a slower rate by having the individual produce a syllable or a word per tap; give frequent instructions to slow down and reinforce the desired rate
 - Reinforce improved articulation of speech sounds; systematically model and reinforce correct articulation in words, phrases, sentences, and conversational speech
 - Use the Intelligibility Drills while turning your back to the client; have the person read aloud a list of words; when you do not understand a word, turn around to face the individual, analyze the errors, give suggestions on correct productions, and reinforce them
 - Use the Phonetic Placement Method to teach correct placement of articulators in producing target sounds
 - Teach exaggerated production of consonants to improve speech intelligibility; ask the individual to fully articulate consonants, especially in medial and final positions of words
 - Use the Minimal Contrast Method to further improve articulation
- **Modify prosodic problems**
 - Model and reinforce improved pitch, stress, and rhythm to reduce abnormal prosodic features
 - Model different levels of pitch and ask the individual to discriminate pitch changes

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- Ask the individual to prolong an /a/ with lower and higher pitch
- Have the individual read printed sentences that indicate higher and lower pitch (arrows drawn above or below words)
- Model different and appropriate patterns of intonation in phrases and sentences and have the individual imitate them
- Monitor appropriate pitch in conversational speech and reinforce or verbally correct unacceptable pitch variations
- Teach the individual to chunk utterances into syntactic units; reinforce pauses (and inhale if necessary) at appropriate junctures in sentences
- Use Contrastive Stress Drills

Hyperkinetic Dysarthria. Target prosodic disturbances, including prolonged intervals, variable rate, monopitch, loudness variations, inappropriate silences, imprecise consonants, and distorted vowels; work with a physician who may prescribe medications to control neurological symptoms; note that medications are not effective in all persons and do not replace behavioral treatment.

- Use a Bite Block (a small plastic cube the individual bites down on) to inhibit or reduce interfering jaw movements during speech in individuals with mandibular Dystonia
- Teach easy onset of speech, as this can help reduce involuntary movements that disrupt laryngeal movements, especially in individuals with mild hyperkinetic dysarthria
- Teach slower rate and increased vocal pitch when appropriate
- Teach Habit Reversal to reduce tics; teach the client to voluntarily exhibit a behavior that occurs involuntarily to gain control over it and thus reduce it (e.g., the person who feels an imminent tic of the eyelids, blinks slowly and voluntarily; keeps the eyes closed for a few seconds)
- Try relaxation therapy to control involuntary movements

Hypokinetic Dysarthria. In treating this type of dysarthria typically associated with parkinsonism (with resting tremor and other neurological symptoms), target voice, articulation, and prosodic problems that include monopitch, monoloudness, reduced stress, imprecise consonants, variable rate of speech, increased speech rate in more cases than a slower rate, short rushes of speech, inappropriate silences, and harsh and breathy voice; work with a physician who may prescribe L-dopa and anticholinergic drugs to control neurological symptoms, and deep brain stimulation to improve muscle function.

- **Increase respiratory support for speech**
 - Model and reinforce deeper inhalations before speaking
 - Reinforce speaking as soon as the inhalation begins
 - Teach slow and controlled exhalation
 - Teach the client to stop from talking well before exhausting the air supply
 - Reinforce gradual increases in the number of words spoken per breath
- **Modify phonatory problems**
 - Use voice therapy techniques to increase vocal loudness and to decrease breathiness; use various biofeedback instruments (e.g., the VisiPitch)
 - Use pushing and pulling while phonating to increase the movement range of laryngeal muscles (e.g., having the individual push down on the arm of the chair while phonating)
 - Use portable voice amplifiers to increase loudness
 - Consider teaching initiation of phonation with hard glottal attacks
 - Use the intensive Lee Silverman Voice Treatment program; model and reinforce imitations of louder speech; fade modeling to maintain louder speech
- **Modify speech articulation problems**
 - Reduce the speech rate to improve intelligibility in individuals who speak rapidly; use hand or finger tapping to cue-in production of syllables or words; use Delayed Auditory Feedback; induce Metronome-Paced Speech; use a Pacing Board or an Alphabet

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Board if necessary; none may be effective in all individuals, so experiment; fade the use of all such devices

- Use Intelligibility Drills
- Teach correct articulation by Phonetic Placement Method
- Improve speech intelligibility by asking the individual to produce word-medial and final consonants with a certain degree of exaggeration
- Use the Minimal Contrast Method
- **Modify prosodic problems**
 - Note that a slower rate can improve the individual's prosody
 - Teach proper intonation through printed sentences that show rising and falling pitch by hand-drawn arrows
 - Use Contrastive Stress Drills
 - Teach appropriate chunking of words according to syntactic units (e.g., pausing at the end of a grammatic clause and a sentence)

Mixed Dysarthria. In treating this type of dysarthria with multiple neurological causation (e.g., multiple sclerosis, Amyotrophic Lateral Sclerosis [ALS], Wilson's disease, etc.) and a mixture of different symptoms, consider the following:

- Identify the dominant type, if any, and target the major speech problems for intervention
- Select speech targets that, when treated, will immediately improve communication
- Treat those targets like you would in the case of pure dysarthrias
- Treat the most severe problem first if multiple problems exist in a single category (e.g., prosody); find out the individual's preference to determine which problems should be addressed first in treatment
- Recommend augmentative communication devices mostly for individuals whose dysarthria is due to ALS

Spastic Dysarthria. In treating this type of dysarthria often due to strokes, Amyotrophic Lateral Sclerosis,

Traumatic Brain Injury, and Multiple Sclerosis, target the assessed deficits, including imprecisely produced consonants, monopitch, monoloudness, reduced stress, hypernasality, slow rate, strained-strangled-harsh voice, pitch breaks, and breathy voice:

- Consult with the individual's physician about medically controlling pathological crying, which might interfere with treatment
- Consider behaviorally modifying crying in treatment sessions by reinforcing noncrying (alternative) behaviors
- Do not teach pushing or pulling exercises that only aggravate hyperadduction
- Use relaxation and stretching exercises with caution because their effects on speech have not been documented
- Note that modification of respiratory behaviors is typically not a major concern in individuals with spastic dysarthria; any apparent respiratory problems may be due largely to such phonatory problems as hyperadduction of vocal folds
- **Modify phonatory problems**
 - Note that efforts to reduce hyperadduction of vocal folds have not been especially successful; nonetheless, these efforts may be made with caution
 - Teach the individual head and neck relaxation by instruction, modeling, and manual guidance; stand behind the individual, take the individual's head between the two hands, and gently tilt it back, move it forward, and to the sides; ask the individual to move the head in the same manner without manual guidance
 - Teach easy onset of phonation to an individual who has learned to relax the neck and head muscles; model soft glottal closure and ask the individual to imitate it; begin with an exhaled sigh and add a prolonged /a/; shape this relaxed production of /a/ into words, phrases, sentences, and spontaneous speech
 - Teach the yawn-sigh motion before starting gentle phonation; ask the individual to inhale through open

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mouth, exhale, and begin phonation; shape the phonated speech into words, phrases, sentences, and spontaneous speech; see [Voice Disorders](#) for more on the yawn-sigh method.

- **Modify resonance problems**
 - Increase vocal loudness to control the extent of hypernasality, as louder speech tends to be perceived as less nasal
 - Reduce the rate of speech to increase speech intelligibility
 - Discuss the usefulness of [Pharyngeal Flap Operation](#) with the individual's physician or a [Palatal Lift Prosthesis](#) with a prosthodontist
- **Modify speech articulation problems**
 - Avoid tongue and lip stretching exercises that have not been shown to be effective in controlled studies
 - Use the traditional articulation treatment; use [Intelligibility Drills](#) in which the individual reads a list of words; judge the accuracy of production solely on the basis of phonetic cues and reinforce and verbally correct
 - Use [Phonetic Placement Method](#)
 - Teach the individual to produce the medial and final consonants in words in an exaggerated manner
 - Use [Minimal Contrast Drills](#)
- **Modify prosodic problems**
 - Ask the individual to vary pitch on a prolonged vowel production (e.g., /a/); if necessary, model pitch variations as you prolong the target vowel; reinforce correct imitations; fade modeling and ask the individual to vary vocal pitch
 - Teach the individual to vary intonation in sentences; use printed sentences that indicate rising or falling intonation by hand-drawn arrows or word underlining; model if necessary and fade modeling; move on to conversational speech and reinforce varied intonation
 - Use [Contrastive Stress Drills](#); teach appropriate stress patterns in conversational speech

- Teach the individual to chunk utterances into syntactic units by modeling and reinforcing pauses at appropriate junctures in speech (e.g., at the end of grammatic clauses and sentences); ask the individual to inhale at such junctures

Unilateral Upper Motor Neuron Dysarthria. In treating this type of dysarthria, target the dominant articulation problems (e.g., imprecise production of consonants); treat other speech and resonance disorders if they are serious enough to affect communication.

- Treat the associated language deficits (aphasia) and apraxia on a priority basis if warranted
- **Modify speech articulation problems**
 - Use traditional methods to treat articulation disorders
 - Use Intelligibility Drills
 - Use Phonetic Placement Method to teach correct placement of articulators
 - Teach the individual to produce the medial and final consonants in words in an exaggerated manner
 - Use Minimal Contrast Drills

Chiaromonte, R., & Vecchio, M. (2021). Dysarthria and stroke: The effectiveness of speech rehabilitation. A systematic review and meta-analysis of the studies. *European Journal of Physical Medicine and Rehabilitation*, 57(1), 24–43.

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Kim, Y. et al. (2024). Smartphone-based speech therapy for post-stroke dysarthria: Pilot randomized control trial evaluating efficacy and feasibility. *Journal of Medical Internet Research*, 26, e56417.

Page, A. D., & Yorkston, K. M. (2022). Communicative participation in dysarthria: Perspectives for management. *Brain Sciences*, 12(4), 420.

Dysfluencies or Disfluencies. Behaviors that interrupt fluency; measured in diagnosing and treating stuttering and cluttering, especially stuttering; treatment target

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is to reduce such specific forms as repetitions of sounds, syllables, words, and phrases; prolongations of sounds and articulatory postures; inter- and intralexical pauses; interjections of syllables, words, and phrases; revisions; and incomplete phrases; see Stuttering for treatment procedures.

Dysphonia. A general term that means disordered voice; any voice disorder with the exception of Aphonia; for treatment procedures, see Voice Disorders.

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Echolalia. To control parrot-like repetition of what others say, a major characteristic of autism spectrum disorder, see Autism Spectrum Disorder (ASD).

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Effectiveness of Treatment. Assurance that treatment, not some other factor, was responsible for the positive changes documented in individuals under treatment; requires controlled experimental evidence; data should show that treatment was better than no treatment; may use a group experimental design or a single-subject experimental design to establish this; not the same as Improvement or Functional Outcomes; see Evidence-Based Clinical Practice for research methods and treatment selection criteria.

Effortful Closure Techniques. Use these behavioral treatment techniques to promote laryngeal adduction for individuals with unilateral or bilateral vocal fold paralysis or weakness often associated with dysarthria and in some individuals with voice disorders without dysarthria; teach the individual to grunt, cough, push, lift, and pull while trying to phonate; the muscular effort improves adduction.

Electronic Communication Systems. Use these methods of augmentative communication with persons who have limited or no oral speech; may be of high or low technology; systems use various electronic devices, including smartphones, tablets, and dedicated systems; see Augmentative Communication—Aided.

Electronic Device for Alaryngeal Speech (Electrolarynx). Use these handheld electronic instruments that generate sound to teach alaryngeal speech to persons who have undergone laryngectomy; for procedures, see Laryngectomy.

Electropalatography (EPG). A biofeedback tool, EPG is used in treating speech sound disorders found in children as well as those with Apraxia of Speech; requires the fabrication of an electronic pseudopalate, similar to an ortho-

odontic retainer, but embedded with electrodes that record the contact between the tongue and the palate; properly interfaced with a computer or a handheld instrument with special software, the pseudopalate provides visual feedback to the client during speech sound production training.

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Elicited Aggression. Aggressive behavior directed against any object or person when an aversive stimulus (as in saying “No” to a child’s wrong response) is delivered; not necessarily directed against the clinician; a potential undesirable side effect of response reduction procedures.

- A child may kick the table when “No” is said to a wrong response
- A child may sweep the stimulus materials off the table when a token is taken away in the Response Cost procedure
- To reduce elicited aggression in treatment:
 - Use response reduction procedures sparingly
 - Offer more positive reinforcement than verbal correction
 - Use differential reinforcement to replace nonaggressive responses with alternative, desirable behaviors
 - Role-reverse and ask the child to play the role of a therapist for a few minutes; for example:
 - Exhibit the same kinds of error responses for which the child was told “No” and have the child say “No” to you a few times
 - Simulate stuttering and ask the child to take away a token each time you stuttered
 - Reinstate the procedure for the child

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.). Plural Publishing.

Escape Extinction. Use this procedure to reduce negatively reinforced undesirable behaviors some clients, especially children, exhibit during treatment (e.g., crawling under the table during treatment); in this procedure, block escape behaviors that are negatively reinforced because it provides relief from aversive treatment.

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- Prevent the occurrence of an undesirable response (e.g., crawling under the table) exhibited to escape from aversiveness (e.g., training trials)
- Physically restrain the child every time an attempt is made to leave the chair to prevent reinforcement of escape behavior
- Reduce the aversiveness of treatment (e.g., simplifying the target skills or using more powerful reinforcers)

Ethnocultural Variables in Treatment

- Consider an individual's culture, ethnicity, dialect, language (mono- or bilingual), and other personal variables that may affect treatment of communication disorders; see under the main entry for each disorder (e.g., [Aphasia](#), [Speech Sound Disorders](#), [Language Disorders in Children](#)) for suggestions on treating individuals with varied ethnocultural backgrounds; see the cited sources for details; follow these general guidelines in treating all disorders of communication in ethnoculturally diverse individuals.
- Make an assessment that is appropriate to the client's ethnocultural background; diagnose a disorder that is consistent with the client's dialect or the bilingual status, not on the basis of Mainstream American English.
- Note that there are no treatment procedures that are unique to each ethnocultural group; ethnocultural considerations are important in selecting what to teach, not how to teach them.
- Select speech-language targets that the client and the family members agree to be taught or modified; generally do not target the unique features of a dialect for modification (e.g., the dialectal features of African American English or English influenced by a primary language of a bilingual person); teach Mainstream American English features only when the client and the family opt for it.
- Select treatment stimuli that are ethnoculturally appropriate and are familiar to the client; in the case of children, find out their preferences in selecting toys, pictures, line drawings, objects, and other stimulus materials; seek family members' input and, if possible, select treatment stimuli from the individual's home environment (e.g.,

ask children to bring their favorite toys or books to treatment sessions and use them as treatment stimuli)

- Evaluate treatment research studies for their subject selection criteria; most often, treatment researchers, especially those who use a group research design, do not adequately sample ethnoculturally diverse populations so the researched method may or may not apply to them
- Apply otherwise supported treatment methods with caution; generally effective treatment procedures, including modeling, demonstrations, instructions, positive reinforcers, verbal corrections, fading, extinction, differential reinforcement, shaping, and so forth, have been researched with good generality; nonetheless, expect to modify them to suit individuals of diverse background
- Measure individual performance data in all treatment sessions
- Modify treatment procedures in light of the performance data and the individual's ethnocultural background

Hegde, M. N. (2022). *A coursebook on aphasia and other neurogenic language disorders* (5th ed.). Plural Publishing.

Kohnert, K. et al. (2020). *Language disorders in bilingual children and adults* (3rd ed.). Plural Publishing.

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Evidence-Based Clinical Practice. Use of treatment procedures whose effects have been demonstrated through experimental and other research procedures; rejection of procedures that lack such evidence; evidence may be generated either through group treatment research designs or single-subject treatment research designs; replications of a study by either the original or other investigators to support generalized use of a procedure.

Group Treatment Research Designs. To demonstrate the effects of a treatment procedure using a group experimental design, researchers treat individuals in one group (the experimental group) and do not treat individuals in another, comparable group (control group); designs help demonstrate that treatment was better than no treatment; the following two are the basic designs:

- **Two-Group Experimental Designs.** In these designs, researchers:
 - Identify a population (a large number of subjects with known characteristics) of participants with the disorder for which the treatment to be evaluated has been designed
 - Randomly select a sample of participants from the population
 - Randomly assign the participants to an experimental and a control group
 - Match participants in the two groups on relevant variables if random selection and assignment are not possible
 - Administer pretests of the disorder (or measure the target skills) in the two groups
 - Treat participants in the experimental group; withhold treatment from the control group
 - Administer posttests of the disorder (or measure specified skills) in the two groups
 - Compare the performance of the two groups on the pretest and the posttest
 - Conclude that the treatment was effective if the experimental group improved while the control group did not
- **Multiple-Groups Experimental Design.** In this design, researchers:
 - Use the same two-group experimental design procedures to select, assign, pretest, and posttest participants, but
 - Randomly form two or more experimental groups to evaluate two or more treatments; offer a single treatment to each experimental group
 - Compare the results of experimental groups to determine the relative effects of researched treatments

Single-Subject Treatment Research Designs.

To demonstrate the effects of a treatment procedure using single-subject research designs, researchers demonstrate contrasts between conditions of no treatment, treatment, withdrawal of treatment, reinstatement of

treatment, and other control procedures; all study participants receive treatment and go through control conditions; three commonly used designs include:

- **ABA Design.** In this design, researchers:
 - Baseline the target behavior to be taught
 - Apply the treatment under evaluation to the target behavior
 - Withdraw treatment when the target behavior increases
 - Chart the outcomes to show that the results for the baseline and withdrawal conditions were similar but those for the treatment condition were different
- **ABAB Design.** In this design, researchers:
 - Baseline the behavior to be taught (the first A condition)
 - Apply the treatment under evaluation to the target behavior (the first B condition)
 - Apply the treatment to another behavior or simply withdraw treatment (the second A condition)
 - Reapply the treatment to the target behavior (the final B condition)
 - Chart the outcomes to show that the two no-treatment conditions were convincingly different from the two treatment conditions
- **Multiple Baseline Designs.** Three variations of this design include:
 1. *Multiple baseline across behaviors.* Investigators baseline multiple target behaviors of the same person in the same setting and sequentially treat the behaviors one at a time, repeating the baselines on all untreated behaviors before treating any behavior; conclude that the treatment was effective only when the treated behavior showed positive changes and untreated behaviors did not
 2. *Multiple baseline across subjects.* The investigators baseline the same target behavior in multiple participants in the same setting and sequentially treat participants one at a time, repeating the baselines of behaviors across all untreated participants and before treating any participant; conclude that

the treatment was effective if the target behavior changed only in treated participants and remained unchanged in the untreated

3. *Multiple baseline across settings.* Investigators base rate a target behavior in three or more settings, treat the behavior sequentially in the baserated settings, and repeat the baserates in all untreated settings and before initiating treatment in any setting; conclude that the treatment was effective if the target behavior changed only in the treated setting and remained unchanged in the untreated settings

Treatment Evaluation and Selection. Treatment procedures may be popular, commonly used, used for a long time, but may still lack evidence to support it; to engage in evidence-based practice, clinicians should evaluate treatment procedures for the kinds of evidence that support them; a single experimental group or single-subject study may demonstrate that the method is effective, but replications are needed to demonstrate generality; evidence is quantitative and hierarchical, falling into several levels; may be scant to substantial; clinicians may select procedures that have the highest level of evidence available for it and avoid procedures with lower level of evidence or no evidence.

- Level 1. Expert Advocacy. Some techniques are simply advocated by experts; no evidence of any kind is offered; may sound logical and appealing, and the advocate may be influential or well known; reject all such procedures in favor of one with some empirical evidence
- Level 2. Unreplicated, uncontrolled case studies. Some techniques may have support from case studies without using an experimental design and without replication; clients have improved under treatment (see Improvement), but there is no assurance that treatment was effective; clinicians may select this type of treatment and use it with caution, provided that no technique with better evidence is available to treat that particular disorder

- Level 3. Replicated, uncontrolled case studies. A technique has been applied more than once and multiple case studies have been published; no controlled experimental evidence yet; clients have improved in replicated studies; clinicians may use such a technique without claiming effectiveness for the procedure
- Level 4. Unreplicated, controlled experimental studies. The technique has been experimentally evaluated with a control group or a single-subject experimental design with control conditions; treatment was shown to be effective by ruling out extraneous variables, but the generality and applicability of the technique is unknown because it has not been replicated in different settings by different clinicians; clinician can use such a technique
- Level 5. Replicated, controlled experimental studies. A technique has been shown to be effective in experimental research and then shown to have generality by repeating its application in different settings, by different clinicians, in treating a variety of clients; a procedure at Level 5 is supported by the highest level of evidence and preferable to procedures at all lower levels

Hegde, M. N., & Salvatore, A. P. (2019). *Clinical research in communication disorders: Principles and strategies* (4th ed.). Plural Publishing.

Executive Functions (EF). Vague and variable definitions make it difficult to specify teachable EF skills (e.g., one of the EF is *judging information and ideas in one's mind*; another is *looking again, in a brand new way*; neither are measurable or teachable); there is no general agreement on what are EFs; basically, the term means *planning* and *executing* various actions of daily living; a person with traumatic brain injury who cannot cook, pay bills, or complete work assignments is said to have an executive function disorder.

Intervention for EFs /Cautionary Guidelines

- Reported correlation between EFs and certain speech-language skills does not justify teaching EFs; it is still better to teach directly the communication skills

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- EFs are an ever-expanding category of skills; one investigator claims that EFs are important for every aspect of life; such overly inclusive descriptions are not clinically useful or practical
- Several skills that were defined traditionally have been redefined as *executive* (e.g., *working memory* and *attention*); operationally defining the skills and directly teaching them will be more productive
- Occupational therapists and psychologists, among others, teach most executive skills
- SLPs may consider targeting verbally based executive skills, especially planning, problem solving, self-monitoring of errors, and performing certain daily living skills; verbally-based EF skills may be better integrated with conversational and social communication training, thus saving time and effort
- Interventions for memory, attention, and inhibition of inappropriate responses, also considered EFs, are described under specific disorders (e.g., Aphasia, Autism Spectrum Disorder, Traumatic Brain Injury) for which they are relevant and are not repeated here.

Teach Planning Skills

- Select common, functional, and client-specific activities; consider the client's age and life circumstances to select the targets that may be efficiently taught during conversational training
- For each target, prompt correct responses when the person hesitates; verbally reinforce correct sequences; prompt the missing steps; generally, ask the person to describe:
 - the steps involved in getting a hotel room out of town
 - how one could get an airline ticket
 - the procedure to be followed in planning an entire vacation out of the state; prompt airline tickets, hotel room booking, day trips to be taken, and so forth
 - the steps involved in organizing a surprise birthday or anniversary party

- how to get son, daughter, or grandchild admitted to a college
- how to go about selecting an assisted living facility in town
- how to sell a house in town and buy a new one in another town

Teach Problem Solving Skills

- Target client's common daily activities; consider the following examples and find other client-specific problems to solve; begin with a simpler problem and progress to more difficult problems that require elaborate responses; say or ask:
 - "You find an extra charge on your telephone bill that you don't recognize; how would you resolve the problem?" Prompt correct sequence of steps
 - "You have a leaky faucet; how do you fix it or getting it fixed?"
 - "You want to shop around for a new central home air conditioner; where do you start and how do you select one?"
 - "Your apartment lease has run out and the owner is not renewing it. What do you do and how do you do it?"
 - "You are not fired from your job, but you want to find a better job. What is the first thing to do and what are other steps you need to take to find a new job?"
 - "You have returned to your job after rehabilitation. You still need assistance to do your job. How do you seek assistance, and from whom?" Questions like this may be appropriate for individuals with traumatic brain injury or right hemisphere disorder.

Teach Execution of Daily Activities

- Ask for descriptions of several client-specific routine activities; ask for details, prompt or question about missing items of information; for instance:
 - "How do you change a flat car tire?"
 - "Tell me about your favorite meal and how you would prepare it. First, please list the ingredients you need."

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- “Write a list of things you typically buy at a grocery store every week.”
- “Describe what you buy at a grocery store every week.” Ask this after the person successfully writes a list
- “How would you wash a load of laundry and get your clothes dried?”
- “What is your favorite smoothie and how do you make it?”

Teach Silent and Overt Verbal Rehearsal Prior to Actions

- Also described as *inner-speech*, *covert speech*, or *self-directed speech*, *talking to oneself*, but *silent verbal rehearsal* is more specific to execution of an act; it is essentially self-instruction instead of instruction from other people; it may be better to teach verbal rehearsal before teaching other skills (planning, problem solving, daily task execution; there is no guidance from research); to promote silent and overt verbal rehearsal:
 - give the client a written direction of how to perform a task (e.g., fixing a salad or changing a car tire)
 - ask the client to silently read the directions a few times to memorize the critical steps
 - ask the client to silently and verbally rehearse the steps needed to complete the action the client signals readiness to orally describe the steps
 - ask the person to orally describe the steps to verify the accuracy of silent verbal rehearsal because it is not possible to give feedback on silent self-directed speech
 - positively reinforce the overtly and correctly described sequence of actions
 - verbally correct the sequencing errors
 - continue training until the individual’s description is complete and correct in sequence
 - teach multiple exemplars of verbal rehearsal of everyday executive functions (actions)
 - assess generalized verbal rehearsal of untrained actions by asking the client to describe the steps without a written script or covert rehearsal

Teach Self-Control (Self-Monitoring)

- Self-control is a skill that helps minimize errors in executing any kind of task; individual who continuously monitor their actions and errors tend to successfully execute their tasks; consider the following suggestions:
 - Have the client count one's own errors and correct responses
 - Ask the client to stop when an error is made and self-correct immediately
 - Ask the person to state why an error occurred
 - Ask the person to contrast correct and incorrect responses
 - Suggest that the client practice self-correction in natural settings
 - Integrate trainings of Social Communication Skills and Self-Control (Self-Monitoring) skills
- Teach family members and other caregivers to prompt and praise self-correction
- See Self-Control (Self-Monitoring) for additional suggestions and strategies

Jeffay, E. et al. (2023). INCOG 2.0 Guidelines for cognitive rehabilitation following traumatic brain injury, Part III: Executive functions. *Journal of Head Trauma Rehabilitation*, 38(1), 52–64.

Kapa, L. L., & Erickson, J. A. (2020). The relationship between word learning and executive function in preschoolers with and without developmental language disorder. *Journal of Speech, Language, and Hearing Research*, 63, 2293–2307.

Morra, S. et al. (2018). Which tasks measure what? Reflections on executive function development and a commentary on Podjarn, Kamawar, and Andrews (2017). *Journal of Experimental Child Psychology*, 167, 246–258.

Mueller, J. A., & Dollaghan, C. (2013). A systematic review of assessments for identifying executive function impairments in adults with acquired brain injury. *Journal of Speech, Language, and Hearing Research*, 56(3), 1051–1063.

Exemplar. An example of a target behavior being taught in a treatment session; for example, the word *soup* is an exemplar if used in teaching the production of /s/ or /p/; thus, all words, phrases, and sentences used in teaching

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correct production of phonemes are exemplars; the phrase *Two hats* is an exemplar in teaching the plural morpheme; the sentence *The boy is running* is an exemplar in teaching the noun phrase, the auxiliary, or the main verb plus *ing*; thus, all phrases and sentences used in teaching language structures are exemplars; pictures and objects used in treatment also may be considered exemplars; to develop treatment exemplars:

- Select multiple exemplars for each target behavior to be taught (e.g., 20 words to teach the /p/ in initial positions; 20 sentences that contain the present progressive *ing*)
- Teach multiple exemplars of target behavior to promote generalized production
- Probe with untrained stimuli after teaching a few (four to six) exemplars to see if generalized productions emerge
- Teach additional exemplars when the probe criterion (e.g., 90% accuracy of the target phoneme production in a set of 10 untrained words) is not met
- Note that typically, generalized production often results with 10 or fewer exemplar training; if this did not happen, reexamine your treatment procedure

Expansions. An individual's incomplete or telegraphic utterances are expanded into grammatically more complete productions; for procedures, see [Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs](#).

Experiment. A means of establishing cause-and-effect relationships; it includes a controlled condition in which an independent variable (such as treatment) is manipulated to produce changes in a dependent variable (production of speech or language); experiments are a means to establish treatment effects; see [Evidence-Based Clinical Practice](#).

Experimental Group. The group that receives treatment in a group design study that evaluates the effects of a treatment. see [Evidence-Based Clinical Practice](#) and [Control Group](#).

Expiratory Board. A prosthetic device designed to improve respiratory muscle strength for speech; controlled evidence of its effects on speech is limited; clients pull a board attached to wheelchair toward their abdomen and lean against its padded surface to stabilize the muscles; often used in improving breath support for speech production in Dysarthria.

Expressive Aphasia. Not a type of Aphasia, but an older and general term for nonfluent aphasia whose main characteristic is difficulty producing language and speech with no significant impairment in auditory comprehension; see Aphasia and Aphasia, Specific Types: Broca's Aphasia.

Extension. A language treatment method in which the clinician makes comments on the child's utterances to add additional meaning; for procedures, see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Extinction. Terminating reinforces to reduce or eliminate undesirable responses; similar to *ignoring* in everyday life; one of the Direct Methods of Response Reduction; extinction takes time to reduce a behavior; responses heavily reinforced and for longer durations take more time to extinguish than those that have undergone prior extinction but have resurfaced; in using extinction:

- Educate parents of young children about extinction before you use it because they find it difficult to watch a behavior under extinction (e.g., crying); explain that other methods of temporarily stopping an undesirable behavior (e.g., picking up a child crying in a treatment session) strengthen that behavior and become more frequent and troublesome in future sessions
- Find out what seems to reinforce the undesirable behavior to be extinguished and withhold it
- Positive reinforcers (e.g., attention, smile, approval, and getting picked up and consoled) may reinforce crying and other kinds of undesirable behaviors in young children; if so, promptly withdraw such reinforcers

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- Negative reinforcers are events that terminate an aversive state of affairs; for instance, a child in a treatment session may leave the chair to terminate aversive therapy trials, thus getting negatively reinforced for leaving the chair; if so, physically prevent such escape behaviors
- Automatic reinforcers are sensory stimulation that results from a behavior; for instance, banging on the therapy table may be automatically reinforced by its auditory sensory consequence; if so, eliminate the sensory stimulation by using a padded table that dampens the sound
- Do not use extinction to control aggressive, self-destructive, and generally disruptive behaviors because extinction is a slow process, and the response is allowed to be made with no reinforcers; allowing such responses to be made may result in injury to others and self; physically prevent the occurrence of such responses if possible; preferably, use one of the Differential Reinforcement procedures to encourage alternative, desirable behaviors that will replace the undesirable
- Promptly extinguish again if the extinguished responses recover
- Expect *extinction burst*, which is the intensification of an undesirable behavior at the onset of extinction; for instance, a child's crying behavior may intensify when extinction is begun; do not give in at this point; stay calm and do not look at the child; praise the child as soon as the crying stops
- Combine extinction with positive reinforcement for a clearly stated alternative, desirable behavior; for instance, you may tell the child that *as soon as you stop crying, we can go out and see your mother*; and follow through

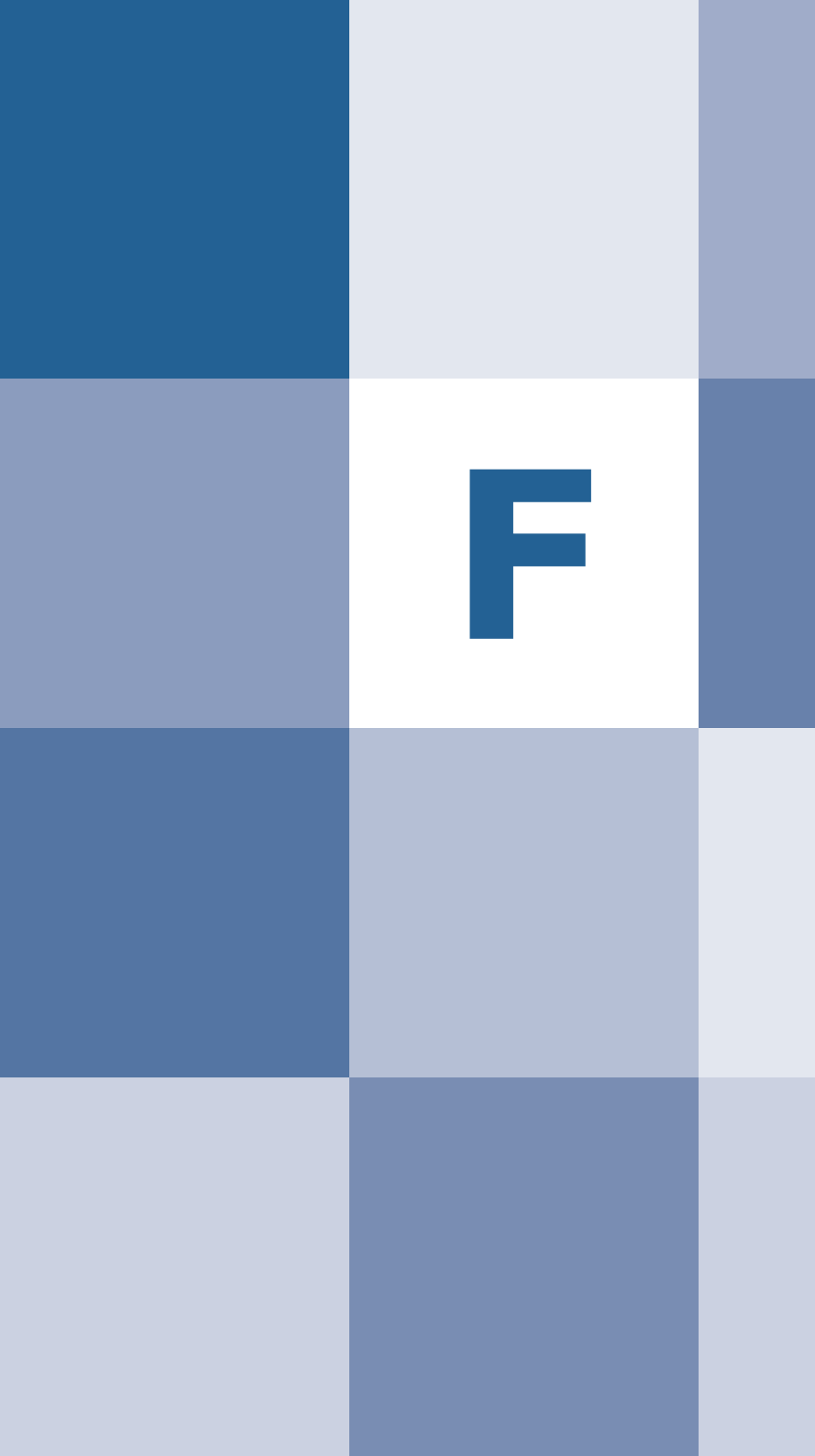
Extraclinical Settings (Training In). Training given in such nonclinical settings as a playground, classroom, home, and other places; an essential part of Maintenance Strategy; training is less formal and targets spontaneous social and functional communication; such significant oth-

ers as teachers, family members, and friends may also be trained to praise and correct the client in a subtle manner.

Eye Contact. Looking at the listener's face during conversation; a social communication intervention target; subject to ethnocultural variations, as in some cultures, eye contact between certain conversational partners (e.g., between a student and a teacher) may not be appropriate; target only with the parental approval; for procedures, see Social Communication Skills.

Eye Gaze. Teach this nonverbal method of communication to individuals with severe neuromuscular impairments; in this method of Augmentative and Alternative Communication (AAC), teach the individual to gaze at a word, a phrase, a symbol, or an object to convey a message.

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Fading. Use this method to gradually reduce and finally eliminate modeling and prompting while still maintaining the target responses; essential to teach more spontaneous speech and language skills.

- Reduce the frequency of modeling gradually until it is completely stopped
- Reduce the intensity of the stimulus (e.g., present Prompts in a progressively softer voice until it is no longer provided)
- Present Partial Modeling
- Make nonverbal prompts progressively more subtle until it is eventually stopped (e.g., make the hand gesture given to slow down the speech of a person who stutters less and less conspicuous)
- Make a mechanical stimulus nonfunctional (e.g., turn off a microphone that the individual still holds, or turn off a computer screen that remains in front of the individual)
- Increase the distance from the individual and the special stimulus in graded steps (move the microphone or the computer screen away from the individual)

Falsetto Voice (Register). Very high-pitched voice, often seen in adolescents past puberty; see Voice Disorders for intervention.

Flaccid Dysarthria. To treat this type of motor speech disorder caused by damage to the motor units of cranial or spinal nerves that supply speech muscles (lower motor neuron involvement), see Treatment of Dysarthria: Specific Types under Dysarthria.

Fluency Disorders. Flowing and smooth speech without an undue amount of dysfluencies or muscular effort is a treatment target in individuals who have Stuttering, Cluttering, and Neurogenic Stuttering; may also be a target in Nonfluent Aphasia.

Fluent Aphasia. To treat this type of aphasia, characterized by flowing but mostly meaningless speech, see the following varieties of fluent aphasia under Aphasia: Specific

Types: Wernicke's Aphasia, Transcortical Sensory Aphasia, Conduction Aphasia, and Anomic Aphasia.

Follow-Up. Take this necessary step following treatment to evaluate maintenance of skills established in treatment; do it according to a schedule (such as 3 months after dismissal or at 6-month intervals).

- Set up a schedule with decreasing frequency (e.g., twice in the first 6 months of dismissal, the next follow-up after 1 year, the next after 2 years)
- Record a speech-language sample during a follow-up
- Measure the frequency of the target behaviors (production of clinically established speech sounds, language features, naming skills, fluency or dysfluency, vocal qualities, etc.)
- Calculate the percent correct use of the clinically established target behaviors
- Give Booster Treatment if the target behaviors are below the previously set criterion (such as 90% accuracy)

Forward Chaining. Break down a skill into its component parts and begin teaching the first component, add the second and third components, and finally teach the full target response. See Shaping.

Frontotemporal Dementia (FTD). To treat communication disorders associated with this neurodegenerative condition, see Dementia; see also Primary Progressive Aphasia, also classified as a type of FTD; consider a few special features in treating the behavioral variant of FTD:

- In communication treatment sessions, manage the varied behavioral problems including deteriorated social conduct, depression, jocularity and euphoria, compulsive behaviors, and delusions; seek help from a psychologist or a psychiatrist
- Verb and action naming and defining word meaning may need greater attention than grammar; sentence comprehension also may need treatment
- Abstract and Figurative Language also may need intervention
- Reading and writing skills may be treatment targets

Functional Equivalence Training. Use this indirect method of reducing an undesirable behavior by reinforcing a desirable behavior that serves the same function as the undesirable behavior; for instance, teach a verbal request to a minimally verbal child who whines to get adult attention; the verbal request serves the same function as whining and thus is reduced in frequency; the same as the Differential Reinforcement of Alternative Behaviors (DRA).

Functional Outcome Measures of Treatment.

During and subsequent to treatment, measure these generalized, broader, and socially and personally more meaningful consequences of treatment that go beyond counting the correct production of specific target behaviors; measures that justify treatment outcomes by global, more general, nonclinical measures such as the following:

- Qualitative measures of not only changes of the clinical target behaviors but also those of related behaviors (e.g., a child who has been taught language skills may begin to exhibit more cooperative behaviors on the playground)
- Measures of changes in the individual's overall quality of life (e.g., persons with aphasia who have regained communication skills through treatment may increase their participation in family conversation, may become socially more active; improved social or academic performance of a child who has received language treatment; an adult's return to work after receiving rehabilitation services following traumatic brain injury; improved dating skills of a treated person who stuttered))
- Measures of functional communication skills as against grammatical accuracy or linguistic complexity (e.g., a clinically treated person may now be able to meet personal communication needs in social situations with whatever the means)
- Document generalized and effective production of clinically established communicative skills in social, personal, family, educational, and occupational contexts; note that minimally, treated target skills should generalize to the home, school, work, and social settings

- Assess functional communication in natural settings (e.g., effective expressions of needs and wants by persons who have received treatment for aphasia; managing daily communication needs by a child who has received augmentative or alternative communication therapy; effective use of a hearing aid in social situations and documented benefits derived by a person who has received aural rehabilitation)
- Measure expanded skills in natural settings (e.g., new and longer sentences produced in natural settings by an individual who has received language treatment)
- Individual's communicative effectiveness in natural contexts as rated or reported by caregivers and family members

The image features a 4x4 grid of squares. The top-left square is a dark blue. The top-middle square is a very light blue. The top-right square is a medium blue. The middle-left square is a medium blue. The center square (middle-middle) is white and contains a large, bold, blue capital letter 'G'. The middle-right square is a medium blue. The bottom-left square is a medium blue. The bottom-middle square is a light blue. The bottom-right square is a very light blue.

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Gender Identification Therapy (Elective). Communication modification programs, designed for individuals who, at some point in their lives, assume a different gender identity; these individuals may be known as transsexual, transgender, and gender-nonconforming individuals; all modifications are entirely elective, and the goals are typically set by the individual seeking services; there is no communication disorder to be treated; aspects of communication typical of individuals with a certain gender identity are changed to aspects that are typical of a different gender identity; SLPs may be asked to help achieve voice, language usage, and nonverbal communication patterns that are consistent with the newly identified gender; generally, more males opt to assume the female gender identity than females opting for male gender identity; therefore, much of the literature emphasizes male-to-female gender identification therapy; to develop a specialized practice in gender identification therapy, follow the standards of care established by the World Professional Association for Transgender Health (<https://www.wpath.org>).

***Gender Identification Therapy:
General Guidelines***

- Get specialized training in gender identity communication intervention before offering services; attend workshops, seminars, and lectures that offer training and most recent practice guidelines
- Visit the websites of the American Speech-Language-Hearing Association, the World Professional Association for Transgender Health, and other agencies to learn more about the guidelines on offering gender identity therapy
- Work with such other professionals as voice coaches, theater professionals, and singing teachers who may be helpful in designing appropriate communication intervention programs
- Use the person-preferred pronouns and currently accepted terms; avoid all discriminatory terms and stereotypes

- Understand the consequences of surgical and hormonal treatments, especially on voice
- Understand the ethnocultural issues that affect gender identification therapy; make a culturally sensitive assessment of the communication needs of the transgender individual; note that the male and female verbal and nonverbal communication patterns may differ across cultures; nonetheless, personal preferences may override cultural expectations or stereotypes

Assist clients in selecting their intervention targets

- Offer counseling about the typical or traditional differences in the male and female communication patterns; note that your clients may have researched this information or may have specific communication targets to be achieved; if needed, direct them to useful resources (e.g., articles, books, and websites that describe gender differences in communication and etiquette)
- Ask the individual to view male and female TV news reporters and anchors to study the gender differences in communication, including nonverbal communication
- Ask the individual to watch lectures given by women and men to study the gender differences in formal communication
- Suggest the range of skills that could possibly be targeted for potential change, but have the individual set the target skills
- Draw attention to both common (even stereotypic) masculine and feminine communication characteristics as well as significant individual differences; some women may have a few stereotypically masculine communication characteristics (e.g., may not ask many questions of their conversational partners), whereas some men may have a few stereotypically feminine communication characteristics (e.g., may ask many questions)

- Assess the extent to which the individual would want to change or retain the previous feminine or masculine communication patterns even though the retained features may be stereotypically uncharacteristic of their new gender identity; for instance:
 - A former woman who has assumed a new male gender identity may wish to speak with more facial expressions, smile as before, and maintain good listening skills even if they are not considered to be stereotypical masculine communication characteristics
 - A former man who has assumed a new female gender identity may wish to retain a more objective or factual speaking style and eschew free and frequent expressions of emotions and feelings, even though the former are not common feminine communication characteristics
- Finalize the features for modification in light of the individual's education, occupation, and stated preferences; a transgender woman who is a business executive may set different goals than a transgender woman who is a nurse or a teacher; to the contrary, a transgender man who is a nurse may set different goals than a transgender man who is a business executive
- Assess the extent to which transgender individuals wish to generalize their new communication patterns that are in congruence with their new identity; many may refrain from using their new style (e.g., their "new voice") with their family members, especially with their parents, spouses, and children; they may code-switch as they find it appropriate
- Assess the factors that determine whether your clients will want to use the new communication style at work, including the employment responsibilities (more or less verbally oriented job), whether they already have "come out" or not at the workplace, the degree of understanding and support they receive from supervisors and colleagues, and so forth

- Promote good transition at the workplace by offering to work with the human resources department of the clients' workplace and the supervisors; let the clients decide on how to proceed
- Refer your clients to a clinical psychologist for managing any emotional problems of the client and especially those involving family members

Communication Modification for the Male-to-Female Transgender Person/General Guidelines

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- Have the woman list aspects of her communication that she would like changed
- Offer voice therapy for a transgender female because the medical treatment to achieve an overall female pitch in the male vocal mechanism has been unsatisfactory
- Teach healthy vocal behaviors that will replace vocal abuse and misuse and help prevent negative consequences of trying too hard to change the vocal characteristics
- Show videotapes of women talking to each other or giving formal lectures to sensitize the individual to feminine communication patterns, including nonverbal communication patterns and body language
- Recommend textbooks on women's studies courses at universities
- Have the transgender woman watch movies and imitate the desired vocal pitch of characters

Teach feminine vocal and speech characteristics

- Identify a pitch that is appropriate, comfortable, and free of laryngeal tension in the individual's own vocal range; increase the vocal pitch level by modeling and systematically reinforcing progressively higher pitch levels; if a female pitch range (220 Hz, plus or minus 20) cannot be achieved, target a gender-ambiguous range of about 170 Hz; advise against raising pitch with increased laryngeal tension
- Use a computerized program to give visual feedback on the individual's practice of new vocal pitch

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- Teach feminine intonation patterns (e.g., rising intonation at the end of sentences, which characterizes female speech)
- Discourage Falsetto Voice (very high frequency pitch)
- Teach speaking with more mouth openness
- Teach softer speech than before, although vocal intensity goals are secondary to other goals
- Decrease such masculine behaviors as throat clearing and coughing
- Teach slightly faster speech rate, increased pause durations, more precisely articulated speech sounds, and elongated vowels
- Model and reinforce easy and smooth initiation of vowels and production of consonants with light articulatory contacts
- Encourage speech initiation with a slightly breathy voice
- Model and reinforce smooth linking or blending of words

Teach feminine language and general communication characteristics

- More indirect rather than direct requests (e.g., “Do you mind doing . . .” instead of “Do this” type of command)
- Expressions of feelings, emotions, and human relationships in speech as against facts
- Openness about self-disclosure during conversation
- Production of tag questions (e.g., “I did a good job, *didn’t I?*”)
- Expression of tag clauses (e.g., “I did fine on the exam, *I think*”)
- Production of conditional clauses at the end of sentences (e.g., expressions that qualify statements by such terms as *unless, whenever, providing, depending on*)
- Production of full forms instead of contractions
- Increased production of pronouns
- Expressions that connect with each other (e.g., the conjunction *and*)

- More elaborate expressions
- Increased production of questions
- Listening skills

Teach more feminine nonverbal communication patterns

- Smile more often during conversation
- Be facially more expressive, including more expressive eyes during conversation
- Touch the conversational partners
- Move hands from the elbow, not from the shoulders
- Maintain more even eye contact
- Maintain a conversational posture that is closer to the conversational partner

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Communication Treatment for the Female-to-Male Transgender Person/General Guidelines

- Assess the transgender male's retained feminine communication patterns and have the man list aspects of his communication that he would like changed
- Assess whether female-to-male voice change has already occurred because of hormone treatment that will thicken the vocal folds and lower the pitch to suit the new male gender identity
- Show videotapes of men talking to each other or giving formal lectures to sensitize the individual to masculine communication patterns, including nonverbal communication patterns and body language
- Ask the individual to watch TV shows or news broadcasts that feature male reporters or anchors to learn an overall masculine style of communication

Target the more masculine vocal characteristics

- A pitch that is appropriate, comfortable, and free of laryngeal tension
- A lower pitch characteristic of the male voice (120 Hz, plus or minus 20 as against 220 Hz, plus or minus 20 for the female voice); if a typical male pitch range cannot be achieved, target a gender-ambiguous pitch (about 170 Hz)
- Pitch and loudness changes with a computerized program that gives feedback

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- Masculine intonation patterns (e.g., falling intonation at the end of sentences that characterize the male speech)
- Limited variation in intonation (a bit more monotonous than the better-intoned female voice)
- Increased stress or loudness in emphasizing a word in a sentence
- Natural-sounding speech, as the process of learning specific pitch, stress, or intonation may introduce unnatural-sounding speech
- Healthy vocal behaviors to reduce the chances of such vocally abusive behaviors as trying too hard to change the vocal characteristics

Target more masculine speech characteristics

- Slightly slower speech rate without much pause durations
- Less precisely articulated speech sounds without elongated vowels
- Firmer articulatory contacts
- No breathiness in initiating speech
- No linking or blending words as women do

Target more masculine language and general communication characteristics

- Fewer indirect speech expressions and more direct requests (e.g., “Do this” type of command instead of “Do you mind doing . . .” type of request)
- Restrained expression of feelings, emotions, and human relationships
- More factual and observational information
- Restraint in talking about self or being open about personal experiences or family matters
- Decreased production of tags at the end of questions (e.g., “I did a good job, *didn’t I?*”)
- Decreased expressions of tag clauses at the end of sentences (e.g., “I did fine on the exam, *I think*”)
- Decreased production of conditional clauses at the end of sentences (e.g., teach the individual not to frequently qualify statements with such terms as *unless, whenever, providing, depending on*, etc.)

- Increased production of contractions
- Restraint in the use of pronouns
- Reduced frequency of connecting expressions that include the conjunction *and*
- Less elaborated expressions
- Asking fewer questions

Target more masculine nonverbal communication patterns

- Less frequent smiles and less prominent facial expressions
- Reduction in touching the conversational partners
- Moving hands from the shoulders, not from the elbow
- Slightly decreased frequency of eye contact
- Maintaining a conversational posture that is somewhat distant

Adler, R. K. et al. (2018). *Voice and communication therapy for the transgender/transsexual individual: A comprehensive clinical guide* (3rd ed.). Plural Publishing.

Coates, J. (2016). *Women, men, and language* (3rd ed.). Longman Publishing.

The World Professional Association for Transgender Health. (n.d.). *Standard of care for the health of transsexual, transgender, and gender nonconforming people, Version 7*. <http://www.wpath.org>

Wood, J. (2019). *Gender lives: Communication, gender, and culture* (13th ed.). Wadsworth.

Generality of Treatment. Applicability of a treatment procedure researched in one study to other clients, administered by other clinicians, and used in varied clinical settings; demonstrated through Replication of treatment research; treatment procedures with greater generality are preferred over those with lesser or no generality.

Generalization. Production of clinically established skills in untrained situations, verbal contexts, verbal forms, and so forth; includes several varieties; all forms of generalization are essential for successful intervention that benefits the individual and improves the quality of life; measured through Probes on which generalized responses are not reinforced; an intermediate goal of treatment, with

maintenance of treatment gains being the final goal; see Maintenance Strategy to promote lasting treatment effects; use the following suggestions to promote the various types of generalization:

Audience Generalization. A treatment target; production of unreinforced responses in the presence of persons not involved in training; to promote this important type of generalization:

- Invite persons not involved in training to treatment sessions conducted in later stages of therapy
- Evoke and reinforce the target behavior (e.g., fluency, correct production of speech sounds, naming, appropriate vocal qualities) in the visitor's presence
- Have the visitor engage the individual in conversational speech
- Have the visitor reinforce the target communication skills

Physical Setting Generalization. Production of trained responses in a setting not used in training; to promote this type of generalization:

- Hold brief and informal training sessions in varied nonclinical settings
- Have parents reinforce the clinically established target skill productions in and around home
- Have teachers reinforce the target skills in the classroom and other school settings

Physical Stimulus Generalization. Production of trained responses in the presence of untrained stimuli that are similar to trained stimuli (e.g., the child's generalized response "cup" when shown a new, untrained cup of different color or size); to promote this kind of generalization:

- Evoke the target response with varied stimuli (e.g., cups of different size and color to teach the word cup)
- Use pictures, objects, line drawings and so forth to evoke the same response

Response Mode Generalization. Production of unreinforced responses in a mode not involved in training; to promote this kind of generalization:

- Train skills in a certain mode (e.g., fluency in oral reading)
- Probe to assess generalized production (fluency in conversational speech)
- Train the skills that do not generalize to a different but desirable mode

Verbal Stimulus Generalization. Production of unreinforced verbal responses when untrained verbal stimuli are presented; to promote this kind of generalization:

- Present the same physical stimulus as used in training (e.g., the picture of *two books* used in teaching an exemplar of the plural morpheme)
- Ask a question other than the one used in training to evoke the response (e.g., if you asked, “What do you see?” on training trials, ask, “What are these?” on probe trials)
- Do not model and do not reinforce or verbally correct
- Record the response
- Present at least 10 trials, each involving a different picture and probe question (different from the training question)

Genetic and Congenital Syndromes Associated With Communication Disorders. A constellation of signs and symptoms that are associated with morbid genetic expressions, causing developmental and communication difficulties; may be congenital or inherited; see the cited sources for syndromes associated with communication disorders.

General Guidelines in Treating Individuals With Genetic or Congenital Syndromes

- An understanding of congenital or genetic syndromes (CGS), their etiologic factors, and varied phenotypic expressions is essential to diagnose and treat communication disorders
- There are no special treatment procedures to improve the communication skills of individuals with CGS; their speech, language, voice, and fluency problems

are treated with the same procedures described under the main entries

- Prognosis for improvement with treatment depends on the syndrome, its severity, and medical prognosis for stabilized, improving, or deteriorating condition
- A team of specialists, including genetic counselors, medical specialists, psychologists, special educators, speech-language pathologists, social workers, occupational therapists, and other health care professionals, may work with individuals affected by a genetic syndrome
- To treat speech production problems, see Speech Sound Disorders and Cerebral Palsy
- To treat language disorders, see Language Disorders in Children; Apraxia and Dysarthria if relevant
- To treat fluency disorders, see Stuttering and Cluttering
- To treat deviant vocal characteristics, see Voice Disorders
- Consider the unique patterns of impairments of each client; the same syndrome may express itself differently or with different severity across individuals
- Make adjustments in treating communication disorders depending on the medical needs of the child; note prosthetic or surgical needs if the child has cleft palate, hearing aids or group auditory trainers if the individual has a hearing loss, orthopedic rehabilitation if the individual has a need for it, and so forth
- Train family members to promote generalization and maintenance
- Integrate academic and literacy goals with speech and language treatment; see Language Disorders in Children and Language and Literacy Skills

Jones, K. L. (2005). *Smith's recognizable patterns of human malformations* (6th ed.). W. B. Saunders.

National Institutes of Health. (n.d.). <https://rarediseases.info.nih.gov/diseases>; key word search with the name of a syndrome is productive.

Zajac, D. J., & Vallino, L. D. (2016). *Evaluation and management of cleft lip and palate*. Plural Publishing.

Gentle Phonatory Onset. A treatment target for people who stutter as well as those who initiate their voice with hard glottal attacks; teach initiation of voice in a gentle, soft, easy, and relaxed manner to reduce hard glottal attacks and some dysfluencies that may be associated with them; for procedures, see [Stuttering: Specific Techniques or Programs](#) and [Voice Disorders](#).

Gestural Communication. Use this method of communication that supplements oral communication with a variety of other facial expressions and body movements, including shoulder shrugging, hand movements, pantomime, pointing, and head nodding or shaking to individuals with limited oral communication skills; it is a part of typical oral communication but enhanced in cases of individuals who need it; gestural communication may be unaided, as in smiling or hand movements, or aided, as in gestures combined with a communication board; procedures described under [Augmentative Communication \(Unaided\)](#) and [Augmentative Communication \(Aided\)](#).

Gradual Increase in Length and Complexity of Utterances (GILCU). A component of the [Monterey Fluency Program](#); for procedures, see [Stuttering: Specific Techniques or Programs](#).

Grammatical Morphemes. Language treatment targets, especially for children who have language disorders; deficient grammatical morphemes are a hallmark of child language disorders, resulting in telegraphic utterances; see [Language Disorders in Children](#) for treatment details; to teach the various grammatical morphemes:

- **Select the grammatical morphemes the child has not mastered**
 - The present progressive *ing*
 - Prepositions *in*, *on*, *under*, and *behind*
 - Regular plural morphemes and their allomorphic variations
 - Irregular plural words

- Regular past tense morphemes and their allomorphic variations
- Irregular past tense words
- Possessive morphemes and their allomorphic variations
- Conjunctions
- Uncontractible and contractible copula
- Uncontractible and contractible auxiliary
- Regular and irregular third person singular
- Negation
- Reflexive pronouns
- Comparatives/superlatives
- **Teach the target grammatical morphemes**
 - Write about 20 target words, phrases, and sentences that include a specific target morpheme
 - Baseline the production of each morpheme in the context of 20 stimuli; use discrete trials
 - Teach each word, phrase, or sentence on Discrete Trials to begin with; model the target response and reinforce correct imitative responses
 - Fade modeling and start teaching the target responses on evoked trials (see Fading)
 - Probe for generalized productions after having trained four to six exemplars and using the remaining untrained stimulus items
 - Offer additional training when the 90% correct probe criterion is not met
 - Alternate training and probe trials until the probe criterion is met for the morpheme being trained at a certain level (e.g., words, phrases, sentences)
 - Shift training to the next higher level or select another morpheme for training
 - Reinforce all morpheme productions in naturalistic conversation

Hegde, M. N. (2005). *Treatment protocols for language disorders in children, Volume I: Essential morphologic features*. Plural Publishing.

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Habit Reversal Training. Training awareness and having the client repeatedly practice an incompatible behavior to eliminate an undesirable behavior; repeatedly exhibiting the same behavior to be eliminated is called *negative practice* (e.g., Van Riper's voluntary stuttering method); examples of better-researched habit reversal include the following:

- To increase awareness of the behavior to be eliminated, describe the behavior to your clients, and train them to measure their frequency; have the clients watch on recorded videos their own behaviors targeted for elimination
- To control the treatment-interfering hand tremors associated with Parkinson's disease, the clinician may ask the person to keep the hands firmly pressed on the arms of the chair (incompatible response)
- To control such vocal tics as grunting and coughing (often in persons with Tourette syndrome), the clinician may teach the client to practice diaphragmatic breathing for 1 minute before or soon after the occurrence of tics
- To reduce interjected speech disfluencies such as *um* and *you know*, the client may be trained to practice a 3-second pause either before or soon after the disfluency

Hard Glottal Attack. Treat abrupt voice initiation, a form of vocally abusive behavior, in individuals with voice disorders; may be a treatment target in some persons who stutter; in all cases, instruct, model, encourage imitation, reinforce, and fade modeling.

- Teach gentle, relaxed, easy onset of phonation
- Teach the individual to blend words initially
- Teach gentle onset in persons with stuttering by instructions and modeling
- Use the Chewing technique, Whisper-Phonation, the Chant-Talk, and the Yawn-Sigh, all described under Voice Disorders, Specific Normal Voice Facilitating Techniques to reduce hard glottal attacks

Harshness. To treat this voice quality that results from excessive laryngeal tension, effort, and constriction by

instruction, modeling, reinforcement for imitation, and fading of modeling:

- Use relaxation to reduce vocal tension
- Teach soft, easy contact of the vocal folds
- Teach gentle onset of phonation
- Use a combination of Specific Normal Voice Facilitating Techniques described under Voice Disorders

Hearing Aids. Electronic devices that amplify sound, prescribed for individuals with hearing loss; mostly digital; types include body-worn, behind-the-ear (BTE), in-the-canal (ITC), in-the-ear (ITE), or completely in-the-canal (CIC).

Hearing Aid Orientation. An aural rehabilitation procedure implemented by an audiologist; a set of procedures in which the use and care of a hearing aid is taught to a person with hearing loss who has just purchased one.

Hearing Assistive Technology Systems. Devices other than the traditional hearing aids that are more or less technologically sophisticated; designed to help persons with hearing loss comprehend speech; used in situations in which the traditional hearing aids are less effective; most are capable of converting speech and environmental sound signals directly from the source to auditory (amplified), visual, or vibratory signals that the person with hearing loss receives; also known as assistive listening devices (ALDs) or hearing assistive technology systems (HATS); most systems have detached microphones that may be placed near the source of sound or speech signal; the signal is transmitted to an amplifier and the device the individual wears; see also Aural Rehabilitation; systems include the following:

- **Frequency-modulated (FM) auditory system** that offers wireless connection between a speaker and a listener; consists of a transmitter with a microphone and a receiver with earphones; signal is transmitted to the receiver through radio frequency and is unaffected by noise in the room, a problem not solved by traditional hearing aids; may be used in group aural rehabilitation

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- **Infrared listening systems** that include transmitters that send messages on pulses of light and receivers worn by individuals to receive and decode those messages; useful in large listening environments such as concert halls, theaters, houses of worship, and classrooms; smaller transmitters can be used in homes to view television and listen to music
- **Alerting systems** that include mechanisms to amplify telephone ringing; flashing lights that alert a person with hearing loss to incoming phone calls, smoke or fire alarm, a baby's cry, ringing of a doorbell, and so forth; vibratory mechanism that wakes up a person with hearing loss from sleep
- **Television or radio amplifiers** that include a small amplifier, a microphone that is plugged into the television or radio, and a receiver the individual needing extra amplification of signals wears; a transmitter sends the signal to the receiver; volume may be adjusted on the amplifier, thus allowing louder signals only for the person using the system
- **Telephone amplifiers** that help individuals with mild hearing loss who do not use a hearing aid; individuals may fit an amplifying device on the telephone itself
- **Telecommunication devices for the deaf (TDDs)** that allow two persons communicating over a phone to type their messages; also called text telephones; allow display of typed messages on a small screen on the telephone; may be a part of a relay service that allows a hearing person without the special equipment to speak to a staff person who types in the message, which is transmitted to the TDD used by person with hearing loss

Hearing Loss. Oral speech and language disorders are a common concomitant of significant hearing loss; treatment of communication disorders associated with hearing loss requires an audiological examination to determine the level and type of hearing loss and an aural rehabilitation program in which the individual has been fitted with a hearing aid; hearing loss is commonly classified as follows:

- Mild hearing loss: 25–40 dB HL

- Moderate hearing loss: 41–70 dB HL
- Severe hearing loss: 71–90 dB HL
- Profound hearing loss: 90 dB and higher

Intervention for Communication Disorders Associated With Hearing Loss/General Guidelines

- Treatment procedures for Language Disorders in Children, Speech Sound Disorders, and Voice Disorders are applicable to children and adults with hearing loss
- Note that parents of deaf children in the United States may or may not opt for oral communication training; most deaf children learn the American Sign Language or other forms of a manual communication system and become competent communicators; parents of children with mild to moderate hearing loss are likely to opt for oral communication training
- Review the audiological reports; consult with the child's audiologist to understand the child's type and level of hearing loss and its educational implications
- Before beginning a home-based language stimulation program or the clinic-based formal teaching program, make sure that the child has been trained well in the use of the hearing aid and that an auditory training program has been completed
- Begin a speech and language stimulation program as early as possible
- Have the child under appropriate medical and audiological management
- Get the family involved from the beginning in speech and language stimulation activities
- Have the child fitted with an individual hearing aid; at the beginning of each session, check the child's hearing aid to ascertain that it is turned on and working properly
- Work closely with educators and special educators, especially the educator of the deaf
- Select treatment targets to support the child's academic work and success in the classroom; consult with the child's regular and special education teachers on

relevant communication targets, especially the language targets

- Integrate literacy targets with communication training, as children with hearing loss tend to have significant reading and writing problems; work closely with the child's teacher to select reading and writing targets that may be a part of the communication training
- Train parents to start a home language stimulation program; impress on the parents that it is inappropriate to withhold normal verbal interaction and special verbal stimulation on the mistaken assumption that it is "of no use because the child cannot hear"

Teach Oral Language

- **Teach functional words, phrases, and simple sentences**
 - Begin clinic-based oral language training as early as possible
 - Use visual cues in all training sessions
 - Begin teaching a set of everyday functional words; ask parents to introduce new but everyday words regularly
 - Teach simple verbs that are especially difficult for children with hearing loss
 - Teach phrases and simple sentences formed out of words the child has mastered
 - Correct immediately any non-English word order that you observe in the child's speech
 - Teach Grammatical Morphemes, as they are especially difficult for children with hearing loss
 - Teach longer, complex, and compound sentences; coordinate teaching various syntactic features with the classroom instruction of language skills; teach a variety of syntactic structures
- **Teach social communication skills**
 - Teach topic initiation, topic maintenance, turn taking, conversational repair strategies, and narrative skills; see Social Communication Skills for procedural details

- Instruct how to give sufficient background information before saying something and reinforce and refine the child's attempts
- Model correct stress and intonation to improve prosodic features of speech; reinforce correct imitations; withdraw modeling
- **Teach specialized academic terms**
 - Teach abstract terms, terms with dual meanings, and the meaning of proverbs, idioms, similes, and synonyms and antonyms, as they are especially difficult for children with hearing loss
 - See Language Disorders in Older Students and Adolescents for teaching abstract, advanced, and academic terms
 - Refer to specialists who can teach such nonverbal communication systems as American Sign Language if the individuals, families, or both prefer

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Teach Speech Production Skills

- Set improved intelligibility as the main speech treatment goal
- Use the treatment procedures described under Speech Sound Disorders with the following modifications:
 - Give ample visual cues in teaching speech sound production
 - Use an FM auditory trainer if working in small groups
 - Use the Phonetic Placement Method
- Target the following speech production tasks and teach them with intensive trials:
 - Teach fricatives, stops, and affricates, as these are especially difficult for children with hearing loss
 - Teach voiced and voiceless sound distinctions; eliminate substitutions of voiced consonants for voiceless consonants
 - Teach discrimination between diphthongs and vowels (e.g., /a/ and /ai/ distinctions)
 - Reduce nasalized production of vowels and oral consonants

- Reduce unnecessarily extended duration of vowels
- Reduce breathiness before vowel productions
- Teach sounds that are omitted in word-initial, and especially word-final, positions
- Teach more precise and definitive production of all speech sounds, especially consonants
- Reduce or eliminate additions of unnecessary sounds (e.g., s_{atop} for *stop*)
- Reduce the frequency of inappropriate release of final consonants (e.g., cap^h)

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Treat Voice and Resonance Disorders

- Use the standard techniques described under Voice Disorders
- Give visual feedback with such instruments as VisiPitch
- Modify such abnormal voice qualities as breathiness, harshness, hoarseness, stridency, and monotone
- Reduce the vocal pitch if it is judged too high
- Teach appropriate loudness of speech if judged too soft
- Modify resonance disorders; modify both hypernasality and hyponasality

Teach Typical Prosodic Features

- Teach smooth flow of speech
- Reduce pauses that may be too frequent and placed inappropriately
- Teach normal intonation to reduce a monotone
- Teach appropriate breath control to improve phrasing
- Judge the effects of pitch and loudness modification on speech prosody

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Tye-Murray, N. (2022). *Foundations of aural rehabilitation* (6th ed.). Plural Publishing.

High-Probability Behaviors. A method of reinforcing a target skill by giving access to a highly desired and frequently exhibited behavior when the child exhibits a stated number of target responses; a behavior that is exhibited often can reinforce the one exhibited less often if access to the more likely behavior is made contingent on first

exhibiting the less likely behavior; an effective method to increase low-frequency treatment targets.

- Identify behaviors your client exhibits frequently (e.g., listening to music, looking in a picture book, listening to stories you tell, drawing on the blackboard, watching television, an ice cream treat, skiing)
- Design a method by which you in the treatment sessions control (e.g., listening to your stories) and the family members at home can control (e.g., ice cream treat or skiing opportunities)
- Give tokens in treatment sessions for producing the target responses that are initially of low frequency (e.g., correct production of speech sounds, language features, fluency)
- Let the individual accumulate the tokens and exchange them for opportunities to engage in the high-probability behaviors (e.g., brief periods of listening to music or stories in treatment sessions, watching TV at home, going on ski trips)

H

Hoarseness. Voice quality that results from leakage of air and aperiodic vibration of the vocal folds; pitch may be too low; any condition that changes the mass and size of the vocal folds, including vocal nodules, may cause hoarseness of voice; treat hoarseness with Specific Normal Voice Facilitating Techniques, described under Voice Disorders.

Huntington's Disease. To treat speech impairments in individuals with this degenerative neurological disease, see Dysarthria and Hyperkinetic Dysarthria under Treatment of Dysarthria: Specific types; to treat individuals in advanced stages, see Dementia; in the late stage, consider simple, no-technology Augmentative and Alternative Communication, including alphabet boards, communication boards, eye-gaze, and Yes-No responses

Hyperadduction. To treat this laryngeal abnormality causing voice problems, reduce the excessive force and tension associated with vocal fold closure.

- Teach laryngeal relaxation
- Teach breathy onset of phonation

- Teach gentle, relaxed, easy phonatory onset
- Massage the larynx
- Use such other specific normal voice facilitation techniques as the [Yawn-Sigh Method](#) and the [Chewing Technique](#) described under [Voice Disorders: Specific Normal Voice Facilitating Techniques](#)

Hyperkeratosis. Keratotic lesions in the pharynx or larynx; pinkish, rough lesions with horny growth; associated with [Voice Disorders](#); nonmalignant in the initial states; may be a precursor to malignant lesions; due to continued irritation of tissue, including chronic smoking; treatment is to stop behaviors that result in continued irritation (e.g., cessation of smoking).

Hyperkinetic Dysarthria. To treat this motor speech disorder, see [Hyperkinetic Dysarthria](#), under [Treatment of Dysarthria: Specific Types](#).

Hypernasality. To treat this resonant disorder in children and adults, see [Velopharyngeal Dysfunction](#) for procedures; see also [Voice Disorders](#) for general information on resonance problems.

Hypertonia. To treat excessive muscle tone or tension associated with neurogenic communication disorders, see [Cerebral Palsy](#) and [Dysarthria](#).

Hypokinetic Dysarthria. To treat this type of motor speech disorder, see [Dysarthria](#) and [Hypokinetic Dysarthria](#) under [Treatment of Dysarthria: Specific Types](#).

Hyponasality. Treatment of reduced or absent nasal resonance on nasal sound productions is typically medical because it is often due to nasal or nasopharyngeal obstruction from nasal tumors, nasal polyps, nasopharyngeal infection leading to edema, overgrown adenoids, surgically created pharyngeal flaps that are too broad, too tight an obturator bulb, and so forth; cases with functional (behavioral) causes are rare but do exist; in some individuals,

hyponasality may persist even after a physiological cause (e.g., chronic allergies) has disappeared; speech-language pathologists may treat functional hyponasality following a medical examination that rules out organic causes; treatment research on hyponasality is limited; a few treatment suggestions follow.

- Refer the individual to an otorhinolaryngologist for an examination before starting behavioral treatment for hyponasality
- Informally test the airflow through the nose to make sure there is no obstruction (exhale through the nose, through only one nostril, and then the other nostril)
- Describe and model nasal and oral speech sound productions
- Select a set of words, phrases, and sentences that contain the three nasal sounds and another set with only oral sounds to provide contrast to the individual
- Model the production of utterances with and without nasal sounds and point out the contrast
- Ask the individual to alternatively imitate utterances with and without nasal sounds
- Experiment with nasal sounds placed in different phonetic contexts, as some contexts may facilitate nasal resonance better than other contexts
- Reinforce the individual with verbal praise for even a slight increase in nasal resonance on nasal sounds; verbally correct denasalized production of nasal sounds; reinforce oral resonance of oral sounds

Hypotonia. To clinically manage reduced vocal tone or tension, see [Dysarthria](#).

I

Iconic Symbols. Symbols that look like the object they are supposed to represent in teaching Augmentative Communication, Aided; easier to teach and learn than Non-iconic Symbols.

Ideographic Symbols. Symbols that are more abstract than pictographic symbols to teach a type of nonverbal communication in which the ideas are expressed through graphic representation; may be line drawings; for details on teaching methods, see Augmentative Communication—Aided.

Idiosyncratic Phonological Patterns. Phonological errors not found in the typical speech of children; target these error patterns for elimination; see Speech Sound Disorders in Children for details.

Imitation. Use this mode of response in the early stage of treatment for all disorders; it is a response that follows a modeled stimulus and takes the same or similar form of its stimulus; essential to establish all communicative target skills in individuals of all ages; Modeling is the preceding stimulus that evokes imitation; to teach imitative responses:

- Instruct and demonstrate the target response
- Model the correct response for the individual to imitate
- Place extra vocal emphasis on the specific target behavior in a modeled utterance (e.g., “Say, two cups,” with an emphasis on the grammatic morpheme in teaching the regular plural productions)
- Reinforce the individual for imitating the modeled response
- Reinforce approximations initially, especially in the treatment of correct speech sound productions
- Require greater match to the modeled response in successive trials
- Fade modeling and transition to evoked (more spontaneous) responses

Imposition of Work. One of the Direct Methods of Response Reduction in which an undesirable behavior is

reduced by immediately imposing work designed to reduce or eliminate the negative effects of that behavior; also known as overcorrection; has two components: restitution and positive practice.

Restitution

- Immediately following an undesirable behavior (e.g., throwing toys around), ask the child to neutralize the effects of that behavior (pick up the toys)
- Ask the child to go beyond neutralizing the effects of behavior by improving the situation (ask the child to put the toys on a shelf and then clean up the mess)

Positive Practice

- Ask the child to practice an incompatible, appropriate behavior repeatedly without reinforcement (ask the child to organize the stimulus materials on your desk)

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.). Plural Publishing.

Improvement. Documented positive changes in the individual's target behaviors under routine treatment compared with initial assessment or baseline performance; needed to justify treatment; what a clinician can claim when routine treatment is offered; not the same as Effectiveness of Treatment, which requires controlled experimental evidence that demonstrates that treatment was better than no treatment; to document improvement:

- Define the target skills in operational (measurable) terms
- Establish reliable baserates of target skills before initiating treatment
- Measure the frequency of target skills in all treatment sessions
- Compare the baserate and treatment response rates to show systematic positive changes in the skills under treatment

Incidental Teaching Method. Use this naturalistic language teaching method that targets functional communication skills in the context of everyday verbal interactions; for procedural details, see Language Disorders in

Children; Treatment of Language Disorders: Specific Techniques or Programs.

Incompatible Behaviors. Behaviors that cannot be produced simultaneously; use this behavioral strategy in which you teach a desirable action that is incompatible with an undesirable action; for example, to indirectly reduce a child's off-seat behavior, systematically reinforce sitting behavior; the strategy avoids aversive methods of response reduction; see Differential Reinforcement for procedural details.

I **Independent Variables.** Hypothesized or demonstrated causes of events that scientists investigate; the same as treatment methods clinicians use; anything a clinician does that affects the individual's behavior, including instructions, modeling, demonstration, positive reinforcement, verbal corrections, and biofeedback, are potential independent variables of positive changes in the individual; contrasted with Dependent Variables.

Indirect Language Stimulation. A collection of somewhat varied, naturalistic, unstructured, or minimally structured language stimulation procedures based on play activities with no specified target behaviors or systematic reinforcement; generally, direct methods of treatment (not just stimulation) have better evidence than indirect stimulation; see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Indirect Methods of Response Reduction. Use these highly effective and desirable means of indirectly reducing undesirable behaviors by increasing desirable behaviors; indirect because nothing is done directly to decrease the undesirable behaviors; more desirable than the direct methods of response reduction because they avoid negative side effects of aversive procedures; see Differential Reinforcement for specific procedures.

Individualized Education Programs (IEPs). Child-specific intervention programs designed for children with disabilities or special needs served in public grade schools.

Individualized Family Service Plans (IFSPs).

Special education programs designed for children with disabilities in the age range of birth through 2 years and their family members.

Initial Consonant Deletion. Teaching the correct production of initial consonants in word positions to eliminate this phonological error pattern; see [Speech Sound Disorders](#) for procedural details.

Informative Feedback. Use this method of reinforcing target skills by providing information to your clients on their performance levels; information feedback may be verbal or mechanical.

- Give *verbal informative feedback*, periodically telling your clients how well they are doing (e.g., “You have improved to 85% today; it was 76% in the last session”); show and describe charts and graphs that depict increases in target skills across sessions
- To give mechanical informative feedback, display positive changes and improvement data on computer monitors and other display devices

Infrared Systems. To use this system in the rehabilitation of persons with hearing loss, see [Hearing Assistive Technology Systems](#).

Inhalation Method. Teach this method of air intake to produce esophageal speech; a method of producing speech without larynx; for procedures, see [Laryngectomy](#).

Inhalation Phonation. Use this method of voice therapy to evoke true vocal fold vibration in individuals who are aphonic; for procedures, see [Voice Disorders](#), [Specific Normal Voice Facilitating Techniques](#).

Injection Method. Teach this method of air intake to produce esophageal speech; for procedures, see [Laryngectomy](#).

Initial Response. The first, simplified element of a target response used in [Shaping](#).

Instructions. Use these common teaching procedures at the beginning of intervention and whenever needed throughout the treatment phase; they are verbal stimuli that promote the production of target responses; combined with Demonstration, Modeling, and Manual Guidance (as in Phonetic Placement Method).

- Design instructions that describe the target behavior for the client
- Simplify your instructions and tailor them to the individual client
- Write your instructions and practice their delivery, but deliver them naturally; do not read them
- Repeat instructions until the individual understands them
- Combine them with demonstrations, modeling, and manual guidance

Intellectual and Developmental Disabilities (IDD).

Intervention for individuals with intellectual, developmental, social, and adaptive behavioral deficiencies is a multifaceted task; communication problems are a significant aspect of these disabilities and are often the early diagnostic markers.

Intervention Programs for IDD/General Guidelines

- Children and adults with IDD need a comprehensive educational, social, and clinical intervention program because of the academic, social, and occupational challenges they face
- Intervention programs are best developed by a team of specialists, including teachers, special educators, psychologists, speech-language pathologists, audiologists, neurologists, and other specialists depending on the complexity of the child's disabilities
- The team should consider the suggestions and expectations of adult clients or the child's parents
- Early intervention is essential for better outcome
- Communication intervention may include speech, language, fluency, and voice problems; presence of

hearing loss and physical disabilities may complicate treatment efforts; see the cited sources for details; the treatment procedures for [Language Disorders in Children](#) and those for [Speech Sound Disorders](#) are applicable with the following special considerations:

Train Family Members to Stimulate Speech and Language Skills

- Conduct a few trial treatment sessions to find out what works best for the child before you recommend a home language stimulation program
- Ask the parents of young children to socially reinforce cooing, babbling, other forms of early vocalization; the first few words; and production of phrases and simple sentences at different stages of intervention
- Ask parents to observe your treatment sessions
- Have parents participate in the treatment sessions by presenting stimuli, modeling the target language skills, and reinforcing the child
- Have parents videorecord sample home language stimulation sessions and review them for accuracy; give constructive feedback to the parents to improve their implementation of the program
- Recommend clinic-based intervention; almost all of them need it
- See [Language Disorders in Infants, Toddlers, and Preschoolers](#) for details

Design a Comprehensive Communication Treatment Program

- Select speech and language targets that help meet the child's academic and social demands
- Include voice and fluency targets if warranted
- Select target behaviors that are functional (effective) in educational, occupational, family, and social situations
- Select emergent or regular literacy skills that may be integrated with communication treatment; see [Literacy and Literacy Skills in Children](#)
- Have the parents review the selected target skills and modify the targets in light of their suggestions or comments

- Select the initial training targets that will be most effective in natural settings and in classrooms in the case of children
- Select articulation training before language training if this leads to improved communication sooner
- Select language training if teaching functional words will help improve social communication
- Select functional signs before verbal expressions in the case of minimally verbal children

Sequence the Target Behaviors

- Start with functional words or phrases if the child already produces a set of words
- Build on the existing skills, skills just mastered; expand the functional verbal responses from words to phrases and phrases to sentences
- Keep social and naturalistic communication as the final target
- Use small-step increments; shape responses whenever the individual cannot imitate modeled responses
- Model the target responses frequently
- Reinforce approximations of modeled responses in the initial stages of treatment
- Require better approximations as the skill becomes more reliably produced
- Require an exact match, if that is practical, as the child's proficiency increases
- Fade modeling in gradual steps; use prompts to fade modeling

Structure the Sessions

- Use a tight structure initially, but keep the structure flexible
- Use objects and events more than pictures as treatment stimuli
- Teach multiple exemplars of each target behavior
- Have parents bring the child's books and toys that may be used as treatment stimuli to promote better generalization to the child's home environment

- Use the Discrete Trial procedure to initially establish the skills unless the child can engage in more naturalistic interactions; give massed treatment trials
- Use primary reinforcers initially, but always pair them with verbal praise
- Fade primary reinforcers as soon as possible to maintain the correct responses on verbal praise
- Consider using tokens for correct responses that you exchange for a small gift at the end of the session; pair the tokens with verbal praise to fade the tokens
- Use food and drink to teach mands (e.g., such mands as “I want some juice, please” cannot be reinforced with verbal praise)
- Stretch the reinforcement schedule gradually; start with continuous reinforcement and shift gradually to such other intermittent schedules as fixed ratio 3, 4, or 5 (reinforcement for the third, fifth, or seventh correct response)
- Include parents, siblings, or peers in treatment with a view to have them prompt and reinforce target skills in natural settings
- Probe for generalized production of trained responses with untrained words, phrases, and sentences; terminate training on target behaviors only when they meet the probe criterion (e.g., 90% correct on probe trials or conversations)

Loosen up the Treatment Sessions

- Shift training from the discrete trial procedure to more conversation-oriented procedures
- Use such procedures as Mand-Model and Incidental Teaching Method described under Language Disorders in Children

Teach Social Communication Skills

- Teach conversational skills; start with such easily learned skills as eye contact and turn taking in conversation; teach topic initiation, topic maintenance, and narrative skills
- See Social Communication Skills for details

Implement a Maintenance Program

- Train parents, older siblings, and teachers in prompting and reinforcing newly acquired speech and language skills
- Train skills in conversational speech
- Train in varied settings, including the child's classroom, cafeteria, and other settings
- Train in the presence of peers and siblings
- Reinforce only socially and occasionally
- See Maintenance Strategy for additional suggestions
- Follow up and arrange for booster treatment

Consider Nonverbal Means of Communication

- Teach a nonverbal means of communication when warranted
- Consider an appropriate form of Augmentative and Alternative Communication

Kaderavek, J. N., & Henbest, V. H. (2024). *Language disorders in children: Fundamental concepts of assessment and intervention*. Plural Publishing.

Seager, E. et al. (2022). A systematic review of speech, language, and communication intervention for children with Down syndrome from 0 to 6 years. *International Journal of Communication Disorders*, 57, 441–463.

Intelligibility. A treatment target for individuals who cannot produce speech sounds accurately; to improve impaired speech intelligibility and teach the correct production of phonemes; see Speech Sound Disorders in Children and Dysarthria for procedural details.

Interfering Behaviors. Leaving the seat, asking irrelevant questions during treatment, crying, wiggling in the chair, and inattentiveness are among the many interfering behaviors that the clinicians need to control in treatment sessions; a priority because speech-language behaviors cannot be taught unless such behaviors are reduced or eliminated.

- Withhold reinforcement for all interfering behaviors (e.g., ignore such irrelevant questions as “Are we done yet?” or “You know what?”)

- Use one of the Differential Reinforcement procedures to increase the alternative desirable behaviors, which will then reduce the interfering behaviors

Intermediate Response. Responses other than the initial and the final responses targeted in shaping complex skills.

Intervention. Manipulation of external variables to affect the course of a disorder, disease, or problem behavior to replace them with outcomes; the same as *treatment*.

In-the-Ear Hearing Aid. A small hearing aid that fits within the ear canal and concha.

I

Intonation. A treatment target for individuals with prosodic impairments; a normal aspect of speech and a treatment target for those with Dysarthria or Stuttering who have been taught to speak with prolonged syllables; a treatment target for any individual with a monotonous speech.

Isolation Time-Out. Removal of persons as soon as they exhibit aggressive or dangerous behaviors from a reinforcing environment and placing them in a physically isolated non-reinforcing environment (e.g., an isolated booth); not to be used in communication treatment; misuse of this technique is common in educational settings; prefer the Nonexclusion Time-Out.

J

Jargon. Reduction of undesirable and excessive use of somewhat meaningless terms the individual has invented is a treatment target in aphasia; see [Aphasia](#) and [Wernicke's Aphasia](#) for details.

Jitter. To treat a cycle-to-cycle variation in frequency of vocal fold vibrations that exceeds 1%, heard as a pitch disorder; see [Voice Disorders](#).

Joint Attention. Target this language and general communication skill to teach two or more people to pay attention to the same event or object simultaneously; to teach joint attention in child language treatment, ask the child to look at what you are looking at, and reinforce the child who immediately complies with your request.

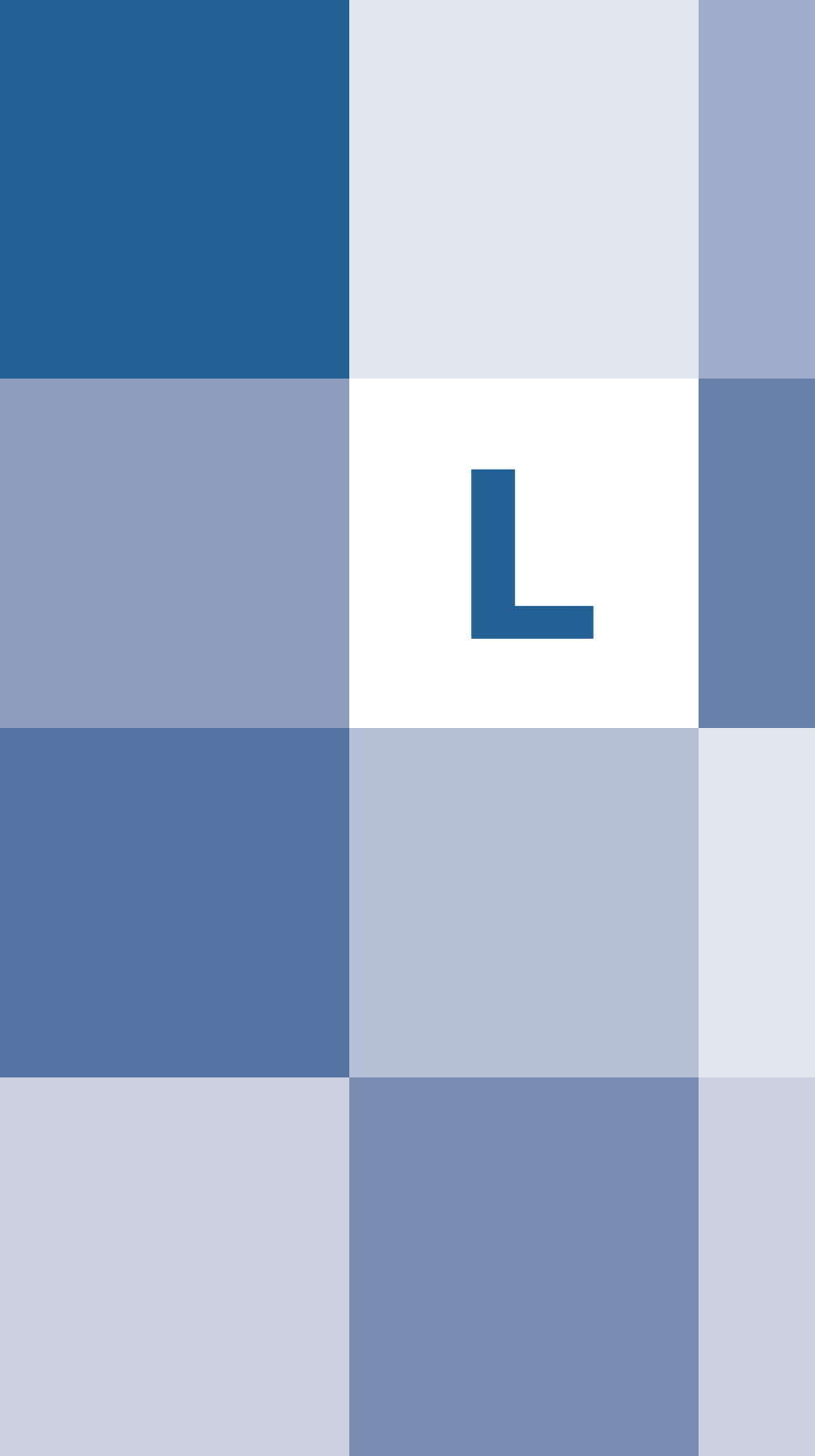
Joint Reference. Target this language skill that is the same as [Joint Attention](#), so that you and the client are looking at the same event or stimulus while teaching specific language skills; ask the child to look at the stimulus item you just presented before evoking a speech or a language response.

K

Key Word. A word in which a generally misarticulated sound is correctly produced because of coarticulation facilitation. Correct production of a target sound in a specific key word is contrasted with incorrect production of the same sound in other, non-key words; see Speech Sound Disorders in Children.

Keyboard. Use these electronic devices to teach nonoral communication skills to individuals with limited oral skills; a keyboard is a device that gives input to the computer with different layouts of letters, numbers, and other command inputs; may include key depression, touch membrane, or touch screen surfaces; see Augmentative and Alternative Communication.

Keyguard. Use a plastic or Plexiglas material to cover the standard keyboard while teaching nonoral communication to individuals who are minimally verbal; a keyguard has holes for each key; the AAC user slides a pointer over the cover; helps prevent miscommunication by accidental hitting of irrelevant keys.



Language Delay in Children. Treat language delay as you would Language Disorders in Children; generally, the two terms mean the same except for the questionable connotation that children with language delay are slow in learning and that they may catch up with their typically progressing peers; language disorders in children tend to persist, especially without treatment.

Language Disabilities in Children. For treatment procedures, see Language Disorders in Children, preferred term.

Language Disorders in Adults. To treat language disorders and other communication disorders in adults, see Aphasia, Dementia, Right-Hemisphere Disorder, and Traumatic Brain Injury in Adults.

Language Disorders in Children. Treatment of language disorders in children may be multifaceted because the disorders may be associated with other clinical conditions; terms keep proliferating to refer to the same condition including *specific language impairment*, *primary language disorder*, *language delay*, *idiopathic language disorders*, *language disability*, *language learning disability*, *late talkers*, *late language emergence*, and *developmental language disorders (DLD)*; the newer term *DLD* is gaining popularity; *SLI* is by definition language disorders with no associated clinical conditions, but other subtle impairments have been reported; none of the terms that are alternatives to *language disorders in children* are free from disagreement, controversy, and unwarranted theoretical assumptions; the new and popular name (*DLD*) implies (biological?) *development* of language and an unsupported assumption of a *developmental disorder* that may resolve itself over time; language disorders are unlikely to remediate themselves without professional intervention; therefore, a theoretically neutral term that makes no assumption about etiology or the course of the disorder, *language disorders in children (LDs)*, is preferred in this guide; *LDs* in children pose significant limitations in social commu-

nication and academic performance; the treatment procedures described under this entry apply to children with LDs, regardless of whether the disorders are considered *specific, idiopathic, developmental, delay*, or some other; to treat children's LDs associated with additional clinical conditions, see Autism Spectrum Disorder, Cerebral Palsy, Intellectual and Developmental Disabilities, Hearing Loss, and Genetic and Congenital Syndromes Associated With Communication Disorders.

Treatment of Language Disorders in Children: General Guidelines.

Issues relevant to the target behavior selection, intervention sequencing, and session structure include the following:

- **Strategies of target behavior selection**
 - In the normative strategy, target behavior selection is based on language developmental norms; skills that are missing according to the age-based norms are the treatment targets
 - In the client-specific strategy, target behavior selection depends on the educational and social demands the child faces and the family communication patterns; this view is more consistent with the functional view of language treatment and uniqueness of each individual; the normative and the client-specific strategies may be combined
- **Sequencing the language targets for treatment**
 - In the normative strategy, target behaviors are taught in the sequence in which children normally acquire them; although some behaviors may be more efficiently taught this way, there is no compelling evidence that this is always the best approach to sequencing target behaviors, rarely put to the test because of the belief that it is the best
 - In the client-specific strategy, clinicians may experiment with different sequences of teaching the target behaviors and then use the one that produces the best results; the data may show that normative sequence is indeed the best or other sequences are

better or just as good; there is emerging evidence that teaching more complex skills ahead of their normative sequence may be effective and efficient

- **Structure of language treatment sessions**
 - Tightly structured sessions are preferred when the skills are taught on massed discrete trials; the clinician controls the stimuli and response consequences; tend to have clearly defined target behaviors for teaching; measurement-oriented
 - Loosely structured sessions tend to be play-oriented; stimuli and consequences may not be directly controlled and may not have clearly specified target behaviors that are taught systematically
 - The two options are not a matter of either-or choice; earlier sessions need to be highly structured to efficiently establish the skills; the later sessions must be more relaxed and naturalistic to promote generalization and maintenance; the final conversational sessions should be more naturalistic in structure, stimuli, and reinforcement
- **Language treatment efficacy**
 - Clinicians should avoid techniques that are based exclusively on expert advocacy or speculative theories in favor of those that are shown to be effective in controlled single-subject or group experimental design studies
 - Most behavioral treatment procedures described in this recommended comprehensive program have been experimentally tested; see [Evidence-Based Clinical Practice](#) for guidelines on evaluating treatment research and selecting treatment procedures

Treatment of Language Disorders in Children:

A Comprehensive Procedure. Use this self-sufficient procedure to treat children with any kind of LDs; all procedures needed to (1) select target behaviors, (2) establish baselines, (3) sequence treatments, (4) teach skills from the very low level of single word productions to the most complex social communication skills, (5) probe for generalization, and (6) imple-

ment maintenance procedures including parent training are described; most specific techniques or programs described in a subsequent section are not as comprehensive as this program; although the program is self-sufficient, clinicians may integrate selected, well-defined, evidence-based technique (e.g., mand-model, expansion, time delay, enhanced milieu teaching) described under specific techniques or programs

- **Steps in developing a language treatment plan for a child**

- Select target behaviors for training
- Plan a sequence of treatment
- Select stimulus materials
- Establish baselines
- Write a treatment and maintenance plan
- Implement the treatment plan
- Implement the maintenance plan

- **Language intervention targets**

- Are child specific and ethnoculturally appropriate; see [Language Disorders in Ethnoculturally Diverse Groups](#) in a later section
- Are useful in natural settings
- Can make an immediate and socially significant difference in the child's communication skills
- Help meet the academic and social demands the child faces
- Help expand communication skills into conversational speech in natural settings
- Are within the child's reach as judged by current performance (words, phrases, sentences, conversational speech)

- **Design a flexible sequence of treatment**

- Begin with modeling-imitation sequence and move through the planned sequences:
 - More spontaneous or typical productions
 - From words to phrases
 - Phrases to controlled (less spontaneous) sentences
 - Controlled sentences to spontaneous conversational speech

- Treatment in clinical settings to treatment in more naturalistic settings
- More structured sessions to progressively less structured sessions
- Continuous reinforcement to intermittent reinforcement
- Primary reinforcers to social reinforcers
- Social reinforcers to natural consequences inherent in communication
- **Prepare treatment stimuli**
 - Select 20 words, phrases, or sentences to teach grammatical morphemes (e.g., contain the plural *s*, the present progressive *ing*, or the past tense *ed*)
 - Select 20 basic, functional, culturally relevant words to teach a child with a minimal verbal repertoire
 - Select ethnoculturally appropriate, client- and target-specific stimuli that are colorful, attractive, and realistic; prefer objects to pictures
 - Obtain stimuli from the child's home (the child's favorite books, toys, and objects)
- **Prepare a Response Recording Sheet**
 - Write 20 selected exemplars of a target behavior (e.g., 20 words that contain the plural morpheme) on the recording sheet
 - Use the same sheet to record correct and incorrect responses in all types of sessions (baseline, treatment, probes, and follow-up)
- **Establish baselines of target behaviors**
 - Record repeated conversational language samples that help reliably document the occurrence of language targets
 - Administer a set of modeled discrete trials and a set of evoked discrete trials that (a) help capture the production of specific language targets that may not be adequately sampled in conversational speech, (b) are necessary in case of children with no or minimum conversational skills, (c) help measure the frequency and percentage of correct and incorrect responses, and (d) will allow for a comparison of response frequency before, during, and after treatment

- **Administer baseline modeled trials**
 - Place a stimulus picture or one or more objects in front of the child (e.g., a picture of two books or two books)
 - Ask a question to evoke the target response (“Johnny, what do you see?”)
 - Model the response (“Johnny, say, *I see two books*”)
 - Do not reinforce or verbally correct
 - Record the response on the recording sheet
 - Present the next picture (e.g., that of two cups); repeat the procedure until all the 20 exemplars
 - Use the same procedure to base rate target words, phrases, sentences, or any other targets
- **Administer baseline evoked trials**
 - Place a stimulus picture or object in front of the child
 - Ask a question to evoke the target response (“Johnny, what do you see?”); do not model the response
 - Do not reinforce or verbally correct
 - Record the response on the recording sheet
 - Repeat the procedure for all target exemplars (grammatical morphemes, basic vocabulary items, phrases, sentences each with 20 exemplars as noted)
 - Calculate the percentages of correct baseline responses separately for the modeled and evoked trials
- **Calculate the percentages of correct responses in conversational speech**
 - Use the *obligatory context method* to score the responses
 - Score the response as correct if produced in an obligatory context (e.g., adding the plural morpheme when the stimuli are plural); incorrect if absent
 - Count the number of obligatory contexts for a grammatic feature in a language sample and the number of correct productions in those contexts to calculate the percentage correct responses in conversational speech
- **Write a treatment and maintenance plan**
 - Specify the response complexity level at which you will begin training (typically, selected individual words or grammatical features at the word level)

- Describe how you would present the treatment stimuli
- Describe how you would instruct the child at the beginning of treatment and at the beginning of treatment trials
- Describe how you would demonstrate the target skill execution
- Describe the question you would ask before you model (e.g., “Johnny, what is the boy doing?”)
- Describe the model you would present (e.g., “Say, *the boy is walking.*”)
- Describe the acceptable response form (complete, appropriate, imitated, typical unimitated)
- Specify the treatment trial format (typically discrete trials in the beginning)
- Describe the reinforcers and how you deliver them (verbal praise, tokens, high-probability behaviors, primary reinforcers in special cases)
- Describe the verbal corrections and how you will deliver them; describe other potential response reduction procedures (e.g., token withdrawal, extinction, time-out)
- Describe possible shaping, prompting, and fading procedures
- Describe the sequence from the initial treatment through the different levels of training, including the final maintenance and follow-up procedures
- Specify performance criteria for levels of training, dismissal, and follow-up

Teach Functional Words

- Begin treatment by teaching a set of functional words to a nonverbal or minimally verbal child; select at least 20 functional words; most of them may be among the first words children acquire (e.g., such words as *mommy* and *daddy*; food items like *milk*, *candy*, *juice*, *apple*, *banana*; clothing and personal items like *socks*, *shirt*, *pants*, *shoe*; simple adjectives like *big*, *little*; animals like *kitty*, *doggie*; household objects like *spoon*, *chair*, *book*, *pen*); select child-

specific words (e.g., names of siblings, playmates, and pets) after consulting with the parents; establish the target words with the discrete modeled trials:

- Place a stimulus picture or object in front of the child
- Ask a question to stimulate the target response (e.g., “Johnny, what is this?”)
- Model the response (“Johnny, say *sock*”)
- Positively reinforce the correctly imitated response by verbal praise; accept an approximation of the correct response
- Verbally correct wrong response (e.g., Say, “No, that is not correct; it is a *sock*, not a ____.”)
- Move on to the next trial if the child did not respond
- Move the stimulus picture or item toward you to signal the end of a trial
- Record the child’s response on the recording sheet (correct, incorrect, or no response)
- Wait for a few seconds, draw the child’s attention, and present the next trial
- Stop modeling and present evoked trials when the child correctly imitates the target word on five consecutive trials
- Follow the same procedure as before: Present the stimulus, ask the question (“What is this?”), but do not model the response
- Reinforce the correct response and verbally correct the wrong response
- Reinstate modeling when the child gave three or four wrong responses on the first introduction of evoked trials; again withdraw modeling or fade modeling with Partial Modeling (e.g., This is a . . .) when the child correctly imitates the word on five consecutive trials
- Continue training in this manner until the child gives at least 9 correct responses out of 10 evoked (no modeling) trials (a 90% accuracy in producing the word without modeling)

- Reinforce continuously in the initial stages; in gradual steps, reduce the amount of reinforcers by switching to intermittent schedules
- Give social reinforcers for all responses, even when using tangible reinforcers
- Fade tangible reinforcers if used
- Train four to six exemplars (words at this level) to a training criterion of 90% correct on a set of 10 trials before you probe
- Conduct a probe by presenting the same trained words on the probe trials, ask the same question, and record the response; do not model, reinforce, or verbally correct; just ask the questions and record the response
- Consider the probed words tentatively trained when the child meets the 90% correct probe criterion
- Train four to six more (new) words if the probe criterion is not met; train the new words with the same procedure; administer another probe (no modeling, no reinforcement, no verbal correction)
- Continue to train the selected functional words (some 30 to 50); make sure that the words trained include nouns (*car, cup, sock, ball, kitty, candy, man, woman, baby*), adjectives (*big, small, blue, red, thin, thick, smooth, rough, happy, sad, hot, cold*), and main verbs (*walking, sleeping, eating, running, jumping, smiling*)

Teach Two-Word Phrases

- Shift training to the phrase level when a set of core vocabulary is established
- Form two-word phrases out of already trained words that have met the probe criterion (e.g., *big car, red sock, small kitty; man sleeping, woman walking, doggie eating*)
- Begin training the phrases with the same modeled discrete trials; fade modeling and move on to evoked discrete trials
- Probe the phrase production without modeling, reinforcement, or verbal correction when you have

trained four to six phrases; if the probe criterion (90% correct) is met, begin training on new phrases; if not, continue training on the original set of phrases

- Shift training to the level of grammatical morpheme and sentence training when about 20 trained phrases have met the probe criterion

Teach Grammatical Morphemes and Sentence Varieties

- Note that one grammatical morpheme (e.g., the present progressive *ing*) may have been taught in the context of main verbs (e.g., *walking, eating*); similarly, irregular plural words (*men, women, children, teeth, feet*) also may have been trained as words
- Note also that some grammatical morphemes can be trained only in sentences (e.g., the auxiliary *is*, as in *He is running*), whereas others may be trained in words or phrases (e.g., the regular plural *s*, as in *books* or *two books*)
- Select grammatical morphemes for initial training; informally experiment to see if a child responds correctly at the simple sentence level (e.g., *I see two books; They are two cats*); if not, train them in phrases (e.g., *two books; blue blocks*); in addition to the regular plural inflections *s*, teach the plural *z* (e.g., *four bags, two dogs, red balls, long trains*) for the initial training; other morphemes in phrases or sentences include the irregular plurals (*women, men, teeth, feet*), prepositions *on* (*on the table, on the bed*) and *in* (e.g., *in the hat, in the box*), possessive inflection (e.g., *mommy's hat, doggie's tail*), and so forth
- Use the discrete trial procedure to establish the morphemes; show a picture, ask a question, and model the word, phrase, or sentence with an emphasis on the grammatical morpheme (e.g., “Johnny, What are these? Say, two book**s**” or “What do you see? Say, I see two book**s**”); fade modeling and train with evoked trials

- Train the morphemes in four to six exemplars (phrases or sentences) each to the training criterion of 90% correct on a set of 10 evoked trials
- Probe for generalized production of the trained morpheme in initially baserated but untrained phrases or sentences; if the 90% correct probe criterion is not met, train three or four more phrase or sentence exemplars with the same grammatical morpheme; use at least 10 untrained stimulus items; administer probes as follows:
 1. Present an untrained stimulus (a picture, a single object, or multiple objects) that was baserated but not used on training trials (the *probe stimulus*); ask a question; do not reinforce or verbally correct; record the response as correct or incorrect
 2. Present another probe stimulus; use the same procedure as in #1
 3. Present all 10 or more untrained probe stimuli with the same procedure
- Calculate the percent correct probe response rate (e.g., if the child gave five correct responses out of 10 untrained stimulus presentations, the probe response rate is 50%)
- Train additional exemplars in phrases or sentences when the child does not meet the probe criterion of 90% correct production of the trained grammatical morpheme in untrained phrases or sentences
- Shift training to the next level when the child meets the probe criterion at each level:
 - Expand trained phrases into short sentences (e.g., *I see three cats; I see big men*)
 - Expand trained simpler sentences into more complex sentences (e.g., *I see three white cats; this is mommy's red hat*)
 - Alternatively, shift training to another language target; begin at the complexity level the child can handle, probe, and shift training to more complex levels or other language targets; a few examples of typical sentence targets include:

- Auxiliaries *is, are, was, were*, and so forth in such sentences as *girl is reading, boy is running; women are writing, dogs are chasing; he was sleeping, she was smiling; they were working, men were digging*
- Copulas *is, was, are, and were* (e.g., *girl is nice; boy was happy; turtles are slow; plants were green*)
- Regular past tense inflection (e.g., *he mailed; he mailed yesterday; he painted; he painted yesterday; she walked; she walked yesterday*)
- Irregular past words (e.g., *he came, she went; he came home; she went to the store*)
- Articles *a* and *the* (e.g., *the girl, the boy; the girl is reading, the boy is running*); note that phrases that were trained earlier to induce the production of another phoneme may be used to add additional grammatic elements
- Pronouns *he, she*, and *it* (e.g., *he is playing, she is riding, it is hopping*)
- Conjunction *and* (e.g., *milk and cookies; I like milk and cookies; lions and tigers; I like lions and tigers*)
- Comparative *-er* and superlative *-est* (e.g., *dark; this is darker; this is the darkest; warm; this is warmer; this is the warmest*)
- Adjectives *big* and *small* (e.g., *big house, small house; this is a big house, that is a small house*)
- Tacts or simple active declarative sentences (e.g., *This is a nice house; that is a pretty flower*)
- Mandates or requests (*Juice, please; Crayon, please; Please give me that ball; I want that book*)
- Questions (e.g., *What is this? Where is it? Who is it? When do you eat breakfast?*)
- Negation (e.g., *she is not sleeping, he is not eating, they are not walking*)
- **Train each new target with the same procedure**
- Baserate with 20 exemplars

- Train initially with modeling-imitation-reinforcement sequence
- Fade modeling
- Probe for generalized production
- Train other exemplars until the probe criterion is met
- Shift training to a higher complexity level or different language targets

Strengthen the Mastered Skills in Conversational Speech

- Strengthen generalized production of trained features in conversational speech, newer and longer utterances, and in different language contexts
- Do not use the discrete trials procedure
- Loosen the structure of treatment sessions
- Structure conversation to maximize opportunities to evoke the language targets that were previously taught
- Target such conversational speech skills as eye contact, topic initiation, topic maintenance, turn taking, and narrative skills in subsequent sessions, described in the next section
- Monitor or train only a few previously trained language features (few grammatical morphemes, two- or three-sentence variety) in conversational mode at any one time; it is difficult to monitor too many target features in conversational speech; repeat the procedure until all the initially trained features are strengthened in conversational speech
- Evoke speech in naturalistic contexts; if essential to use pictures, toys, or books, use them only to stimulate conversation and fade them out as soon as possible; preferably use the typical conversational method with no clinically manipulated stimuli; keep the interactions as natural and spontaneous as possible
- Engage the child in conversation
- Reinforce the production of trained language structures naturally (e.g., smiling, head nodding, hand-

ing a requested object, answering questions, making such verbal statements as *I agree, I had the same feeling, you are right about that, you said it well, etc.*)

- Use an intermittent reinforcement schedule; reinforce perhaps every other or every third correct response
- Verbally correct all errors (continuous schedule)
- Administer a conversational speech probe without reinforcers or verbal corrections to see if the skills are likely to be maintained in natural settings
- If the probe criterion of 90% correct production of the trained features in obligatory contexts produced in conversational speech is not met, revert to more explicit reinforcement and verbal correction in conversational speech
- Alternate conversational training and probe until the probe criteria for the targeted features are met

Teach Social Communication Skills. Shift training to social communication skills when the previously trained individual language targets (e.g., words, grammatical morphemes, descriptions [tacts], requests [mands], negation, question, etc.) are reliably produced in conversational speech; teach these skills during the final (and most complex) level of intervention for children with language disorders, except for eye contact which may be targeted from the beginning; essentially, instruct, model when feasible, prompt, and reinforce the following skills.

- Teach eye contact in all treatment sessions
- Teach conversational topic initiation
- Teach topic maintenance
- Teach conversational turn taking
- Teach requests for clarification
- Teach ways of clarifying their unclear expressions
- Teach narrative skills
- Use the procedures described under Social Communication Skills

Implement a Skill Maintenance Program

- Have family members, teachers, caretakers, peers, and others observe the treatment sessions
- Train the significant others, especially the family members and teachers, in evoking, prompting, and reinforcing target behaviors
- Have family members conduct informal therapy sessions at home and bring recorded evidence for evaluation
- Give a list of language skills taught to the child's teacher; ask the teacher to provide opportunities for communication in the classroom, prompt specific language skills, and praise the child for producing targeted and other language skills
- Hold informal training sessions outside the clinic room, outside the building, in other parts of the school or campus, and at other settings to the extent practical
- Use intermittent reinforcement schedule and natural, social reinforcers
- Delay reinforcement in later stages of treatment; increase the delay in gradual steps
- Take training to the natural conversational level
- Teach Reinforcement Priming by training the child to draw attention to one's own newly acquired communicative skills
- Teach Self-Control (Self-Monitoring) by training the child to recognize and measure self-generated right and wrong responses
- Continue treatment until the language skills stabilize in the natural environment
- Provide Follow-Up and Booster Treatment
- See Maintenance Strategy for more details

Telepractice Considerations

- Although telepractice is gaining increasing acceptance in assessing and treating communication disorders, there are significant gaps; tele-assessment or treatment of child language disorders in routine clinical settings, especially in public schools, remains

limited; much of the telepractice work is limited to research studies; studies are limited to simpler language structures such as word productions

- There is some evidence that parents of children with language disorders may be trained to do home-based intervention through online training and videoconferencing; see the cited Mettler study
- Many school-based SLPs may not have access to telepractice infrastructure in their work settings; but if they do, they may consider telepractice intervention for children with language disorders

Quality of Life Considerations in Child Language Treatment

- Research on assessing Quality of Life (QoL) following treatment of young children is extremely limited
- Parents or other caregivers have provided proxy information on the young child's QoL; reliability of this information has been difficult to establish; parents may over- or underestimate QoL in their children treated for language disorders
- An open-ended interview (more like a conversation) with the child and the caregiver about how the child is meeting social (friendship-related), academic, and family communication challenges following treatment may identify potential problems
- More often, deficient communication skills of certain kinds (e.g., academic or abstract language) or in certain situations (communicating with teachers or older siblings) may be the cause of lower QoL
- Clinicians may address those deficiencies in booster treatment sessions
- See [Quality of Life After Treatment](#) for additional suggestions that apply across disorders

Camarata, S. et al. (2024). Naturalistic developmental behavioral interventions for developmental language disorder. *American Journal of Speech-Language Pathology*, 33(2), 627–641.

Finestack, E. (2018). Evaluation of an explicit intervention to teach novel grammatical forms to children with developmental language disorders. *Journal of Speech, Language, Hearing Research*, 61, 2062–2075

Hegde, M. N. (2005). *Treatment protocols for language disorders in children: Volume I. Essential morphologic skills*. Plural Publishing.

Hegde, M. N. (2005). *Treatment protocols for language disorders in children: Volume II. Social communication*. Plural Publishing.

Mettler, H. M. et al. (2023). Vocabulary acquisition and usage for late talkers: The feasibility of a caregiver-implemented telehealth model. *Journal of Speech, Language, Hearing Research*, 66, 257–275.

Roberts, M. Y. et al. (2019). Association of parent training with child language development: A systematic review and meta-analysis. *JAMA Pediatrics*, 173(7), 671–680

Treatment of Language Disorders: Specific Techniques or Programs.

Most programs are not comprehensive, and none may replace the treatment outline described in the previous section; most programs help teach a specific skill at a certain stage in language treatment but do not specify all treatment targets and procedures to teach them from the beginning stage to the end stage (e.g., baserate, treatment, probes, maintenance, etc.); some have better evidence than others; clinicians should check the evidence before selecting a technique.

Delayed Stimulus Presentation. A procedure in which the clinician delays modeling for about 15 seconds to see if the child responds without such stimuli; provides the stimuli only when the child does not respond within the time limit.

- Establish joint attention regarding a stimulus (e.g., hold an object; establish eye contact with the child; look questioningly or expectantly)
- Do not speak for 15 seconds when the child approaches you or looks at you
- Model a mand (e.g., “I want that”) after the 15-second delay
- Give the object when the child imitates your modeling
- Give the object anyway when you have modeled three times and the child has not yet imitated

Direct Language Treatment Approaches. Clinician-implemented intervention with specific language targets, specific physical stimuli, modeling, prompting, and manual guidance; uses explicit reinforcement contingencies; expects the child to imitate or produce the target responses upon stimulation; moves through a planned sequence of treatment stages; similar to the comprehensive treatment program described previously; contrasted with Indirect Language Stimulation.

Enhanced Milieu Teaching (EMT). A collection of child language intervention procedures that emphasize natural, functional, and conversational communicative contexts for teaching language; based on milieu teaching research; includes three components: (1) environmental arrangement, (2) responsive interaction (RI), and (3) milieu teaching (MT); appropriate for children in the early stages of language learning; also applicable for children with autism spectrum disorder and those from low-income families with a high risk for language disorders; mostly parent administered with 24 to 36 training sessions, although may be clinician-administered as well; the method uses modeling, prompting, shaping, and positively reinforcing specific language targets evoked in natural settings; caregiver training follows the Teach-Model-Coach-Review model, described under Language Disorders in Infants, Toddlers, and Preschoolers: Specific Programs; may also include the Mand-Model and Time-Delay procedures; supported by controlled evidence.

- **Environmental rearrangement by parents**
 - Ask parents to select manipulable toys and other stimulus materials the child is interested in
 - Have them arrange the materials to evoke language responses from the child; limit the number of stimulus materials; keep some out of reach or in containers
 - Let them manage the materials to maintain the child's interest and interaction; control the access to the materials; give only a few at a time

- **Teach parents responsive interaction**
 - Listen to the child and respond quickly, positively, and warmly; stay at the child's eye level
 - Let the child lead the interaction; initiate language
 - Engage the child in conversation; talk about the child's play
 - Ask for clarification when the child is unclear
 - Expand the child's productions
- **Milieu teaching**
 - Model responses at the child's language level (e.g., if the child looks at a toy car, the mother might model, "Say 'want car'"; if the child said, "Car," the mother might model, "Say, 'want car'")
 - Expand the child's production (e.g., parent might say, "You want the red car," and hands it to the child)
 - If the child did not imitate, model the response and give the reinforcer (e.g., if the child did not respond, the mother might model, "Want car," and give the child the car)
 - Prompt no more than one a minute; stop the prompts when the child is no longer interested

Kaiser, A. P., & Roberts, M. Y. (2013). Parents as communication partners: An evidence-based strategy for improving parent support for language and communication in everyday settings. *Perspectives on Language Learning and Education*, 20(3), 96–111.

Expansions. Expanding a child's incomplete or telegraphic statements into grammatically more complete productions; may be used in both clinician-administered or caregiver-administered intervention.

- Arrange a play situation that provides opportunities for language production
- Engage in parallel play with the child or take part in the child's activities
- Expand the child's structurally incomplete productions into more complete sentences (e.g., when a boy says "baby cry" as he looks at a picture, expand it into "The baby is crying")

- In direct treatment sessions, ask to imitate the expanded production; in some indirect formats, skip this requirement for imitation only to notice a slower rate of learning

Extensions. Comments on the child's utterances to add additional meaning; part of Indirect Language Stimulation.

- Arrange a play situation that provides opportunities for language production
- Engage in parallel play with the child or take part in the child's activities
- Extend the child's semantically limited productions into semantically richer, structurally complete sentences through comments (e.g., when a girl says "play ball" as she plays with a ball, you extend it to include additional information: "Yes, you are playing with a big ball"; "You are playing with a blue ball")
- Require and reinforce correctly imitated extensions

Focused Stimulation. A technique of language intervention in which the clinician repeatedly models a target structure to stimulate the child to use that structure; usually a part of play activity; appropriate for toddlers and preschoolers and early school-age children; repeated modeling without requiring correct imitations and withholding systematic feedback to the child are potential weaknesses.

- Design a play activity to focus on a particular language structure (e.g., the plural morpheme *s*)
- Collect various stimulus materials (books, cups, hats)
- Talk about the materials and repeatedly model the plural constructions (e.g., "I see two *books* here. The *books* have pictures. Here are two red *cups*. You can drink out of these *cups*. There are some *hats*. The *hats* are big")
- Do not correct the child's wrong productions
- Respond to the child's nontarget responses without insisting on the correct response (e.g., the child says,

“The *book* is nice”; the clinician says, “Yes, the books are nice”)

- Continue until the child begins to produce the target structure

Wolfe, D. L., & Heilmann, J. (2010). Simplified and expanded input in a focused stimulation program for a child with expressive language delay (ELD). *Child Language Teaching and Therapy*, 26(3), 335–346.

Incidental Teaching Method. A naturalistic child language teaching method that uses typical, everyday verbal interactions to teach functional communication skills; the child often initiates an interactional episode; the clinician turns such episodes into opportunities to teach language; effective when the child and the teacher interact for extended periods of time and in natural settings (e.g., in special education classrooms, institutions for the intellectually disabled, and for those with autism spectrum); an effective method for parents to learn and use at home; a part of Milieu Teaching and Enhanced Milieu Teaching.

- Select certain functional communicative skills for teaching (e.g., requests)
- Arrange therapy situations such that the child is likely to initiate a conversational exchange (e.g., place attractive toys on a shelf the child can see but cannot reach; arrange a child’s clothing items or some desirable food items)
- Stay close to the arranged materials and give non-verbal cues to speak (focused attention on the child and a questioning look)
- Give a verbal cue (e.g., ask “What do you want?”) only if the nonverbal prompts fail to evoke a response from the child
- Give cues that evoke more complex responses (e.g., “Ask me in a sentence” if the child gives only single-word responses)
- Reinforce the child with natural consequences (e.g., “Good, here is the car,” “Very good, here is your sock,” “Fine, have some juice”)

- Arrange as many such teaching episodes as possible in a day

Blackwell, W., & Hall, N. (2021). Incidental teaching of conversational skills for students with autism spectrum disorder. *Teaching Exceptional Children*, 54(2), 116–123.

Indirect Language Stimulation. A collection of language stimulation procedures that are part of a play-oriented approach to teaching language disorders; also called Child-Centered Approach; less structured and more naturalistic; needs more controlled evidence to support this approach.

- Arrange a play situation that provides opportunities for language production
- Choose the play materials that are relevant to the targeted responses
- Let the child lead the interaction
- Engage in parallel play with the child or take part in the child's activities
- Talk about what the child is doing, looking, playing, or talking
- Describe what you do (e.g., "See, I am drawing a face; I am drawing")
- Describe or comment on what the child is doing (e.g., you say, "You are drawing; you are drawing a face; you are making a nose")
- Imitate the child's production (do *not* ask the child to imitate)
- Use Expansions (described earlier in this section; expand the child's telegraphic speech into grammatically more complete sentences; e.g., the child says "Mommy hat"; you expand this to "That is Mommy's hat")
- Use Extensions (described earlier in this section; comment on the child's utterances to add additional meaning; e.g., the child says, "Mommy hat," and you say, "Yes, it is a big blue hat")
- Recast (described later in this section) the child's utterances (expand the child's utterance type into a different kind of sentence; e.g., the child says,

“Mommy hat”; you ask a question, “Is this Mommy’s hat?” or make a statement “This is not Mommy’s hat”)

- Do not ask the child to imitate; do not target specific language structures; do not explicitly reinforce correct productions

It Takes Two to Talk: The Hanen Program for Parents. A language intervention program in which parents are trained to facilitate language development in their toddlers and preschoolers under naturalistic communication contexts; the target population is late-talking toddlers in the age group of 18 to 30 months and their parents; also applicable to preschool children and their parents; clinicians who wish to implement this program should get trained from the program developers.

- **Structure the program**
 - Conduct an initial orientation meeting
 - Assess the child
 - Conduct group parent training sessions
 - Offer three individual video feedback sessions for a total time commitment of 30 hours
- **Train the parents**
 - Arrange natural and interactive occasions with the child (e.g., any daily situation, including hand washing or brushing at a sink, eating, playing with toys, bedtime reading, sibling interactions)
 - Pause and wait for the child to initiate an utterance
 - Imitate (repeat) the child’s utterance, comment on it, or interpret it (e.g., during hand washing, the child might say such words as *water*, *cup*, and *drink* and the mother might repeat each word several times to emphasize it)
 - Expand the child’s utterances, extend the topic of the child’s utterances, and ask questions to teach turn taking in conversational exchanges

Pepper, J., & Weitzman, E. (2004). *It Takes Two to Talk®: A practical guide for parents of children with language delays* (2nd ed.). The Hanen Centre.

Joint Book Reading. May also be called shared book reading or interactive book reading; systematic use of storybook reading to teach or stimulate language in children; allows for repetitive use and practice of the same phrases and concepts; helpful in establishing joint attention as well.

- Select storybooks that are ethnoculturally appropriate for children under treatment
- Select books with colorful pictures
- Read the same story several times within a few sessions so that children memorize it
- Use prosodic features frequently to draw attention to specific language structures
- When the children know the story well, pause at points containing target language structures and prompt the children to supply the words, phrases, or sentences
- During different readings, pause at different junctures so the children produce different language structures
- Manipulate and vary pause locations that prompt progressively longer utterances from the children
- Ask a child to “read” (recite from memory, but looking at the text and the pictures) and pause
- Let the other children supply the words, phrases, and sentences

Dowdall, N. et al. (2020). Shared picture book reading interventions for child language development: A systematic review and meta-analysis. *Child Development*, 91(2), e383–e399.

Mand-Model. A part of the Incidental Teaching Method, although the method may be used independently to teach mands (e.g., requests, questions); uses typical adult–child interactions in a play-oriented setting to teach language; supported by controlled evidence.

- Select a variety of attractive toys, pictures, and other stimulus materials
- Design a naturalistic interactive situation
- Establish a joint clinician–child attention to a particular material (such as a toy); if necessary, direct the child’s attention to a stimulus

- Mand a response from the child (e.g., say, “Tell me what you want,” “Tell me what this is”)
- Model the correct, complete response if the child fails to respond or gives a limited (e.g., single-word) response
- Prompt if the child does not imitate the whole sentence you modeled (e.g., “Tell me the whole sentence”)
- Praise the child for imitating or for responding correctly without modeling
- Give the material the child wants as you praise

Venn, M. L. et al. (1991). Effects of teaching preschool peers to use the mand-model procedure during snack activities. *American Journal of Speech-Language Pathology*, 2(1), 38–46.

Milieu Teaching. A collection of child language intervention procedures that emphasize natural, functional, conversational communicative contexts for teaching language; uses natural consequences as reinforcers; includes the Mand-Model and Incidental Teaching Method and Time Delay; supported by controlled evidence; for its updated version, see Enhanced Milieu Teaching.

Peer Modeling. A child language intervention method in which the peers are trained to model the target skills for the child client.

- Select a peer who agrees to help and is acceptable to the child client
- Conduct a demonstration treatment session and have the peer observe your treatment sessions
- Describe the target behaviors, modeling and imitation sequence, and reinforcement procedures
- Ask the peer to model and reinforce the child’s productions in your presence
- Refine the peer’s skills in modeling the target behaviors
- Ask the peer to submit recorded language samples that document appropriate modeling outside the clinic

- Periodically assess the results of peer modeling and provide additional training to the peer

Goldstein, H., & Strain, P. S. (1988). Peers as communication intervention agents: Some new strategies and research findings. *Topics in Language Disorders*, 9(1), 44–59.

Recast. Clinician's production that changes some or all of the words in a child's incorrect or inadequate utterance into a more accurate or more complete utterance while retaining the child's presumed meaning; recast also may just add new information, without using the child's words; if the child is not asked to imitate the clinician's recast, the method is Indirect Language Stimulation; if the clinician asks the child to imitate the recast and reinforces or corrects the child, then it is a part of a more efficient direct intervention strategy; evidence supports the method, but there are variations; in a treatment session (see the cited sources), typically multiple language structures may be targeted in a restricted language context, but in its enhanced version, a single target may be recast in multiple contexts (e.g., the plural s morpheme inflected with multiple object names).

- Arrange play activity designed to enhance opportunities for language production
- Play with the child
- Expand the child's utterance into a sentence type that may be different from the child's presumably intended sentence type (e.g., the child says, "Big ball"; you recast it into a negative sentence, "No, it is not a big ball"; or expand it into a question form, "Is this a big ball?")
- Recast the child's grammatically wrong utterance into a correct utterance (e.g., the child says, "Him want cookie," and you recast it as "He wants cookie")
- Recast to add new information (e.g., the child says, "He eat a lot," and you recast it as "Yes, he is hungry")

Cleave, P. L. et al. (2015). The efficacy of recast in language intervention: A systematic review and meta-analysis. *American Journal of Speech-Language Pathology*, 34(2), 237–255.

Kapa, L. H. et al. (2020). Predictors of treatment response for preschool children with developmental language disorder. *American Journal of Speech-Language Pathology*, 29(4), 2082–2096.

Time Delay. A language intervention procedure in which the stimulus to evoke the target response is presented, but modeling or prompting may be progressively delayed from 1 to 9 seconds to promote more naturalistic (spontaneous) responses without models or prompts; a method of fading the two procedures; both caregivers and clinicians may use this procedure.

- Select highly preferred stimulus items (e.g., the child's favorite toy car)
- Establish joint attention—hold the stimulus item and draw the child's attention to it
- Immediately model the request for the child to imitate ("I want that car"); no time delay
- If the child correctly imitated the model, give the object to the child
- If the child did not correctly imitate, say, "No" and withdraw the object
- Train correct imitation on at least three trials before introducing time delay
- Introduce time delay; show the object and do not model for 1 or 2 seconds; if the child did not request it, model after the delay
- Progressively increase the delay duration (up to 8 or 9 seconds) before modeling; in most cases, the child is likely to request before long; a constant time delay may be equally good
- Use the same time delay procedure with more expanded productions (e.g., "I want to play with that red car")
- Any time there is no response, model the response, and reinforce the child's imitation; reintroduce time delay
- Use the same procedure for prompts

Walker, G. (2008). Constant and progressive time delay procedure for teaching children with autism: A literature review. *Journal of Autism and Developmental Disorders*, 38, 261–267.

Language Disorders in Infants, Toddlers, and Preschoolers.

Treatment of young children who face high risk for language (and speech) problems or who are already showing signs of language delay requires a two-pronged approach: caregiver-implemented home treatment program and the clinician-implemented formal treatment; research evidence supports both approaches.

Training Caregiver to Implement Home-Based Interventions

- Throughout this entry, the term *caregivers* includes parents, grandparents, older siblings, babysitters, and preschool professionals, although more typically, it will be the parents who will conduct home-based programs. Trained older siblings may play an important role in home treatment programs. If the child spends most of the time with a babysitter or in a preschool, training these caregivers will be essential. Generally, the clinician may recruit as many individuals as possible to support the young child's communication skill learning; not everyone may be involved to the same extent, but any help they can offer will be additive.
- In-person, face-to-face training of caregivers may be preferred, although telepractice with live or remote sessions may be more practical in some cases; a hybrid program that began with a few initial in-person caregiver training involving the child and then transitioned to an online program may be useful; working parents may prefer demonstrative videos and other descriptive text resources that they can access at their convenience; videos placed online should be those of the initial in-person training sessions involving the child as well as the parents; see the cited resources.
- All caregiver training session should begin with the clinician conducting a few treatment sessions in which the caregivers participate; the goal of these sessions should be to demonstrate and model all skills (described in following sections for each age group) the caregivers are expected to exhibit during home intervention sessions.

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- Clinicians may use a caregiver training package; see [Babble Boot Camp \(BBC\)](#), [Teach-Model-Coach-Review](#), and [Vocabulary Acquisition and Usage for Late Talkers \(VAULT\)](#) under [Language Disorders in Infants, Toddlers, and Preschoolers: Specific Programs](#) following this main entry; clinicians also may use [Enhanced Milieu Teaching \(EMT\)](#) described under [Treatment of Language Disorders: Specific Techniques or Programs](#).

Clinician-Implemented Formal Treatment

- Generally, use the same procedures you recommend to the caregivers
- When the child is old enough to participate in clinic treatment sessions, use the procedures, with suitable modifications, described under [Language Disorders in Children](#)

Treatment of Infants, Toddlers, and Preschoolers: General Guidelines

- Be a member of an interdisciplinary team and design a comprehensive communication intervention program that will include home-based as well as clinic-based treatment
- Test the home treatment program with a few sessions before you recommend it to the family
- Have the early home-based language stimulation program implemented in naturalistic communication and child care contexts
- Ask caretakers to model target words throughout the day and reinforce with smile, touch, gentle tickle, and so forth
- Monitor and refine the caretakers' work; frequently assess the child's progress
- Design and implement a formal language treatment if the caretaker's work did not produce desired results in a matter of a few months; target words, phrases, and sentences
- Include the caregivers in the formal treatment sessions to continuously refine their skills in implementing the home-based program

- Consider the child's need for aural rehabilitation if the child has a hearing loss; work with an audiologist to develop an aural rehabilitation program with an emphasis on speech-language stimulation that you monitor or implement

Intervention During the First 6 Months. Target is to increase babbling and responsiveness to verbal stimuli.

- **Train the caretakers to increase babbling**
 - Ask the caretakers to present various kinds of environmental and human sounds, including the caretaker's voice throughout the day
 - Train the caregivers to establish a *feedback loop* as soon as the baby begins to coo and babble; when the baby makes a vocal sound, have the caretakers make their smiling faces visible to the baby, imitate the same sound, touch and gently tickle the baby to establish a feedback loop in which the baby produces sounds, caretakers imitate them, and the baby in turn imitates the caretakers
 - Ask them to vocalize while attending to the physical needs of the child (e.g., bathing, feeding, changing)
 - Have the caregivers draw the baby's attention to their smiling faces
 - Let them gently tickle and touch the baby as the baby looks at the face, turns in the direction of the mother's voice, smiles, or vocalizes
 - Suggest calling the baby's name out frequently
 - Teach them to hold the infant at eye level, keep smiling, maintain an expressive face, and talk to the child in a soft, affectionate, and pleasant voice
 - Train them to exhibit contingent smiling and laughing (reaction to similar behaviors in the infant)
 - Ask them to avoid quiet play with the child and play with speech and vocalizations; vary the tone of speech and vocalizations; give a contingent response to the child's reactions
- **Train the caretakers to target more speech-like vocalizations**

- Reinforce any vowel-like sounds, sound vocalizations, and increased range of vocalizations the infant produces spontaneously; touch, gently tickle, pick up, and smile when the baby is looking at the face
- Reinforce marginal babbling (consonant-like sounds in babbling) and imitate the baby's babbling
- Model frequent vocalizations if the infant's level of vocalizations is low; ask the parents to get closer to the infant, draw the infant's attention, and model babbling-like sounds and other kinds of vocalizations
- Reinforce all kinds of imitative or spontaneous vocalizations the infant makes

Intervention During 6 to 12 Months of Age

- **Train the caregivers to expand verbalizations**
 - Target consonants, consonant-vowel combinations, and word-like utterances
 - Model and reinforce consonant-vowel syllable productions like *bababa* or *mamama*
 - Model and reinforce production of sounds produced in the front of the mouth (e.g., /p/, /t/, /d/)
 - Take turns in vocalizations (e.g., the baby and a caregiver may vocalize in an alternating fashion); reinforce baby's vocalizations
 - Vocalize additional and varied consonant-like sounds for the infant to imitate
 - Model adult vocalizations or gestures for the infant to imitate
 - Reinforce the child for differential responding in any form; the infant may produce specific sounds in specific situations (e.g., one vowel sound to suggest a desire for an object and another sound to suggest disapproval)
 - Vary the intonations of modeled vocalizations or verbalizations for the infant to imitate
 - Utter the names of family members when they are around; reinforce with smiles, gentle touch, and appropriate vocalizations when the baby looks around or looks at the person whose name is called

- Name objects that are present in the baby's environment (e.g., toys, food items, clothing) and reinforce the baby for looking at the named object or for trying to reach it
- Present different sounds associated with the baby's toys and reinforce the baby for looking at the toy when the sound is heard
- Take note of the child's response to "No" (e.g., cessation of activity, hesitancy, unpleasant facial expression) or to scolding (e.g., frowning, crying, unpleasant facial expression) to continuously assess the child's reactions and improvements to different kinds of verbal stimuli
- Teach the baby to imitate various kinds of gestures (e.g., clapping when clapped, touching the body parts when modeled)
- Draw attention to an object and reinforce the infant for joint attention
- Keep written notes on the kinds and number of words the baby understands at home to informally assess whether there is improvement due to such stimulation activities
- Reinforce signs of intentional communication between 9 and 10 months of age (e.g., specific gestures, pointing, and vocalizing while making eye contact)

Intervention During 12 to 18 Months of Age

- **Train the caretakers to teach simple words and word approximations**
 - Reinforce the baby for responding—verbally or non-verbally—to familiar names and objects
 - Reinforce the baby's response to names of absent objects and persons (e.g., when said "ball," looking at the place where a ball was kept but is now absent; looking in the direction of a sibling's room when the sibling's name is uttered)
 - Model simple and everyday words for the child to imitate; reinforce vocal approximations or appropriate nonverbal responses

- Begin to model such constructions as possessor + possession (e.g., *Mommy's shoes*) for the child to comprehend (not produce)
- Give simple commands and reinforce the child (e.g., "Clap your hand," "Pick up the car," "Show me your nose")
- Reinforce increased production of gestures and vocalizations to obtain objects, draw attention, and regulate the behavior of caregivers
- Reinforce the approximations of the first words that should appear during this time; if not, frequently model simple everyday terms like *Mommy*, *Daddy*, names of the siblings and pets, child's food and clothing items
- Keep track of the baby's progress

Intervention During 18 to 24 Months of Age

- **Train the parents to target production of varied words and simple phrases**
 - Evoke the baby's response to absent objects more frequently; talk to the baby about siblings not present, objects not in the view; reinforce baby's vocal or nonvocal responses (e.g., imitating the name of the sibling, vocalizing noises associated with toys)
 - Model and reinforce productions of two-word combinations; model phrases constructed out of the words the child has learned to produce
 - Make simple requests or give commands (e.g., "Give me the pencil," "Find the ball") and reinforce correct responses
 - Model words in differentiated categories (nominals, action words, modifiers, or adjectives); reinforce the child's approximations
 - Model 10 to 15 selected functional names of objects while showing the objects; reinforce the child's approximations
 - Begin teaching two-word phrases, constructed out of the single words the child already produces; model and reinforce the correct productions or approximations

- Model speech sounds for improved articulation while teaching simple words; target 50% intelligibility of the child's speech to the caregivers
- Reinforce a sentence-like intonation, even if the child's production is an unintelligible strings of syllables

Intervention During 24 to 36 Months of Age

- **Train the caretakers to target more complex language**
 - Teach correct nonverbal responses (comprehension) to three-word sentences; reinforce the actions
 - Teach the child to respond correctly to two-, three-, and four-element commands and reinforce correct nonverbal actions
 - Make progressively more complex requests and reinforce correct responses
 - Ask questions involving *what*, *who*, and *where* and reinforce correct responses
 - Model and reinforce the production of simple sentences
 - Model three- and four-word sentences for the child to imitate; model five-word sentences toward the end of this period
 - Model and reinforce the production of such questions as "What is that?" or "What are you doing?"
 - Continue to work on more precise production of speech sounds to increase intelligibility of speech (up to 75%)
 - Continue to work on expanding the child's vocabulary by introducing new words for the child to imitate and produce spontaneously; frequently reinforce the production of new words (at 24 months, just under 300 and at 36 months, 900 to 1,000 words are expected in a typical child)
 - Reinforce any form of functional communication, including words, phrases, sentences, gestures, and other forms of nonverbal communicative behaviors while continuing to stimulate verbal expressions
 - Withdraw reinforcement gradually for gestures and undifferentiated signs for which the child can substitute verbal responses

- Encourage interactive play to teach social communication; play with the child, include a sibling or a peer, and teach cooperative play; have the playmates exchange toys or other materials first by requesting them; name objects involved in play; have the child describe the activities; model requests and descriptions of activities; comment upon and expand the child's utterances
- Take part in the child's constructive activities (e.g., painting, block building, art or craft activities) and induce as much speech as possible

Carson, L. et al. (2022). A systematic review of interventions for late talkers: Intervention approaches, elements, and vocabulary. *American Journal of Speech-Language Pathology*, 21, 2861–2874.

Finestack, L. H. et al. (2022). Caregiver implemented communication interventions for children identified as having language impairment 0 through 48 months of age: A scoping review. *Journal of Speech, Language, Hearing Research*, 65(8), 3004–3055.

Pak, N. S. et al. (2023). Long-term effects of early communication interventions: A systematic review and meta-analysis. *Journal of Speech, Language, Hearing Research*, 66, 2884–2899.

Quinn, E. D. et al. (2021). Hybrid telepractice delivery of enhanced milieu teaching: Effects on caregiver implementation and child communication. *Journal of Speech, Language, Hearing Research*, 54, 3074–3099.

Language Disorders in Infants, Toddlers, and Preschoolers: Specific Programs.

Most specific programs or techniques tend to target one or a few skills.

- Among the few procedures that have received research support, consider Enhanced Milieu Teaching, Focused Stimulation, It Takes Two to Talk—The Hanen Program for Parents, and Recast, all described under Treatment of Language Disorders: Specific Techniques or Programs, because the programs are flexible enough to be used with younger or older children.
- To train caregivers to provide home-based interventions to young children, consider the following training packages:

- **Babble Boot Camp (BBC).** Helps train caregivers to increase language skills in 2- to 24-month-olds; caregivers may be trained to make eye contact, imitate the child's vocalizations, increase the amount and quality of babbling by reinforcement, reinforce dyadic exchanges, increase social interactions, pair verbal and nonverbal cues, increase receptive and expressive vocabulary, build expressive syntax, and so forth. Access the full document at the Open Science Framework entry for the BBC (<https://osf.io/yzht4/>)
- **Teach-Model-Coach-Review Instructional Approach.** This approach may be used to train caregivers in providing different kinds of services, although it has been used extensively in training caregivers to provide home-based language intervention, especially the Enhanced Milieu Teaching (EMT); the model's Teach component includes information and video demonstrations, the Model component includes modeling the methods of teaching language to children, the Coach component includes sessions in which the caregiver practices intervention methods with the child, and the Review component includes discussing the session and planning the home intervention sessions.

Roberts, M. Y. et al. (2014). Effects of teach-model-coach-review instructional approach on caregivers use of language support strategies and children's expressive language skills. *Journal of Speech, Language, Hearing Research*, 57, 1851–1869.

Language Disorders in Older Students and Adolescents. Language skills change more gradually in children older than 6 than younger children; the disorders are more subtle; treatment targets are higher-level language skills, including more academic and abstract language; see the cited sources for details.

Treatment Objectives/General Guidelines

- **Language treatment targets for older students and adolescents**
 - Deficiencies in basic language skills that may have persisted from early childhood

- Mastery and expression of scientific, technical, literate, academic, logical, and discipline-specific terms
- Word retrieval, word definitions, word relations, and abstract language skills
- Advanced syntactic structures
- Advanced narrative and discourse skills
- Reading and writing skills
- Treatment is oriented toward social communication and academic demands and performance
- Treatment needs to be fully coordinated with the teacher's regular or special education programs for the student

Ethnocultural Considerations

- See Language Disorders in Ethnoculturally Diverse Groups in the previous section and a separate entry, Ethnocultural Variables in Treatment
- Pay special attention to the student's academic standing and social communication patterns and to the family's communication patterns
- Check whether the student is bilingual; if so, assess the student's dual-language proficiency to design appropriate intervention; consult with family and teachers

Treatment of Language Disorders in Older Students and Adolescents. Teach deficient basic skills as revealed by your assessment (e.g., basic words, sentence structures, use of grammatical morphemes, etc.); use the procedures described under Language Disorders in Children

Teach Advanced Semantic Skills

- Select comprehension and correct production of academic, scholarly, learned, and lexicon after consulting with the teacher and the academic curricula; see Abstract and Figurative Language
- Write precise definitions of academic terms selected for teaching; use the student's textbooks or other classroom materials and have the teacher define the target terms

- Ask the student to define the terms on the list; prompt correct definitions and reinforce approximations to begin with
- Require progressively more precise and complete definitions of terms
- Ask the student to produce sentences with those terms
- Ask the student to write sentences that include the terms
- Ask the student to contrast the meaning of terms, shape correct responses, and reinforce approximations initially and require more precise definitions of selected terms (e.g., *fact* and *opinion*, *inference* and *assumption*, *illusion* and *allusion*, *further* and *farther*, *affect* and *effect*)
- Teach both comprehension and correct production of figurative language; make a list of common proverbs, metaphors, and idioms and ask the student what they mean (e.g., *a stitch in time saves nine*, *a penny saved is a penny earned*, *bury the hatchet*, *off the wall*, etc.)
- Prompt correct interpretations of the selected figurative language exemplars
- Contrast the literal and figurative meanings of the exemplars and ask the student to state the contrasts
- Ask the student to write sentences or short paragraphs, including figurative language exemplars; reinforce or verbally correct; show published writing samples for modeling

Teach Word Retrieval Problems in Conversational Speech

- Prompt the correct words at the earliest sign of false starts, pauses, revisions, repetitions, and beating around the bush—all signs of word retrieval problems
- Teach the student to pause and think of the correct words instead of beating around the bush or getting stuck with false starts

- Teach the student to use specific words and avoid vague terms; prompt specific words and withdraw prompts

Teach Word Relation Problems

- Have the student define and contrast synonyms and antonyms
- Ask the student to explain why the meanings are similar across synonyms and different across antonyms
- Teach the student to write brief essays in which relational words are correctly used; give corrective feedback and ask the student to rewrite

Teach the Production of Precise Terms in Conversation and Narrative Tasks

- Reduce the frequency of such expressions as “this,” “that,” “you know what I mean,” “this thing,” “that stuff”
- Prompt the student for more precise words when the student uses just general terms
- Withdraw prompts and reinforce the student for pausing and thinking of more precise words during conversation
- Have the student write short essays for precise words; ask the student to avoid the general words in writing the essays; evaluate the student’s writing and give feedback

Teach Advanced Syntactic Skills

- Use the assessed and analyzed speech and language samples, narratives, and writing samples to target syntactic skills for teaching; generally, teach the production of low-frequency constructions the student may have difficulty with:
 - Passive sentences (e.g., The examination *was thoughtfully constructed* to make sure that most students would pass it)
 - Modal auxiliary verb (e.g., They *should have* studied harder)
 - Perfect aspect (e.g., He *had been* spending money recklessly)

- Appositives (e.g., John Laughlin the *famous comedian* left the audience laughing hysterically)
- Elaborated subjects (e.g., *National parks such as the Yellowstone, Yosemite, Grand Canyon, and Bryce Canyon* are treasures of the American West)
- Post-modifications (e.g., The next person *to pull the handle* could win the jackpot)
- Prompt and teach the use of cohesion devices or connectives (e.g., such expressions as *therefore, as a result of, consequently, subsequently, because of*)
- Teach correct agreement (e.g., noun-verb agreement) in both connected speech and writing samples
- Teach the correct use of pronouns to avoid ambiguous referents; prompt and model correct pronoun usage

Teach Social Communication Skills. Generally, eye contact, topic initiation, topic maintenance, conversational repair strategies, narratives, and discourses are among the most often taught social communication skills; [See Social Communication Skills](#) for teaching these skills; in addition:

- **Teach the use of correct register**, which is language appropriate to the context, situation, and the conversational partner; using the role-playing technique, teach the student to switch from one kind of register to the other:
 - You play the role of a peer and talk to the student
 - You play the role of a teacher and discuss a formal topic with the student
 - You play the role of a parent and talk to the student
 - Model, prompt, and reinforce correct form of speech that fits the role the student is asked to play; stop and verbally correct when the student uses incorrect registers (e.g., lay language while discussing a formal academic topic with the teacher, production of slang used among peers when talking to the parent)
 - Modify inappropriate gestures the student may exhibit during conversation

Teach Reading Skills

- Ask the student to read aloud a printed passage from grade-level text; follow the student's reading on a copy of the passage; verbally correct misreading
- Teach a slower rate of reading to promote more fluent reading
- Take note of the words the student does not know (hence struggles to read) and teach the meaning of those words
- Teach syntactic structures that seem to give difficulty in reading
- Have the student learn to spell words that are difficult to read
- Ask questions about the read material to assess reading comprehension
- Ask the student to reread and ask you questions about material not understood
- Give reading assignments and have the student write brief reports on the read material; give feedback to the student

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Teach Writing Skills

- Dictate a passage for the student to write
- Ask the student to write a paragraph or two on a personal experience
- Ask the student to copy a brief printed passage
- Ask the student to bring an extended writing assignment submitted to the teacher and analyze the various writing samples for problems in writing and:
 - Give edited (written) feedback on spelling errors, poor formation of letters, general organization and neatness, and overall quality of handwriting
 - Give edited (written) feedback on errors in punctuation (the use of comma, semicolon, period, dash, quotation marks, and parenthetical constructions)
 - Discuss with the student the writing sample for its content; in case of the extended piece of writing (e.g., an essay submitted to the teacher), give feedback on the adequacy, depth, appropriateness, and details of information offered

- Correct the errors of syntax in the written samples; identify wrong or incomplete sentences; suggest varied syntactic constructions
- Discuss with the student the appropriateness of the language to the topic of writing; correct errors in technical word usage, pro and con arguments elaborated, conclusions offered, and support for views advocated
- Give feedback on overall organization and neatness of writing

Implement a Maintenance Program

- Periodically follow-up the client to assess skill maintenance
- Record samples of conversational speech, a narrative, and a brief discourse
- Analyze the samples for loss of clinically established skills or for newly identified deficits
- Arrange for booster treatment to remediate the deficiencies
- See Maintenance Strategy for additional suggestions and strategies, including training significant others to support language skills

Consider Quality of Life Assessment Following Treatment

- Research on quality of life assessment following treatment of language disorders in older students and adolescents is extremely limited
- Parent- or other caregiver-supplied information may not fully reflect the communication successes or challenges the student faces, especially at school and larger social milieu
- Assessment instruments specific to language disorders without associated conditions (e.g., autism spectrum) in older students and adolescents are lacking
- Asking students *how happy* they are (as is done in some tools) may not help assess quality of life limitations due specifically to language skill deficits
- An open-ended and conversation-like interview in which specific areas of life (e.g., social-friendship,

family, academic) that are effected by language skills may reveal relevant information

- See Quality of Life for additional information

Fahy, J. K., & Browning, D. K. (2021). Adolescent language therapy: Syntax and semantics for reasoning and planning. *Perspectives of the ASHA Special Interest Groups*, 6, 1327–1342.

Larson, C. (2024). Exploring self-reported quality of life in developmental language disorder. *Perspectives of ASHA Special Interest Groups*. ePub Ahead of Issue, https://doi.org/10.1044/2024_PERSP-24-00122

Lowe, H. et al. (2019). The effectiveness of classroom vocabulary intervention for adolescents with language disorders. *Journal of Speech, Language, and Hearing Research*, 62(8), 2829–2846.

Nippold, M. A. (2020). *Language sampling with adolescents: Implications for intervention* (3rd ed.). Plural Publishing.

Spencer, T. D., & Petersen, D. B. (2020). Narrative intervention: Principles to practice. *Language, Speech, and Hearing Services in Schools*, 51, 1081–1096.

Language Disorders in Ethnoculturally Diverse Groups.

There are no unique treatment procedures demonstrated to be essential for treating language disorders (LDs) in ethnoculturally varied children; use instructions, demonstrations, modeling followed by imitation, shaping, verbal praise, and verbal correction, among other procedures described under Treatment of Language Disorders in Children: A Comprehensive Procedure; critical in treating children of varied cultural and language background is the correct diagnosis of a communication disorder and selection of target behaviors that are consistent with the child's primary language or different dialects of Mainstream American English (MAE); see Ethnocultural Variables in Treatment for general guidelines; consult the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for the unique speech-language features that contrast with those of MAE and yet are not targets of intervention unless the parents request it.

Pretreatment Considerations

- Diagnose a language disorder only when the child's language is inconsistent with the characteristics of

the child's particular variety of English (e.g., Spanish-influenced English) or a particular dialect of American English (e.g., African American English [AAE])

- Compile and keep a list of the unique features of AAE and bilingual children's English varieties influenced by their primary language (e.g., Spanish, an Asian language)
- Assure the parents that a diagnosis of language disorders was made on the basis of the child's and the family's language background and not on the basis of MAE
- Consider what the child, parents, and teachers expect from speech-language intervention; generally treat the disorder in the child's language (e.g., Spanish-influenced English) or dialect (African American English); if the child and the parents prefer, however, teach the missing features of MAE; encourage code switching (e.g., the child may speak MAE in the school and continue to speak AAE at home)
- Finalize the target skills and general intervention approach after reaching a consensus with parents and educators; let the child's academic, personal, family, and social communication needs suggest intervention targets
- Select stimulus materials and treatment activities that are culturally appropriate and relevant to the child and the family; request parents to supply stimulus materials from home (books, toys, pictures, objects, activities the child enjoys at home)
- Have a collection of ethnocultural storybooks, fairy tales, ethnic holiday celebration ideas, unique ways of celebrating birthdays, or other kinds of family events; incorporate such resources into language treatment sessions
- Ask parents to periodically observe your treatment sessions and offer suggestions on treatment stimuli and treatment activities interesting to the child
- Make your clinic display boards, activity boards, and bulletin boards culturally rich and varied; include

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different ethnic materials to represent varied cultural backgrounds

Language Treatment Targets for African American Children

- Ascertain whether the child and the family speak AAE at home; not all do; if the child is an MAE speaker, select target behaviors as you would for any other child
- Do not select the unique features of AAE for modification unless opted for by parents and the child; a few examples:
 - Omission of plural morphemes. Forms of to be (e.g., *is* and *are*), auxiliary *are*, present or the past tense forms of the auxiliary *have*, the plural morpheme, and so forth
 - Replacement of certain MAE features with AAE features. Auxiliary or copula *is* for *are*, past tense *was* for *were*, *do* for *does*
 - Additions within the same sentence of expression. Double auxiliaries *might* and *could*, double negatives *never* and *no*, *at* at the end of *where* questions, *done* before a past tense verb
- Select language treatment targets that are consistent with AAE and the goals of the family, and help meet the child's educational and social demands
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support

Hegde, M. N., & Pomaville, F. (2021). *Assessment of language disorders in children: Resources and protocols* (4th ed.). Plural Publishing.

Stockman, I. J. et al. (2013). Toward validation of a minimal competence core of morphosyntax for African American children. *American Journal of Speech-Language Pathology*, 22(1), 40–58.

Language Treatment Targets for Children of Asian Cultures

- Ascertain whether the child is bilingual and, if so, the language features of the child's first (home) language; because of the enormous diversity of Asian languages

that influence English, it is not possible to list the features; consult the cited sources

- If the child of Asian background is predominantly an English speaker, treat the language disorder as you would in any other child
- Select treatment targets for a bilingual child that are consistent with the child's primary language skills or the features of MAE if the parents and the child prefer
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support

Hegde, M. N., & Pomaville, F. (2021). *Assessment of language disorders in children: Resources and protocols* (4th ed.). Plural Publishing.

Shekar, S., & Hegde, M. N. (1996). Cultural and linguistic diversity among Asian Indians. *Topics in Language Disorders*, 16(4), 54–64.

Language Treatment Targets for Hispanic Children.

Most guidelines offered here are relevant to all children whose primary language is other than English; because of the varieties of Spanish that influence bilingual English-speaking children, it is not possible to list the language features; consult the cited sources.

- Ascertain whether the child and the family speak a primary language that is other than English at home; some ethnoculturally diverse children may acquire English as their first language even though the parents speak a different language at home
- Ask whether the child and the parents want to have the mainstream English language patterns taught as well
- Select language treatment targets that are consistent with the child's primary language and supportive of educational goals and the wishes of the family
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support

Goldstein, B. A. (2022). *Bilingual language development & disorders in Spanish-English speakers* (3rd ed.). Paul H. Brookes.

Hegde, M. N., & Pomaville, F. (2021). *Assessment of language disorders in children: Resources and protocols* (4th ed.). Plural Publishing.

Language Treatment Targets for Native American Children.

Most children of Native Americans tend to acquire English as their only language, as the Native American languages are on the decline within their communities; if intervention for English is justified for a child who speaks one of the several Native American languages, use procedures appropriate for any bilingual child.

- Ascertain whether the child and family speak a Native American language at home and that English is a second language
- Ask whether the child and the parents want to have the mainstream English language patterns taught as well
- Select language treatment targets that are consistent with the child's Native American language, the goals of the family, and the child's educational and social demands
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support

Hegde, M. N., & Pomaville, F. (2021). *Assessment of communication disorders in children* (4th ed.). Plural Publishing.

Language Impairment. Generally the same as *language disorder*; see Language Disorders in Children for treatment procedures.

Language-Learning Disorders. Generally the same as language disorders; links language disorder to a general learning disorder that negatively affects academic learning; often used in special educational contexts; see Language Disorders in Children for treatment procedures.

Language Stimulation by Parents. Activities parents implement at home to stimulate language in infants and toddlers; may be exclusively parent-administered after receiving training or may supplement or parallel clinicians'

treatment; may use the telepractice format; supported by some evidence; see [Language Disorders in Infants and Toddlers](#); to develop a parent-implemented language stimulation program:

- Assess the child's current language status; assess the parents' education, sophistication, time commitment, and motivation to conduct regular activities at home
- Design a language stimulation program for the child; consult with parents to set the target behaviors for intervention
- Test the program in the clinic for a few sessions to make sure it works
- Have parents observe your sessions
- Train parents in the effective methods; model the methods frequently
- Have parents conduct a session or two in the clinic in your presence; refine their skills
- Train them to keep records of therapy that you can evaluate
- Give parents simple, clear written instructions
- Give parents videorecorded samples of treatment techniques
- Periodically assess the child and the parents' sessions at home
- Suggest needed modifications and movement to higher levels of training
- Initiate formal clinical treatment when assessment results warrant it
- Consider using a parent-administered language stimulation program such as [It Takes Two to Talk—The Hanen Program for Parents](#), [Enhanced Milieu Teaching](#), [Focused Stimulation](#), [Responsive Education](#), or [Conversational Recast](#)

Laryngeal Cleft. A cleft between the larynx (cricoid cartilage) and the esophagus; caused by a failure of dorsal fusion of the cricoid lamina; may be accompanied by other congenital anomalies, including feeding and respiratory problems soon after birth; treatment is surgical closure of the cleft.

Laryngeal Cysts. Formation of small, fluid-filled sacs on the larynx, especially in the ventricle; often congenital, but also may be acquired later in life; mucous gland blockage or phonotrauma are potential causes; symptoms include swollen false (ventricular) folds; if enlarged, can result in hoarseness; treatment is surgical removal of the cyst with follow-up voice therapy if needed.

Laryngeal Hyperkeratosis. A thickening of the laryngeal mucosa resulting from an abnormal growth of the epithelium due to cigarette smoking, heavy alcohol use, environmental pollutants, and strained and tense speaking habits; usually occurs on the true vocal folds; may sometimes be premalignant.

- Modify the client behaviors that are causative (e.g., reduced use of alcohol and tobacco)
- Counsel the parents to reduce exposure to environmental factors that lead to laryngeal hyperkeratosis

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Laryngeal Leukoplakia. Appearance of white patches on the laryngeal mucosa; may be premalignant; voice may be hoarse, but behavioral therapy may not be effective.

Laryngeal Stoma. An opening made into the trachea (in front of the neck) to allow for breathing in patients with laryngectomy.

Laryngeal Web. Growth of a thin membrane across portions of the vocal folds; may be congenital or induced by trauma later in life; negatively affects respiration; treatment is surgical removal of the web.

Laryngectomy. A person who has had a partial or total Laryngectomy.

Laryngectomy. Laryngeal surgery due to disease or trauma; communication intervention requires an understanding of laryngeal surgeries and their consequences for communication and communication training options; speech-language pathologists are involved from the preop-

erative stage to postoperative rehabilitation, and their main task is to teach a new mode of communication; for assessment procedures, see the cited sources and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.

Preoperative Evaluation and Counseling

- Work as a member of the rehabilitation team
- In consultation with the surgeon, counsel the patient and the family about the effects of the planned medical treatment on communication
- Invite and answer all questions from the patient and the family members; give answers that are consistent with advice from other professionals on the team
- Obtain a sample of the patient's speech and writing; make an assessment of the client's communication skills
- Give an overview of the different methods of speaking without a biological larynx; discuss communication options that may be preferable to the client; be consistent with the surgeon's preferences and recommendations
- Reassure the patient that with training, communication is possible with new techniques
- Have the patient meet and speak with a rehabilitated Laryngectomee who has mastered a new method of communication
- Assist the patient in acquiring an intraoral electrolarynx for functional communication immediately following surgery; train the patient in its use
- Provide training in the use of the intraoral electrolarynx

Postoperative Management

- Review the information provided during the preoperative counseling
- Discuss methods of alaryngeal speech (described later in this section)
- Demonstrate how electronic speech aids work
- If a tracheoesophageal puncture (TEP) has been done, measure the fistula, select and place an appropriate size and style voice prosthesis, and educate the patient on its care and maintenance

- Discuss the patient's rehabilitation plan; be cautious in making prognostic statements
- Give written information on rehabilitation plans and possibilities for the patient to read later
- Arrange a visit from a rehabilitated laryngectomee to encourage the patient

Teaching New Methods of

Communication: General Principles

- Select an appropriate method of communication judged to be efficient, practical, and the individual prefers
- Teach the client to use the new method of communication
- Recommend a neck-type electrolarynx to the individuals who prefer other methods of communication; this may serve as a "backup" method, secondary to the primary method
- Begin to teach the use of a neck placement electronic device only after the neck and throat areas recover from swelling and tenderness and the surgical suture lines heal
- Teach tracheoesophageal speech if the patient is surgically prepared for it and a voice prosthesis is in place
- Consider both individual and group therapy sessions
- Determine the frequency of treatment sessions based on the patient's physical condition
- Consider daily sessions in the beginning if the patient's physical stamina permits them
- Hold at least one weekly session
- Get family members involved in training sessions
- Let the patient's performance and progress dictate the pace of therapy
- Ask the client to practice the new method of communication at home

Teach Alaryngeal Speech With Electronic Devices

- Help the patient select an electronic larynx; transoral devices, held in hand, carry the sound into the mouth via a tube; transcervical types transmit the sound via skin into the mouth when held against the neck

- Recommend a transoral device soon after the surgery because the patient urgently needs a device for functional communication; the neck model should not be used until the surgical wound is healed
- Recommend an adapter to convert a neck-type to a transoral type; the adapter, affixed to the head of the device, delivers the sound to the mouth via a tube
- Recommend a transoral device soon after the surgery, as the unit is not held against the neck
- Demonstrate how speech produced with it sounds like
- Experiment with the neck device to find the best position, usually under the jaw; let the head of the device make good contact with the skin without pressing it
- Manipulate the button for sound production and ask the patient to count aloud
- Ask the patient to clearly shape the words with the mouth; esophageal speech involves mild overarticulation of the speech sounds, particularly final consonant sounds
- Ask the patient not to exhale forcefully
- Teach the patient to handle the device
- Instruct the patient to coordinate sound and speech and to turn off the sound when not talking
- Reduce the patient's rate of speech to increase intelligibility
- Teach the patient to increase articulatory precision by practicing words that begin with voiceless consonants
- Shape progressively longer utterances
- Ask the person to practice phrasing with appropriate timing (voice on and voice off) of the electric larynx
- Teach the client to maintain eye contact with the listener

Teach Esophageal Speech

- Note that tracheoesophageal speech is preferred over this traditional esophageal speech, which is now used much less frequently
- Begin esophageal speech training soon after the patient starts eating food orally
- Describe the anatomy and the physiology of esophageal speech production

- Describe esophageal sound production to the patient
- Use diagrams to explain esophageal speech
- Teach the client the production of esophageal sound
- Try various procedures and settle on the one most effective with the client
- Teach the patient to use the injection method of taking air into the esophagus
 - Ask the patient to press the tongue tip against the alveolar ridge to push the air back toward the esophagus without the tongue making contact with the pharyngeal wall (glossal press)
 - Ask the client to press the tongue tip against the alveolar ridge and to move the tongue back to make contact with the pharyngeal wall; thus, push air back into the esophagus (glossopharyngeal press)
 - Ask the patient to keep the velopharyngeal port closed
 - Ask the client to inject the air in an audible manner, producing the sound called the “klunk”
- Teach the person to use the inhalation method of taking air into the esophagus if necessary
 - Teach the individual to synchronize the air intake through the stoma with air intake through the mouth into the upper esophagus
 - Ask the persons to relax the P-E segment to create a negative pressure there to move air into the upper esophagus
- Ask the person to produce plosive consonants to stimulate esophageal sound
- Instruct the patient to say ta-ta-ta or sta-sta-sta
- Ask the patient to use easy injection of air and say a series of ta-ta-ta or sta-sta-sta
- Reinforce a likely emergence of esophageal sound
- Teach the patient to puff the cheeks out and move the air trapped in the mouth from one side to another; instruct the patient to move this trapped air quickly into the esophagus
- Ask the patient to produce words that typically trigger sound production: *church*, *stop*, *skate*, *scotch*, and *scratch*

- Use consonant-vowel (C-V) combinations initially, opening the mouth wide during the vowel production
- Move on to single-syllable words
- Increase response complexity
- Ask the patient to slow down the rate of speech

Teach Tracheoesophageal Speech

- Select a Voice Prosthesis for a patient who has undergone the Tracheoesophageal Puncture (TEP) procedure
- Insert the voice prosthesis into the fistula that is properly healed; check for leakage of fluid around or through the prosthesis
- Ask the patient to inhale, occlude the stoma with a finger, and exhale
- Ask the client to produce sound as the air from the lungs enters the P-E Segment through the voice prostheses
- Have the patient practice sound production
- Shape the sound into speech
- Increase the length of utterances
- Reinforce all correct or approximate responses and verbally correct inappropriate responses
- Experiment with “hands-free” speaking valves and train the patient to utilize them, if appropriate

Sapientza, C., & Hoffman, B. (2021). *Voice disorders* (4th ed.). Plural Publishing.

Stemple, J. C. et al. (2024). *Clinical voice pathology* (7th ed.). Plural Publishing.

Laryngitis. Irritated and swollen vocal folds; causes include vocally abusive behaviors and infection; may be temporary or chronic; causes Hoarseness, low pitch, and vocal tiredness; may lead to vocal nodules or polyps.

- Suggest vocal rest without whispering
- Do not recommend voice treatment other than vocal rest for temporary laryngitis following an episode of vocally abusive behavior
- Reduce vocally abusive behaviors; see Treatment of Vocally Abusive Behaviors under Voice Disorders

Laryngoplasty. Surgical treatment to improve phonation in people with vocal cord paralysis or weakness; involves medial displacement of vocal cords with the help of implant materials to promote better approximation.

Left Neglect. To treat reduced awareness of stimuli in the left visual field in individuals who have sustained right brain injury, see [Right Hemisphere Disorders](#).

Levodopa. A neurotransmitter whose deficiency causes movement disorders; drugs containing this neurotransmitter may be prescribed for patients to control their movement disorders (e.g., those with Parkinson's disease).

Lewy Body Dementia (LBD). Some symptoms of LBD may be similar to those found in [Parkinson's Disease](#); see [Dementia](#) for general treatment procedures and consider the following unique features:

- Note that cognitive disorders precede movement disorders typical of Parkinson's
- Target confabulation, reduced speech fluency, difficulty comprehending longer and complex sentences, and narrative skills
- Consider muscular rigidity and slow movements that might affect speech production
- Work with other professionals to control hallucinations, paranoid thoughts, depression, anxiety, agitation, restlessness, pacing, and so forth

Lipreading. Understanding speech by watching the mouth of the speaker; gaining cues from the movement of the lips and the tongue; a skill that may supplement limited comprehension of speech through residual hearing; see also [Speechreading](#).

Literacy and Literacy Skills in Children. To promote academic success of students, SLPs may teach literacy skills in the context of speech and language treatment; literacy skill intervention targets emergent literacy skills,

reading, and writing; a parent-conducted home literacy training is essential; literacy and speech-language interventions may be integrated; as a first step, assess home literacy environment and implement a home literacy promotion program; see the cited sources at the end of this main entry.

Home Literacy Promotion Program

- Train family caregivers to enrich the home literacy environment
- Ask parents to supply literacy resources for the child (e.g., books, supplies, a work area, a chair and desk to read and write)
- Ask parents and older siblings to read and write in the target child's presence to provide literacy models
- Suggest the literacy support systems the family could use (e.g., access to a public library; libraries and bookstores that offer story reading sessions to children; organizations that may supply free books for children)
- Inform about children's storybooks from the town or community libraries and used book markets to get children's books at low cost
- Supply the writing materials to the child (e.g., crayons, pencils, pens, paper, eraser, a child's desk and chair)
- Read stories to the child in an interactive manner; ask the reader to let the child see the words and the lines being read; move the finger along the line; let the reader ask questions and invite comments from the child
- Encourage the child to draw, copy, and color
- Teach the child to write the letters of the alphabet
- Obtain educational toys (e.g., magnet letters, map puzzles, counting beads)
- Provide small gifts and inexpensive treats for listening to stories, tracing letters, drawing, and printing the letters of the alphabet

Teach Emergent Literacy Skills to Preschool Children

- Select intervention materials that are at the child's grade and are appropriate for the child's cultural, language, and ethnic background

- Use modeling, prompting, manual guidance, verbal praise, and verbal correction in all sessions; target and teach:
 - Print awareness to preschool children
 - Alphabet printing during oral language teaching (ask the child to print the first letter of words that you use as stimuli for oral language teaching)
 - Pointing to the words and letters as you read the story aloud; reinforce the child for looking at the printed material
 - Telling the meaning of commonly encountered logos and signs (e.g., a stop sign or signs in supermarkets)
 - Picking up a book among several displayed; reinforce the child for picking up a book and looking into it; offer to read a story from a book the child picks up
 - Counting in the context of speech or language treatment
 - Reciting the alphabet, nursery rhymes, days of the week, or names of the months
 - Pointing to single words and sentences on printed passages
 - Showing the front and the back of a book
 - Correcting the orientation of a book handed upside down; if the child fails to turn it around, guide the movement manually
 - Pointing to the beginning and the end of a story in a simple storybook; teach if necessary
 - Naming the letters of the alphabet when pointed to each on a printed sheet (e.g., “Can you find the letter s? The letter b?”)

Teach Early Reading and Writing Skills

- Naming the letters of the alphabet from A through Z; give prompts
- Naming letters presented randomly; give prompts
- Making the sound a letter represents (point to printed letters and ask, “How does this sound?”); model the missed sounds and reinforce imitations
- **Teach early reading skills**
 - Sounding out simple printed words

- Reading a passage from their grade level; correct misreading of words; prompt when the child hesitates, and draw attention to omitted words
- Reading slightly faster; reinforce small increments in faster but fluent reading; measure the number of words read per minute
- Reading a short passage silently; ask questions about the material; reiterate what the child missed; have the child reread for better comprehension
- **Teach elementary writing skills**
 - Simple drawing skills; give verbal corrections and ask the child to redraw
 - Tracing the letters; prompt when necessary
 - Printing all the letters of the alphabet; prompt correct sequencing
 - Writing a few sentences; suggest a topic to write on, and correct spelling mistakes
- **Teach more advanced reading skills**
 - Reading aloud a longer and more complex piece of printed material; select the material from the child's grade level; prompt to minimize errors
 - Answering questions about the read material; correct errors in comprehension; have the child reread the material; ask questions again and reinforce correct comprehension
 - Distinguishing between literary genres (e.g., an essay, a poem, a novel, nonfiction); describe the differences to the child and have the child restate the differences
- **Teach more advanced writing skills**
 - Writing a brief essay on a topic the student chooses (e.g., a vacation experience, a book recently read); evaluate the writing for word usage, spelling, letter formation, capitalization, punctuation, organization, logical sequence, sentence variety, grammaticality, and other features of good writing; give written feedback; discuss orally the strengths and weaknesses; give suggestions on how to overcome the weaknesses

- Repeating the writing exercise until the writing skills improve to an acceptable level; assess and give suggestions at each stage

Teach Abstract Language Skills

- Defining selected terms found in the child's textbook; prompt correct definitions; reinforce progressively more accurate definitions
- Giving examples of concepts described in the grade-level books; help the child refine the examples or correct errors
- Stating the meaning of selected proverbs, idioms, and slang; model correct responses or prompt them; teach progressively more complex and abstract expressions
- Defining selected antonyms and synonyms sampled from the child's grade book; prompt when the child hesitates; have the child restate the meanings of words missed or misstated
- Describing how a camping trip should be planned; prompt and reinforce details and comprehensive descriptions
- Retelling a heard story; correct errors; prompt details and correct sequencing of story elements; have the child tell multiple stories
- Stating the moral of the story the student tells; correct any errors
- Talking on a topic or social issue (earthquakes or floods, school lunch program); model and prompt as found necessary

Culatta, B. et al. (2013). *Systematic and engaging early literacy*. Plural Publishing.

Hegde, M. N., & Pomaville, F. (2016). *Assessment of communication disorders in children: Resources and protocols* (3rd ed.). Plural Publishing.

Lombard Effect. Increased vocal intensity in noisy environments or under auditory masking with white noise; typically reflexive, but may be modified with instruction and reinforcement.

Loudness. To increase vocal loudness in individuals who speak with excessively soft voice, see [Voice Disorders](#).



M

Maintenance Strategy. Use a variety of strategies to promote the sustained production of clinically established skills in natural environments; plan these strategies from the beginning of treatment because maintenance is affected by selected treatment stimuli and target behaviors, the reinforcement schedule used, the treatment setting, and support for the clinically established skills in the natural environment.

Select Treatment Stimuli That Promote Maintenance

- Select client-specific stimuli; in teaching speech and language skills, select common and functional stimulus items, preferably objects; let children bring stimuli from home (e.g., a girl could bring her toys to serve as stimuli in speech or language training); let your adult clients help select stimuli that are personally relevant
- Select ethnoculturally appropriate stimuli; in the case of children, let the parents suggest culturally relevant stimuli that the child is familiar with
- Select colorful, unambiguous, and realistic pictures
- Select simple and common verbal stimuli that are used to evoke the target responses
- Vary the conversational partners because they are also stimuli for the target behaviors; have family members and other persons participate as conversational partners in treatment sessions

Select Functional Targets That Are Likely to Be Maintained

- Select client-specific and functional responses for treatment targets
- Select ethnoculturally relevant and appropriate treatment targets
- Select target behaviors that are likely to be produced at home
- Select target behaviors that can easily be expanded into more complex communicative behaviors

Train to Promote Generalization

- Train multiple exemplars of each target skill at each level of response complexity

- Take training to complex levels of target skills
- Give sufficient training at the conversational level
- Teach Self-Control (Self-Monitoring) Procedures (e.g., counting one's target behaviors)
- Give treatment for a sufficient duration

Use Reinforcement Schedules and Specific Reinforces to Promote Maintenance

- Reinforce every correct response (continuous reinforcement schedule) in the initial stages of treatment to establish the target skills
- Thin out reinforcement gradually by reinforcing every other, every third, and every fourth correct response (intermittent schedule) in the latter stages of training to strengthen the skills
- Use conditioned reinforcers (tokens with backup reinforcers), but pair them with verbal praise
- Delay reinforcement in the latter stages of training
- Use only occasional and natural reinforcers (e.g., smile, verbal praise or approval, expression of agreement)
- Reinforce generalized responses in the clinic (responses given to untrained stimuli) unless you are measuring such responses on probe trials
- Teach Reinforcement Priming to your clients (e.g., teach children to draw attention to their production of target behaviors at home so the ignoring parents can pay attention and reinforce the client)

Vary the Treatment Settings

- Walk with the client outside the clinic and reinforce the target skills in conversational speech
- Hold informal training sessions in natural environments (e.g., school cafeteria, playground, classroom, a young child's home if practical)
- Conduct informal sessions with individuals not involved in treatment in nonclinical settings (e.g., ask a teacher to prompt and reinforce the targeted responses from a child)

Increase the Environmental Support for the Clinically Established Skills

- Let the family members and others watch treatment sessions so they can better understand the treatment targets and teaching methods
- Train caregivers to evoke, model, and prompt the target skills at home and other natural settings
- Train caregivers to socially reinforce the target behaviors and do that in a subtle manner at home and in other natural settings
- Train caregivers to offer more positive comments than critical statements

Follow Up and Offer Booster Sessions

- Schedule regular Follow-Up sessions to assess response maintenance
- Arrange for Booster Treatment if the skills have deteriorated

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.). Plural Publishing.

M

Mand-Model. A naturalistic child language treatment method; uses typical adult-child interactions in a play-oriented setting to teach functional communication skills to children with language disorders; includes the Incidental Teaching Method; for procedures, see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs; supported by experimental evidence.

Mands. A class of verbal behavior caused by a state of motivation; includes requests, commands, demands, and many kinds of questions; need to create a state of motivation to teach mands; several kinds of mands are reinforced only with primary reinforcers.

- Create a state of motivation:
 - Arrange treatment around lunch or breakfast time so food may be used as a reinforcer (hunger is the state of motivation)
 - Hold food in front of the child until the child asks for it

- Place attractive toys on a high shelf to encourage the child to request it; ask a question (e.g., “Do you want that car?”), and if there is no response, model the request for the child (e.g., “Say, I want that car”)
- Offer a food item the child does not like to encourage the child to verbally refuse it; model the response if the child fails to respond verbally (e.g., “Say, I don’t like it”)
- Eat something the child is fond of without offering it (the child should request it; model the request if necessary)
- Give a tightly closed jar with candy in it (the child should ask you to open it; model the request if necessary)
- Reinforce promptly with the displayed or held-back item; remove promptly an aversive item presented when the child makes an appropriate response

Manual Communication. Sign language, finger spelling, gestures, and other forms of nonoral communication taught to children and adults with severe oral language difficulties; when taught to young nonverbal children, manual modes of communication may promote verbal communication; for general procedures, see [Augmentative and Alternative Communication](#).

Manual Guidance. Physical guidance provided to shape a response; the [Phonetic Placement Method](#) is similar to manual guidance; needed when the client cannot imitate a response; used in teaching practically all types of communication skills.

- Use your fingers to shape articulators while teaching speech sound productions
- Use tongue depressors to move the tongue to desired positions for correct articulation
- Take the client’s hand and make it touch the target picture while training comprehension of words
- Apply slight digital pressure to the laryngeal area to lower a client’s pitch

- Apply slight pressure on the chin of a child who does not readily open the mouth to achieve oral resonance
- Fade manual guidance to promote more typical production of target responses

Masking Noise. Delivery of noise through headphones to reduce or eliminate auditory sensitivity of pure tones or speech during hearing assessment; typically induces the Lombard Effect, an increased vocal intensity; used to induce stutter-free speech in people who stutter, although not recommended because of only short-term gains in unnatural-sounding fluency; one of the Specific Normal Voice Facilitating Techniques described under Voice Disorders.

Matching. A scientific method of selecting and assigning similar (matched) participants to experimental and control groups in treatment research; an alternative to random selection and assignment of participants.

M

Matrix Training. A well-researched language training method that is known to enhance recombinative generalization of trained verbal responses; may be useful in treating language disorders in general but known to be especially effective in teaching generalized productions to preschoolers and individuals with autism or intellectual disabilities; to implement this procedure:

- Select nouns and words the individual already produces; if not, teach them in isolation
- Organize the language targets in a matrix as shown (from the cited source); the matrix shows the noun-verb training items and multiple untrained probe items to assess recombinative generalization:

	Verb 1	Verb 2	Verb 3
Noun 1	Noun 1– Verb 1 Diagonal Training Item	Noun 1– Verb 2	Noun 1– Verb 3
Noun 2	Noun 2– Verb 1	Noun 2– Verb 2 Diagonal Training Item	Noun 3– Verb 3
Noun 3	Noun 3– Verb 1	Noun 3– Verb 2	Noun 3– Verb 3 Diagonal Training Item

- Train the diagonal training items with modeling, praise, and verbal correction; probe for recombinative generalization of untrained noun-verb combinations

Frampton, S. E. et al. (2016). The use of matrix training to promote generative language with children with autism. *Journal of Applied Behavior Analysis*, 49, 869–883.

Maximal Contrast Method. A speech sound (articulation) training method in which word pairs that phonemically contrast the most are used to train target phonemes; contrasted with Minimal Pair Contrast Method, in which word pairs differ by one phoneme; in maximal contrast pair, the words may differ by several features or phonemic contrasts; for procedural details, see Speech Sound Disorders: Specific Techniques or Programs.

Mechanical Corrective Feedback. A method to reduce incorrect responses in treatment; also known as Biofeedback; feedback is presented soon after an incorrect response is made; includes such feedback as provided on a computer monitor for incorrect responses (e.g., undesirable vocal pitch or intensity) and electromyographic feedback

on muscle tension; helpful in treating [Voice Disorders](#) and neuromuscular disorders associated with [Dysarthria](#).

Melodic Intonation Therapy. An aphasia treatment program that uses musical intonation, continuous voicing, and rhythmic tapping to teach verbal expressions to patients with severe nonfluent aphasia with good auditory comprehension; for procedural details, see [Aphasia; Treatment of Aphasia: Specific Technique or Programs](#).

Memory Impairments. Impairments in remembering, recalling, or acting on the basis of remote or recent experiences; impairment in learning or remembering current or recent events; typically described in such unhelpful mentalistic or mechanistic terms as storage and retrieval and a variety of presumed mental or neurological patterns; almost always measured in terms of present or absent behaviors; found in many persons with a variety of neurological and psychiatric disorders; speech-language pathologists are unlikely to treat memory problems found in individuals with no communication disorders (psychiatrists and psychologists may); therefore, treatment of memory problems concern speech-language pathologists only when they work with individuals who have any form of [Dementia](#), including [Alzheimer's Disease](#), [Traumatic Brain Injury in Adults](#), and, to some extent, individuals who have [Aphasia](#); [Attention Disorders](#) also are relevant to memory management; see those main entries for procedures to manage problems.

Memory Log Books. A memory aid for patients with memory impairments; consists of written material that helps sustain skills or prompt actions; recommended for patients who have retained at least a basic level of reading and writing skills; not useful for patients who are confused, have left-sided neglect, or have severe uncorrected visual defect.

- Design a simple log book that will contain only essential information
- Design a colorful cover for the book so it is easy to locate

- Designate a regular place where the book will always be kept; let the health care staff or family members know the place
- Select functional information to be included in the log book; consult with the patient, staff, and the family members in making this selection
- Include the patient's biographical information (name, age, address, phone number, family members' names); the current month, year, and the name of the health care facility; names of main health care workers and individual clinicians treating the patient
- Train the patient to take the book to all appointments, including treatment sessions
- Include in the book pictures and names of therapists and family members; train the client to frequently consult this information to facilitate memory for their names and faces
- Designate a single health care worker who will help maintain, update, and modify the book as found appropriate
- Divide the book into easily identifiable sections, preferably of different colors; organize information in the sections for easy consultation (e.g., a section each on treatment session appointments, daily activities, medications and their schedules, recreational activities)
- Train the client to consolidate all written forms of memory aids into the log book and not have multiple and odd pieces of information strewn around
- Teach the patient to write down information and then to periodically review the written information to act on it

Metronome-Paced Speech. In this method, clients are asked to pace a syllable or a word to each beat of a metronome to slow down their rate of speech; may be used in the treatment of stuttering, cluttering, and certain forms of dysarthria; induces monotonous speech, so it should be faded soon; see [Dysarthria](#) and [Stuttering, Specific Techniques or Programs](#).

Minimal Pair Contrast Method. Use this method of treating [Speech Sound Disorders in Children](#) and speech

production problems in adults with Dysarthria; use word pairs that differ only in one phoneme (e.g., *pat*, *bat*); for procedural details, see Speech Sound Disorders in Children: Specific Techniques or Programs; see also Maximal Contrast Method.

Mixed Dysarthrias. To treat this type of motor speech disorder that is a combination of two or more pure types of dysarthria, see Treatment of Dysarthria: Specific Types under Dysarthria.

Mode of Response. Manner or method of a response; includes imitation, spontaneous production, conversational speech, narratives and discourse, and oral reading; useful in sequencing treatment targets:

- Teach a target behavior first in the imitative mode if necessary; model the target response
- Fade the modeling and teach a target behavior in evoked (more spontaneous) mode
- Teach the target behavior in conversational and other modes (e.g., narratives and discourse)
- Teach the target behaviors in oral reading mode if judged useful

Modeling. Clinician's production of a target behavior for the client to imitate; needed when instructions and requests fail to prompt the response; used frequently in treating communication disorders; there is much experimental evidence to support its use in treatment; clinician may model at different stages of treatment; for procedural details, see Baseline Modeled Trials under Baselines, Treatment Modeled Trials under Treatment Discrete Trials, and Probe Discrete Trials.

Moto-Kinesthetic Method. An articulation treatment method; similar to Phonetic Placement Method; emphasizes awareness of kinesthetic movement involved in articulation.

- Consider using the technique in the initial stages of treatment

- Manipulate the client's articulators with your fingers ([Manual Guidance](#))
- Provide visual stimulation of the movements with the help of a mirror

Motor Speech Disorders. To treat a group of speech disorders associated with impaired motor control of speech muscles or motor programming of speech movements, see [Cerebral Palsy](#), [Dysarthria](#), and [Apraxia of Speech](#).

Multi-Infarct Dementia. To clinically manage this type of vascular dementia, see [Dementia](#) and [Vascular Dementia](#).

Multiple Phoneme Approach. To use this behavioral method of teaching children with multiple misarticulations, see [Speech Sound Disorders in Children: Specific Techniques or Programs](#).

Multiple Sclerosis (MS). To treat communication disorders associated with this neurological disease, see [Dysarthria](#).

Mutational Falsetto. To treat the undesirable continuation of prepubertal, high-pitched voice after attaining puberty:

- Have medical confirmation of laryngeal maturation
- Establish a lower-pitched voice; use techniques described under [Voice Disorders](#); [Treatment of Disorders of Loudness](#) and [Pitch](#).

Mutism. Treatment of mutism requires a knowledge of various clinical conditions that lead to, or are associated with, lack of oral speech; treatment of mutism may be a part of treating various kinds of neurological diseases that cause dysarthria and dementia that may terminate in mutism:

- See [Psychiatric Communication Disorders](#) for treating a type of mutism not associated with neurophysiological problems (called *selective mutism*)

- See [Aphasia](#), [Apraxia of Speech](#), [Dysarthria](#), and [Traumatic Brain Injury in Adults](#) for treatment of associated mutism
- See [Voice Disorders](#) for aphonia, which should not be confused with mutism
- No treatment may be needed for transient mutism in patients with aphasia or traumatic brain injury
- [Augmentative and Alternative Communication](#) choices may be considered for patients who have mutism but cannot benefit from oral communication training (e.g., individuals with severe apraxia of speech or severe global aphasia)

McInnes, A., & Manassis, K. (2005). When silence is not golden: An integrated approach to selective mutism. *Seminars in Speech and Language*, 26(3), 201–210.

Wintgens, A. (2005). Selective mutism in children. *Child Language Teaching and Therapy*, 21(2), 214–216.

N

Narrative Skills. To treat language skill in describing events in a sequential, chronologically correct, and logically consistent manner, see [Language Disorders in Children](#) and [Language Disorders in Older Students and Adolescents](#).

Nasal Assimilation. A phonological pattern in which nasal consonants are substituted for oral consonants (e.g., /n/ for /d/); see [Speech Sound Disorders in Children](#) for treatment approaches.

Nasal Emission. Audible escape of air through the nose during speech; often found in children with cleft palate; see [Cleft Palate](#) and [Velopharyngeal Dysfunction](#) for treatment procedures.

Natural Settings. Nonclinical settings where clients communicate for the most part; communication in natural settings is always a final treatment target; in the case of infants and toddlers, treatment may be implemented in natural settings; extending treatment to such settings is essential to promote response maintenance.

N

Natural-Sounding Fluency. A stuttering treatment target following treatment with delayed auditory feedback, metronome-paced speech, and prolonged speech that induce unnatural-sounding speech; see [Stuttering; Treatment of Stuttering: Specific Techniques or Programs](#).

Neck Brace. A brace around the neck used to stabilize the weakened neck muscles; often used in treating clients with dysarthrias.

Negative Reinforcers. Aversive events that are removed, reduced, postponed, or prevented; responses that accomplish these tend to increase in frequency; less useful than positive reinforcers in teaching speech-language skills; avoidance behaviors seen in people who stutter, for example, may be negatively reinforced; prevention of avoidance is effective in eliminating negative reinforcement.

Neglect. To treat unilateral neglect of visual stimuli, see Right Hemisphere Disorder with which the problem is most often associated.

Neologism. To reduce the frequency of new but meaningless words that patients with aphasia create, see Aphasia.

Neural Anastomosis. Connecting a branch of an undamaged nerve to a damaged nerve; a surgical treatment for certain persons with dysarthria; a branch of the intact cranial nerve XII may be connected to the damaged cranial nerve VII to restore function and appearance.

Neurogenic Stuttering. Treatment of stuttering with an adult onset and a demonstrated neurological basis has lagged behind descriptive and diagnostic information; there is little or no research on treatment effects; treatment may not be necessary for some individuals in whom the problem is transient; suggested treatment procedures are based on reported clinical experiences and on the reasonable but untested hypothesis that stuttering treatment procedures may be used with suitable modifications; clinicians should evaluate the results of selected procedures carefully and abandon those that are ineffective.

- Develop a comprehensive treatment program for the neurogenic stuttering as well as associated disorders such as aphasia, hypokinetic dysarthria, or any other neurogenic communication disorder
- Delay treatment to see if stuttering due to drug toxicity will improve when the toxicity is treated effectively
- Develop a symptomatic behavioral treatment program to reduce stuttering or stuttering-like behaviors; use such techniques as the following, but carefully document and evaluate the course of the disorder and improvement in fluency:
 - Use the rate reduction strategy to minimize the frequency of dysfluencies; see Stuttering for procedural details
 - Do not use pacing devices, delayed auditory feedback, and auditory masking to treat individuals with

neurogenic stuttering along with slow and effortful speech because they are ineffective

- Consider relaxation and biofeedback to reduce speech muscle tension; evaluate the results carefully

Bloodstein, O. et al. (2021). *A handbook on stuttering* (7th ed.). Plural Publishing.

Nonexclusion Time-Out. Use this procedure of response-contingent arrangement of a brief duration of time in which all interaction is terminated to reduce a variety of undesirable behaviors, including stuttering; do not remove the individual from the clinical or educational setting; one of the Direct Methods of Response Reduction.

- Give a response-contingent signal to start time-out (e.g., saying “Stop” as soon as a dysfluency occurs); do not let the client talk during time-out
- Turn your face away from the client
- Stay motionless for 5 seconds
- Turn toward the client, and continue the interaction

Nonfluent Aphasia. To treat this type of aphasia, see Aphasia; Treatment of Aphasia: Specific Types.

N

Noniconic Symbols. Use these geometric shapes that do not look like what they suggest to teach a form of non-verbal communication; establish the meaning of such shapes by training; more difficult to learn than Iconic Symbols but more flexible; plastic chips or various shapes are an example; used in teaching Augmentative and Alternative Communication—Aided.

Nonpenetrating (Closed-Head) Injury. To treat communication disorders associated with this type of head injury in which the skull may or may not be fractured or lacerated and the Meninges remain intact, see Traumatic Brain Injury in Adults.

Nonverbal Oral Apraxia. To treat difficulty in executing oral movements that are typically involved in speech,

see [Apraxia](#) and [Apraxia of Speech](#), with which it may be associated.

Nonverbal Communication. Modes of communication that do not involve spoken speech; use of gestures, signs, symbols, printed material, electronic display, communication boards, and so forth to communicate; also includes such fully developed nonoral languages as [American Sign Language](#); see [Augmentative and Alternative Communication](#).

Nonverbal Corrective Feedback. A method used to reduce incorrect responses in treatment by giving immediate feedback for them; includes various forms of gestures, hand signals, and facial expressions that suggest to the client that the response was wrong (e.g., the stereotypic sad face); a form of [Corrective Feedback](#); often paired with [Verbal Correction](#).

Normal Prosody. Prosodic features include socially acceptable rhythm, stress, intonation (pitch variation), intensity, transition between words and phrases, correct phrasing and pausing at appropriate junctures, and acceptable rate of speech; teach them to persons with apraxia of speech, cluttering and stuttering following fluency shaping, dysarthria, foreign accent reduction, hearing impairment, voice disorders, and so forth; prosodic impairments are a side effect of treatment in which the speech rate is slowed down to either improve intelligibility or reduce stuttering.

- Select a particular aspect of prosody for treatment (e.g., pitch variations or increased speech rate)
- Model the target behavior (e.g., a higher pitch)
- Demonstrate the target on a computer screen, if possible
- Audiorecord the model and play it
- Ask the client to match the live or recorded model (imitate the higher pitch)
- Shape the target behavior in successive and progressively more complex steps
- Reinforce any movement in the direction of the model

- Set a higher level of response (e.g., sentences) when the target (a certain pitch or intensity) is achieved at a lower level (e.g., phrases)
- Give maximum feedback, including auditory and visual feedback
- Teach other prosodic features in the same manner (e.g., loudness variations and correct pausing)

Normative Strategy. An approach to selecting treatment target behaviors for clients, especially children, based on age-based norms and developmental sequences; often used in selecting target speech sounds and language structures for children; some evidence contradicts this traditional strategy; see Complexity Approach under Speech Sound Disorders in Children; also contrasted with Client-Specific Strategy.

- Assess speech-language skills of the child to determine potential treatment targets
- Select behaviors the child should already have acquired based on the age-based norms but has not
- Teach the selected behaviors in the normative sequence in which they are acquired
- Note that this is not the only approach to target behavior selection and sequencing
- Do not hesitate to experiment with different sequences that may not conform to the normative sequence; within limits, it is possible to teach language skills outside of their normative sequence

O

Objectivity. Agreement among different observers who measure the same event in the same manner; important in treatment research so that different observers agree that a treatment had an effect; helps replicate treatment procedures by other clinicians.

Obturator. A structure that helps close an opening; a prosthetic device that helps close a cleft of the hard palate; the device has a plate that covers the cleft.

Omission. Missing speech sounds; see Speech Sound Disorders in Children for teaching methods.

Omission Training. A behavioral method of reinforcing a person for not exhibiting an undesirable behavior to indirectly decrease it and to increase the reinforced desirable behavior; the same as Differential Reinforcement of Other Behavior.

Open-Head Injury. To treat communication disorders associated with this type of head injury, also called Penetrating Head Injury, see Traumatic Brain Injury (TBI) in Adults.

O

Operant. A behavior that is affected by its consequences; most voluntary behaviors, including communicative behaviors, are operants; behaviors that can be increased or decreased by arranging differential consequences for them.

Operant Aggression. A type of aggression directed against the source of aversive (unpleasant) stimulation; clinicians may be at the receiving end of this kind of aggression; contrasted with Elicited Aggression.

- A child may say “I hate you” when the clinician says “No” for an incorrect response
- A child may fight the clinician’s attempt to take a token away in a response cost procedure
- To reduce operant aggression in treatment, use aversive response reduction procedures sparingly and use more positive reinforcement

- Use Differential Reinforcement to indirectly control undesirable behaviors by reinforcing and increasing desirable behaviors

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.). Plural Publishing.

Operant Conditioning. Skinnerian conditioning; method of selecting and strengthening behaviors of an individual by arranging reinforcing consequences; evidence-based and commonly used procedures to teach speech-language skills.

Operational Definitions. Definition of variables in measurable terms; in treatment, it is essential to define target behaviors in measurable terms so the progress or the lack of it can be objectively documented:

- Specify the topographic aspects of the target behavior (e.g., production of /s/ in word-initial positions, phrases, sentences)
- Specify the mode in which the response will be measured (e.g., single-word productions, conversational speech, reading)
- Specify the stimuli and settings (e.g., when shown pictures, in the clinic, at home)
- Specify the accuracy criterion (e.g., 90% correct)

Oral Apraxia. Deficits in making movements unrelated to oral speech; see Apraxia and Apraxia of Speech.

Oral Language. Articulated speech and language; the most typical form of communication in most societies; disorders of oral language are the most commonly treated.

Orofacial Examination. An element of assessment; an examination of the structures of the face and mouth to detect their overall integrity and any deviations that may be present; see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for procedures.

Overarticulation. Exaggerated articulatory movements that may improve speech intelligibility; may be a treatment target in individuals with Dysarthria.

P

Pacing. A speech treatment procedure in which the rate of speech is reduced with rhythmic external stimulus to cue the production of syllables or words; used in the treatment of Aphasia and Dysarthria.

Pacing Board. A wooden board that has a series of colored slots that are separated by ridges; used in reducing the speech rate of clients with motor speech disorders; the speaker touches one slot for each word spoken; see Dysarthria.

Paired-Stimuli Approach. An articulation treatment method that uses a Key Word in which the sound is correctly produced but misarticulated in other words (contexts); for the procedures, see Speech Sound Disorders in Children: Specific Techniques or Programs.

Palatal Lift Prosthesis. Constructed by a prosthodontist in consultation with a speech-language pathologist, this device helps achieve velopharyngeal closure to reduce hypernasality in clients with flaccid dysarthria; anchored to the teeth, the device has a plate that covers the hard palate; the rear end is custom-shaped to fit the patient's oropharynx; it pushes the soft palate up and back to make contact with the posterior pharyngeal wall.

P

Palate Reshaping Prosthesis. An intraoral device that lowers the palatal arch by artificially increasing its bulk; may be designed with teeth to replace the missing teeth of the patient; helps the tongue with limited vertical movement to make contact with the hard palate to chew food.

Palilalia. A speech disorder in which a word, a phrase, or a sentence is repeated with increasing speed and decreasing intelligibility; it is a treatment target in some neurogenic communication disorders; often a symptom of Parkinson's Disease.

Palatoplasty. Surgical repair of the clefts in the palate; see Cleft Palate.

Pantomime. A method of communication in which the speaker acts out a message by gestures and bodily movements; a target communication skill for some minimally verbal individuals who can use gestures and bodily movements; unlike in other gestural systems, uses whole- as well as part-body movements; often more concrete and easier to understand than other gestures; used in teaching Augmentative Communication—Unaided.

Papillomas. Wart-like growths on the larynx; thought to be of viral origin; may be life-threatening if they block the airway; may be a recurring condition; treatment is laser surgery, which needs to be repeated; may need voice therapy to make the best possible use of the compromised larynx.

- Teach the client to achieve appropriate pitch and loudness; use techniques described under Voice Disorders: Treatment of Disorders of Loudness and Pitch.
- Teach proper respiratory control; treat any other voice symptom with Specific Normal Voice Facilitation Techniques (described under Voice Disorders)

Paraphasia. Treatment of unintended word or sound substitutions that result in fluent but somewhat meaningless speech is a treatment target in Aphasia.

Paraplegia. Paralysis of both legs, often due to spinal cord injury or disease; see Cerebral Palsy.

Parent or Caregiver Training. Preparing parents, other family members, and professional caregivers to conduct informal treatment at home and to perform maintenance activities to sustain treatment gains in natural settings; see Language Disorders in Infants, Toddlers, and Preschoolers and Maintenance Strategy.

Parkinson's Disease. To treat speech disorders associated with this neurological syndrome, see Dysarthria and Hypokinetic Dysarthria under Treatment of Dysarthria: Specific Types; to treat communication disorders in advanced and progressive conditions, see Dementia.

Partial Modeling. Use this procedure to withdraw modeling of complete sentences in gradual steps; a method of Fading to promote more spontaneous production of clinically established skills.

- Initially model complete sentences for the client to imitate (e.g., “The book is on the table”)
- Drop the last word when it is time to fade modeling (e.g., “The book is on the . . .”)
- Drop additional words, one word at a time, on subsequent trials (“The book is on . . .”; “The book is . . .”; “The book . . .”; etc.)

P-E Segment. Pharyngeal-esophageal segment; a part of the pharynx and the esophagus; muscle fibers from the cricopharyngeus, esophagus, and inferior constrictor blend at this site to create a sphincter that can reduce the cross-sectional area of the esophagus.

Peer Modeling. Use this naturalistic child language intervention method in which peers are trained to model the target skills for the child client; see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs; briefly:

- Select a peer who agrees to help and is acceptable to the client
- Have the peer observe your treatment sessions
- Describe the target behaviors, modeling and imitation sequence, and reinforcement procedures
- Let the peer model and reinforce the child’s productions in your presence
- Refine the peer’s skills in modeling the target behaviors
- Ask the peer to submit recorded language samples that document appropriate modeling outside the clinic
- Periodically assess the results of peer modeling and provide additional training to the peer

Peer Training. Training peers of clients to evoke and reinforce target behaviors in natural settings is an important treatment as well as maintenance strategy; it has been

well researched with adequate experimental support; see also Maintenance Strategy.

- Ask the peers to initially observe your treatment sessions
- Describe the target skills the client is being taught
- Let the peers count the occurrence of the skill along with you
- Give them feedback on their counting
- Train the peers to prompt, evoke, model, and reinforce the target communication skill
- Have peers conduct a session in your presence
- Give peers feedback and refine their skills
- Give peers simple, clear written instructions
- Give peers a sample of videorecordings of treatment procedures
- Ask peers to monitor the target skills in natural settings
- Ask peers to audiorecord a monitoring session outside the clinic or submit data recorded on paper
- Review submitted data
- Assess the client who is taught by the peers on an ongoing basis
- Initiate clinical treatment if peer training is not effective or their training cannot be improved

Goldstein, H. (2002). Communicative interaction for children with autism: A review of treatment efficacy. *Journal of Autism and Developmental Disabilities*, 35(5), 373–396.

Penetrating (Open-Head) Injury. To treat communication disorders associated with brain injury in which the skull is perforated or fractured and the Meninges are torn or lacerated, see Traumatic Brain Injury in Adults.

Pervasive Developmental Disorders. See Autism Spectrum Disorder.

Pharyngeal Flap Operation. A surgical procedure designed to reduce hypernasality in persons with repaired cleft or in those with weak or paralyzed soft palate (as in flaccid dysarthria); to improve velopharyngeal closure, the surgeon:

- Cuts a flap of tissue from the posterior pharyngeal wall
- Brings the flap down or raises it up (depending on how the flap is cut)
- Attaches the flap to the velum to provide extra muscular mass that helps achieve velopharyngeal closure
- Leaves an opening on either side of the flap to allow breathing, nasal drainage, and production of nasal sounds

Phonetic Derivation. Use this Shaping procedure (progressive approximation) to teach correct speech sound production to clients who do not imitate modeled productions; in using this procedure:

- Break the target sound production into its simpler components (e.g., teaching the production of /m/ by first having the client put the lips together, a simplified component of the total response)
- Teach the next component that will move the sequence in the right direction (e.g., add humming to the closed-lip posture)
- Teach other response components to achieve the total response (e.g., opening the mouth while humming through the nose, resulting in *ma*)
- Have the client practice the integrated response (e.g., *mommy*)

P

Phonetic Placement Method. As a part of a comprehensive speech sound treatment program, show the child the placement of articulators necessary to produce different speech sounds; do it when the client cannot imitate the modeled sound production.

- Describe how the target sound is produced
- Demonstrate how the sound is produced
- Show (model) the placement of articulators
- Give maximum visual feedback; use a mirror and a drawing of articulatory placements; use palatograms and breath indicators
- Show the differences between correct and incorrect productions of the same sound
- Help position the tongue of the client with tongue blades

- Use your fingers to manipulate and correctly position the client's lips and jaw
- Let the client feel the presence and absence of laryngeal vibrations
- Reinforce correct responses and give verbal corrections

Phonological Awareness. Many clinicians recommend phonological awareness training to children with speech sound disorders; such skills as rhyming, alliteration, phoneme isolation, sound blending, syllable identification, sound segmentation, and invented spelling are thought to be prerequisites for effectively treating speech sound disorders and to prevent reading disorders; the assumption is based on correlations between phonological awareness and speech and literacy problems; although popular, there is no evidence supporting the assumption that phonological awareness training is essential to treat speech and literacy problems; there is evidence contradicting the assumption; it is more effective to directly teach speech production and literacy skills than to spend time on presumed prerequisites; if still preferred, teach the following skills:

- **Rhyming.** Present selected words to the child and ask the child to produce words that sound alike or rhyme with each word you present; present rhyming and non-rhyming words, each printed on a card, and ask the child to sort them out
- **Alliteration.** Ask the child to name words that begin or end with certain sounds
- **Phoneme isolation.** Present printed words or present words orally and ask the child to tell whether a specific sound occurs in the beginning, end, or middle of a word
- **Sound blending.** Orally present words but pause in between the word productions and ask the child to say the word so produced
- **Syllable identification.** Orally present words that contain a different number of syllables, and ask how many syllables are in each word you present
- **Sound segmentation.** Orally present words and ask the child to specify the number of phonemes in each of the words presented

- **Invented spellings.** Orally present words and ask the child to spell each word phonetically

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of articulation and phonological disorders in children* (4th ed.). Pro-Ed.

Phonological Disorders. To treat multiple errors of articulation that form phonological rule-based patterns, see Speech Sound Disorders and Phonological Patterns; the treatment target is to eliminate phonological patterns by teaching the correct production of misarticulated sounds within patterns.

Phonological Patterns. Multiple ways in which children simplify adult production of speech sounds; become *error patterns* when they persist beyond certain age limits; see Speech Sound Disorders in Children, Treatment of Speech Sound Disorders: Specific Techniques or Programs; note that phonological pattern analysis includes only omissions and substitutions and entirely misses all distorted speech sounds; by teaching specific sounds within a pattern, eliminate the following patterns:

Assimilation Patterns. Productions of dissimilar phonemes that sound more alike; eliminate the following:

- **Alveolar assimilation:** Substitution of an alveolar sound by a nonalveolar sound (e.g., substitution of /d/ for /p/)
- **Devoicing:** Substitution of a voiceless final sound for a voiced (e.g., /k/ for /g/ in final positions)
- **Devoicing of final consonants:** Substitution of a voiceless final consonant for a voiced (e.g., /t/ for /d/)
- **Labial assimilation:** Substitution of a labial sound for a nonlabial (e.g., /b/ for /d/)
- **Nasal assimilation:** Substitution of a nasal consonant for an oral sound (e.g., /n/ for /d/)
- **Postvocalic voicing:** Substitution of a voiceless sound for a voiced sound that follows a vowel (e.g., /t/ for /d/)

- **Prevocalic voicing:** Substitution of a voiced sound for voiceless sound preceding a vowel (e.g., /b/ for /p/ in prevocalic positions)
- **Reduplication:** Repetition of a syllable, resulting in substitution of one for another (e.g., wawa for *water*)
- **Velar assimilation:** Substitution of a velar consonant for a nonvelar (e.g., /g/ for /d/)

Deletion Patterns. One or more consonants or a syllable in a word is deleted or omitted; teach the missing phonemes to eliminate the patterns:

- **Cluster reduction:** One or more consonants are deleted in a cluster of consonants (e.g., bu for *blue*, sike for *strike*)
- **Initial consonant deletion:** Omission of an initial consonant of a syllable (e.g., og for *dog*, ink for *sink*)
- **Final consonant deletion:** Omission of a final consonant (e.g., ha for *hat*, goo for *good*)
- **Unstressed syllable deletion:** Omission of a syllable (e.g., medo for *tomato*, nana for *banana*)

Substitution Patterns. Substitution of one class of sounds for another; eliminate such patterns by teaching specific phonemes:

- **Deaffrication:** Substitution of a fricative for an affricate (e.g., /t/, /s/, or /k/ for /tʃ/; /d/ or /z/ for /dʒ/)
- **Denasalization:** Substitution of an oral consonant for a nasal consonant (e.g., /d/ for /n/)
- **Gliding:** Substitution of a glide for a liquid (e.g., /w/ for /r/)
- **Stopping:** Substitution of a stop for a fricative or an affricate (e.g., /p/ for /f/; /p/ for /v/; /t/ for /s/)
- **Velar fronting:** Substitution of an alveolar for a velar (e.g., /t/ for /k/; /d/ for /g/; /n/ for /ŋ/)

Physical Prompts. Use these effective techniques that include visual signs or gestures that you give before a response is produced to increase its probability of occurrence (e.g., showing a lifted tongue tip just before the child attempts to produce a /t/; showing lip closure to prompt the production of a bilabial sound).

Picture Communication Symbols. A large collection of transparent pictures that represents words, phrases, sentences, and social exchanges used in Augmentative and Alternative Communication.

Picture Exchange Communication System (PECS).

A communication system used extensively in teaching communication skills to children with autism spectrum disorder; may be used as an Augmentative and Alternative Communication (AAC) method with any child who is minimally verbal; no verbal prompts or instructions are needed; a transitional system eventually leading to verbal communication training because PECS does not inhibit verbal learning but stimulates it (like all AAC systems do); no receptive training is involved; supported by experimental evidence.

- Display pictures that depict items the child might potentially want
- When the child looks at a picture or touches one, physically assist the child in picking up the picture
- Let the child drop the picture into your open hands
- Immediately give the item the picture depicts
- Let the child interact with the item (e.g., may play with the ball for a few seconds or eat a portion of something edible)
- Make a picture binder of essential items and let the child go to the book, pick a picture, and hand it to the therapist
- Train in different contexts, with different persons (e.g., family members or other caregivers), across all daily activities, and with different reinforcers
- Teach discrimination between pictures; display a highly desirable item with paired with a least desired item; give the selected item; if the child rejects it, physically prompt the selection of the desired item
- See the cited sources for additional steps to teach verbal skills (e.g., *I want*, *I see*, and so forth)

Bondy, A. S., & Frost, L. A. (2001). The Picture Exchange Communication System. *Behavior Modification*, 25, 725–744.

Picsyms. A set of symbols associated with printed words and line drawings used to teach nonoral expression of

nouns, verbs, prepositions, and so forth; clinicians can add their own picsyms to these open-ended sets; used in teaching Augmentative Communication—Aided.

Pictographic Symbols. Pictorial representation of objects and events; easier to learn than abstract symbols; used in teaching Augmentative Communication—Aided.

Polyps. Protruding, soft, fluid-filled growths on the inner margin of the vocal folds; result of vocal abuse, may be from a single abusive episode; typically unilateral; may be *sessile* (broad-based) or *pedunculated* (the mass of the polyp is connected to the vocal fold by a stalk-like structure); associated with hoarseness and breathiness; surgically removed; voice specialists reduce the vocally abusive behaviors; see Voice Disorders: Treatment of Vocally Abusive Behaviors.

Positive Reinforcers. Events that, when presented immediately after a response is made, increase the future probability of that response; an effective method to select and increase the frequency of target communication skills; extensively researched with a variety of clinical populations; essential in communication training.

- Select potential reinforcers after consulting with the client, the family, or both
- Present the potential reinforcer immediately after the correct response is made
- Reinforce all correct responses initially and intermittently in later stages; see Reinforcement Schedules
- Prefer Conditioned Generalized Reinforcers (e.g., Tokens) to Primary Reinforcers
- Offer verbal praise (even when you use other kinds)
- Use a different event when the one selected does not increase the response rate
- Consider an event a reinforcer only when it increases a response rate; if it does not, change the reinforcer

Posttests. Measures of behaviors established after completing an experimental or routine teaching or treatment

program; compared with Pretests; in a group design study, posttests help establish changes in treated target behaviors.

Pragmatic Language Skills. See Social Communication Skills to teach such conversational skills as eye contact, narrative skills, topic initiation and maintenance, and so forth.

Prephonation Airflow. A target behavior for people who stutter and those who exhibit hard glottal attacks; includes a slight exhalation before initiating phonation; for procedures, see Stuttering and Voice Disorders.

Premack-Type Symbols. Plastic shapes or tokens to teach communication to chimpanzees; Noniconic Symbols that may be used to teach communicative skills to individuals with limited oral communication skills; may be used in teaching Augmentative Communication—Aided.

Pretests. Measures of behaviors established before starting an experimental or routine teaching program; in a group design study, pretests may be compared to Posttests to document changes in treated target behaviors.

Pretraumatic Memory Loss. To treat loss of memory for events *preceding* brain injury, see Traumatic Brain Injury in Adults.

Primary Progressive Aphasia (PPA). Considered a type of frontotemporal dementia (FTD), PPA is an atypical form of aphasia that is due to progressive neurodegenerative disease terminating in a type of dementia; to treat PPA in early stages, see Aphasia; to treat it in the latter stages, see Dementia; but in all cases, target the assessed symptom complex as there may be variations across individuals; consider the following specific variables in treating the three varieties of PPA:

- **Nonfluent variant of PPA.** In treating this variant, target agrammatic, effortful, slow, nonfluent, and halting speech as you would in nonfluent Aphasia; especially

Broca's; also target apraxia of speech (inconsistent articulation errors), a slow speech rate, and prosodic deficits; mutism may be the final stage

- **Semantic variant of PPA.** Target noun naming and category naming because loss of word meaning is the main feature of this variant; other targets include comprehension of low frequency words and mutism in the final stage
- **Logopenic variant of PPA.** May be considered different because of its association with Alzheimer-type neuropathology in addition to temporal lobe atrophy; initial-stage treatment targets include the features of nonfluent aphasia, although agrammatic speech may not be a target; later-stage treatment targets include paraphasic and empty speech as well as memory deficits

Primary Reinforcers. Reinforcers whose effects do not depend on past learning; often fulfill biological needs; contrasted with Conditioned Reinforcers, Secondary Reinforcers, or Social Reinforcers; also known as unconditioned reinforcers. Combine them with social reinforcers and withdraw them as early as possible. Use primary reinforcers while teaching:

- Infants, toddlers, and other children who do not respond well to Social Reinforcers (e.g., verbal praise)
- Individuals with intellectual disabilities, minimal verbal skills, autism spectrum disorder, and brain injury (especially in the initial stages of treatment)

Principles of Treatment. Empirical rules from which treatment procedures are derived; procedures may be modified (e.g., ineffective reinforcers may be replaced with effective ones), but experimentally supported principles do not change (e.g., the principle of positive reinforcement remains valid).

Probes. Procedures used to assess generalized production of clinically established responses without modeling, prompting, reinforcement, or verbal correction; administered every time a few exemplars are trained to assess

generalized productions; probes should be conducted initially in discrete trials and later in conversational speech.

- **Discrete Trial Probes.** To consider that a target behavior (e.g., production of a grammatical morpheme or a target phoneme in sentences) has been trained, clients should meet a probe criterion (not just a training criterion); typically, 90% correct on probe trials involving untrained stimulus items that were baselined; procedure is similar to baseline evoked trials.

- Train four to six exemplars of a target behavior before conducting a probe
- Present only the untrained stimulus items; for instance, of the 20 exemplars baselined, present 14 to 16 untrained exemplars
- Present an untrained physical stimulus to evoke a response (e.g., a picture, an object)
- Ask a question that evokes a response (e.g., “What is this?”)
- Do not model, reinforce, or verbally correct responses
- Move the stimulus away from the individual
- Record the response as correct, incorrect, absent, and so forth on a recording sheet
- Wait for a few seconds to mark the end of a trial
- Re-present the next untrained stimulus and start a new trial
- Calculate the percent correct probe responses

- **Conversational Probes.** Conduct this probe to measure generalized production of clinically established communication skills in naturalistic conversation when systematic reinforcement and verbal corrections are absent; this is an ultimate test of treatment success; the procedure is similar to that used in establishing baselines of communication skills.

- Conduct a conversational speech sample in as naturalistic a manner as possible
- Have toys, pictures, books, and other materials to evoke speech with children; engage the child in conversational speech in a play-oriented session; target the speech or language features to be measured (e.g.,

drawing the child's attention to actions you perform to evoke the *ing*)

- Hold a conversation with adults on their favorite topics
- Measure the correct and incorrect productions of the target behaviors in the sample
- Calculate the percent correct baseline response frequency to be compared with the same under treatment and baseline

Procedures of Treatment. Technical operations the clinician performs to effect changes in client behaviors; actions of clinicians; contrasted with Treatment Targets; in describing treatment procedures:

- Specify the target communication skills
- Specify what you ought to do to achieve those target skills
- Describe the stimulus conditions you need to arrange
- Specify the kinds of positive reinforcement and verbal corrections you should give to the client under the differing conditions of correct, incorrect, and lack of responses
- Clarify how you measure the skills during treatment to document progress
- Describe how you plan to promote generalized productions and maintenance over time and across situations
- Specify the follow-up and booster treatment procedures

Production Training. Treatment designed to teach a client to produce a specified speech or oral language target; contrasted with Auditory Discrimination Training; emphasis is on what the client ought to say rather than just listen or respond nonverbally.

- Model the target skills and ask the client to imitate your productions
- Fade modeling when imitation is established
- Evoke the target skills with appropriate questions and other devices

Prognosis. A statement about the future course of a disorder under specified conditions, which typically include treatment or no treatment; good prognosis implies that

the patient will recover from the clinical condition (with or without treatment, usually specified).

Program of Treatment. An overall description of target behaviors, treatment variables, measurement procedures, generalization measures, maintenance strategies, follow-up, and so forth.

Progressive Assimilation. A phonological pattern in which a sound takes on the properties of a preceding sound; see [Phonological Patterns](#) and [Speech Sound Disorders](#).

Prolonged Speech. To use this method of slowing down the speech of people who stutter, see [Stuttering](#), [Treatment of Stuttering: Specific Techniques or Programs](#).

Prompts. Verbal or nonverbal stimuli that increase the probability of a response; often called *cues* in aphasia treatment literature, but they are the same; see [Aphasia](#) for more examples.

- Prompt promptly when the client hesitates (e.g., in treating naming in a client with aphasia: “What is this?” “The word starts with a /t/”)
- Prompt more frequently in the beginning to reduce errors
- Prefer a subtle or short prompt to ones that are loud or long (e.g., in treating a person who stutters to speak slowly: “Slower” instead of “Speak at a slower rate”)
- Prefer a gesture to a verbal prompt (e.g., in treating a person who stutters to speak slowly: make a hand gesture to suggest a slower rate)
- Use [Partial Modeling](#) as a prompt
- Fade prompts as the responses become more consistent

Prosthesis. A device fashioned for individual clients and fitted to compensate for a deficient or deformed structure to improve its function; see [Cleft Palate](#) for some examples.

Pseudodementia. To understand the clinical management of apparent dementia-like symptoms in people who

are depressed, see [Psychiatric Problems Associated with Communication Disorders](#).

Psychiatric Problems Associated with Communication Disorders. To treat certain communication disorders effectively, clinicians need to know the associated psychiatric problems; while treating communication disorders in persons with dementia and neurodegenerative diseases, specific psychiatric symptoms need to be dealt with; in other cases, the clinician may encounter a communication disorder that is entirely psychiatric in nature; stuttering that is due to malingering or faking, certain forms of mutism, a type of aphonia, and a type of dementia may all be entirely a psychiatric problem without a neurological basis; in yet another set of patients, communication disorders may be a part of a major psychiatric disorder; patients with schizophrenia are an example; to assess these problems, see the cited sources; consider the following guidelines in clinically managing communication impairments of psychiatric origin or those associated with diagnosis of a psychiatric disorder:

Treatment of Psychiatric Disorders:
General Guidelines

- With appropriate specialization, speech-language pathologists (SLPs) may treat communication disorders that may be psychiatric in nature; a person with malingered stuttering, SLPs need not refer to a psychiatrist or a psychologist who may lack expertise in diagnosing the typical stuttering, differentiating it from malingering, and treating either or both of them
- If the treatment of a communication disorder of possible psychiatric origin (e.g., functional aphonia) is successful and satisfactory, there may be no need to refer the client to a psychiatrist or psychologist unless the client still has additional, independent, and significant emotional problems (e.g., individuals who remain depressed or anxious)
- Communication disorders of a psychiatric nature may spontaneously recover in some but may need intervention in others

- Persons who receive both behavioral therapy to change their maladaptive behaviors and treatment for their communication disorders are more likely to sustain improvement than those whose communication disorder is changed but their overall behavior pattern remains maladaptive; a certain number of patients with psychiatric speech or voice disorders may need to change the way they react to stress
- In some cases, a psychological or behavioral explanation, found during an analysis of the problem, may help diminish the symptoms; in other cases, the symptoms may diminish without any analysis or explanation
- Most persons with psychiatric communication disorders believe that their disorder is organically based; the SLP should handle this belief in a serious and respectful way, not in a dismissing manner; the topic may be discussed at various stages of treatment and the person may be more inclined to take the explanation seriously when the treatment has produced positive changes
- Improvement may be so rapid in a few cases that treatment may be initiated in the diagnostic session and completed in the next few sessions
- When a neurological disorder is also present along with a psychiatric communication disorder, it is essential to make it clear to the person the different bases of their different symptoms
- A medical report that clears any organic basis for the disorder to be treated will help establish the credibility of behavioral management
- A careful case history, taken with a view to establish the conditions of onset, is the initial strategy in treating the client; lack of a physical cause for the disorder may be established during the interview when negative medical reports are reviewed; the attention can then be turned to behavioral history, stressful situations the client has been facing, the client's typical reactions to stress, possible consequences of the disorder, and so forth

Abulia. SLPs may treat abulia in individuals with an extreme lack of motivation to do anything, including talking; it is a form of mutism called *akinetik mutism*; this psychiatric disorder may have a neurological basis; treatment of akinetic mutism involves several considerations:

- Defer treatment if the person is just emerging from a mute state in the process of recovering from strokes and trauma; when the physiological state is stabilized, motivation to talk may also improve; address this type of mutism in the context of language treatment; use strong reinforcers to increase motivation to talk
- Treat an alert and attentive individual who does not talk like you would a functionally aphonic individual (see the next entry); digitally manipulate the larynx to reduce tension and reinforce vocalization that may be associated with coughing
- Prompt vocalizations in a person who whispers; digitally manipulate the larynx to identify “accidental” vocalizations, shape them into more sustained vocalizations, and subsequently into typical speech production

Aphonia—Functional. Treatment of functional aphonia (without a neurological basis) should follow a medical examination that rules out a physical (organic) basis for the disorder (e.g., bilateral vocal fold paralysis with a wide-open glottis); treat this as a psychiatric or behavioral aphonia:

Explore the Factors That Surrounded the Onset of Aphonia

- Use the interview and case history taking to identify possible stress factors in the client’s life that surrounded the onset of aphonia
- Were there any traumatic or stressful situations surrounding the onset of aphonia?
- Had the individual faced stress-related difficulties in the past? What were the reactions?
- Is there any significant tension in the laryngeal structures that may have triggered functional aphonia?

Explore the Consequences that Followed the Loss of Voice

- Was there a reassignment of job responsibilities that reduced the client's work-related stress? Did a student escape from stressful oral presentations because of aphonia? If so, aphonia in both cases may be stress related, and it is negatively reinforced because the disorder led to a reduction in stress
- How have the listeners reacted to the client's whispering? Perhaps the client may have been, and still is being, complimented for effective whispering and gesturing; if so, the client's aphonia is likely maintained by positive reinforcement
- Discuss the possible connection between stress, aphonia, and reduced work-related stress or academic stress that followed the onset of the problem; be objective and nonjudgmental; this may help induce insight and hint at the direction of treatment

Explore Potential Vocalizations the Individual May Be Unaware of

- Ask the person to cough and clear the throat and crack some jokes to induce laughter
- Quickly point out that you heard the voice; do not make the person feel that they were "caught"; instead, use such observations to reassure the patient that there is no physical problem that is preventing voice production and to induce hope that improvement is possible
- Repeat the procedure with throat clearing, coughing, or laughing
- Digitally manipulate the laryngeal area (laryngeal massage) to reduce tension and to induce voice production
- Lower the larynx by gently kneading or massaging the laryngeal muscles because a high larynx is a tensed larynx; if pain is felt, reassure the patient that it will pass and that the manipulation is essential to relieve tension; to do this, encircle the hyoid bone

- with the thumb and middle finger; make circular and up-and-down motions with gentle pressure
- Ask the patient to prolong vocalization associated with throat clearing, coughing, or grunting
- Reinforce positively even the slightest improvement in phonation or its duration
- Fade out the digital manipulation; shape the brief or faint phonation the patient has achieved into stronger phonation of progressively longer duration
- Continue treatment with words, phrases, and sentences produced with phonation; reinforce voice in oral reading as well
- Use the yawn-sigh approach described under [Voice Disorders](#) to reduce tension and increase the likelihood of normal voice production

Delusions and Hallucinations. The need to treat delusional speech associated with other types of communication disorders may vary across individuals; most SLPs will not treat *delusional disorder*, a specific psychiatric problem psychiatrists and psychologists manage; in various ways, SLPs may address coexisting delusions, hallucinations, or both in their clients who have communication disorders associated with neurodegenerative diseases and dementia; clinical management of such delusions and hallucinations may be a part of an overall interventional strategy, often implemented by a team of specialists including psychologists; use the following general management strategies:

- Discuss your strategies of handling delusions, hallucinations, confusion, and other psychiatric disorders in your clients with psychologists or psychiatrists who may also be treating the same clients
- Do not positively reinforce utterances that contain reference to delusions and hallucinations; ignore them completely (extinction); such utterances increase in frequency when people pay attention to them or ask the person to elaborate on them; they decrease when ignored

- Use time-out if extinction does not work; as soon as the person begins to express undesirable speech, cease all interaction for 5 seconds; turn your face away from the person; reestablish eye contact and ask a question that will prompt a different response
- Tell the individual that you need to work on other aspects of communication and that it would be helpful if you did not exhibit “that kind of speech,” referring to speech that is full of delusional thoughts and hallucinatory experiences; do not keep repeating this instruction, which may actually reinforce the patient; once stated, ignore the patient’s inappropriate speech
- Educate professional staff and family caregivers not to reinforce speech that contains reference to delusions and hallucinations
- Educate staff to pay attention to speech that is more reality oriented and free from reference to delusions and hallucinations
- Consider such strategies for patients with Alzheimer’s disease, frontotemporal dementia, Parkinson’s disease, Huntington’s disease, HIV-associated neurocognitive disorder, and patients with other kinds of dementia who may have speech that refers to delusions and hallucinations

P

Depression. Communication treatment of individuals with mild and severe forms of depression (e.g., in stroke patients with aphasia) may require efforts coordinated with psychiatrists and other specialists; moderate to severe depression, but not mild depression, is likely to be medically treated; SLPs do not treat depression but may have to modify their communication treatment procedures to maximize their effects on individuals with depression:

- Conduct a few trial treatment sessions with depressed individuals to evaluate the usefulness of communication intervention; defer treatment to those who are severely depressed until psychiatric treatment improves their mood and responsiveness

- Use the most colorful and realistic stimulus materials that may cheer up the person; build into your treatment sessions the client's favorite activities, hobbies, and interests; physical activities may improve both the mood and the speech response rate
- Find functional reinforcers; most may not respond to verbal praise; listening to music or a story read aloud to them may increase treatment participation; experiment with different kinds of reinforcers to find the effective ones
- Give more time to respond when you ask questions or show pictures; frequently prompt the correct responses
- Give frequent but short breaks during treatment
- Find a suitable time to conduct communication treatment; ask the caretakers if the patient is more alert and responsive at certain times during the day than at certain other times

Euphoria and Mania. Treatment of an exaggerated feeling of well-being (euphoria) and an extreme degree of hyperactivity (constantly on the go) combined with an unrealistic sense of well-being (mania) is left to the psychiatrist or psychologist; to supplement or support such treatment, SLPs may do the following during communication training:

- Ignore excessive jocularity and statements of exaggerated self-esteem (in patients who have frontotemporal dementia); euphoric-like false sense of superiority (e.g., in patients with Huntington's disease) or outright euphoria (e.g., in patients with dementia caused by repeated head injury, vascular dementia, and Creutzfeldt-Jakob disease)
- Suspend communication treatment when the patient is extremely euphoric or in a state of mania, who is always jumping from one activity to another, without finishing any (e.g., patients with HIV-associated neurocognitive disorder)
- Suspend treatment when the patient is in an extremely depressed state, as euphoria or mania often alternates with depression

Factitious Disorders. Treatment of communication disorders that are factitious in nature may be an occasional clinical challenge SLPs face; factitious disorders—those that are simply invented by the patient—typically take the form of physical diseases that require hospitalization and sometimes serious and painful treatment; occasionally, they may take the form of stuttering or a voice disorder; there is very little research on treating factitious communication disorders; some general suggestions include the following:

- Note that a differential diagnosis of a factitious disorder is half the treatment effort; consult the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for specific signs that help make a diagnosis; among others, document:
 - A desire to play the role of a sick person
 - No evidence of gain and actually evidence of pain, inconvenience, or extraordinary expenses
 - Mysterious and frequently treated illnesses in the past; self-inflicted injuries; extensive and frequent travels to distant places to consult many specialists; a history of many ineffective treatments
 - A tone of dramatic flair in describing the symptoms
- Discuss the results of your analysis with the client; there are differences between the typical stuttering or aphonia and their factitious forms; point them out to the client
- Offer trial treatment; for instance, for a person suspected of factitious stuttering, use Delayed Auditory Feedback (DAF) or masking noise; use such other procedures as shadowing, choral reading, metronome speech, and extremely slow speech; if there are no beneficial effects—a likely outcome if the disorder is factitious—discuss with the patient that the problem may not be the kind of stuttering you would typically treat
- Note that whether factitious communication disorders can be eliminated with confrontation is not clear; factitious disorders of other kinds have been difficult to treat

Malingering. Treatment of malingering can be challenging because a symptom complex is being faked (feigned) by a client to gain a certain advantage; the clinician has to scrutinize not only the symptoms and the conditions of their onset but also the motivation of the client for exhibiting the symptoms; the conscious and well-planned symptoms may resemble a variety of disorders; SLPs occasionally see malingered stuttering, voice disorders (especially mutism or aphonia), or cognitive dysfunction (especially faked memory loss) that is disproportionate to the extent of traumatic brain injury, movement disorders with no neurological basis, and hearing loss; more common in medical and psychiatric practice; treat malingered aphonia as you would functional aphonia, described previously; treat malingered stuttering as follows:

Treatment of Malingered Stuttering

- Scrutinize the total symptom complex carefully; the person consciously faking a disorder has studied its symptoms to adopt them but still not an expert in it; there will be discrepancies between actual stuttering and feigned stuttering
- Offer experimental treatment with such devices as DAF and masking noise; offer trial therapy with shadowing, choral reading, extremely slow speech induced by instruction and modeling; these methods reduce the typical stuttering but may have no effect on malingered type; if the procedures have no effect, discuss this and point out the discrepancies
- Counsel the client about stressful life events that may include a history of marital problems, traumatic divorce, various kinds of tragedies in personal or family life, death of loved ones, failure in meeting employment responsibilities, feelings of frustration and anger because of chronic unemployment or sudden loss of employment, unjust accusation of wrongdoing, strong religious differences with family members, and so forth; raise the possibility that such traumatic events may cause fluency problems

(not necessarily implying that the client is feigning the disorder); there is some clinical evidence that an understanding of the relationship between stressful and conflicting experiences and stuttering (or other disorder of communication) may help resolve it

- Discuss any features of the client that are at odds with the known characteristics of the disorder, only to impress the client that the stress, not some other organic or genetic factor, is responsible for stuttering
- Discuss any potential gain that may have accrued to the client; while not suggesting that the client is faking the disorder to obtain the benefits, point out in an objective tone potential positive reinforcement (financial or other kinds of gain) or negative reinforcement (avoidance of difficult situations) as possible mechanisms of the genesis of the disorder; do not force the connection between the gains and the disorder, but suggest the idea for the patient to think about
- Continue treatment methods that have worked in the initial stages, as the client who feigns stuttering needs a face-saving strategy; to give a feeling that the improvement is due to treatment, not to the disclosure of the underlying problems; treatment needs to be continued for at least a few sessions

Mutism—Selective. Treatment of selective mutism—a psychiatric variety of mutism—requires a differential diagnosis from mutism due to paralysis of the vocal folds or an advanced stage of neurodegenerative disease; rule out physical causes for mutism before you treat it as selective and functional (psychiatric in nature); see the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* to make the differential diagnosis; if confirmed as selective mutism, use the following suggestions to treat it:

- Find out from the case history and interview the situations in which the client speaks because selective mutism is situation specific
- Get an audiorecorded sample of speech from the situation in which the client spoke; ask the client to repro-

duce speech that is evident on the recording; reassure the client that phonation is not lost

- Observe limited verbalizations that may occur during treatment; they may be characterized by monotonous or altered voice; tell the client that this would be the starting point of regaining a typical voice
- Ask the client to produce the monotonous or altered voice on repeated trials
- Reinforce such voice or phonated speech; reassure the client that there is no physical barrier to voice production and that it is possible to shape normal voice and speech out of even the slightest phonation
- Shape and systematically reinforce progressively more typical voice and speech production
- Use additional suggestions or strategies described under Aphonia-Functional under this main entry

Pseudodementia. Symptoms of psychiatric depression that resemble dementia is pseudodementia; such symptoms should not be diagnosed as dementia and should not be so treated; however in some older persons depression may be a dominant symptom of true dementia, and they should be treated for dementia; to offer the right kind of intervention, clinicians should first determine if an individual has depression without dementia or has dementia with depression; for assessment and diagnosis, see sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*; there is very little research on treating pseudodementia; consider the following general suggestions:

- Do not offer communication treatment to individuals diagnosed with pseudodementia in depressed individuals; let the psychiatrist or psychologist treat depression
- Reassess the client for memory impairments when the patient is no longer depressed
- Treat the communication deficits if they still exist, an unlikely event, with procedures described under Dementia

- Reassess the client because persistent symptoms of dementia when the individual is no longer depressed suggest that the initial diagnosis of a pure pseudodementia was wrong; the person perhaps had dementia *and* depression
- Use suggestions offered under Depression within this main entry to treat if there are still lingering signs of depression but it is evident that the person needs communication treatment as well
- Take note of more rapid progression of cognitive impairment than is found in generally slowly progressing dementia associated with neurodegenerative diseases
- Assess visual memory tasks on which the depressed individuals with symptoms of dementia may do better than those with dementia without depression
- Take note of a history of psychiatric problems in the past
- Work closely with behavioral health professionals to diagnose communication problems associated with depression and dementia

Schizophrenia. Speech-language pathologists typically do not treat aberrant speech patterns associated with schizophrenia; it is thought that psychiatric treatment of schizophrenia, when successful, will reduce or eliminate schizophrenic speech and language; nonetheless, modification of schizophrenic speech and language characteristics may be a significant part of the overall treatment plan for persons with schizophrenia; if successful, reduced frequency of schizophrenic speech will enhance social appropriateness; behavioral methods of reducing schizophrenic speech and language are known to be effective; to reduce these abnormal speech and language patterns, generally use the strategy that was outlined under Delusions and Hallucinations; in managing schizophrenic speech and language:

- Work with behavioral analysts to develop an overall behavior modification program; reducing speech and language deviations may be effective only when all

therapists and all health care workers take the same steps in modifying both the verbal and nonverbal aberrant behaviors

- Do not try to reason with persons that their language does not make sense, because such attempts are ineffective
- Do not reinforce aberrant speech and language expressions; ignore them completely
- Redirect the conversation to something concrete, specific, and present in the situation when you hear aberrant speech
- Reinforce appropriate speech and language behaviors
- Have the person describe pictures, narrate personal experiences, or retell stories; differentially reinforce only the appropriate behaviors
- Design and implement a social language intervention program; ignore frequent and abrupt changes in the topic of discussion or conversation; redirect the person to the topic under discussion
- Stop the client from speaking on topics or ideas that are remotely or loosely related to the point of discussion
- Ignore irrelevant answers to questions and inappropriate comments; model relevant answers and comments and reinforce the client for producing them
- Consider using such behavior reduction procedures as a time-out for meaningless schizophrenic speech
- Teach appropriate stress patterns and other prosodic features by modeling and reinforcement
- Reinforce eye contact; frequently draw attention to your face

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders, 5th ed.*—DSM-5.

Bass, C., & Halligan, P. (2014). Factitious disorders and malingering: Challenges for clinical assessment and management. *The Lancet*, 383(9926), 1422–1432.

Brody, H., & Connors, M. H. (2020). Pseudodementia, pseudopseudodementia, and pseudodepression. *Alzheimer's Dementia*, 12(1), e12027.

Duffy, J. R. (2020). *Motor speech disorders: Substrates, differential diagnosis, and management* (4th ed.). Elsevier Mosby.

Pure Agraphia. To treat writing problems with no reading or other associated language problems due to recent brain injury, see [Agraphia](#) and [Aphasia](#), in which such problems are typically treated.

Pure Alexia. To treat reading disorders without writing problems associated with brain injury, see [Alexia](#) and [Aphasia](#), in which such problems are typically treated.

Pushing Approach. A voice therapy procedure to promote better vocal fold approximations in clients who have weakened or paralyzed folds; for procedures, see [Voice Disorders: Specific Normal Voice Facilitating Techniques](#).

Q

Quality of Life (QoL) After Treatment. Although it is a matter of assessment, improved QoL is the most desirable and ultimate effect of treatment; because some treatments may be more effective in promoting better QoL than other treatments, several variables must be built into the treatment itself; better quality of life is a function of useful and effective communication skills maintained in everyday living and working environments; consider the following variables in designing a communication treatment program to increase the chances of better QoL for individuals as a consequence of treatment:

Select Functional Target Behaviors. QoL considerations begin with the right target behavior selection; all skills may be trained, but some skills are more useful to clients than other skills; generally, an excessive concern with purely linguistic structures and theoretical targets is not the best strategy.

- Let the clients and their family members (especially the parents in case of young children) list the target skills they consider the most desirable
- Select functional targets that enhance everyday family and social communication
- Select target behaviors that are essential for academic success for school-age children
- Select target behaviors that are essential to succeed in work settings for working adults
- Select natural conversational speech skills produced in everyday situations

Give Sufficient Training at the Advanced Skill Level. Insufficient training, especially at the advanced skill level, is a reason for poor QoL following treatment.

- Give sufficient training to establish the skills more firmly
- Do not prematurely dismiss the clients from treatment
- Move treatment to higher skill levels, including sentences, connected speech, and social conversation, academic or occupational discourse, and reading and writing skills when warranted; train sufficiently to stabilize these advanced speech-language skills

Train for Skill Generalization. Newly learned communication skills that do not generalize to living and working environments do not improve QoL; see [Generalization](#) for procedures to promote the following kinds of generalization:

- to other conversational partners
- to nonclinical physical settings
- to untrained stimuli that evoke the skills
- to untrained modes of responses
- to untrained verbal stimuli

Train for Maintenance. Skills that initially generalize may not be maintained; maintenance is also part of the treatment plan; clinically established skills that disappear in natural environments of clients cannot enhance their QoL.

- Consider maintenance as a treatment matter
- Plan for maintenance from the beginning by selecting functional targets
- Establish prompts and reinforcers in natural environments
- Teach self-control
- Assess maintenance and give booster therapy when needed
- See all procedural details under [Maintenance Strategy](#)

Assess Quality of Life

- See the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for procedures of QoL assessment
- If the individuals are not satisfied with their QoL, analyze the reasons; most often, the reason is lack of maintenance of clinically established skills, especially functional communication skills, in all communication contexts and situations the individuals face
- Give additional therapy to address the deficiencies
- Depression and psychiatric problems may also lower the QoL; neurologically based communication disorders, including aphasia, traumatic brain injury, dementia, and so forth may be associated with depression and other psychiatric problems; if you suspect these problems, refer the client for appropriate specialists

Question. Interrogative verbal form that is typically a request; treatment targets for clients with language disorders; set up situations and descriptions to teach questions; a few examples:

- Describe the scene such as this one to the child: You meet a new student in the school cafeteria. You don't know the person's name.
 - Ask the child, "What do you ask?"
 - If no response, model the question: "Say, 'What is your name?'"
 - Reinforce the correct imitation; repeat the trials a few times
 - Fade the modeling and just ask the question ("What do you ask?")
 - Reinforce the correct response ("What is your name?")
- Show the child a familiar picture or object and ask, "What is this?" in the same manner
- Show photographs of unfamiliar persons to teach "Who is this?" with the same procedure
- Show pictures of a cat on a rooftop to teach, "Where is it?" with the same procedure
- Use similar strategies to teach other forms of questions (*which, when, whose, why, and how*)



R

Random Assignment. A method of assigning participants selected for an experimental study to either the experimental or the control group without the experimenter bias; used in treatment research; part of the [Group Design Strategy](#).

Random Selection. A method of selecting participants to evaluate the effects of new treatments; theoretically, each potential participant has an equal chance of being selected for the study, hence no experimenter bias in participant selection; need a large number of potential participants for the method to work; part of the [Group Design Strategy](#).

Rate Reduction. See [Stuttering](#), [Cluttering](#), and [Dysarthria](#) to slow down the speech to reduce stuttering or increase speech intelligibility.

Recast. To use this child language intervention procedure, see [Language Disorders in Children](#); [Treatment of Language Disorders: Specific Techniques or Programs](#).

Recurrent Laryngeal Nerve Resection. A surgical treatment procedure for adductor [Spasmodic Dysphonia](#); the recurrent laryngeal nerve is unilaterally resected to paralyze one of the folds to prevent hyperadduction; voice therapy may be needed following this procedure.

Reinforce. To strengthen and increase behaviors by arranging immediate consequences for them; an important action clinicians perform in teaching target responses to children as well as adults; see [Reinforcers](#).

Reinforcement. A method of selecting and strengthening behaviors of individuals by arranging consequences under specific stimulus conditions; widely used in the treatment of communicative disorders; see [Reinforcers](#).

Reinforcement Priming. Seeking reinforcers for one's own behaviors; useful strategy for the client to learn in getting parents, teachers, peers, and others to notice the

production of clinically established behaviors in natural settings and thus get reinforced; part of Maintenance Strategy.

- Teach family members and other caregivers to reinforce the client for the production of target behaviors established in the clinic
- Teach the clients to draw attention from others to their own desirable communicative behaviors; for instance:
 - Teach the child who stuttered but has mastered fluency skills to make such statements as “Mom, did you hear my smooth speech?” “Dad, did you notice that I didn’t stutter once?” and so forth
 - Teach the child who has mastered specific language features to make such statements as “Mrs. Lyndner [the teacher], did you notice that I said *boys are* and not *boys is*?” “Dad, did you hear my long sentence?” and so forth
- Verify that others are indeed reinforcing the client when attention is drawn to the production of target behaviors

Reinforcement Schedules. To correctly reinforce the target responses, use an appropriate reinforcement schedule, which specifies a pattern of reinforcement; a schedule is a relation between the number of responses and reinforcers; each schedule has its advantages and disadvantages and correct and incorrect applications as noted:

- **Continuous.** In this schedule, reinforce every correct response; effective in the initial stages of treatment to establish the target behaviors but not good for response maintenance; begin with continuous reinforcement and gradually thin the reinforcement out by using an intermittent schedule; state the schedule rule to the client (e.g., “Every time you say it right, you will get a token.”)
- **Fixed Ratio (FR).** In this intermittent reinforcement schedule, some responses go unreinforced; you reinforce a response after a stated number of responses go unreinforced; you may reinforce every second (FR2), third (FR3), or fourth (FR4) correct response; after continuous reinforcement, use a lower schedule, such as an FR2, and

gradually increase the ratio; not efficient for establishing the responses but effective in strengthening once established; always state the schedule rule to the client (e.g., “I will give you a token every time you give three correct responses.”)

- **Variable Ratio (VR).** In this intermittent reinforcement schedule, the number of responses that go unreinforced varies from occasion to occasion; you may once reinforce every third response and the next time you may reinforce every fifth response; the client cannot easily predict the occurrence of reinforcement; generates stronger responses than the continuous or the FR; not preferred for establishing the target responses, but preferred to generate stronger responses that may be sustained longer with little or no reinforcement; state the schedule rule (e.g., “You will get a token sometimes for two correct responses, sometimes for three or four, but always after you give a correct response.”)
- **Fixed Interval (FI).** In this intermittent reinforcement schedule, you reinforce a response that occurs after a fixed interval of time (e.g., 2 minutes), measured from the previously reinforced response; after you have reinforced the first correct responses, wait for a minimum of 2 minutes before you reinforce another response; state the rule to the child (e.g., “You will get a token every 2 minutes, but you have to give a correct response to get it”); note that the correct response may or may not occur precisely at the end of the 2-minute interval; its application in communication treatment is limited
- **Variable Interval.** In this intermittent schedule, you vary the time duration between reinforcement delivery; after the initial reinforcement delivery, you may wait for 2 minutes to deliver the next and wait for only 1 minute to deliver the following reinforcer; state the schedule for the child (e.g., “Sometimes you will get a token in 2 minutes, and at other times it may be sooner or later, but always after you have given a correct response”); inefficient in establishing responses, but generates a strong response rate; of limited usefulness in communication treatment sessions

Reinforcement Withdrawal. Taking away reinforcers to decrease a response; one of the Direct Methods of Response Reduction; includes Response Cost and Time-Out; effective in reducing various kinds of general undesirable, noncooperative behaviors as well as specific disorders of communication, including stuttering and specific speech sound misarticulations.

Reinforcer Selection by Free Operant Procedure.

Also known as *stimulus preference assessment*, the procedure is useful to select reinforcers when clients (e.g., children with minimal verbal skills due to autism or brain injury) cannot state their preferences; it is possible to select effective reinforcers in just 5 minutes of assessment; to use this procedure:

- Present an array of 10 or more items (e.g., toys, food items, activities) that may be potential reinforcers for the child; parents or care staff might suggest items
- Allow free access to the array; wait for the client to select one or more items (e.g., a child might grab and use crayons to draw or pick up a toy and start playing with it)
- Use the selected item or items as reinforcers (e.g., after 10 correct responses, you may allow the child to draw or play for 2 minutes)

Roane, H. S. et al. (1998). Evaluation of a brief stimulus preference assessment. *Journal of Applied Behavior Analysis*, 31, 605–620.

Reinforcers. Events that follow behaviors and thereby increase the future probability of those behaviors; widely used in treating communicative disorders; select one of the several kinds of reinforcers:

- Use the Reinforcer Selection by Free Operant Procedure to choose client-specific reinforcers when it is difficult to judge what reinforces an individual, especially a child
- Select the Primary, Secondary, Conditioned Generalized, Informative Feedback, or High Probability Behaviors to reinforce clinical targets
- Reinforce initially on a Continuous Reinforcement schedule described under Reinforcement Schedule

- Switch to an Intermittent Reinforcement schedule described under Reinforcement Schedule
- Always use verbal (conditioned) reinforcers (e.g., verbal praise) in conjunction with other types
- Teach the client's significant others to reinforce the skills you establish

Reinke's Edema. Also known as polypoid degeneration; sausage-shaped, grayish-pink, or red projection-like growth on vocal fold margins associated with severe and persistent hoarseness.

- Treatment is surgical removal of the growth; performed on one fold at a time; if normal voice is restored, the second operation may be avoided
- Voice treatment involves modifying smoking and drinking behaviors and any residual voice problems

Reliability. Consistency of repeated measures of the same event; important in clinical work and clinical as well as nonclinical research; includes inter- and intraobserver reliability; for reliability associated with standardized test measure, see the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.

Interobserver Reliability. The extent to which two (or more) observers agree in measuring an event.

- Measure a behavior of interest with its location identified for a unit-by-unit analysis (count not only the behaviors being measured but also their locations in transcribed speech samples)
- Have another trained observer measure in the same manner (unit-by-unit analysis)
- Score the total number of locations for which both of you agreed for an Agreement count (A)
- Count the total number of locations for which only one of you, not both of you, scored the behavior (stuttering, pitch breaks, articulatory error) for a Disagreement count (D)
- Calculate the unit-by-unit Agreement Index by using the following formula: $A / (A + D) \times 100$

Intraobserver Reliability. The extent to which the same observer repeatedly measures the same event consistently.

- Measure the behavior of interest using the unit-by-unit method
- Measure again by the same method
- Calculate the Agreement Index using the same formula as given under Interobserver Reliability

Hegde, M. N., & Salvatore, A. P. (2019). *Clinical research in communication disorders: Principles and strategies* (4th ed.). Plural Publishing.

Replication. Repeated experimental evaluation of the same treatment procedure to show that it works with different clients, in different settings, and when different clinicians apply it; important in treatment efficacy research; includes direct replication and systematic replication; both designed to show treatment Generality.

Direct Replication. The same investigator repeats the same treatment experiment in the same setting but with different participants who have the same characteristics as the original participants.

- Initially, an investigator shows that a treatment works with some participants
- Selects different participants who share the same personal (age, gender, health) and clinical characteristics (severity, age of onset) as the original participants
- Repeats the treatment experiment
- Analyzes the results to evaluate generality of the treatment method

Systematic Replication. The same or different investigators repeat a treatment experiment in different settings, with participants who have different characteristics than the original participants; may even include individuals with totally different diagnoses.

- Initially, an investigator shows that a treatment is effective with a sample of clients
- The same or a different investigator repeats the treatment research with another sample, with different

personal (gender, age, health) and clinical characteristics (severity, age of onset), and in a different setting than the original

- The investigator analyzes the results to evaluate the broader generality of the treatment method

Hegde, M. N., & Salvatore, A. P. (2019). *Clinical research in communication disorders: Principles and strategies* (4th ed.). Plural Publishing.

Response Class. A group of responses created by the same or similar contingencies; functionally, but not necessarily structurally, similar responses; good treatment targets because teaching a few responses (exemplars) within a class can cause generalized production of other responses within the same class (e.g., the plural morpheme *s* trained in some words may generalize to other, nontrained words); exemplars that belong to different classes are discriminated (not generalized).

Response Complexity. Different topographic levels of a target behavior; structural complexity of speech-language skills typically create a sequence of treatment.

- Teach words before phrases
- Teach phrases before sentences
- Teach sentences before conversational speech
- Do not assume, however, that simpler behaviors must always be taught before teaching more complex behaviors (e.g., phrases may be taught without teaching individual words in the phrases)
- Experiment with different sequences (generally, if a more complex behavior can be taught first, its simpler version may be produced without training or with fewer trials)

Response Cost. A direct response reduction strategy in which the production of each response scheduled for reduction results in the loss of a reinforcer; effective in treating Stuttering in young children and Speech Sound Disorders in Children; also effective in reducing various behavior problems children exhibit during any kind of treatment session. There are two varieties: (1) earn and lose and (2) lose-only

Earn and Lose

- Let the child select a backup reinforcer for the session (typically a small gift)
- Give a token for correct responses (to be exchanged for a backup reinforcer)
- Take a token away each time the client produces an incorrect response
- Exchange the tokens the client still possesses for a backup reinforcer at the end of the session

Lose-Only

- Let the child select a backup reinforcer for the session (typically a small gift)
- Give a certain number of tokens at the beginning of a session
- Take a token away each time the client produces an incorrect response
- Exchange the tokens the client still possesses for backup reinforcers at the end of the session

Response Recording Sheet. A prepared sheet for recording correct, incorrect, and no responses given in treatment sessions.

- Design and use a response recording sheet similar to the one below; modify as necessary

Name of the Client	Treatment Target
Clinician	Date
Response Recording Sheet	
Stimulus Items	Responses: + (Correct), - (Incorrect), 0 (No responses)
1.	
2.	
3.	
4.	

Response Reduction Strategies. To use one of several methods of reducing undesirable behaviors during treatment, see Differential Reinforcement, Direct Methods of Response Reduction, and Indirect Methods of Response Reduction.

Response Substitution. Increase in an undesirable behavior or the emergence of a new undesirable behavior when another behavior is reduced; exemplified by increased frequency of wiggling in the chair when a child's disruptive hand movements are reduced by a response reduction method.

- Apply a response reduction strategy to the newly emerged undesirable behavior
- Apply such strategies sequentially if new undesirable behaviors keep emerging
- Emergence of new undesirable behaviors is likely to stop when a few new responses are reduced

Restitution. An element of overcorrection in which the persons eliminate the effects of their own misbehavior and then improve the situation; described under Imposition of Work.

- Ask the child who disorganizes your stimulus materials to reorganize them
- Next, ask the child to organize the toys on the floor when the disorganized toys were not the child's making

Retrograde Amnesia. To treat memory loss for events that preceded brain injury, also known as pretraumatic memory loss; see Traumatic Brain Injury in Adults.

Right Hemisphere Disorder. A constellation of symptoms due to brain injury sustained in the right cerebral hemisphere and characterized by various attentional, emotional, and communication deficits; varying degrees of functional involvement depending on the site, nature, and extent of damage; see the cited sources for etiology and symptomatology of right hemisphere disorder.

Treatment: General Considerations

- Most of the treatment suggestions are based on clinical experience with little or no research support; use them with caution; collect data during treatment to continuously evaluate improvement; keep abreast of treatment research
- Counsel the family about communication impairments and their treatment soon
- Target specific skills for teaching instead of presumed underlying, often mental or cognitive processes; when underlying processes are treated, skills may not improve
- Begin treatment as soon as it is practical
- Prioritize social communication skills and effectiveness over grammatical accuracy
- **Select the client-specific treatment targets**
 - That are functional in everyday living
 - Result in the most improvement in family, social, and vocational communication
 - That are building blocks for more advanced communication skills
 - Those that the client, family members, and professional caregivers agree with
- **Develop stimulus materials**
 - They should range from simple to progressively more complex and from fewer to greater number of elements
 - They should be clear, unambiguous, and relatively concrete
 - Preferably familiar, meaningful to the client, attractive, and ethnoculturally appropriate
- **Establish the target behaviors**
 - Provide extensive and intensive practice
 - Avoid computerized cognitive rehabilitation programs because they are yet to be proven
 - Structure treatment sessions initially and loosen them as the client becomes more proficient in producing the target responses
 - Use instructions, models, and prompts and eventually fade them

- Shape the target behaviors
- Reinforce and verbally correct
- Work with the family members to promote generalization and maintenance

Treatment Targets and Procedures

Treat Awareness Deficit

- Note that there is little or no controlled research that supports direct work on increasing a person's awareness of deficits; repeatedly pointing out clients' deficits to increase deficit awareness is a negative approach; do not spend too much time on this strategy
- Increase awareness as you teach positive behaviors; give immediate verbal feedback on errors in communication, left neglect, and other deficiencies while teaching positive target behaviors that replace those errors; deficit awareness increases as skills improve
- Give visual feedback on errors whenever possible (e.g., point to the neglected left side of a printed page the client reads aloud); do this while positively reinforcing more attentive reading
- Audiorecord and replay the speech to the client and discuss the errors
- Teach the client Self-Control (Self-Monitoring) skills

Treat Impaired Attention

- Do not target attention as an independent skill; there is no compelling evidence that supports teaching such nonfunctional skills as letter cancellation (crossing out a target letter printed among other letters on the same page) and pressing a certain key on the computer keyboard when certain dots appear or disappear against a background of dots
- Regulate attention while teaching target communication skills
- Use the shaping procedure to establish sustained attending behaviors
- **Reinforce specific attending behaviors during communication training**

- Reinforce the individual for paying attention to treatment stimuli and activities
- Reinforce maintenance of eye contact during conversation
- Reinforce extended talk on a given topic (concentration on a topic of conversation); stop the person when the topic diverges inappropriately
- Reinforce the client for talking about a stimulus picture and stop the client who begins to talk about extraneous stimuli
- Reinforce appropriate responses to questions you ask and requests you make
- Reinforce both the communication target behaviors and attending behavior
- Specify why you are reinforcing what action (e.g., “Mrs. Bobcat, thanks for looking at the picture!” “That was a nice description of the picture!”)
- **Structure the treatment sessions to suit the individual**
 - Limit the number of stimuli and response choices in the beginning
 - Conduct treatment in a quiet and distraction-free room
 - Watch for signs of fatigue
 - Give frequent, brief breaks in the initial phase of treatment
 - Reduce the number and duration of breaks gradually as the client’s attention and performance improve
 - Conduct treatment in more naturalistic settings after the attending behaviors improve
- **Prompt and give attention-getting cues**
 - Before presenting the treatment stimuli, before giving instructions, or before asking the client to perform an action
 - Say, “Look at me” before modeling a response
 - Say, “Look at the picture before you say something about it”
 - Say, “Talk only about this part, not any other part of this picture” and point out the target stimulus part

- Say, “Get ready, here comes the next picture” before you present the next stimulus picture
- Say, “Please listen, I am going to tell you something”
- Say, “Be careful now! I want you to do something different!” and then describe the new action required of the client
- Touch the client before presenting a treatment stimulus to draw attention
- Give a nonverbal cue to draw attention (e.g., point to your face or to the stimulus picture to keep the client focused)
- Fade all special stimuli to draw attention
- **Use effective stimuli**
 - Vary the treatment stimuli
 - Drop unattractive stimuli that do not help sustain the client’s interest or attention
 - Use clear, colorful, and forceful stimuli

Treat Visual Neglect

- **Reinforce attention to the neglected side**
 - Force attention to the neglected side of the visual field by giving frequent verbal instructions (e.g., “look to the left”)
 - Reinforce the client for paying attention to the stimuli on the neglected side
 - Note that mostly occupational therapists manage visual neglect, but SLPs may work on neglect while treating communication skills, reading, and writing
- **Draw attention to printed stimuli on the left side of the page**
 - Teach the person to keep a finger on the left margin while reading and track back to it before beginning a new line
 - Color the entire left-side margins to draw attention
 - Draw a colored line through the margin
 - Place a few colored dots on the left margin
 - Point to the left margin with a pencil or a ruler as the client reaches the end of the right margin

- Tell the client to “Look to the left” when the end of the right margin is reached
- Fade all such special stimuli used to draw attention
- Design reading materials with large print and progressively smaller print and ask the client to read them aloud
- Reinforce the person for reading aloud the words on the left side
- Interrupt reading each time the client misses words on the left side
- **Conduct group sessions**
 - Teach communication partners on the side the client tends to neglect to draw attention to themselves before speaking (e.g., “Listen, Mr. Showcase, I want to tell you something”)
 - Prompt the client to look at different persons before they begin to talk
 - Sit on the neglected side and monitor the client’s responses, first drawing attention to yourself and then to specific individuals in the group (e.g., “Look at me, Mrs. Whitewater. Now I want you to look at Mr. Merriweather, who is talking to us”)

Control Impulsive and Inappropriate Behaviors

- **Give nonverbal signals to delay impulsive responses**
 - Instruct the client to respond only when you give a hand signal to respond
 - Give a hand gesture that suggests *wait*
 - Give a gesture that suggests *think*
- **Give verbal signals to delay responses**
 - Say, “Where do you live? Please wait for a few seconds before you answer”
 - Say, “Please wait until I am finished”
 - Say, “Please respond when I say *now*”
 - Say, “Please listen when someone is talking”

Teach Social Communication Skills

- **Teach conversational skills**
 - Topic initiation and topic maintenance during conversation

- Conversational turn taking
- Eye contact
- Request for clarification
- Response modification when requested to clarify
- see Social Communication Skills for procedures
- **Teach typical prosodic features**
 - Model and reinforce typical speech rate that will reduce dysprosody
 - Model and reinforce typical patterns of intonation in sentences
 - Model and reinforce acceptable loudness of speech
 - Model and reinforce acceptable stress patterns within utterances
 - Discourage hypermelodic speech
- **Teach socially appropriate speech**
 - Stop the client as soon as you hear irrelevant, tangential, or inappropriate comments or jokes
 - Give hand signals to stop when the person rambles or talks excessively
 - Consider the PACE program (Promoting Aphasics' Communicative Effectiveness; described under Aphasia; Treatment of Aphasia: Specific Techniques or Programs) for teaching social communication

Teach Executive Function Skills

- Note that a variety of actions, including planning skills, have recently been included under executive functions, most of which are not measurable because they are not operationally defined; executive function often refers to various kinds of actions that an individual may either perform or describe; executive functions are treatment targets in individuals who have communication disorders due to brain injury or disease; other professionals including occupational therapists may teach executive skills; for an SLP, verbal description of planning skills, problem solving, and execution of daily activities might be appropriate treatment targets.

- **Teach planning skills**
 - Ask how the person might plan for a vacation that involves flying to a destination and getting a hotel room
 - Ask how the person might plan for a surprise birthday, wedding anniversary, or other kinds of parties at home
 - Reinforce when the client describes the steps in a logical manner
 - Stop and question the client when the steps described are unreasonable, illogical, potentially unproductive, or out of sequence
- **Teach Problem Solving and Execution of Daily Activities**
 - Ask, “How do you resolve a billing problem with your credit card company?”
 - Ask, “What do you do when you run out of gas while driving on a highway?”
 - Ask, “How do you prepare a sandwich for lunch?”
 - Reinforce and provide corrective feedback as necessary
 - Teach multiple exemplars and test for generalized description of untrained skills
 - See Executive Functions (EFs) for additional targets and strategies, including self-monitoring and verbal rehearsal of skills

Treat Abstract and Figurative Language Comprehension and Production

- Make sure the selected treatment stimuli and other materials (e.g., pictures or drawings) are appropriate for the age of the client and the ethnocultural background of the client and the family; proverbs, idioms common statements, jokes, and humorous sayings are culture-specific; consider the following intervention targets:
 - Impaired inference
 - Impaired recognition of absurdities
 - Impaired comprehension of proverbs, metaphors, or idioms

- Impaired comprehension of humor
- See Abstract Language for procedural details

Teach Functional Reading and Writing Skills

- **Reestablish premorbid reading skills**
- Teach reading skills in the context of treating left neglect
 - Give feedback on all kinds of reading errors as well as left neglect
 - Treat writing skills only if the client needs them
 - Ask the client to write a paragraph and give feedback as the client writes (e.g., repetition of letters, poor letter formation, confused spacing, crowding of the letters on the right side, failure to give the left margin)
 - Dictate a brief passage and give feedback on errors

Teach Compensatory Strategies

- **Teach the use of functional memory aids**
 - Teach the client to make lists of things to do and appointments to keep
 - Teach how to write a checklist of things to do before leaving the house (e.g., switching the lights and the stove off)
 - Suggest keeping related things together in one place and the same place (e.g., the house key and the wallet)
- **Teach self-control (self-monitoring) skills**
 - Teach the client to stop and self-correct as soon as an error is made
 - Make a few socially appropriate and inappropriate statements and ask the client to evaluate them
 - Stop the individual when you hear a socially inappropriate statement and ask the client to self-evaluate the appropriateness of speech
 - Teach the client to rephrase inappropriate comments to make them more appropriate
 - See Self-Control (Self-Monitoring) for additional suggestions

Quality of Life (QoL) Considerations in Treatment

- There is no substantial amount of research on QoL in persons with RHD
- Administer the *Quality of Communication Life Scale* (ASHA QCL); get it from asha.org
- Through the Ethnographic Interview, assess the effects of treatment on the individual's QoL; ask the individual if the treatment was satisfactory or not
- Ask family members and other caregivers about any improvement in the individual's personal, social, and family communication patterns; ask if they are satisfied with the treatment outcome
- Analyze the reasons for lower QoL; offer additional treatment to remediate lingering deficits
- See Quality of Life After Treatment for additional strategies

Blake, M. L. (2017). *The right hemisphere and disorders of cognition and communication: Theory and clinical practice*. Plural Publishing.

Hegde, M. N. (2022). *A coursebook on aphasia and other neurogenic language disorders* (5th ed.). Plural Publishing.



S

Sample. A smaller number of individuals selected from a larger population for a research study; participants included in a treatment research study; in a group design study, some receive treatment (the experimental group) and others do not (control group).

Satiation. Temporary termination of a drive or need because it has been satisfied; a potential problem with Primary Reinforcers (e.g., food) that lose their effect because of satiation; to control this problem in treatment sessions:

- Give only small amounts of food or drink to reinforce responses
- Let the clients accumulate food that they can eat later
- Arrange treatment sessions, especially with infants and toddlers who need primary reinforcers, around breakfast or lunchtime

Scripts and Script Therapy. Written descriptions of frequently occurring, sequenced events, episodes, or personal experiences used in teaching advanced language skills, including narrative skills; a description of *baking cookies* or *running a lemonade stand* is a script; it has a beginning and an end, actions people take, or roles people play; for procedures, see Language Disorders in Children.

Secondary Reinforcers. Conditioned reinforcers whose effects depend on past learning, frequently used in speech-language intervention; appropriate for all kinds of verbal responses except for certain kinds of mands that request food and drink; include Social Reinforcers, Conditioned Generalized Reinforcers, Informative Feedback, and High-Probability Behaviors.

S

Self-Control (Self-Monitoring). A behavior that monitors and modifies other behaviors of the same person; a necessary skill in maintaining clinically learned communication skills; see Maintenance Strategy; teach such skills as the following to promote self-control or self-monitoring of newly learned skills; a few examples follow:

Teach Clients to Discriminate Their Correct and Incorrect Responses

- Model a child's phrases or sentences that do not contain required grammatic morphemes and then model the correct response (e.g., "Two cup," "Two cups")
- Play back a recorded sample of a client's speech and point out undesirable voice quality
- Contrast easy and hard glottal attacks of a voice client or a fluency client by modeling the two kinds of phonatory initiations
- Contrast incorrect and correct productions of phonemes by modeling them
- Contrast correct and incorrect utterance forms (e.g., "Two cup," "Two cups")
- Contrast the desirable lower pitch with the client's excessively high pitch by modeling the two pitch levels or by audio recording the client's voice that catches both levels
- Ask them to contrast their own correct and incorrect responses

Teach Clients to Measure Their Behaviors

- Have your clients measure their correct and incorrect target behavior productions along with you in treatment sessions; note that children, too, can learn to make a tally mark to measure correct productions of sounds, grammatic morphemes, voice qualities, and so forth; for instance, ask clients to chart, along with you
- Verbally correct incorrect responses; reinforce correct responses
- Give them informative feedback ("You and I agreed on three of the five instances"; "This time, you and I agreed on four of the five instances")
- Monitor the progress to improve the clients' skill in measuring their behaviors

Teach Clients to Pause and Self-Correct Mistakes

- Teach fluency clients to pause briefly after each stutter and say the same word fluently (self-time-out)

- Teach clients with voice disorder to pause soon after producing a hard glottal attack and start again with a softer attack
- Teach children in articulation therapy to stop as soon as they produce a target sound incorrectly and correct themselves
- Teach clients with apraxia of speech to stop as soon as a wrong articulatory gesture is initiated and then initiate the correct gesture
- Give corrective feedback for every incorrect response to begin with
- Tell clients that you may not verbally correct and that they should correct their own wrong responses
- Verbally correct clients who do not immediately self-correct; remind them that they should self-correct
- Continue the procedure until the client reliably self-corrects; reduce the frequency of verbal corrections further while maintaining self-corrections

Teach Clients to Correct Each Other in Group Therapy Sessions

- Note that monitoring other participants' skills may enhance self-monitoring
- In group therapy for children who stutter, teach each child to monitor other children's speech rates and to praise a slower rate
- In aphasia group therapy, teach each to remind the other to practice specified skills (e.g., self-cueing words with the first phoneme of the word)
- In child language group therapy, teach each child to praise or verbally correct target productions in other group members

Teach Clients to Monitor Their Behavior in Natural Settings

- Ask clients with aphasia to write down the names of family members and frequently consult the list before engaging in conversational speech
- Ask clients to wear electronic devices that remind them of appointment, to slow down the speech, to speak more softly or loudly, and so forth

- Teach a person who stutters to practice saying “Hello” with appropriate fluency skills (slower rate, deep breathing) when the telephone starts ringing
- Teach a person with aphasia to rehearse the names of people who are expected to visit shortly
- Teach a voice client with vocal abuse to reduce talking before a scheduled and unavoidable speech to be given to a group
- Teach another voice client to drink more water to ward off vocal problems related to vocal dryness
- Ask fluency clients to keep a record of their use of slower speech rate or gentle phonatory onset
- Ask a voice client with excessively soft voice to keep a record of the number of social situations in which the vocal loudness was adequate
- Ask a woman with aphasia to keep a record of the number of times she successfully recalled her husband’s name on a given day

Sequence of Treatment. Stepwise movement within a treatment program from the beginning to the end, from a simple to a more complex level; see [Treatment of Communication Disorders: General Sequence](#).

Shaping. A method of teaching nonexistent (zero baseline) responses that are not even imitated; also known as *successive approximation* or *forward chaining*; supported by experimental evidence; essential in teaching a variety of skills.

- Select a terminal target response (e.g., the production of /m/ in word-initial positions)
- Identify an initial component of that target response the client can imitate (e.g., putting the two lips together)
- Identify intermediate responses (e.g., humming or other kinds of vocalizations, opening the mouth as humming is continued)
- Teach the initial response by modeling and immediate positive reinforcement for correct imitations (e.g., putting the lips together on several trials)
- Teach the intermediate responses in successive stages (e.g., adding humming when the lips are closed; opening

the mouth when the humming is continued; adding other sounds to form words)

- Continue until the terminal response is taught

Single-Subject Experimental Design Strategy.

Methods of demonstrating treatment effects by showing contrasts between conditions of no treatment, treatment, withdrawal of treatment, and other control procedures; typically, all participants receive treatment because the method does not use a control group of individuals who are denied treatment; see [Evidence-Based Practice](#) for details.

Social Communication Skills. Also known as *pragmatic language skills* and *conversational speech skills*, social communication may be treatment targets in individuals with different diagnoses, including [Children with Language Disorders](#), [Autism Spectrum Disorders](#), [Traumatic Brain Injury](#), [Cleft Palate](#), and others as noted under main entries; teach social communication skills (except for eye contact) after establishing more basic skills including sufficient number and variety of words, phrases, a variety of sentence types, and grammatical morphemes that are essential to hold a conversation or narrate an event; the following is a generic outlines to teach social communication skills; modify the wording, instructions, stimuli used, the level of skill complexity, manner of reinforcement, and so forth to suit the individual; take into consideration the person's age, educational level, the health status (e.g., in TBI), existing language skill level, and so forth; the skills to be targeted in treatment include:

- Eye contact during conversation
- Conversational topic initiation
- Conversational topic maintenance
- Conversational turn taking
- Requests for clarification
- Clarifying one's own unclear expressions
- Narrative skills
- Discourse skills

- **Teach eye contact during conversation**

- Ascertain first that eye contact is culturally acceptable or parent-opted in case of children
- Target eye contact from the beginning of treatment and especially during conversational speech training
- Prompt the client by saying, “Look at me” before you present stimuli, when you begin to talk, and when the client begins to talk
- Hold the stimulus parallel to your face so the client looks at the face and the stimulus simultaneously
- Praise the person for maintaining eye contact
- Track progress by measuring the duration for which eye contact was maintained at the beginning (baseline) and throughout the treatment phase (improvement)
- Fade the prompts or other special cues used

- **Teach conversational topic initiation**

- Target this social communication skill in starting a conversation with new topics; individuals who cannot initiate new topics will have limited social participation
- Arrange a variety of stimuli that could trigger a new topic: in case of children, objects, pictures, storybooks, topic cards, toys, structured play situations such as a kitchen, and so forth; in case of adults, verbal prompts may be adequate; if not, use topic cards and age-appropriate pictured sequences
- Introduce one of the stimulus items or situations and draw the client’s attention to it (e.g., a picture of a family setting up a tent in a park; stimuli like this could work with younger and older clients); suggest a topic to older clients (e.g., “I would like you to tell me something about your work [your school]”)
- Wait for the person to initiate conversation about the picture and the story
- Instruct to say something about the picture if there is no response
- Prompt the story beginning when the client does not initiate (“They are setting up a . . .”)
- Praise the person for saying anything related to the topic depicted or suggested

- Accept statements that are remotely connected to the topic at hand; gradually demand more relevant responses
- Do not interrupt or overly correct the forms of responses
- Train to use topic cards to initiate new topics
- Prompt or suggest new topics to talk about (“You can talk about your summer vacation”)
- Withdraw or fade such prompts, cues, cards, pictures, and other special stimuli to make topic initiation more spontaneous
- Train parents, spouses, or other caregivers to use your techniques so they can continue intervention at home or in a professional setting
- **Teach topic maintenance**
 - Teach this advanced social communication skill in staying on a topic of conversation for an extended duration, instead of abruptly switching topics
 - Target topic maintenance when treatment has advanced to a conversational speech level
 - Ask the person to select topics of interest for talking; prompt the previously trained topics to establish topic initiation; suggest new topics if necessary
 - Set a realistic duration for which you want the client to talk on a single topic, or set a flexible number of target words to be produced on a topic
 - Increase the duration or the number of target words in gradual steps
 - Use such devices as *Tell me more. What about that? What happened next? Who said what? Where was it? When did that happen?* and so forth to stimulate more speech on the same topic
 - Praise the client for maintaining the topic
 - Stop the client when you hear abrupt switching of topics
 - Ask the client to continue to talk on the target topic (e.g., “You were telling me about ____”)
 - Give training on a few topics and then probe with untrained topics to see if the skills have generalized

- Train on additional topic exemplars if the skills have not generalized
- **Teach conversational turn taking**
 - Target this advanced social communication skill of alternatively playing the role of a speaker and a listener; you need to discourage interrupting the speaker and encourage to take turns to speak during conversation
 - Teach turn taking when treatment has advanced to conversational speech
 - Baseline the number of interruptions and failure to take cues to talk
 - Design verbal or nonverbal signals for the client to talk (e.g., “Your turn” or a hand gesture to suggest *you speak*)
 - Design a signal that says *do not interrupt, do not talk, it is not your turn* (e.g., finger on your lips)
 - Use a real or toy microphone that you exchange with the client; the one holding the microphone talks and the other listens
 - Reinforce talking only when signaled or while holding the microphone
 - Follow the same rule that you impose on the client (e.g., talk only when you hold the microphone)
 - Teach the person to say, “It is your turn”
 - Reinforce for yielding the floor
 - Teach turn taking until the client meets a performance criterion (e.g., no errors of turn taking in three consecutive conversational exchanges)
 - Fade the signals or other special discriminative stimuli (e.g., the microphone) used as prompts
 - Probe turn taking in brief conversational exchanges without signals or special stimuli
 - Train until a probe criterion is met (at least 90% accuracy in turn taking without models or prompts and with only such naturalistic reinforcers as smiling, nodding, agreeing)
- **Teach requests for clarification**
 - Also called conversational repair strategy, it is what a person does when something said is not understood

- Target this conversational repair skill to request clarification when faced with unclear statements
- Make ambiguous statements (e.g., say, “Give me the car” when you have displayed several toy cars; say, “Point to the square” when the picture shows multiple squares of different colors)
- Wait for a request for clarification
- Model a response when the client fails to request clarification (e.g., “When you are not sure, ask me ‘What do you mean?’”)
- Make a series of ambiguous statements, model different requests for clarification (e.g., “What do you mean?”; “I don’t get it”; “Which one?”); reinforce imitative responses
- Prompt to withdraw modeling (e.g., “What do you ask me?”)
- Reinforce the client for asking for clarification (e.g., “What do you mean?”)
- Introduce varied ambiguous statements and reinforce varied responses
- Fade modeling and prompting
- Train family members in prompting the persons to request clarification
- Probe the generalized repair skill by presenting untrained and ambiguous messages
- Continue training until a probe criterion is met (such as 90% accuracy in responding with the target skill)
- **Teach individuals to clarify their unclear expressions**
 - Teach clients to vary their expressions when a speaker does not understand them
 - Play the role of a listener who does not fully understand the expressions of your client during conversation
 - Use a variety of strategies to prompt clarifications from the client (e.g., “Could you please repeat?”; “What do you mean?”; “I don’t understand”)
 - Negate a client’s utterance to prompt clarification by assertion (“You did not go on the roller coaster 20 times, did you?”; the child might say, “No, I went on it two times”; “You didn’t go to Italy nine times,

did you?"; the client may say, "No, I said I went three times")

- Model the clarified statement by modifying what the client said ("You mean you went on the roller coaster two times, right?")
- Rephrase the client's utterance into a question and say it with a rising intonation ("You went to Italy three times?")
- Model different ways of saying the same thing to make it clearer; reinforce varied phrases or sentences
- Periodically stop responding to the client's request so as to prompt a rephrase
- Train parents, spouses, or other caregivers to prompt the person to vary expressions and to reinforce compliance
- Probe the generalized repair skill by presenting untrained messages
- Continue training until a probe criterion (such as 90% accuracy in responding with the target skill) is met
- **Teach narrative skills**
 - Telling stories, talking about events, describing personal experiences in a logically consistent, cohesive, temporally sequenced manner is typically a target for older children, but judge the suitability of this target for the child under treatment; even preschoolers may tell a brief story or narrate an experience
 - Use more or less complex narrative scripts to suit the client; see Language Disorders in Older Students and Adolescents; use Scripts (written description of events) to teach narrative skills
 - Play such scripts as grocery shopping, eating in a restaurant, birthday parties, camping trips, personal vacations, job-related stories, description of work or college experiences, playing certain games and sports, and so forth
 - Play daily routine scripts in case of children
 - Read repeatedly or tell the same stories so the children or severely impaired older clients can memorize the words, temporal sequences, characters, and events
 - Ask children to narrate experiences as they play out scripts and assume different roles

- As you retell stories, pause before important phrases or critical descriptions so that the individuals may supply them
- Prompt the phrases and descriptions as the clients hesitate; fade the prompts
- Ask the client to tell the stories or narrate events with the help of pictures or slides
- Ask the client to tell stories or narrate events without scripts, pictures, or slides
- Ask the client to narrate new events or experiences (not rehearsed or scripted) to probe for generalization
- Verbally reinforce all attempts to narrate in the beginning; reinforce progressively better narratives in later stages
- **Teach discourse skills**
 - A target mostly for older students and adults
 - Suggest an academic topic or a social issue and ask the person to give a speech on it (e.g., Electoral College, civil rights, school discipline, cell phone technology, healthy school lunch program)
 - Ask the client to give a brief speech on a chosen topic
 - Encourage the client to give detailed descriptions and coherent arguments, represent opposing views, be objective and state evidence, draw conclusions, and use technical or academic language
 - Model advanced sentences with abstract words when necessary
 - Record the speech and replay the recording to give feedback

Nippold, M. A. (2020). *Language sampling with adolescents: Implications for intervention* (3rd ed.). Plural Publishing.

Social Reinforcers. A variety of conditioned reinforcers frequently used in treatment sessions; include verbal praise, attention, touch, eye contact, and facial expressions; resistant to satiation effect; may not work with nonverbal clients.

- With children who are nonverbal, with severe intellectual disabilities, and the very young (infants and toddlers), pair social reinforcers with primary reinforcers

- Eventually, fade the primary reinforcers and maintain the responses on social reinforcers only
- Do not use to reinforce requests for food and drink; such mands are reinforced only with what is requested (food and drink)

Spasmodic Dysphonia. Treatment of this voice disorder, characterized in most cases by severe overadduction of vocal folds and strained or choked-off voice quality (in the adductor type) or excessively breathy and irregular voice (in the abductor type), has been difficult; traditional voice therapy techniques have been unsuccessful in producing lasting treatment effects; some clinicians combine voice therapy with medical management in a team effort; current medical management includes:

- For the abductor type, injection of Botox (botulinum toxin A) into one or both of the vocal folds to induce paralysis of the folds; generally, the outcomes have been unsatisfactory
- For the adductor type, recurrent laryngeal (laryngeal adductor) nerve sectioning, botulinum toxin type A injection has been used to paralyze one or both vocal folds to eliminate vocal spasms; results have been mixed
- Voice clinicians may try several more or less traditional approaches, including inverse phonation (phonation on inhalation), slightly elevated pitch, whispered speech, breathy voice, singing and humming that are shaped into speech; relaxed phonation, relaxation of the laryngeal muscles and the general body; expect only moderate effects; see Voice Disorders for procedures

Sapienza, C., & Rudy, B. F. (2021). *Voice disorders* (4th ed.). Plural Publishing.

Stemple, J. C. et al. (2024). *Clinical voice pathology* (7th ed.). Plural Publishing.

Spastic Dysarthria. To treat this type of motor speech disorder, see Treatment of Dysarthria; Dysarthria: Specific Types: Spastic Dysarthria.

Specific Language Impairment (SLI). Language disorders in children who do not have any other significant neural, behavioral, or intellectual disabilities; by definition, the impairment is limited to language, although studies have suggested other subtle impairments; SLI has been recently renamed *developmental language disorder* (DLD); because both the terms—SLI and DLD—pose theoretical and clinical problems, a more descriptive and theoretically neutral term, *language disorders in children*, is preferred in the *PocketGuides*; for treatment procedures, see [Language Disorders in Children](#).

Speechreading. Understanding speech by watching the face of the speaker as well as using the residual hearing; while [Lipreading](#) involves watching mostly the lips and tongue, speech reading uses all available visual and auditory cues.

Speech-Generating Devices. Various electronic devices, including handheld computers, smartphones, tablets, and dedicated augmentative and alternative communication (AAC) devices that display words, phrases, symbols, icons, pictures, drawings, and so forth for the AAC user to select individual displays to express basic needs or combine them to form more complex communicative expressions. See [Augmentative and Alternative Communication \(AAC\)](#).

Speech Sound Disorders in Children. Difficulty in producing a few sounds with no pattern or derivable rule is often treated as an *articulation disorder*; multiple errors that can be grouped on some principles or characteristics and thus form patterns are typically treated as *phonological disorders*; see the sources cited at the end of this main entry for the epidemiology, ethnocultural factors, symptomatology, etiology, and theories of speech sound disorders.

Treatment of Speech Sound Disorders: General Guidelines

- ***Treatment targets.*** Treatment targets may be the correct production of phonemes in error or the elimi-

nation of phonological error patterns; either way, you teach individual sounds; correct production of phonemes in conversational speech produced in natural settings is the final treatment target

- **Sequencing the target sounds.** Select certain sounds for initial training, others for subsequent training; consider the following varied recommendations:
 - Select sounds that are *functional* for the child for immediate training; sounds that are important from the standpoint of social and academic performance (e.g., even though /r/ may not be an initial target for all 4-year-old children who misarticulate that sound, it may be for a child named Robert Roberson, who refers to himself as *Wobert Woberson*; in this case, /r/ is *functional* for the child)
 - Select misarticulated sounds that, when corrected, will produce the greatest improvement in intelligibility
 - Select for immediate training sounds the child imitates correctly (stimulable sounds) when modeled; contrarily, select the sounds that are most difficult for the child, the sounds that are not even imitated because when taught, they have the most beneficial generalization effect; two opposing views
 - Select *high-frequency* sounds for immediate training; correct production of those will improve intelligibility faster and more noticeably than correcting sounds that occur less frequently; contrarily, select sounds of low frequency that tend to be more difficult sounds; correction of such difficult sounds may produce better generalization
 - Teach first sounds that are normally *acquired earlier* because they may be easier to teach; there is very little evidence to suggest that sounds need to be taught in their normative sequence; contrarily, select sounds that are *normally acquired later*, not earlier, because when corrected, they produce the greatest change in the child's sound system; evidence supports this
 - Select for immediate training sounds that the child does produce on assessment and baseline trials,

even though at a low frequency, because they may be taught more easily; contrarily, select sounds the child does not produce at all because they tend to be more complex or difficult, and when taught, they produce a greater degree of generalization

- Select for elimination phonological patterns that are less frequent in the child's speech, less complex, and more common across children, because the sounds in those patterns may be easier to teach first; contrarily, target first the elimination of phonological patterns that affect the greatest number of sounds, patterns that are idiosyncratic, and patterns whose occurrence is 100% because such an approach yields better and quicker speech intelligibility
- Evidence is emerging that better generalization and intelligibility is achieved by addressing the more difficult sounds and patterns first
- Regardless of the strategy used, always select sounds that are appropriate for the child's ethnocultural and language background; select sounds that are in the child's dialect; select sounds that are not in the child's dialect only when the child or the parents demand their teaching
- Select academically useful words to teach specific sounds; consider the academic demands made on the child; consult with teachers in selecting words, phrases, and sentences
- **Sequencing response complexity.** Use the following guidelines in sequencing response topography:
 - Teach the target sounds at the highest level of response complexity if the child can imitate modeled productions
 - Do not begin teaching at a lower level if the child can succeed at a higher level (e.g., do not train a sound at the syllable level if the child can imitate its production at the word level)
 - Begin training a sound minimally at the word level, if practical at the phrase level, or even at the sentence level: drop to a lower level when the child cannot imitate at a higher level

- End training at the level of conversational speech produced in more naturalistic settings

Ethnocultural Considerations in Treating Speech Sound Disorders. See Treatment of Speech Sound Disorders in African American Children and Treatment of Speech Sound Disorders in Bilingual Children in later sections; see also Ethnocultural Variables in Treatment for general guidelines.

A Comprehensive Treatment Program for Speech Sound Disorders. Use the following outline to treat correct production of speech sounds regardless of whether it is diagnosed an articulation or phonological disorder; it includes all steps needed to establish baselines, treat the speech sounds in error, probe for generalized production, and implement a maintenance procedure.

- **Select treatment targets and their sequence of training**
 - Consider previous guidelines
 - Sequence the training of selected target sounds for the early, middle, and final phases using previous guidelines
 - Group multiple misarticulations according to phonological patterns (e.g., elimination of the final consonant deletion pattern)
- **Specify measurable objectives for target speech sounds**
 - The response to be taught (e.g., The production of /s/ . . .)
 - The complexity level at which the response will be taught initially (e.g., The production of /s/ in words . . .)
 - The position of the target sound in words (e.g., The production of /s/ in word-initial positions . . . and similar statements for other word positions)
 - The performance criterion to judge success (e.g., The production of /s/ in word-initial positions at 90% accuracy in a set of 10 training words . . .)
 - The correct response context and setting (e.g., The production of /s/ in word-initial positions at 90%

accuracy in a set of 10 training words measured across three consecutive speech samples or treatment sessions); for initial training, the setting is clinical, and for final training, it may be more naturalistic

- Write similar target statements for different levels of training (initial, medial, and final word positions; sounds in phrases, sentences, and conversational speech)
- Write similar target statements for sound production in the clinic, classroom, or home settings (e.g., production of /s/ at 90% accuracy in conversational speech exhibited in the clinic, home, and classroom)
- **Prepare stimulus materials and a recording sheet**
 - Prepare 20 words, phrases, and sentences for each target sound in each word position (i.e., 20 words *each* for initial, medial, and final word positions; 60 target words for each sound to be trained); select picturable words to the extent possible
 - Select pictures, drawings, objects, or a combination of these to evoke each target word, phrase, or sentence; preferably, select colorful and realistic pictures from popular magazines
 - Test the stimulus materials with the child to make sure that the child is familiar with them; show the materials to parents and get their suggestions; replace any ethnoculturally inappropriate stimulus materials with those that are appropriate, familiar to the child, and approved by the parents
 - Prepare a treatment response recording sheet on which you can record correct, incorrect, and no responses for each trial
- **Write training and probe criteria**
 - **Imitative accuracy criterion:** When the child gives five consecutively correct imitated responses, modeling will be discontinued
 - **Modeling reinstatement criterion:** When the child gives three incorrect responses on evoked (without modeling) trials, modeling will be rein-

stated and then withdrawn after five correct imitated responses as before; reinstated and withdrawn in this manner

- **Shaping criterion:** When the child fails to correctly imitate the target sound in the first word selected for training on 10 consecutive trials, a different word will be selected for training; if the child fails to imitate the sound in the second word in 10 trials, the training of the sound will be initiated in syllables or at the level of isolated sounds; when the child gives five correct responses at the phoneme level, the training will shift to syllable level; five correct responses at the syllable level will move training to the word level
- **Training criterion:** A 90% accurate production of a sound at each level of response complexity (i.e., 90% accuracy in sound production at the word, phrase, and sentence levels evoked on a block of 10 trials and 90% accuracy in conversational speech with at least 20 productions of the target sound)
- **Probe criterion:** A 90% accuracy in the production of sounds in untrained words presented on a series of probe trials with at least 10 untrained words
- **Probe timing and frequency criterion:** A probe will be administered when a child meets the training criterion on four to five stimulus items or exemplars; for example, when the child's correct response rate on 4 words—*soup*, *soap*, *sun*, and *seed*—reaches 90% accuracy on a block of 10 evoked training trials, a probe will be conducted with at least 10 untrained words interspersed with trained words; the same criterion for all word positions, phrases, and sentences
- **Reinstatement of training criterion:** Treatment on the same sound in the same word position will be reinstated if the probe criterion is not met (similar criteria for phrases and sentences); after training the sound in four new words, a probe will be conducted again; training and probes will be alternated until the probe criterion of 90% accuracy in untrained words is achieved

- **Establish baselines of target sounds**
 - Measure the correct production of target sounds in conversational speech; use the initial speech sample recorded during assessment; repeat the procedure to obtain another sample in the first intervention session; if the initial assessment sample and the first intervention-session sample show different percentages of errors for the same sound, repeat the sample
 - Measure the production of target sounds in words with the procedure of Discrete Trials; administer one set of trials with modeling and one without; do not reinforce or correct any responses:
 1. Place a picture or an object in front of the child
 2. Ask a relevant question (e.g., “What is this?”)
 3. Model the correct response (e.g., “Say **soup**”); emphasize the target sound
 4. Wait a few seconds for the child to imitate
 5. Move the picture toward you and record the sound production as correct, incorrect, or no response
 6. Re-present the same picture; ask the same question, but do not model the response; record the response as before
 7. Administer all target words once with modeling (modeled trial) and once without (evoked trial)
 8. Calculate percent correct imitated and evoked response rates for each sound and in each word position
- **Teach the baserated speech sounds on modeled trials**
 1. Start at the word or syllable level
 2. Describe the target sound and how it is produced, demonstrate its production, and point out its Phonetic Placement; if necessary, use Manual Guidance to move articulators to their correct position; see Sound Evoking Techniques for English Speech Sounds
 3. Place a picture or an object in front of the child
 4. Ask a relevant question (e.g., “What is this?”)

5. Model the correct response (e.g., “Say **sun**”); emphasize the target sound
 6. Wait a few seconds for the child to imitate
 7. Praise the child and give a token or a primary reinforcer as soon as the child correctly imitates the sound
 8. Verbally correct errors (“That is not correct; you said _____ ; I want you to say _____ ”)
 9. Move the picture toward you and record the sound production as correct, incorrect, or no response on the prepared recording sheet
 10. Re-present the same picture; repeat the instructions on how to produce the sound; show the phonetic placement for its production; ask the same question; model the response; record the response as before
 11. Continue this training with modeling until the child gives five consecutively correct imitated responses; discontinue modeling
 12. If the child fails to imitate the sound in a word in 10 trials, use Shaping by dropping to the syllable or isolated phoneme production level; when the child correctly imitates the phoneme on five trials, move up to the syllable level; with five correct imitative responses at the syllable level, move up to the word level; with five correct imitations at the word level, discontinue modeling
- See Treatment Modeled Trials
 - **Teach the sounds on evoked trials**
 1. Present the same picture, ask the same question, but do not model
 2. Reinforce the correct responses or verbally correct the incorrect responses; repeat instructions and demonstrate phonetic placement if necessary
 3. Continue this training without modeling until the child gives nine correct responses in a block of 10 trials (90% accuracy); consider the particular word tentatively trained
 4. Select another word with the same target sound and begin training with modeling; discontinue

and reinstate modeling as before; drop to phoneme or syllable level if necessary; move up to the word level; when the child gives nine correct evoked responses on a block of 10 trials, consider this word tentatively trained; follow this procedure until you train correct production on at four to six words

- See Treatment Evoked Trials
- **Probe for generalized production**
 - Conduct the probe with at least 10 untrained (probe) words
 - Record the responses to untrained stimulus items (words) on a probe recording sheet
 - Do not model correct productions because all probe trials are evoked only
 - Calculate the percent correct probe response rate (e.g., if the sound production was correct in 5 of the 10 probe words, the correct probe response rate is 50%)
 - See Probes for additional information
- **Alternate between probe and training**
 - Provide additional training with new training words if the child's probe response rate is below 90%; after training four or six more words, administer another probe; if the probe criterion is not met, train more words and probe again until the child meets the probe criterion
 - Switch training to another target sound or the same sound in a different word position; follow the same procedure as outlined to complete training the sound at the word initial, medial, and final positions; train all target sounds at the word level with the same procedure
- **Shift training to higher levels of response complexity**
 - Construct two- or three-word phrases that contain a word in which the sound is produced correctly (i.e., when a sound meets the probe criterion for a given word position)
 - Train phrases the same way as the words; present, withdraw, and reinstate modeling as before

- Probe and alternate between probe and training trials as before; when the probe criterion is met, shift training to sentences formed with trained words
- Train sentences in the same manner as words and phrases; after presenting the stimulus item, ask a question that would evoke a sentence (e.g., ask “What is the boy doing?” to evoke “He is having some *soup*”); initially, model the sentences and withdraw modeling; probe generalized production of sounds in conversation in the same way as described for words and phrases
- Shift training to the conversational speech level when the probe criterion is met for sentences; use pictures and other stimulus materials to engage the child in conversational speech and positively reinforce correct productions on an Intermittent Reinforcement Schedule; prompt the correct productions; verbally correct incorrect responses
- Conduct periodic conversational probes in which you engage the child in conversation without reinforcement or verbal correction; continue training until the child meets the 90% correct production of all target sounds in conversational speech with little or no prompting and reinforcement
- Integrate components of various programs of articulation treatment described under Treatment of Speech Sound Disorders: Specific Techniques or Programs into this comprehensive treatment program; for instance:
 - Use aspects of the Contrast Approach; in using the Minimal Pair Contrast Method, write minimal pair contrast words for training each sound; for instance, in teaching a child to produce final consonants (elimination of final consonantal deletion), write such word pairs as *bow-boat*, *bee-beet*, *toe-toad*, *pie-pine*; to teach the correct production of a single phoneme such as /b/ for which the child substitutes /p/, write such pairs as *bye-pie*, *bat-pat*, *beat-peat*, *bike-pike*; instead of placing a single stimulus item as previously described, place a pair of stimulus

items in front of the child; model both the words and ask the child to imitate; withdraw modeling and prove evoked trials; move through response complexity levels as described earlier

- Use aspects of the Paired-Stimuli Approach; instead of presenting only the target word in which the child misarticulates the sound, present both the target word and a Key Word in which the sound is correctly produced; ask the child to produce the key word first and then the target word; see Paired-Stimuli Approach for details; when the child meets the training criterion for words, probe for generalization and then train the words in phrases or sentences
- Use aspects of the Multiple Phoneme Approach if the child exhibits multiple errors; instead of training one sound at a time as described, train multiple phonemes
- Use aspects of the Phonological Pattern Approach if the individual exhibits multiple misarticulations with identified patterns; note that the phonological pattern approach does not have a unique treatment *procedure*; it is only a method of error analysis; to eliminate a pattern, individual phonemes still need to be taught as in any other approach; identify patterns in misarticulations; select patterns for elimination; teach the phonemes as described in the comprehensive program; when you teach a few phonemes within a pattern, probe the other phonemes within the pattern to see if they are now produced on the basis of generalization; if they are, move on to the other patterns or to higher levels of response complexity; if they there is no generalization, train additional sounds within the pattern
- **Implement a maintenance program**
 - Always train the correct production of sounds in sentences and in naturalistic conversational speech during the final stages of treatment
 - Conduct informal training sessions in Extraclinical Settings; monitor the child's correct speech sound production outside the therapy room but within

the clinic, outside the clinic but in the surrounding areas, and in such natural settings as the classroom, the school dining hall, and the library

- Train family members, teachers, and peers in reinforcing the correct production of sounds in natural settings; ask the parents or other caregivers to observe your treatment session; educate them about the correct production of target sounds; train them to prompt and reinforce the correct productions and verbally correct the errors at home; train them to hold brief training sessions at home; ask parents to audio-record the session and review the sample to fine-tune their skills
- Teach the child Self-Control (Self-Monitoring) techniques to self-manage the correct and incorrect productions; ask the child to judge the accuracy of productions; give feedback on this judgment; ask the child to mark on a sheet responses as either correct or incorrect; teach the child to pause briefly as soon as an error is produced and then try to produce it correctly
- Follow up with the child periodically and assess the correct production of phonemes in conversational speech; provide booster treatment when the correct production in conversational speech falls below 90% accuracy
- See also Maintenance Strategy

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.). Plural Publishing.

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Treatment of Speech Sound Disorders in African American Children.

In treating speech sound disorders in African American children, assessment and diagnosis of a disorder must be made in the context of African American English (AAE), not Mainstream American English (MAE). See the cited sources for details, and note that:

- The phoneme inventory of children speaking AAE will consist of the same phonemes as in MAE; only some

phonemes will be used differently, substituted for other phonemes, or omitted in certain contexts

- A majority of phonemes are produced in the same way in both AAE and MAE
- Phonological patterns that are accepted in AAE are not to be treated as disorders; a treatable articulation disorder for a child who speaks AAE is a disorder in the context of AAE, not in the context of MAE; see the cited sources for details
- Note that not all African Americans speak AAE; family communication patterns and the parents' preferences will dictate the treatment targets
- Use the following guidelines to plan speech sound treatment programs for African American children who speak AAE and have a justified speech sound disorder:
 - Select the misarticulated speech sounds of AAE (even if they vary from those of MAE English) as treatment goals for a speaker of AAE
 - Select for treatment the following misarticulated phonemes for children in the age range of 33 to 36 months that are typically produced correctly: /m/, /n/, /p/, /b/, /t/, /d/, /k/, /g/, /f/, /s/, /h/, /w/, /y/, /l/, and /r/
 - Accept and reinforce sound patterns that are accepted in AAE (e.g., a child's production of *baftub* for *bathtub* is acceptable in treatment) unless the parents have opted for MAE sound patterns
 - Treat first the phoneme usages that are the same in AAE and MAE (many actually are)
 - Treat next the unique phoneme usages of AAE that the child does not produce correctly; the goal is to teach what is acceptable in AAE, even if it deviates from MAE
 - Change AAE dialectal patterns to MAE patterns only if the individual, the family, or both demand it; in this case, treatment of MAE sound patterns is *elective*
- Select functional speech sounds, child-specific sounds, and sounds that, when treated, will rapidly improve the child's speech intelligibility

- Select stimulus items for treatment from the child's home environment; consult with parents about pictures, objects, toys, and other materials for their socio-cultural appropriateness and child familiarity
- Consult the family members about acceptable reinforcers for the child; find out the child's preferences; but as always, determine that a consequence is a reinforcer only after data show that correct productions have increased during treatment
- Assume that standard treatment procedures (e.g., modeling, phonetic placement, verbal praise, a token system backed up with a variety of culturally appropriate reinforcers) will be effective with African American children; in all treatment sessions, record the response rates systematically to support or correct this assumption

Hegde, M. N. (2026). *Hegde's pocketguide to assessment in speech-language pathology* (5th ed.). Plural Publishing.

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Stockman, I. (2008). Toward validation of a minimal competence phonetic core for African American children. *Journal of Speech-Language-Hearing Research*, 51, 1244–1262.

Treatment of Speech Sound Disorders in Bilingual Children. Bilingual children are a large and varied group, whose primary language may be one of many varieties of Spanish, an Asian language, a Native American language, or an East European language, including Russian. Therefore, it is not possible to list the characteristics of all the various primary languages that influence American English. Follow the guidelines summarized here to develop appropriate speech sound treatment programs for bilingual children:

- Analyze errors in the primary language sample; select these as the treatment targets; note that such treatment requires the working knowledge of the child's primary language; in the absence of such knowledge, refer the child to a speech-language pathologist (SLP) who has the knowledge
- Analyze errors in MAE that are *not* due to the influence of the primary language; these errors too may be

treatment targets; note that a monolingual English-speaking SLP can offer this treatment, assuming that an assessment was made with the help of a bilingual clinician

- Analyze English sound productions that vary from those in MAE but are due to the influence of the child's primary language; these are not normally the targets of speech sound modification (e.g., final consonant devoicing, as in *dose* for *doze*, is not a treatment target because that is a feature of Spanish)
- Select functional speech sounds, child-specific sounds, and sounds that, when treated, will rapidly improve the child's speech intelligibility
- Select stimulus items for treatment from the child's home environment; consult with parents about pictures, objects, toys, and other materials for their appropriateness and child familiarity
- Consult the family members about effective reinforcers for the child; find out the child's preferences, but, as always, determine that a consequence is a reinforcer only after data show that correct productions have increased during treatment
- Assume that standard treatment procedures (e.g., modeling, phonetic placement, verbal praise, a token system backed up with a variety of culturally appropriate reinforcers) will be effective with bilingual children; in all treatment sessions, record the response rates systematically to confirm or correct this assumption
- Modify treatment techniques to suit the individual child; keep good records of such modifications to evaluate whether they were needed because of individual differences or ethnocultural variations
- Refer the child to a bilingual clinician who knows the child's primary language
- See the cited sources for the unique characteristics of the Spanish language that influence English learning; these characteristics are not a basis to offer speech sound intervention
- To treat speech sound disorders in bilingual Spanish-English-speaking children, follow these guidelines:

- Correct the errors in phonemes that are common to the child's primary language and the secondary MAE on a priority basis
- Teach the speech sounds that are misarticulated in Spanish if this is feasible and the treatment plan is to modify a speech sound disorder in Spanish
- Correct MAE phoneme misarticulations if they are not due to an influence of the primary language (this is a true speech sound disorder in English in a bilingual child)
- Correct MAE sounds whose productions are affected by the primary language only if the individual or the family requests such treatment because of the advantage MAE offers in educational, social, and occupational settings; most likely, such a request will be made
- To treat speech sound disorders in other bilingual children (such as those whose primary language is one of many Asian languages):
 - Use the general guidelines already specified for working with bilingual children
 - Determine the phonological characteristics of the specific Asian American child because Asian languages belong to different language families and hence are highly varied in their phonological and other characteristics
 - Use the guidelines offered in the previous section on Spanish-English bilingual children to select and correct speech sounds

Hegde, M. N. (2026). *Hegde's pocketguide to assessment in speech-language pathology* (5th ed.). Plural Publishing.

Kohnert, K. et al. (2020). *Language disorders in bilingual children and adults* (3rd ed.). Plural Publishing.

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Sound Evoking Techniques for English Speech Sounds.

There is no research on the relative effectiveness of sound-evoking procedures; they are based on clinical experience; the core of all sound-evoking techniques is Shaping, which includes instruction, demonstration,

modeling, prompting, manual guidance, and positive reinforcement; see the cited source for detailed instruction on evoking English speech sounds; some general guidelines include the following:

- Use the basic sequence of instruction-demonstration-modeling-reinforcement-verbal correction in evoking each target sound production; note that in the following shaping procedure, you progressively simplify the target response, teach the components of the target in sequence, and put the skills together
- Instruct the child on how to produce a target sound
- Demonstrate the articulatory positions and movements necessary to produce the sound
- Model the target speech sound in short words (e.g., *pie*, *ball*)
- Model just the sound in a syllable (e.g., *pa*, *ba*) if the child cannot produce it in words
- Demonstrate the initial articulatory position for the sound if the child cannot produce the syllable (e.g., demonstrate lip closure for /p/, tongue tip against the alveolar ridge for /t/)
- Add the sound to the articulatory posture or position resulting in a syllable (*pa*, *ta*)
- Add a second vowel or consonant to make a short word (*pie*, *ball*)
- Continue to train the target sound in words, phrases, sentences, and conversational speech

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Speech Sound Disorders: Specific Treatment Techniques or Programs.

There are many packaged programs to treat speech sound disorders, and only a few are highlighted here. Most programs differ on theoretical grounds, some on how the sessions are arranged, others on how the target sounds are selected, and several on how the assessment and posttreatment data are analyzed. Nonetheless, all programs use the same core treatment variables to teach the correct production of speech sounds, including picture stimuli, instructions

on sound production, modeling, shaping, fading, manual guidance of the articulators, visual cues (position, transition, and movement of the articulators), immediate reinforcement for approximations and correct productions, and immediate verbal correction of errors. Programs that describe themselves as phonological, linguistic, or cognitive achieve their goals only by teaching individual sound productions through behavioral methods; phonology, linguistics, neuroscience, and cognitive science offer no treatment procedures, technically defined as the clinician's activities that teach speech sound productions. Therefore, the following descriptions highlight the unique aspects of different approaches; the comprehensive program described earlier may be supplemented with selected useful strategies from these programs.

Cycles Approach. A phonological pattern approach to treat children with multiple misarticulations and highly unintelligible speech; consists of treatment cycles that vary between 5 and 16 weeks; includes auditory stimulation and production practices.

- Assess the child's phonological performance with 50 spontaneous naming responses and continuous speech samples; may use Hodson's *Assessment of Phonological Patterns—Revised*
- Arrange a hierarchy of stimutable phonological patterns that occur in at least 40% of the relevant contexts
- Treat the most stimutable pattern first, then the next most stimutable pattern, and so on
- Target only one phonological pattern in any single session
- Treat each phoneme within a target pattern for about 60 minutes per cycle (one 60-minute, two 30-minute, or three 20-minute sessions) before moving to other phonemes within the pattern or to other patterns
- Review the prior week's production practice word cards (see below); skip this step if introducing a new pattern for treatment

- Begin treatment with auditory bombardment; ask the child to listen attentively for about 2 minutes as you produce 12 words with the target sound and sentences containing those words; do not ask the child to produce the sounds; periodically contrast the correct and the incorrect production of the target sound
- Use five production-practice word cards: Ask the child to first say a target word and then draw, color, or paste the picture of the word on 5×8 index cards; write the word on the card
- Begin production practice; ask the child to name about five target pictures (five words per sound); model the target word; use auditory, tactual, and visual cues; engage the child in conversation; use a game format
- Probe for stimulability of next session's target sounds
- Repeat the amplified auditory bombardment; present the same 12 words as before
- Ask the family members or teachers to read the same 12-word list to the child; ask the child to name the five picture cards used in production practice during the week
- Recycle a pattern that persists in conversational speech

Hodson, B., & Paden, E. (2004). *Targeting intelligible speech: A phonological approach to remediation* (2nd ed.). Pro-Ed.

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Complexity Approach. Not a new treatment procedure, but almost a revolutionary method of selecting phonemes for intervention; supported by experimental evidence; treatment procedures are behavioral, the same as modeling, shaping, positive reinforcement, and verbal correction used in all approaches; an efficient approach because it yields more widespread generalization within and across phoneme categories; does not include auditory discrimination training.

- Select sounds that are not produced during assessment or on baseline trials (nonstimulable, 100% error frequency)
- Select and teach sounds that are learned-later (L-L) in the normative sequence to realize generalized production of earlier-learned (E-L) sounds; for instance, teach L-L affricates to produce generalized production of E-L fricatives, teach L-L fricatives to produce generalized production of E-L stops; note that generalized production may not occur in the opposite direction (from E-L to L-L)
- Select consonant clusters for treatment before targeting singletons; generally, select clusters the child does not produce, but do not select clusters that are considered unmarked, hence easier to produce (e.g., /sp/, /st/, and /sk/); select clusters in which at least one of the sounds is not produced (not stimulable) at all; expect generalized production of singletons when they are treated in clusters
- Develop nonwords with selected target sounds and present them during treatment
- Make the nonwords “meaningful” by assigning them to pictures or drawings of imaginary toys, animals, creatures, objects and so forth
- Teach the target sounds with behavioral techniques as in other programs
- Begin with modeling and imitation sequence, and when the child is 75% correct on imitated trials over minimally three sessions, advance the child to training the target sound productions in spontaneous speech; use a criterion of 90% accuracy of target sound production at this level
- Frequently probe for generalization within and across the sound classes or categories

Gierut, J. A. (2001). Complexity in phonological treatment: Clinical factors. *Language, Speech, and Hearing Services in Schools*, 32, 229–241.

Storkel, H. L. (2018). The complexity approach to phonological treatment: How to select treatment targets. *Language, Speech, and Hearing Services in Schools*, 49, 463–481.

Concurrent Treatment Approach. Unique aspect of this behavioral treatment program is to teach speech sounds at different levels of complexity concurrently, randomly varying the levels in all sessions; supported by experimental evidence. Suitable for small group sessions and for childhood apraxia of speech.

- Develop stimuli at different levels of training; the simplest level of training is imitation of sound in single syllables moving through multisyllables, words, phrases, sentences, and finally evoked conversational speech—the most complex level of training; consult the cited sources for the specific levels because they may be as many as 29
- Establish the target sound in CV and VC syllables to meet the criterion of correct responses on 8 out of 10 blocked trials
- Randomly intermix the simple and complex levels of training tasks and train all exemplar types in a randomized order, and present them at least once (except for the conversational speech segment, which should be presented at least four times); present the randomized levels on as many passes as possible in each session
- Use the behavioral procedures of modeling, shaping, positive reinforcement, tokens with backup reinforcers, and verbal correction
- Extend the same to multiple sound training, with each sound first established in the CV and VC trials and then intermixing simple and complex training tasks

Skelton, S. L. (2004). Concurrent task sequencing in single phoneme phonologic treatment and generalization. *Journal of Communication Disorders*, 37, 131–155.

Core Vocabulary Approach. An approach designed for about 10% of children who produce speech sounds in words sometimes correctly and at other times incorrectly, resulting in poor intelligibility; whole words are trained, with an emphasis on improving speech intelligibility.

- Ask the child to name 25 pictures, each three times; give a score on 1 for words produced inconsistently on at least one out of the three trials; add the scores to calculate percentages of inconsistency; 40% inconsistency qualifies for intervention
- Select 50 to 70 *core vocabulary* words (not phonemes) that are functional and important for the child (names of family, friends, teachers; pets; favorite activities; foods, places, common expressions like *thank you, please, sorry*); consult the child, parents, teachers, and others to select the core vocabulary
- Find pictures for the selected core words
- Train up to 10 words in a session, twice a week, for up to 8 weeks
- Model, give phonetic cues, shape the sounds (sound-by-sound successive approximation), and give repeated trials but do not overuse the modeling-imitation sequence
- Ask the child to produce each word the same way each time to eliminate inconsistent productions; accept developmentally appropriate errors; target best production, not necessarily the fully correct production
- Verbally reinforce consistent productions and verbally correct inconsistent productions of the same word
- Train family members to provide informal therapy at home; the goal is not error-free production but consistent production
- Administer probes to assess consistent and generalized productions

Dodd, B. et al. (2006). A core vocabulary approach for management of inconsistent speech disorder. *Advances in Speech-Language Pathology*, 8, 220–230.

Phonemic Contrast Methods. The contrast methods differ from other methods of treatment only in selecting phonemes that contrast either minimally (a single feature) or maximally (as many feature contrasts as feasible) and present them in pairs of words during treatment; target phonemes are trained with the same

behavioral procedures as modeling, shaping, positive reinforcement, and verbal correction.

- **Minimal Pair Contrast Method**

- Select word pairs that have minimal phonemic contrast (e.g., *bat-pat*, which differ in only one feature—voicing-nonvoicing of the initial sound); difference in one or just a few phonemic contrasts is considered *minimal*
- Analyze the individual's misarticulations
- Write minimal contrast word pairs; for instance, to remediate deletion of final consonants, write such pairs as *boat-bow*, *bee-bead*, and *tee-teeth*; to remediate fronting, write such pairs as *can-tan*, *key-tea*, and *gate-date*; select 8 to 10 word pairs for a phonemic contrast (e.g., to eliminate final consonant deletion, select such word pairs as *bow-boat*, *bee-beet*, *toe-toad*, and *pie-pine*)
- Obtain pictures for words in selected pairs
- Place the pictures representing the word pairs in front of the child
- Model a given word pair (e.g., *bow-boat*) and ask the child to imitate; reinforce correctly imitated productions or approximations
- Provide extensive trials on imitative production of the target and contrast words
- Ask the individual to spontaneously name the picture pairs
- Ask the individual to name the pictures and then sort them into separate piles
- Alternatively, ask the individual to say the target word as you pick the correct picture (the individual says *boat* and you pick up the picture of *boat*; if the individual says *bow*, you pick up the picture of *bow* and then correct the individual)
- Ask the individual to match two pictures by first picking a picture from several displayed and then selecting its minimal pair match
- Arrange controlled play activities in which the target sounds and words are practiced in conver-

sational speech (e.g., a toy *soap* used to pretend wash hands, pots, dolls, and so forth while talking about the activity)

- Move on to more naturalistic conversational speech and reinforce correct productions of target speech sounds
- **Maximal Phonemic Contrast Method**
 - Select word pairs that contrast maximally; for instance, select such word pairs as *chain-main*, *can-man*, and *gear-fear* (the initial phoneme in the first word of each pair is the target of treatment; the initial phoneme in the second word in each pair is the phoneme with maximal opposition because it differs from the treatment target on more than one feature)
 - Use the general procedure outlined for Minimal Pair Contrast Method, in the previous section
- **Multiple Contrast Approach**
 - Treat simultaneously all error contrasts, instead of treating one contrast at a time (something not done in other contrast approaches)
 - Create multiple sets of minimal pairs, but include all error types in each set (e.g., to treat the previously mentioned single-sound substitution for multiple sounds, create such pairs as *tip-sip*, *tip-Kip*, *tip-chip*, *tip-trip*); note that all errors are contrasted in this set
 - Create additional sets to achieve the same multiple contrasts (e.g., *tease-ease*, *tease-keys*, *tease-cheese*, *tease-trees*)
 - Treat all or most of the errors simultaneously, especially in children who substitute a single sound for several sounds (e.g., a child who substitutes /t/ for /s/, /k/, /tʃ/, and /tr/ will substitute a single word such as *tip* for multiple words—*sip*, *Kip*, *chip*, and *trip*)
 - Teach the contrasts simultaneously, using the behavioral methods of modeling, positive reinforcement, and corrective feedback

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Phonological Pattern Approach. Not a treatment method, but a procedure to analyze speech sound errors and patterns of generalization; based on the assumption that teaching only a few individual phonemes within a pattern will be sufficient to realize generalized production of other sounds within the pattern; an assumption with mixed evidence; untrained phonemes within a pattern may or may not be produced.

- Assess phonological patterns with a comprehensive conversational speech sample, use one of the several available protocols of phonological pattern analysis, and describe the Phonological Patterns that account for errors
- Identify all phonemes that are misarticulated within a pattern (e.g., all final consonants a child omits—the *final consonant deletion pattern*)
- Teach only a few sounds within a pattern (e.g., a few deleted final consonants)
- Teach sounds in syllables, words, phrases, and sentences with behavioral methods
- Probe for generalization within and across patterns when a few final consonant productions meet the training criterion
- If there is no generalized production of untreated phonemes, then treat them as well
- If untreated phonemes are produced on the basis of generalization, select another phonological pattern for elimination; teach selected phonemes within the pattern; administer probes to assess generalization within and across patterns
- Schedule maintenance activities

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Programmed Conditioning for Articulation. A treatment program based on behavioral principles and programmed learning concepts; uses a performance

criterion of 10 correct responses in a row; includes an establishment phase and a transfer (generalization) phase; reinforce either on a continuous reinforcement schedule (crs) or an intermittent reinforcement schedule (irs) of 50% or 10%

- **Establishment phase: Teach the target sound**

- In initial, medial, and final positions of nonsense syllables (crs)
- In all three positions of words (irs), in two- or three-word phrases (irs), and in four- to six-word sentences (irs)
- In oral reading (skip if the child is a nonreader [crs])
- Through story narration after silently reading a story (crs), picture description (crs), and conversational speech, first with continuous reinforcement, then at 10% reinforcement
- Administer the criterion test

- Train additional sounds with the same steps/levels

- **Transfer phase**

- Home training: Teach the sound in words, phrases, and sentences, oral reading, picture description, and conversational speech (crs)
- Train in different settings, including outside the clinic, playground, cafeteria, classroom, speaking with the teacher, and speaking to the class (crs)

- **Maintenance phase**

- Conversation during weekly meetings for the first 4 weeks (crs)
- Conversation during one monthly meeting (crs)
- Dismiss the individual

Baker, R. D., & Ryan, B. P. (1971). *Programmed conditioning for articulation*. Monterey Learning Systems.

Traditional Approach. In this early treatment approach, sounds are trained in isolation, in syllables, in words, and in sentences; aspects of this program, developed mostly by Van Riper beginning in the late 1930s, are now part of many existing programs; its main components include:

- **Perceptual training (ear training).** Teach auditory discrimination of speech sounds; model words that contain the child's misarticulated sound as well as their correct productions and ask the child to raise the right hand when the sound is correct and the left hand when it is not
- **Production training: Establishment.** Model correct productions of target sounds in isolation, in syllables, and in words; and ask the child to imitate; use Phonetic Placement and Shaping
- **Production training: Stabilization.** Continue to train the sound in isolation as well as in nonsense syllables; in words, in phrases, in sentences, and finally in conversational speech; vary the sentence lengths and complexity; reinforce correct productions in oral reading
- **Production training: Transfer (generalization).** Give speech assignments the child completes at home; get parental reports; teach self-monitoring, and have the child practice correct speech sound productions in varied settings

Peña-Brooks, A., & Hegde, M. N. (2023). *Assessment and treatment of speech sound disorders in children* (4th ed.). Pro-Ed.

Van Riper, C., & Emerick, L. (1984). *Speech correction: An introduction to speech pathology and audiology* (7th ed.). Prentice-Hall.

Stimulation Versus Treatment. In language intervention, stimulation is indirect, more naturalistic, without specific language targets, without a requirement that the child imitate modeled responses, and often implemented by family members and other caretakers; to the contrary, treatment or teaching tends to be more clinician directed, less naturalistic, with specific language structures as targets, with the requirement that the child imitate modeled responses; clinician reinforces correct responses and verbally corrects incorrect responses; vague stimulation procedures are not as effective as direct interventions.

Stimulus Generalization. Production of a newly learned response to stimuli not used in training; an important goal of intervention; to promote stimulus generalization:

- Use a variety of stimuli during treatment
- Use common stimuli, preferably from the child's home (objects, toys, books, etc.)
- Train multiple exemplars; give sufficient training
- Train in extraclinical situations; train in the presence of other individuals
- Train significant others to promote generalized productions
- Probe to assess generalized productions; train additional exemplars if there is no generalization

Stridency. A voice disorder characterized by an unpleasant, shrill, and metallic-sounding voice; caused by excessive pharyngeal constriction and an elevated larynx.

- Use the Chewing Method
- Model and contrast good vocal production with a strident production
- Lower the pitch; use Glottal Fry
- Teach relaxation
- Teach oral openness
- Use the Yawn-Sigh method

Stuttering. A disorder of fluency in children and adults with three major varieties: stuttering, with early childhood onset; Neurogenic Stuttering of late onset; and Cluttering, in both children and adults. To treat stuttering, see the following two entries: Stuttering in Adults and Older Children and Stuttering in Younger Children.

Stuttering in Adults and Older Children. There are varied forms of treatment for this fluency disorder in adults, but only a few are experimentally supported; fluency shaping with airflow management, gentle phonatory onset, and speech rate reduction through prolonged syllables are common elements across several treatment programs for adults who stutter; but there are procedures that do not include elements of fluency shaping.

Goals of Stuttering Treatment in Adults and Older Children

- Discuss the treatment goals with the client and the family members and come to a consensus; persons who stutter (PWS), including children, seek help from SLPs to get their stuttering under control; if this is confirmed, reduce stuttering or dysfluencies in conversational speech to less than 1% in clinic speech samples and no more than 5% in everyday situations (some clinicians may opt for 3%); define stuttering or dysfluencies in measurable terms
- Establish generalized and maintained fluency in natural settings to promote better Quality of Life
- Assess negative emotional reactions to speech and speaking situations before and after treatment; have the person or parents list typically avoided situations and speaking tasks
- Reduce both avoidance behaviors and negative emotional reactions to speech and speaking situations as a result of clinically significant reduction in stuttering
- Train significant others in the client's life to support fluent speech production at home, school, and workplace
- Assess satisfaction with treatment and improved Quality of Life following treatment
- Follow up with the individual periodically; give booster therapy to further improve generalized and maintained fluency, increase satisfaction with the treatment, and enhance the client's quality of life

General Guidelines on Treating Stuttering in Adults and Older Children

- Note that operationally defined specific dysfluency measures are generally more reliable than such undefined and global measures as *stuttering moments* and *stuttering events*
- Establish a stable base rate of stuttering or specified dysfluencies before starting treatment; take extended and repeated conversational speech samples for reliable base rates; count all stuttering or types of dys-

fluencies, as well as the number of words spoken; calculate the percentage of stuttering

- Discuss treatment options, their advantages, limitations, and expected outcomes with the client and the family members; come to a consensus on a treatment selection and overall goals to be achieved; give an overview of evidence-based techniques that may be appropriate to the client; consider the client's and the family members' views in finalizing the selection
- Answer all questions the individual and the accompanying family members may have about the treatment, ensure that the client and the family members agree with the goals and procedures of selected treatment, confirm that they understand the advantages and disadvantages of the procedure, and appreciate the need for home practice and willingness to do it.

Treatment for Adults and Older Children: Fluency Shaping. Give an overview of the fluency-shaping procedure; let the client and the family members know that initially, the slower rate of speech and other elements of the procedure eliminate stuttering but makes the speech sound unnatural; assure them that the final goal is not noticeably slower speech but fluent speech that sounds typical

- **Teach airflow management first**
 - Model slightly deeper than usual amount of air through the nose, and exhale a slight amount of air through the open mouth as soon as inhaling the air (air not to be impounded in the lungs)
 - Ask the individual to imitate your airflow modeling
 - Reinforce with verbal praise the correctly imitated responses
 - Stop the individual at the earliest sign of mismanaged airflow; point out the error (e.g., too deep an inhalation; impounding the air in the lungs; exhausting the air supply when asked to exhale only a small amount of air); model again and reinforce the imitated response

- Continue until the individual can, on request and without modeling, inhale through the nose and exhale a slight amount of air through the mouth; give several successful trials
- **Introduce gentle phonatory onset**
 - Model gentle voice onset and contrast that with hard glottal attacks the individual may exhibit; use short and simple words (e.g., *hi*, *how*, *bye*, *my*); initiate syllables softly, gently, slowly, and in a relaxed manner
 - Ask the individual to imitate modeled gentle onset; reinforce correctly imitated soft onset productions
 - Continue until the individual can, on request and without modeling, initiate sounds softly while producing several words
- **Combine airflow management and gentle phonatory onset**
 - Model both the component skills—airflow management and gentle phonatory onset—for the individual to imitate; use the same words as before
 - Ask the individual to imitate your modeling of airflow management and gentle phonatory onset; reinforce correct productions
 - Go back to single-skill training if the individual mismanages a component; concentrate on the mismanaged component; combine them again
 - Continue until the individual can, on request and without modeling, inhale and exhale slightly and initiate sounds softly while producing several words; reinforce such evoked (unimitated) productions
 - Stop the individual at the earliest sign of mismanaged airflow, abrupt or tensed onset of phonation, or stuttering (dysfluency); explain what went wrong, concentrate on the missed target if necessary, combine the skills again, and give additional practice
- **Introduce speech rate reduction through syllable prolongation**
 - Prefer the clinician-induced rate reduction; use Delayed Auditory Feedback only if necessary (described later under Treatment of Stuttering: Specific Techniques or Programs); note that instructions,

modeling, reinforcement, and verbal corrections are effective in reducing the rate; avoiding mechanical devices saves the time needed to fade their use

- Model a slow speech through syllable prolongation; use the same words until this stage; stretch the syllable duration; reinforce correct imitation of slow, syllable-prolonged speech
- Stop the individual as soon as you hear the sign of increased rate of speech or stuttering; explain the error, model again, and reinforce correct imitations
- Continue until the individual can, on request and without modeling, stretch syllables in all the words practiced
- **Combine airflow management, gentle phonatory onset, and syllable prolongation into an integrated skill**
 - Model inhalation and slight exhalation, gentle onset, and syllable prolongation using the same words practiced until this point
 - Ask the individual to imitate your modeling of all three target skills in words; reinforce correct imitations
 - Monitor all three skills at this stage and stop the individual for mismanaged targets or production of stuttering
 - Provide training in combined fluency skills with single words initially modeled and later evoked by questions that lead to one-word responses (“What is your first name?” “What is your last name?” and several similar questions to evoke single-word names of family members and friends; “What is the name of the city you live in?” “What is the name of your street?” and several similar questions)
 - Continue until the individual can, on request and without modeling, produce all three target behaviors and with stutter-free speech at the word level with 98% to 100% accuracy
 - Note that within the first one or two sessions, the individual should be producing words (some individuals can be moved to phrases even in the first session) with little or no stuttering; if this is

not happening, perhaps the skills are not managed properly; make a careful analysis of errors and pay close attention to skill execution; be more prompt in reinforcing stutter-free speech and verbally correcting mismanaged skills or imminent stuttering

- **Shift training to the phrase level as soon as possible**

- For the initial phrase training, form two-word phrases with the words already trained (e.g., the first and the last name of the individual and family members; name of the town and that of the city)
- For subsequent training, form phrases with one trained and one untrained word
- Finally, form phrases with both untrained words
- Model the target phrases using all three skills in a smoothly integrated manner
- Do not give a phonatory break between the two words; blend the two words; produce the two words as though they are a string of syllables with no break; stretch all the syllables; emphasize this aspect to the individual
- Ask the individual to imitate the phrase
- Reinforce correct imitations with good airflow, gentle onset, prolonged syllables, and continuous phonation throughout the utterance
- Stop the individual at the earliest sign of trouble (mismanaged airflow, hard glottal attacks, increased speech rate, or a stutter); explain the error and concentrate on the skills that broke down (e.g., failure to exhale before starting phonation) until it is relearned
- When the individual has imitated several phrases with no stuttering and all skills efficiently exhibited, fade modeling; ask questions that can be answered by two-word phrases (e.g., “Tell me your full name”)
- Promptly reinforce the completely stutter-free productions of phrases with inhalation and slight exhalation, gentle phonatory onset, and rate reduction through syllable prolongation

- Stop the individual for mismanagement of any of the targets and production of dysfluencies; describe the error, repeat the skill mismanaged, and continue the training
- **Shift training to the sentence level**
 - Initially, expand already trained phrases into sentences (e.g., “I live in Fresno,” “My name is Kopitron Stratofearopolis”)
 - Model and have the individual imitate sentences if necessary; target completely stutter-free productions with efficient use of the skills in a smooth and integrated manner with no phonatory breaks between words; continue to model until the individual imitates several stutter-free sentences smoothly; use reinforcement and verbal correction as before
 - Fade modeling and ask questions the individual will answer with complete sentences; monitor closely all target skills and the resulting stutter-free speech
 - Stop the individual promptly for mismanagement of any of the targets and production of dysfluencies; make an error analysis and concentrate on the particular skill that broke down
 - Continue until the client speaks in sentences with stutter-free speech by using airflow management, gentle phonatory onset, and rare reduction through syllable prolongation
 - Spend the longest training time at the spontaneous conversational speech level; in fact, most individuals should reach this stage in three to five sessions with little or no modeling, only an occasional verbal praise, and few verbal corrections
 - While treating at the conversational speech level, probe periodically; at the beginning of treatment sessions, conduct 5 minutes of probe; do not model, reinforce, or verbally correct; ask the individual to speak as usual; although the fluency skills may be used, neither encourage nor discourage the individual to do so; record these samples and calculate the percentages of stuttering

- **Shape normal prosodic features**

- Begin shaping normal prosodic features when three successive probes show a stuttering or dysfluency rate around 1% of the words spoken
- Describe normal prosodic features; inform that gradually increased speech rate and typical intonations are the key to normal-sounding speech
- Model a slightly higher speech rate and request imitation
- After a few imitated sentences at the new rate, withdraw modeling and engage the individual in conversational speech maintained at the slightly increased speech rate
- Monitor the skills and dysfluent productions carefully; if stutterings appear as the rate is increased, slow down to a rate that eliminates them; after some practice at this slower level, increase the rate again
- Reinforce increased speech rate and maintained fluency; reinforce stutter-free speech that is moving in the direction of more natural speech; emphasize airflow and gentle onset only when fluency begins to break down
- Reinforce progressively higher speech rate while maintaining stutter-free speech
- Begin training on pitch variations; model a higher or lower pitch in phrases and sentences and let the individual imitate; withdraw modeling and reinforce more natural-sounding spontaneous speech with typical patterns of intonation
- Begin training on appropriate vocal intensity; note that excessively monitored airflow and rate result in too soft speech; model appropriate vocal intensity and let the individual imitate and then talk with sufficient loudness
- Make continuous judgments about speech naturalness and modify speech to approximate normal prosodic features
- Continue to reinforce normal sounding speech for several sessions; stop the individual for excessively slow rate and monotonous speech

- Continue until the speech is judged normally fluent and natural-sounding
- **Measure the frequency of stuttering in all sessions**
 - Measure in all treatment and probe sessions the frequency of stuttering or dysfluency as you defined and baserated
 - Calculate the number of words spoken and the number of stuttering or dysfluencies to derive a percentage for the entire treatment session
 - Record at least a few minutes of treatment-free speech sample (“Speak as you typically do”) in every treatment session, perhaps a 2-minute sample in the beginning (as in the previously described probe procedure) of the treatment session and 2 minutes at the end to calculate mini probe stuttering frequency
 - Check your own reliability by periodically measuring the sample twice; make sure that you measure stuttering frequency with at least 90% reliability
- **Implement a maintenance program**
 - Note that lack of maintenance of treatment gains, especially fluency in everyday speaking situations, is the reason for lower QoL and dissatisfaction with treatment; fluency maintenance program is a critical component of treatment and it consists of several strategies
 - Teach **Self-Control (Self-Monitoring)** skills; have the client count dysfluencies, the production of target behaviors, increase in rate, abrupt phonatory onset, and breath-holding; let the individual chart these behaviors along with you in treatment sessions
 - Ask the individual to judge the appropriateness of airflow, gentle onset, rate reduction, and prosodic variations to encourage self-evaluation of skills
 - Teach the individual to stop talking and to pause briefly when target skills are mismanaged or when a dysfluency occurs
 - Hold informal treatment sessions in naturalistic settings; move treatment out of the treatment room first and eventually out of the clinic environment

- Give subtle signs and gestures to monitor typical-sounding fluent speech in naturalistic settings: give a hand gesture to prompt a slightly slower rate, touch your throat to prompt a gentler onset of voice, and so on
- Train the family members, teachers, friends, colleagues, and other usual conversational partners in prompting and reinforcing fluent speech
- Train teachers and family members to provide opportunities to practice typically fluent speech
- Train family members to hold informal treatment sessions at home
- **Follow up after dismissal**
 - Dismiss the client only when natural-sounding fluency is established in everyday contexts and in conversational speech
 - Informally assess satisfaction with the treatment and ask if there are still any concerns; address those concerns (possible concern may be that the client is still dysfluent in certain situations)
 - Urge the individual and the family to contact you as soon as an increase in stuttering is noticed to schedule a follow-up assessment and booster therapy
 - Give a follow-up schedule to bring the individual back to the clinic periodically regardless of the outcome
 - Take conversational speech samples during a follow-up assessment and offer booster treatment when dysfluency rates reach or exceed 5% of the words spoken
- **Vary the treatment procedures to suit the individual and your experience**
 - Note that the procedures described are highly structured and the steps are specified somewhat rigidly
 - Note that a beginning clinician might find it easier or more efficient to follow a structured format where decision points and criteria are specified; however, with some experience, clinicians may skip steps, move faster or slower, and even skip a particular target skill if that does not make a difference

- See if only a rate reduction would be sufficient for a given individual; add airflow only if it enhances the rate of improvement (more quickly eliminates certain kinds of dysfluencies)
- Deemphasize or stop explicitly monitoring a skill (e.g., airflow management) sooner or later if you did employ all the skills to begin with; probe to see if fluency is sustainable without an explicit monitoring of those skills
- Use oral reading initially if that seems to give a better control on the fluency skills; some individuals do better in oral reading where they do not have to think of what to say
- Start treatment at any level the individual can handle the skills; some need not be started at the word level; phrases might work for others, and with a few individuals, short sentences may be the starting point
- Try moving to the short sentence level briefly if you did start at the word level; if stutter-free speech can be sustained, continue at this level (thus skipping the phrase level)
- Let the individual's performance data dictate the sequence, speed, relative emphasis on skills, and the number of treatment steps needed

Treatment for Adults and Older Children: Pause-and-Talk.

Pause-and-talk, also known as time-out, effectively reduces stuttering while maintaining normal speech rate and prosodic features; hence, it is more efficient and preferable to fluency-shaping procedure; pause-and-talk is not negative because it includes positive reinforcement for fluency; *pause-and-talk* gives a better hint at the procedure than *time-out*.

- **Introduce the procedure and baserate stuttering**
 - Describe the procedure and its effectiveness in words that are appropriate to the adult or the child (e.g., “I want you to try to talk without stuttering; when you stutter, I will look away for a few seconds.

When I look back at you, you talk again.”); point out its advantages (no need to speak slowly, no negative effects on speech rhythm and intonation); select another procedure if the client or the parent do not approve the method (an unlikely event)

- Baserate stuttering, select stimulus materials to evoke continuous speech, and engage in verbal interactions that approximate everyday conversation
- Continue to engage the client in conversational speech; reinforce fluent production with maintained eye contact, smile, nod, and other natural reinforcers inherent to everyday conversations; offer periodic verbal reinforcers as well
- **Implement the pause-and-talk procedure**
 - At the earliest sign of a stutter, say “Stop,” turn your face away to avoid eye contact with the individual, and freeze for 5 seconds; you may look at your watch to count 5 seconds
 - At the end of the pause (time-out) duration, look at the client to reestablish eye contact, smile, and say something that will let the individual continue talking (e.g., “You can talk now,” “You were saying . . . ,” “OK”); the individual will soon learn to begin talking again as soon as you reestablish eye contact
 - Make sure that the client completely ceases talking when you say “Stop”; the method is ineffective if the individual continues to talk during the planned pause duration
 - Measure stuttering and words spoken in all sessions or selected durations (e.g., a few minutes in the middle of a session) of sessions to calculate the percentage of words stuttered
 - Continue the procedure until fluency is stabilized in conversational speech in natural settings
 - Teach Self-Control (Self-Monitoring); ask the client to pause soon after a stutter and delay your instruction to pause; eventually fade the pause-and-talk procedure
 - Conduct periodic probes; let the client talk without the procedure; measure speech rate, and judge pro-

sodic features to make sure they are not negatively affected; if the person has adopted the strategy of speaking slowly to avoid stuttering, shape a rate typical of the individual while still maintaining fluency

- **Promote maintenance**

- Train family members to conduct home treatment sessions
- Encourage the person to use the self-monitoring procedures
- Monitor the home treatment sessions with audio- or video-recorded samples and refine the skills in administering the treatment
- Schedule follow-up and booster treatment sessions as needed

Treatment for Adults and Older Children: Combined Minimal Syllable Prolongation and Pause and Talk.

A novel approach to treating stuttering is to combine pause-and-talk with a slight syllable prolongation that does not adversely affect prosodic features. Clients with a very high frequency of stuttering get timed out so constantly that there is not much speech produced in treatment sessions; if so, pause-and-talk will be ineffective for them. Adding minimal syllable prolongation of only the initial syllable of the first word of utterances will greatly reduce both stuttering and the frequency of pause-and-talk application, enhancing the effectiveness of the procedure.

- Describe the procedure to the individual and the family; tell them you will be using a slight syllable prolongation with pause-and-talk; demonstrate both the procedures with modeling and self-imitation
- Engage the older child or the adult client in conversation using age-appropriate stimuli
- **Teach first the minimal syllable prolongation**
 - Model a slight prolongation of the initial syllable of the first word in an utterance
 - *Do not* prolong all the syllables of all the words as it is done in the full-fledged fluency-shaping technique

- Ask the child to imitate your production; make sure the child imitates a slight prolongation of the initial syllable of only the first word and says all other words with typical rate
- Reinforce the child for slight prolongation of initial syllables with verbal praise
- **Introduce pause-and-talk**
 - When the child has correctly produced the slight syllable prolongation on five to eight trials, introduce pause-and-talk
 - Use the procedure described in the earlier section; if necessary, periodically instruct the child to slightly prolong the initial syllables of words
 - Continue this for two or three sessions to evaluate the effects of this combined procedure
 - Measure the frequency of stuttering or dysfluencies and the number of words spoken; calculate the percentage of stuttered words for all sessions
- **Complete all other procedures**
 - Teach self-monitoring skills
 - Reinforce fluent speech in nonclinical settings to promote generalization
 - Train others to prompt, reinforce, and otherwise support fluency in natural settings
 - Implement a maintenance program
 - Assess Quality of Life after treatment

Stuttering in Younger Children (2- to 6-Year-Olds). There are effective procedures to treat stuttering in preschoolers and younger children and eliminate the consequences of persistent stuttering. The fluency-shaping and pause-and-talk are not recommended for the very young; the two most effective procedures are Fluency Reinforcement and Response Cost.

Fluency Reinforcement for Younger Children.

Systematic positive reinforcement for fluency in very young children is effective; an advantage of an *exclusive* fluency reinforcement program (nothing else is used) is that the parents and the student clinicians may be trained

more easily to use the technique than perhaps any other method; the procedure does not negatively affect the prosodic features of speech; the method should be explained to the parents and the need to conduct home treatment sessions at least three times a week, each session lasting 15 to 20 minutes, must be impressed on them; follow the steps outlined here to use a straightforward fluency reinforcement program:

Baseline Stuttering in Three Consecutive Sessions

- Hold play-oriented conversational speech sessions and audio record the sample
- Count stuttering as defined and the number of words spoken
- Calculate the percentage of stuttering (or dysfluencies)
- Do not treat if the baserates show consistent decline over the three sessions because this may suggest recovery without treatment; schedule a reassessment in three months; ask parents to return to the clinic sooner if the their concern does not mitigate

Begin Fluency Reinforcement Procedure

- Start with conversational speech mode only, although the child's utterances may be words, phrases, or short sentences
- Select stimulus materials the child likes; toys, manipulated play situations, big, colorful pictures, storybooks with attractive pictures, and so forth; consult the parents before selecting the materials; if practical, ask parents to bring the child's favorite toys and storybooks to the treatment sessions
- Ask the parents to observe the first few treatment sessions and then have them join you and the child

Evoke Speech in a Play-Oriented Format

- Use the toys, pictures, and other materials to stimulate speech
- Show a picture and ask the child such general questions as "What do you see here?" and "What is happening here?"

- Show a picture and ask such specific questions as “What is the boy doing here?” “What is the girl eating here?” “What is this mommy doing?” “What is the kitty doing here?” and so forth
- Tell a short story about a picture and ask the child to retell it
- Ask the child to tell you a story
- Let the child engage in controlled play and ask questions about what the child is doing to stimulate speech
- Role-play such situations as cooking or shopping to stimulate speech from the child
- Use hand puppets and carry on a conversation that is appropriate to the characters
- Build simple blocks with the child and pretend difficulty on your part and ask questions about how to do it
- Give slightly more difficult block designs to stimulate questions and requests for help
- Show an array of toys placed on a high shelf and ask the child to describe them and request specific items
- Stimulate speech in the context of play and storytelling with any method but do not allow the child to get lost in play

Reinforce Fluent Productions

- Positively reinforce the child for all fluent productions, be they single-word productions, phrases, or short or long sentences
- Use verbal praise as the main reinforcer (“That was nice speech,” “That was not bumpy! It was smooth,” “I like the way you talk”)
- Ignore all stutterings; do not verbally correct
- Train parents to reinforce the child’s fluent productions
 - Ask the parents to hold informal treatment sessions at home
 - Ask the parents to submit an audio- or video-recorded sample of home treatment sessions

- Listen to the recorded sample along with the parents to fine-tune their skill in immediately and positively reinforcing fluent productions and to completely ignore stuttering
- Measure stuttering as done during the baserate sessions
- Vary the treatment as found appropriate in light of treatment data
- Dismiss the child from therapy when the child has sustained fluency at 95% or better in conversational speech produced in naturalistic settings over at least a 2-week period
- Tell the parents to contact you if there is an increase in stuttering
- Give them a follow-up schedule and conduct follow-up assessments
- If the stuttering frequency exceeds 5% of words spoken during any follow-up assessment session, schedule booster treatment sessions; offer the same treatment for a few sessions, or use response cost, described next

Response Cost for Younger Children. For children under the age of 6 (preschoolers), response cost is an effective strategy. In this procedure, you reinforce fluent productions with a token and withdraw a token for stuttering; describe the procedure to the child and the parents; point out its effectiveness; baserate stuttering, select stimulus materials, and structure the treatment session as described under the previous entry.

Design a Token System

- Consult with parents and the child in selecting backup reinforcers
- Select small toys, stickers, or other inexpensive gift items such as pencils, balloons, and crayons
- Use activities as backup reinforcers because they can be more effective than small gifts; arrange for opportunities to engage in such activities as listening to a story that you read, drawing on the chalkboard, a

brief play activity that the child selects, and listening to music

Begin the Response Cost Procedure

- At the beginning of the session, ask the child to select one tangible reinforcer to be exchanged for the earned tokens at the end of the session; offer only those choices that you have access to
- Tell the child what you will do; the number of tokens needed for a reinforcer is child-dependent (e.g., “You get this token when you talk smooth; I take one of your tokens when your speech is bumpy. When you have 10 of these, you can play for 1 minute; at the end of the session, you give me back your tokens and get your gift. Try not to lose tokens and your gift to your bumpy speech. Keep as many tokens as possible.”)
- Demonstrate the procedure: Model a stutter and immediately remove a token from your own pile; say something fluently and award yourself a token
- Evoke conversational speech (as described in the previous entry)
- Give a token for each fluent production (words, phrases, or sentences) and verbally praise as you give the token (“Great job!”; “That was smooth speech!”)
- Take a token back from the child as soon as a sign of stuttering appears (e.g., a lip puckering or eye blink) or a stuttering occurs
- Verbally correct as you withdraw tokens (“Oho! That was bumpy! Make it smooth!” “Sorry, didn’t sound good! Make it good!”)
- Take the token back in a matter-of-fact manner; do not show unpleasant reactions
- Keep an eye on the tokens the child is accumulating; there should be enough tokens to exchange for an activity during the session or for the gift at the end; if the stuttering frequency is very high and the opportunities for earning the tokens are few, give two tokens for every fluent production

- Give periodic and short breaks and let the child engage in the selected activity if activity is the selected reinforcer (e.g., as soon as the child accumulates 10 tokens stipulated to earn the backup reinforcer, take a break for 1 or 2 minutes and allow the child to draw, listen to part of a story, listen to music, etc.); promptly terminate the activity and return the child to treatment
- At the end of the session, exchange the token for the selected reinforcer (e.g., a sticker or a pencil)
- Continue the procedure until fluency is stabilized in conversational speech in natural settings
- Measure stuttering the way you have defined in all sessions
- Probe fluency and stuttering periodically as described in the previous entry; during the probe, judge the child's speech prosody and rate; neither should be affected; rate may increase slightly when stutterings decrease
- Train parents to conduct home treatment sessions
- Monitor the home treatment sessions with recorded samples and refine the parents' skills in administering the procedure
- Schedule follow-up and booster treatment sessions

Quality of Life (QoL) After Treatment. Although QoL is a posttreatment assessment matter, it is an essential consideration in designing a comprehensive treatment program for persons who stutter; following treatment, QoL and satisfaction with the treatment received—the two closely related variables—may be assessed to find out any deficiencies in treatment outcomes and remediate them; however, effective treatment itself is the most critical variable in promoting improved QoL and the client satisfaction with treatment; some variables to consider:

- Assess QoL before and after treatment, during follow-up visits, and after any additional booster treatment; for procedures, see Quality of Life After Treatment in this volume and the companion volume, *Hegde's*

PocketGuide to Assessment in Speech-Language Pathology; even preschoolers (and definitely their parents) may be asked how the treatment has or not changed the negative aspects of stuttering; older students and adults will respond well to questions asked during interviews

- Select treatment procedures that are known to produce better maintenance; persons who receive fluency shaping are known to relapse because of the slow and unnatural speech; pause-and-talk in adults may promote better maintenance because it does not require a slower speech and unnatural speech prosody; children do well with fluency reinforcement and response cost
- Emphasize generalization and maintenance of fluency in therapy; individuals who still, after treatment, stutter in some social, educational, and occupational situations are unlikely to experience improved QoL; obviously, they will be dissatisfied with treatment
- Teach Self-Control (Self-Monitoring) skills and establish environmental support for fluency (e.g., trained family members) as these are two critical variables in maintenance
- Eliminate or greatly reduce avoidance and negative emotions by establishing strong, generalized, and maintained natural-sounding fluency in everyday settings

Hegde, M. N. (2006). *Treatment protocols for stuttering*. Plural Publishing.

Treatment of Stuttering: Specific Techniques or Programs.

The following are other named techniques; most are not as comprehensive as the treatment outline given in the earlier section; some have no experimental support; some do not target reduction in stuttering at all; evaluate the evidence for the technique you select for your clients.

Camperdown Program. Treatment program designed for older students and adults who stutter; essentially a syllable prolongation method described under Treat-

ment for Adults and Older Children; to use this procedure, clinicians need to get the manual from the authors and follow the directions (<http://www.fhs.usyd.edu.au/asrc/>); a few unique features of the program include:

- Persons who stutter watch video-recorded samples of prolonged speech (PS) at 70 syllables per minute and imitate the skill; clinician gives feedback; in most other programs, the clinician models PS for the client to imitate
- Soft contacts and gentle phonatory onset are not targeted as they are in other fluency-shaping techniques
- Stuttering is not operationally defined for the clients, nor are they measured during the treatment sessions; pre- and posttreatment stuttering is measured in %syllables stuttered; clients are asked to rate their stuttering on a 9-point scale (1= no stuttering; 9 = extremely severe stuttering)
- Three to five individual practice sessions are held; when clients' 3-minute speech sample is stutter-free, they are advanced to an intensive practice day in groups of three, assisted by clinicians; clients first practice the PS and then try to speak as naturally as possible without stuttering; speech-naturalness and satisfaction with treatment are rated
- Clients are helped with generalization strategies

O'Brian, S. et al. (2003). The Camperdown program: outcome of a new prolonged-speech treatment model. *Journal of Speech, Language, and Hearing Research*, 46(4), 933–946.

Counseling Parents to Reduce Stuttering in Children.

Parents are counseled to indirectly reduce stuttering in their children; clinicians talk with the child's parents to change their feelings, attitudes, ideas, and expectations about stuttering and fluency; counseling not combined with direct treatment of the child and parent training in a home intervention program may be ineffective. Generally, the following kinds of advice is part of parent counseling without treating the child:

- Be a sensitive, uncritical, accepting listener of parents' feelings and thoughts about their child's stuttering

- Find out what the parents wish to accomplish through counseling
- Let the parents explore their feelings, emotions, ideas, and expectations relative to their child's stuttering
- Let the parents freely talk about their fears, possible feelings of guilt, and their overt reactions to stuttering in their child
- Let the parents talk about their strategies for dealing with stuttering in their child
- Help the parents find their own solutions by offering professional views and ideas they may not have known or considered
- Express approval of their positive feelings and helpful reactions toward their child
- Help them realize their child's strengths and limitations
- Reduce their negative thoughts and feelings, including potential feelings of guilt, by reassuring them that they may not have created the problem
- Let the parents put their child's stuttering in perspective so that they do not exaggerate its negative effects
- Let the parents realize that no child is fluent all the time
- Explore actions the parents may have taken with positive effects on the child's fluency and encourage them to increase or strengthen them
- Explore actions the parents may have taken that have worsened the child's problems and encourage them to eliminate or reduce them
- Explore the parents' ideas about fluency and stuttering to encourage a more realistic view of them
- Discuss the kinds of communicative demands the parents make and ask them to reduce such demands
- Encourage the parents to create more positive speech experiences for the child by withholding criticism and accepting the child's stuttered attempts at communication

Counseling Persons Who Stutter. Counseling is an indirect and psychological method to treat adults who stutter; the main approach is talking with the individual

to change feelings, attitudes, and expectations; counseling is a part of all stuttering treatment procedures, but done exclusively in the absence of direct stuttering treatment procedures may be ineffective; often successfully combined with direct stuttering treatment (described earlier in this section); successful combination is no evidence for exclusive counseling.

- Be a sensitive, uncritical, accepting listener of what the clients have to say
- Find out what they wish to accomplish through counseling
- Let your clients explore their feelings, emotions, ideas, and expectations relative to stuttering
- Let the clients talk about the difficult speaking situations, listener reactions, and their own emotional reactions
- Help the individuals find their own solutions by offering professional views and ideas the individual may be unaware of or may not have considered
- Discuss your clients' strengths that they may not have realized
- Advise your clients to concentrate on their positive speech experiences to reduce negative thoughts and feelings about their stuttering
- Let your clients put their stuttering in perspective so that they do not exaggerate its negative effects
- Let the individuals realize that no one is fluent all the time
- Explore actions the individuals take that may exacerbate stuttering and encourage the individual to eliminate or reduce them
- Explore actions and situations that enhance fluency and encourage the individuals to increase them or strengthen them
- Encourage the individuals to talk more positively about themselves

Delayed Auditory Feedback (DAF). Hearing one's own speech after a delay introduced by a mechanical device; used in treating persons who stutter and

those who clutter to slow their speech rate; reduces or eliminates stuttering but induces atypically slow and unnatural-sounding speech; once a widely used stuttering treatment technique, but is waning because of the unnatural speech it induces and lack maintenance; still useful in establishing stutter-free speech (described in earlier sections) but needs additional procedures to make the speech sound natural and to make the fluency last over time and across situations; a better alternative is to induce slow and prolonged speech with instructions and modeling as described under Treatment for Adults and Older Children: Fluency Shaping.

- Assess the individual thoroughly and establish base-rates of stuttering and the speech rate
- Select a miniaturized, electronic feedback device that the individual can use in most situations; a brand is not recommended here because most units have not survived in the market
- Ask the individual to wear the portable device (may use a desktop unit with a loss in flexibility)
- Experiment with different delays to set an individual-specific delay that induces stutter-free speech (most individuals are stutter-free at 250 milliseconds of delay)
- Begin by asking the individual to respond to questions that evoke two- or three-word phrases or short sentences
- Drop down to word level if the individual cannot maintain stutter-free speech at the phrase or short-sentence level
- Model slow, prolonged speech if necessary
- Use oral reading to stabilize a slow, prolonged production if necessary (some initially find oral reading under DAF easier than speaking)
- Establish stutter-free speech with the initial delay over a few sessions
- Increase the length of utterances gradually
- Move to more spontaneous conversational speech containing longer and more complex utterances

- Fade the DAF by initially decreasing the delay in gradual steps; reduce it in 50-millisecond intervals that still help maintain stutter-free speech
- Reduce the intensity of DAF
- Increase the rate of speech while still maintaining stutter-free speech; reinforce the individual for speaking at progressively faster rates
- Eliminate the delay altogether, but let the individual wear the unit
- Increase the rate further to move it closer to the pre-treatment, judged normal, or natural-sounding rate
- Shape the normal prosodic features, including normal rhythm, intonation, intensity variations, emotional connotations, and so forth
- Conduct informal treatment sessions in varied non-clinical settings
- Train family members, teachers, and others to reinforce fluent, natural sounding speech in nonclinical settings
- Teach Self-Control (Self-Monitoring) skills (charting one's own stuttering, stopping soon after stuttering or at the earliest sign of increased rate)
- Dismiss only after a natural-sounding fluent speech is established
- Counsel the individual about the possibility of relapse and the need for Booster Treatment
- Follow up and arrange for booster treatment (relapses are common; a strong maintenance program is essential)

Fluent Stuttering. Fluent stuttering is Van Riper's extensive, early, and influential treatment program for stuttering; also described as *stuttering modification therapy*; goal is to teach less abnormal, socially more acceptable stuttering, not necessarily typical fluency; based on the assumption that normal-sounding fluency is unrealistic for most adults who stutter; includes counseling and psychotherapy to change feelings and attitudes; has multiple steps:

- **Teach stuttering identification**
 - Teach clients to identify their stuttering and related problems, including negative feelings, avoidance, word fears, and easy and difficult stutterings, with discussion, demonstration, reading, modeling, and so forth
 - Teach clients to identify their own stuttering and associated problems in everyday speaking situations
- **Desensitize clients to their stuttering**
 - Ask them to be open and honest about their stuttering
 - Ask them to continue to stutter until told to stop
 - Let them face different audiences and stutter voluntarily to learn that most people do not react negatively, and if some do, they can tolerate it
- **Modify stuttering**
 - Ask them to stutter more fluently, easily, and less abnormally
 - Have them start using feared and avoided words, despite stuttering on them
 - Teach them *cancellation*, pausing after a stuttered word and saying the word again with easy and more relaxed stuttering (soft articulatory contacts and slower rate); do not target fluent word production; encourage the use of cancellation outside the clinic
 - Teach *pull-outs*, changing stuttering in its mid-course; let them pull themselves out of stuttering by slowing down and using soft articulatory contacts; ask them to use pull-outs outside the clinic
 - Use *preparatory sets*, anticipating stuttering and stuttering in an easy and relaxed manner
- **Stabilize the treatment gains**
 - Have the clients assign themselves speech tasks that help stabilize the use of cancellations, pull-outs, and preparatory sets (stuttering modification skills)
 - Ask them to practice those skills on difficult words
 - Let them seek out difficult and previously avoided speaking situations

Van Riper, C. (1973). *The treatment of stuttering*. Prentice-Hall.

Gentle Phonatory Onset. Soft, easy, slow, and relaxed initiation of sounds as against harsh, abrupt, and tensed initiation; a target behavior in fluency-shaping technique; may also be used to reduce vocal abuse and hoarseness and harshness of voice; often combined with airflow management and syllable-prolonged slow speech in treating stuttering; see Treatment of Adults and Older Children: Fluency Shaping described in an earlier section.

Indirect Stuttering Treatment. Methods in which clinicians counsel adults who stutter or parents of children who stutter without using any methods to directly reduce stuttering; direct stuttering reduction methods are better researched and supported than indirect methods.

Integration of Stuttering Modification and Fluency Shaping. A dual approach that uses both the Stuttering Modification (essentially Van Riperian) and Fluency Shaping Techniques; this approach is more forcefully applied to advanced stuttering than to beginning stuttering; uses a variety of handouts (e.g., understanding stuttering, how to be open about stuttering, how to use feared words) during treatment sessions.

- **Understand the client's stuttering**
 - Be warm and friendly to the client; describe the treatment program
 - Ask the individual to read a brief description of stuttering, what it is, and how it develops; use the authors' handout "Understanding Your Stuttering"; answer all questions; share and reinforce the individual's insights
 - Catalog all aspects of the client's stuttering to promote better understanding of the problem; model stuttering, and use videotapes or mirrors to demonstrate stuttering to the client
- **Reduce negative feelings, attitudes, and avoidance behaviors**
 - Encourage clients to discuss their stuttering openly with family, friends, and acquaintances; use the

authors' handout "Discussing Stuttering Openly" in the treatment session

- Ask clients to create a hierarchy of feared and avoided words and situations; encourage them to use feared words; enter previously avoided speaking situations freely and frequently; use the authors' handout "Using Feared Words and Entering Feared Situations"
- Teach clients the technique of *freezing or holding on to the moment of stuttering*; when the individual stutters, ask to continue (to repeat, prolong) until you signal to stop; teach the individual to be calm while doing this
- Teach clients Voluntary Stuttering; use the authors' handout "Using Voluntary Stuttering"; model brief, easy repetitions or prolongations for clients to imitate; take them to naturalistic settings where they will stutter voluntarily
- **Teach fluency-enhancing skills and modify the moments of stuttering**
 - Use Delayed Auditory Feedback to teach a slower rate of speech; teach Gentle Phonatory Onset and Soft Articulatory Contacts; use the authors' handout "Using Fluency-Enhancing Behaviors"; fade DAF in gradual steps
 - Stabilize fluency-enhancing skills in conversational speech without DAF
 - Promote generalization of fluency to situations outside the clinic and with an audience other than the clinician
 - Teach easy stuttering; teach cancellation, pull-outs, and preparatory sets described under Fluent Stuttering: Van Riper's Approach; teach your clients to integrate fluency-enhancing skills with stuttering modification
 - Initiate activities to generalize stuttering modification skills to situations outside the clinic and with other audiences
- **Help maintain improvement**
 - Help your clients to become their own clinician; use the authors' handout "Becoming Your Own Clini-

cian”; help the individual learn to design assignments to reduce fear and avoidance (e.g., voluntary stuttering in a difficult situation); encourage the individual to work on stuttering and fluency every day

- Establish long-term fluency goals; help the individual set the goal of spontaneous (unmonitored) fluency whenever possible, controlled (monitored) fluency when it is important to be fluent, and controlled stuttering (mild stuttering with which the person who stutters is comfortable) when it is acceptable

Guitar, B. (2024). *Stuttering: An integrated approach to its nature and treatment* (6th ed.). Lippincott Williams & Wilkins.

Lidcombe Program. Treatment for preschoolers who stutter, administered mostly by trained parents; involves positive reinforcement for fluency and corrective feedback for stuttering; parents and their stuttering children visit the clinic once a week to get training in 1-hour sessions; includes systematic methods to obtain at-home and in-the-clinic measures of stuttering (although undefined) throughout the treatment phases, problem-solving, and maintenance procedures; supportive evidence for effectiveness is available; because the clinicians have to be trained by the authors to use the technique (www.lidcombeprogram.org), only an overview of the procedure is offered here:

- Train parents to calculate the percent syllables stuttered (%SS) in 5- to 10-minute home speech samples before the onset of treatment; continue to collect %SS throughout the treatment phase; rate their child’s stuttering severity in periodically recorded speech samples on a 10-point scale (1 = no stuttering; 10 = extremely severe stuttering)
- Deliver treatment in 5- to 10-minute sessions held at home one or two times a day to induce effortless and stutter-free speech in the child
- Engage the child in structured play with storybooks, picture books, selected toys, and so forth; evoke single words with flash cards from a child with severe stuttering; use more play-oriented activities to evoke

conversational speech from a child with mild stuttering; tell the child: “We will be playing a game and let’s see if you can say a lot of words smoothly.”

- Praise the child for stutter-free speech with such verbal statements as “Good talking!” “Your speech sounds great!” or “Your words are smooth!” and so forth; supplement praise with such tangible reinforcers as stickers or tokens
- React to stuttering in several ways (e.g., ignore stuttering, verbally correct it, or ask the child to repeat the word fluently); deliver fewer corrections than praise; cease verbal correction if the child reacts emotionally; reintroduce correction gradually as stutter-free speech increases
- Decrease the treatment structure as the child becomes more fluent
- Teach the child self-monitoring skills
- Deliver treatment online so the clinicians can monitor the parents’ administration of treatment
- Initiate a maintenance program when the child’s %SS is 1 or less for 3 consecutive weeks
- Decrease the frequency of clinic visits gradually; schedule maintenance clinic visits at regular and decreasing intervals
- Continue monitoring as scheduled

Metronome-Paced Speech. Treatment for stuttering and cluttering; clients utter syllables to the beats of a metronome; syllables or word initiations may be regulated; may be used to slow down or accelerate the rate of speech; immediately reduces or eliminates stuttering but induces timed, rhythmic, and unnatural-sounding speech; poor maintenance of fluency; clients tend to adapt to the beats (no more effective); fluency-shaping procedures are preferred to metronome-paced speech; pause-and-talk may be even better with no negative side effects on prosodic features.

Monterey Fluency Program (MFP). A programmed operant approach to establish, transfer, and maintain fluency in persons who stutter; one of its two methods,

Delayed Auditory Feedback (DAF), is used with older persons who stutter more severely; the other procedure, Gradual Increase in Length and Complexity of Utterances (GILCU), described in this section, is used with younger persons whose stuttering is less severe; in using GILCU:

- Give an overview of the program to the child and the parents; describe the caregivers' role in fluency maintenance at home
- Baseline stuttering in a 5-minute sample of reading, monologue, and conversation; measure stuttering in terms of stuttered words per minute (SW/min)
- Have the child identify and measure stuttering with 75% or better accuracy
- Instruct the child to "Read fluently," starting with one word read fluently; reinforce with verbal praise for fluent production; say "Stop, read fluently" when stuttering occurs; obtain 10 consecutive fluently read words
- Instruct the child to read progressively longer utterances; steps include 2, 3, 4, 5, and 6 fluent words; 1, 2, 3, and 4 fluent sentences; fluency for 30 seconds and 1, 1.5, 2, 2.5, 3, 4, and 5 minutes
- Ask the child to "speak fluently"; engage the child in monologue (first step with a nonreader); use pictures and topic ideas and other necessary stimulus procedures with the same gradually escalating steps
- Engage the child in conversation; use the same gradually escalating steps; reinforce fluent productions with verbal praise and tokens
- Say "Stop, read fluently" or "Stop, speak fluently" when the child stutters
- Model the target response if stuttering persists
- Require 100% fluency (0 SW/min) at each step
- Give a criterion test at the end of the establishment phase (5 minutes of reading, monologue, and conversation with 0 SW/min)
- Implement a fluency transfer and maintenance program; bring unfamiliar persons and parents into the therapy room, move treatment out of the clinic room, train parents to conduct home therapy sessions and

reinforce fluency in natural contexts, conduct informal sessions in the child's classroom, have the child make a presentation to the class, make telephone calls, and so forth; establish a follow-up schedule to assess maintenance; offer booster therapy if there is regression

Ryan, B. P., & Van Kirk Ryan, B. (1995). Programmed stuttering treatment for children: Comparison of two establishment programs through transfer, maintenance, and follow-up. *Journal of Speech and Hearing Research*, 38, 61–75.

Regulated Breathing. A direct stuttering reduction method in which the individual is asked to modify breathing patterns along with the use of such other strategies as thought formulation and relaxation; some clinical evidence supports its use, but the effective component of the eclectic program is not clear; its components, inhalation, and slight exhalation before initiating phonation have been incorporated into the fluency-shaping program described earlier; more effective with older children and adults than with very young children; in this procedure, the clinician teaches the client to:

- Formulate thoughts before speaking
- Inhale and exhale a small amount of air before talking; imitate the clinician's modeling
- Continue to exhale a little even after the last sound is produced
- Pause at natural speech junctures and formulate thoughts again
- Stop soon after a stuttering occurs and relax, especially the chest muscles
- Seek out previously avoided speaking situations
- Practice the new method of speaking daily
- Measure and record stutterings in natural settings
- Maintain phone contact with the individual to schedule follow-up appointments

Azrin, N. H., & Nunn, R. G. (1974). A rapid method of eliminating stuttering by a regulated breathing approach. *Behavior Research & Therapy*, 12, 279–286.

RESTART-DCM. Based on the demands and capacities model of stuttering (see Stuttering Prevention: A Clinical Method), RESTART-DCM works mostly with parents

to reduce fluency demands the child faces and increase capacity for fluent speech so that stuttering is reduced in children (ages 2–6); to administer the program, clinicians need specific training; therefore, only a few salient features are highlighted below; the program includes three phases:

- **Phase I.** Reduce demands for fluency by teaching parents such strategies as modeling a slower articulation while speaking to the child, giving more time for the child to respond, avoiding overlap in conversational turn taking, speaking in a calm and relaxed manner while maintaining eye contact, and avoiding fast and rushed pace of life
- **Phase II.** Increase the child's capacities for fluency by teaching parents to enrich the child's language skills if warranted (e.g., word retrieval and turn taking), use pauses in longer sentences, desensitize the child to stuttering (clinician and the parent may produce dysfluencies, make such neutral comments as *talking can be tough* when the child stutters), and regulate the child's emotions (teach what emotions are; teach the child to recognize emotions)
- **Phase III.** Work more directly on fluency if the parental work proves ineffective or further improvement is needed; the clinician may teach slower and more relaxed speech; slight overarticulation; vary the child's rate, pitch, volume, and melody of speech; encourage less tensed, more relaxed, and less severe stuttering; and fade therapy at home

Franken, M. C., & Laroës, E. (2021). *RESTART-DCM Method*. <https://www.restartdcm.nl>

Shadowing. A stuttering and cluttering treatment technique in which the client, without seeing the text, repeats (shadows) everything the clinician reads aloud from a book; the client stays a few words behind the clinician; typical effect is to reduce the frequency of stuttering; popular in the 1960s and 1970s, especially in Europe; some clinical evidence suggests its effect in reducing stuttering; no research on maintenance of fluency.

- Select a reading material that is suitable for the individual
- Instruct the individual to say everything that you read
- Give practice by reading a few sentences at a time, stopping, and reinstructing, if necessary
- Do not show the text to the individual
- Read normally; do not change the rate, rhythm, or phrasing
- Record the individual's shadowing to measure the frequency of stuttering during treatment sessions

Speech Restructuring. Similar to prolonged speech in fluency shaping; see Treatment for Adults and Older Children: Fluency Shaping.

Stuttering Modification. See Fluent Stuttering approach of Van Riper.

Stuttering Prevention: A Clinical Method. An early treatment program for children who stutter based on the Demands and Capacities Model (DCM) of fluency and stuttering; goal is to reduce demands made on the child's fluency and increase fluency capacities.

- Assess the child's capacity for fluency and the demands the child faces
- Counsel the parents; educate them about stuttering, the treatment program, and prognosis; give an optimistic outlook on improvement with treatment; change attitudes of parents by discussing their negative feelings and possible guilt; ask the parents to speak to the child at a slower rate and in shorter, simpler sentences; withhold negative reactions and punishment; encourage polite turn taking in conversation; demand speech less often
- Modify the child's stuttering and fluency by reducing the tension and struggle behaviors associated with dysfluency; modeling a slower rate, less struggled word and phrase repetitions; asking the child to imitate a slower rate by syllable prolongation; reducing interruptions and demands for verbal performance
- Intervene more directly by using fluency-shaping procedures, including gentle phonatory onset and light

articulatory contacts; time-out contingent on struggle behaviors and self-corrections

- Promote a level of language use that is typical for the child's age and gender
- Dismiss the child only when both the parents' and the child's behaviors have changed

Bloodstein, O. et al. (2021). *A handbook on stuttering* (7th ed.). Plural Publishing.

Hegde, M. N. (2006). *Treatment protocols for stuttering*. Plural Publishing.

Subcortical Aphasia. Aphasia due to subcortical damage, especially to the basal ganglia and surrounding areas; somewhat controversial because some experts suspect that subcortical aphasia involves cortical damage as well; characterized by generally fluent speech, intact repetition, and articulation problems; see [Aphasia](#).

T

Telerehabilitation and Caregiver Training. Distance-connecting with clients, typically in their homes, and conducting assessment, treatment, teaching, or rehabilitation sessions is a growing mode of service delivery; requires a technological infrastructure (computers, tablets, smart phones), highspeed internet connection, dedicated software for videoconferencing and service delivery, a physical setup at the clinic and the client's home, a trained helper at the client's site, and a clinician trained in tele-service delivery; most of the available studies are research investigations conducted at university clinics with adequate facilities; studies on long-term effects and maintenance of telerehab effects are lacking; lacking also are studies of complex communication training; all clinicians and clients may not have access to the needed unfractured, although improvements may be expected.

- Assess the available infrastructure for teleservice delivery in your clinical setup
- Assess the exiting or potentially arrangeable access to similar infrastructures at the client's home
- Get training in using the telerehabilitation procedures
- Conduct a few face-to-face clinical sessions to demonstrate that the selected target behaviors are appropriate, intervention procedures work, and a caregiver can be trained to conduct sessions at home
- Consider home practice of the initially clinically established skills; video-monitor the practice sessions to improve the caregiver skills
- Give separate and specific feedback on performance of the client and the caregiver
- Reschedule clinic-based sessions whenever the need arises
- See specific disorders for considerations that apply to them (e.g., see [Traumatic Brain Injury](#), [Aphasia](#), [Stuttering](#), etc.)

Brunner, M. et al. (2017). Technology and its role in rehabilitation for people with cognitive-communication disability following a traumatic brain injury (TBI). *Brain Injury*, 31(8), 1028–1043.

Time-Out (TO). Time-out from positive reinforcement; also described as pause-and-talk as used in treating [Stut-](#)

tering; effective in reducing several undesirable behaviors; following a response to be reduced, the clinician imposes a brief period (typically 5 seconds) of no reinforcement (cessation of verbal interaction) and resumes verbal interaction at the end of the TO duration

- Use this method with older children and adults
- Target stuttering; rambling, inappropriate, or excessive speech associated with several neurologically based communication disorders; inappropriate turn taking (interrupting speakers); abrupt topic shifts; interrupting treatment by asking irrelevant questions; and so forth
- Engage the client in conversational speech
- Instruct the person to stop talking when you give a time-out signal (e.g., saying “stop” or giving a hand signal to stop talking)
- Begin TO as soon as the undesirable response occurs
- Limit TO duration to no more than 10 seconds; 5 seconds may be ideal
- Do not interact with the client during TO; avoid eye contact
- At the end of the TO duration, smile, reestablish eye contact, and resume interaction

Token Reinforcement. Tokens are conditioned generalized reinforcers; they are earned during treatment and exchanged later for backup reinforcers; effective in teaching a variety of communication skills.

- Use this method with younger children and preschoolers
- Target fluent utterances in children who stutter; correct production of speech sounds; quiet sitting in treatment sessions; paying attention to treatment stimuli and tasks; and so forth
- Always have backup reinforcers the child can exchange the tokens for
- Let the child choose a backup reinforcer at the beginning of each session
- Let the child understand the ratio of tokens to a backup reinforcer (e.g., say, “You can get this toy car at the end of the session if you have ten tokens left with you”)

- Set a low ratio in the beginning and gradually raise the number of tokens needed to receive the backup reinforcer (e.g., a token for every correct response in the beginning; later one for every third response, fifth response, etc.)

Tongue Thrust. A deviant swallow in which the tongue is pushed forward against the central incisors; treatment considerations include the following:

- Experimental treatment research on either tongue thrust modification or speech error modification presumably due to tongue thrust is limited
- Correct productions of /z/ and /s/—typically distorted—are primary articulatory targets in tongue thrust; additional target sounds may include /ʃ/, /ʧ/, /dʒ/, /ʒ/, /t/, /d/, /l/, and /n/; these may be produced with a more anterior tongue position than normal; treatment is concerned with more normal tongue position in producing these sounds
- See Speech Sound Disorders for treatment procedures

Topic Initiation. A social communication skill in initiating conversation on a topic; a language intervention target; see Social Communication Skills for procedures.

Topic Maintenance. A social communication skill in maintaining conversation on a specific topic for socially acceptable time periods; a language intervention target; see Social Communication Skills for procedures.

Topographic Sequence of Treatment. Sequencing treatment based on response complexity; if preferred to begin treatment with simpler targets, increase complexity gradually and incrementally; alternatively, teach first the more complex, difficult, low-frequency skills on baselines as well as those not produced during assessment to promote better generalization; informal experimentation helps design the sequence; in all instances, move to higher levels sooner rather than later and end training at the conversational speech level.

- Begin treatment not necessarily with the simplest responses but at a relatively complex level
- In treating speech sound disorders, begin teaching a phoneme at the word or even phrase level; drop to the syllable level or isolated phoneme level only if the child cannot handle the more complex level; as the client becomes proficient in producing the sound at the successful initial level, shift training to the higher levels
- In language therapy, begin teaching grammatical morphemes in phrases or even sentences (*two cups* or *I see two cups*); shift training to the higher levels
- In fluency therapy, train such skills as gentle onset, prolonged speech, and airflow management initially in phrases and sentences and subsequently in conversational speech
- In voice therapy, begin to modify voice quality at the sentence level and if unsuccessful drop to single word or vowel productions (e.g., /a/) to improve voice quality; shift training to conversational speech

Total Communication. The simultaneous use of multiple modes of expression to enhance communication; includes speech, gestures, informal and formal (e.g., American Sign Language) signs, and facial expressions.

Tracheoesophageal Puncture (TEP). A surgical procedure to help persons with laryngectomy produce laryngeal speech with the help of a voice prosthesis; the tracheal wall is punctured to create a small tunnel (shunt) into the esophagus to allow air into the esophagus through a Voice Prosthesis inserted into the opening; air goes up through the P-E segment and results in the production of sound; see Laryngectomy for additional information.

Stemple, J. C. et al. (2024). *Clinical voice pathology* (7th ed.). Plural Publishing.

Traditional Orthography. Written natural language; a normal form of communication; a teaching target of literacy intervention; also may be a teaching target in

Aphasia, Dysarthria, severe Apraxia of Speech, and other neurologically based communication disorders; a method of nonvocal communication for individuals with limited oral communication skills; used in teaching Augmentative and Alternative Communication—Aided.

Training Criterion. A rule—such as a 90% correct responses—that specifies when a speech or language exemplar (a target skill) has reached the acceptable performance level.

- Specify a training criterion in measurable terms (e.g., 9 out of 10 correct responses)
- Continue training until that criterion is met
- Probe when the training criterion is met
- If the probe criterion (90% correct in untrained contexts) is not met, resume training
- When the probe criterion is met on a given target, either select another target for training or shift training to a higher level of response complexity (e.g., from the word level to phrase level)

Training in Natural Environments. Training in extraclinical contexts to promote generalization and maintenance.

- Initially, hold training in varied settings in and around the clinic
- Next, hold informal training sessions in nonclinical settings
- Train parents and other caregivers to hold training sessions at home
- Take the client to such natural settings as shopping centers and restaurants
- Let the training in natural settings be less conspicuous, involving mostly conversational speech
- Prompt the target responses and deliver reinforcers and corrective feedback in a subtle manner

Transcortical Motor Aphasia. A type of nonfluent aphasia, similar to Broca's Aphasia. See Aphasia; Treatment of Aphasia: Specific Types.

Transcortical Sensory Aphasia. A type of fluent aphasia, similar to Wernicke's Aphasia. See Aphasia; Treatment of Aphasia: Specific Types.

Traumatic Brain Injury (TBI) in Adults. Treatment of communication disorders in individuals with TBI follows the more urgent medical management; TBI may be either Penetrating (Open-Head) Injury or Nonpenetrating (Closed-Head) Injury; immediate concern is medical; long-term concern is rehabilitation; communication treatment is offered when the person's physical status is relatively stable; treatment follows a thorough evaluation of communication and related disorders; note the similarities in treatment goals and procedures for patients with TBI and those with Right Hemisphere Disorder; consider the general guidelines given under Ethnocultural Variables in Treatment while serving diverse persons with TBI; consult with family members to select personally and ethnoculturally appropriate treatment stimuli, target behaviors, and intervention activities; see the sources cited at the end of this main entry.

Treatment of Traumatic Brain Injury: General Principles

- **Work with an interdisciplinary rehabilitation team**
 - Work closely with other professionals on the team to develop and implement an integrated plan of rehabilitation
 - Work with other medical staff who are caring for the patient
 - Plan for long-term treatment and rehabilitation, depending on the severity of injury
- **Select client-specific and family-approved functional treatment targets**
 - Counsel the family members on the effects of TBI on communication skills and give an overview of communication treatment
 - Have the client and the client's family members participate in treatment target selection; ask them to

rate the relative importance of potential communication skills; if in a position to discuss, ask the client to prioritize the treatment goals

- Select targets that help improve immediate communication, orientation to the environment, memory for events and persons, and those that help reduce confusion
- Revise treatment targets as the patient's condition improves (or deteriorates); select goals that are appropriate and practical for the current physical condition of the client; with improved physical status, the individual may be better able to discuss potential treatment targets
- Accept any mode of communication initially (e.g., gestures, words, phrases, or grammatically incorrect or incomplete expressions if they effectively communicate)
- Avoid training isolated nonfunctional skills such as orientation, memory, and attention (e.g., paying attention to printed dots or squares); avoid indirect, underlying, and presumed cognitive or linguistic process training; integrate functional cognitive skills into communication training (e.g., reinforce increasingly longer durations of attention while training eye contact or topic maintenance)
- **Use the evidence-based direct behavioral treatment procedures**
 - Simplify the tasks with shaping procedure; offer immediate positive reinforcement for improved skills
 - Select effective reinforcers; use tangible or primary reinforcers when the patient fails to respond to verbal praise
 - Plan therapeutic activities that are appropriate for each of the different stages of recovery from TBI (acute, postacute, outpatient, and long term)
 - Include behavioral self-management exercises in your treatment program

- Use real-life activities in treatment (e.g., in improving memory skills, use pictures of family members instead of irrelevant geometric shapes)
- **Structure the sessions and treatment sequence**
 - Begin with simple activities and move through a sequence of more complex activities, but always experiment to see if the client can handle more complex skills
 - Hold brief and frequent treatment sessions in the initial stages of rehabilitation; repeat treatment trials; see [Treatment Discrete Trials](#)
 - Increase the duration of sessions as the client's general physical condition improves; hold longer sessions less frequently
 - Eliminate distraction, especially in the initial stages of recovery; loosen the structure gradually to better resemble everyday living conditions
 - Sequence treatment tasks to suit the individual's success in handling them; don't rigidly follow the commonly used simple-to-complex strategy; improvement in the physical status may enable the person to handle more complex tasks
 - Design and implement a community reentry program
 - Consider the educational, social, and occupational environment that the client will enter or reenter after discharge from the rehabilitation facilities
 - Consult with such other persons as the supervisors or colleagues of adult workers or advisors of college students

TBI Treatment Procedures: Initial Stages

- **Improve the client's orientation and attention to surroundings**
 - Simplify the patient's daily activities and routines
 - Decrease variability in activities and stimulation; plan for only a few structured activities
 - Induce consistency in staff care and planned activities and events

- Arrange familiar cues; paste familiar pictures, posters, and objects
- Play favorite music
- Post written signs and lists about the daily routines; train the patient to consult the signs and lists; ask the patient to read them aloud and describe scheduled activities and their timings
- Ask questions about time, place, and people (e.g., “Where are you now?” “What time is it?” “Who am I?” “Who is she?”); prompt and reinforce correct responses
- Model expected responses and reinforce correct imitative responses
- Increase the patient’s attention to the surroundings; draw attention to surrounding events and persons, including health care staff; stimulate speech about the immediate surroundings and people; offer verbal praise and corrections
- Use familiar odors to reorient patient to previously identifiable smells
- Simplify all demands to promote success in speech and other activities
- Use such tangible reinforcers as sweets, music, touching, and massage if verbal praise is initially ineffective
- Keep the treatment sessions brief
- Prompt and assist self-care activities (dressing, eating)
- Reduce gradually the amount of physical help offered
- Have the client participate in group treatment sessions as soon as it is practical
- Shape desired communication targets
- **Target augmentative and alternative communication in the initial stage**
 - In a professional care facility where the person is being treated, organize a simple communication board for basic communicate needs
 - Pair your verbal expressions with gestures and other nonverbal means of communication; train the caretakers to do the same

- Fade the initial nonverbal device (e.g., the communication board) as the person improves physically and begins to be more verbal
- Assess the potential need for long-term use of an augmentative or alternative mode of communication; the need may depend on the severity of TBI
- See Augmentative and Alternative Communication (AAC) for details
- **Target functional communication strategies**
 - Do not overstimulate
 - Use colored objects and pictures in treatment
 - Start with strong cues and fade them later
 - Relate new information to experiences that have occurred in the patient's life
 - Teach *yes* or *no* responses
 - Introduce familiar sounds from the patient's home (e.g., the bark of the family dog)
 - Talk only after you gain the patient's attention (e.g., "Listen, carefully, now"; "I want to say something to you"); educate the clinical staff to do the same
 - Give introduction to new topic (e.g., "I am now going to tell you about . . ."); do not introduce topics abruptly
 - Assess comprehension of spoken speech frequently; ask the patient to restate what was just said or summarize the main points of discussion; prompt correct responses to minimize errors
 - Reinforce nonverbal communication or signs of attention (e.g., eye contact, smiling, nodding); assess comprehension to make sure the patient understands
 - Withhold attention from irrelevant, inappropriate, and tangential responses; without responding, ask a simple question that might evoke a more relevant response; reinforce it; use Time-Out to decrease more serious undesirable behaviors
 - Reduce complexity and rate of your and staff members' speech
 - Use statements instead of questions during the initial stages

- Prompt, gesture, and use verbal instructions to promote comprehension
- Give time to listen to and understand instructions
- Use sentence completion tasks for patients with initiation or inhibition difficulties
- Promptly reinforce correct responses with verbal praise and verbally correct incorrect responses

TBI Treatment Procedures: Intermediate Stages

- **Establish more complex routines**
 - Increase the number of daily routines and activities
 - Assess often to ensure that the increase is gradual, the progress is acceptable, and the individual is not overwhelmed
- **Teach the patient to request information**
 - Ask questions about time, space, or persons (e.g., “What is today’s date?”)
 - Model a request for information (e.g., “I am not sure, can you tell me, please?”) for the patient who cannot answer questions
- **Improve the patient’s memory skills**
 - Make a list of the patient’s scheduled activities and their timings; post it in a prominent place in the patient’s room
 - Post signs and symbols to remind the patient of activities or appointments
 - Teach the patient to periodically consult the lists, and look at signs and symbols; ask the patient to read them, and state their meaning
 - Ask the patient to write a list of activities after consulting the posted list
 - Have the patient silently read the list and describe the activities to be attended to
 - Use pictures of family members to reestablish their names; show the pictures frequently and ask the patient to name them
 - Repeat the names of health care staff throughout the daily routines; ask the patient to write down the names of family members and those of health care staff

- **Repeat treatment trials**
 - Note that repetition is the key to rehearsing the lost skills
 - Repeat the successful trials more than the unsuccessful trials
 - Arrange repeated trials with needed rest pauses in between; see Discrete Trials
- **Improve selective attention and comprehension**
 - Teach picture matching to sentences
 - Reinforce the person for following spoken instructions
 - Model and reinforce retelling a message to another person
 - Model answers to simple questions
 - Withdraw all modeling in gradual steps
- **Work closely with health care workers**
 - Teach them to recognize the client's problems
 - Inform them of the target communication skills you have trained or still training
 - Ask them to respond promptly to positive changes in communication skills
- **Have caregivers and family members participate in treatment session**
 - Teach them to prompt the client who hesitates
 - Model appropriate behaviors
 - Reinforce the behaviors naturally and sustain those behaviors
- **Increase deficit awareness**
 - Use simple explanations to describe the problem to the patient, but do not dwell too much on deficit unawareness, a negative approach
 - Verbally correct problem behaviors
 - Reinforce positive responses (e.g., paying attention to treatment stimuli or giving a correct response to a question) instead of constantly pointing out the problem behaviors
 - Use group therapy to allow the patient to see that others have similar problems
- **Reduce irrelevant, inappropriate, and tangential responses**

- Do not reinforce such responses with attention or verbal response; ignore them (extinction)
- Change the topic immediately; reinforce appropriate responses
- Use time-out (5 seconds of no interaction) to decrease such responses
- **Treat dysarthric speech**
 - Postpone treatment for mild dysarthria if speech intelligibility is acceptable
 - Prioritize the complex speech production targets the patient needs the most (e.g., intelligibility or typical speech prosodic features)
 - Improve respiratory support for speech, reduce hyperadduction by gentle voice onset, improve laryngeal adduction by pulling and pushing exercises, reduce hypernasality, and improve consonantal articulatory precision
 - See [Dysarthria](#) for details
- **Fade special stimuli as performance improves**
 - Post reminders and schedules in less conspicuous places
 - Print reminders and schedules in smaller prints
 - Reduce the frequency of verbal models, prompts, and reminders
 - Reinstate special stimuli when skills deteriorate; fade again after a while
- **Teach self-control (self-monitoring) skills**
 - Teach such skills as pausing soon after a wrong or irrelevant response is made, prompted self-correction, counting errors and so forth
 - Reinforce correct responses
 - Give feedback on such self-assessment
 - Delay corrective feedback to promote self-correction
 - See [Self-Control \(Self-Monitoring\)](#) and [Executive Functions](#) for additional suggestions and strategies
- **Evaluate the need for compensatory skills**
 - Continue the use of initial compensatory strategies (e.g., the use of a communication board)

- Add additional strategies as found appropriate (e.g., the use of smartphones, tablets, and other handheld electronic devices)

TBI Treatment Procedures: Later Stages

- **Train more complex activities that enhance independence**
 - Teach the client to write self-instructions (for such activities as cooking)
 - Teach the client to use smartphones and other handheld electronic devices to keep track of appointments and activities
 - Teach the client to plan the next day's activities, write them down, and consult the list periodically throughout the day
- **Continue to stabilize the skills taught in earlier stages**
 - Continue to treat Dysarthria
 - Continue to teach Self-Control (Self-Monitoring) skills as before
 - Assess and treat reading and writing skills if warranted
- **Teach social communication skills**
 - Target such skills as eye contact during conversation, topic initiation and maintenance, conversational turn taking, conversational repair, narrative skills, and social discourse skills
 - Use instructions, modeling, positive reinforcement, and verbal corrections; see Social Communication Skills for intervention procedures
- **Treat impaired executive functions**
 - Teach planning skills (e.g., planning a vacation)
 - Teach how to perform selected daily activities (e.g., resolving a billing problem, preparing a meal)
 - Teach verbal rehearsal (silently reading and rehearsing directions on how to perform specific actions)
 - See Executive Functions for procedures
- **Teach abstract language**

- Teach both the comprehension and production of abstract language, including metaphors, proverbs, and implied meaning of statements
- See Abstract Language for procedures
- **Integrate work-related communication into treatment activities**
 - Have the client and a family member supply work-related words, phrases, and sentences and research them as well; the work supervisor or a colleague may be sources as well
 - Include work-related activities (e.g., drawing, typing on a computer keyboard, writing assignments, making brief oral presentations on technical topics)
 - Have the client describe the work done in a typical day at the work setting
 - Have the client write a description of the work-related communication goals needed to be achieved
- **Teach additional compensatory strategies**
 - Teach ways of breaking down larger tasks into smaller, more manageable components; teach the patients to write down steps involved in performing an action (e.g., preparing a breakfast)
 - Reinforce requesting information relative to time, date, and so forth
 - Teach the person to request others to modify their speech (e.g., requesting others to speak slowly or to repeat)
 - Teach rehearsal of important information (e.g., self-talk about how to perform such activities as fixing lunch or changing light bulbs)
 - Have the person write down instructions, appointments, important information, and so forth
 - Teach the person to ask for written instructions from people
 - Encourage the use of active instead of passive cues (an alarm instead of a reminder in a diary that may not be consulted)
 - Teach the use of electronic devices (digital watches that display time, day, and date; smartphone alarms)

for activities and appointments); use electronic pill boxes that remind the patient to take medications; use recorders (e.g., tablets or smartphones) to record lectures, instructions, and discourse they can listen to later; use handheld electronic spell checkers; use notebook computers for more complex information management

- Suggest that the person keep possessions at specific and constant places to improve access
- Teach self-cuing for activities, names, places, and appointments
- **Increase general socialization**
 - Increase socially appropriate verbalization
 - Encourage self-evaluation of strengths and limitations
 - Learn from others with similar problems
- **Promote community reentry**
 - Prepare the patient for reentry into the educational, occupational, and social situations in the final stages of treatment; emphasize self-help skills and independent living skills; stimulate the patient's interest in academic, social, occupational, and household activities
 - Educate family members, teachers, and supervisors about the current status of the patient so that they understand the patient's strengths, limitations, the needed compensatory strategies, and necessary support
 - Modify the communication styles of family members and significant others to suit the remaining and possibly permanent deficits (e.g., teach them to draw attention before speaking, speak slowly, repeat often, speak in simpler sentences)
 - Modify the number and nature of demands people make (e.g., the teacher may give reduced amount of work or simplified work; work supervisor may give extra time to complete a task)
 - Teach family members and others to recognize reasons for oppositional behaviors (e.g., oppositional

behaviors may diminish if the demands are modified or tasks are simplified)

Consider Teleintervention

- Several studies have demonstrated the feasibility of teletraining of social communication skills (see the cited Rietdijk paper)
- Assess the availability of teleintervention infrastructure in your work setting
- Assess the client's and the family members' access to the teleintervention technological facilities at home
- Use a commonly used platforms like Skype-based videoconferencing to remotely connect to the individual and the family members
- Hold a few in-person training sessions for the selected target behaviors and have the family members observe the session; train the family member to conduct the sessions at home
- Begin a teletraining program in which the family member holds conversation with the client
- Monitor the session and give feedback to the caregiver-trainer' performance to refine the skills
- Measure progress both online and in-person; seek self-reported and trained family member-reported data on outcome

Quality of Life (QoL) After Communication Treatment for TBI.

Although a matter of assessment, QoL needs to be considered in treatment planning; consider the following:

- Administer ASHA's *Quality of Communication Life Scale (ASH-QCL)*; visit <http://www.asha.org>)
- Administer the *Quality of Life After Brain Injury* scale (see the von Steinbüchel et al. citation)
- Follow-up the client and assess maintenance of treatment gains
- Analyze the reasons for dissatisfaction and lack of skill maintenance; more often than not, lower QoL is due to lingering disabilities, lack of maintenance of communication skills established in the clinic, and possible depression and other behavioral problems

- Offer booster treatment to strengthen functional, occupational, or educational communication skills as found relevant for the client
- Refer the client to other specialists as needed (e.g., to a psychologist for depression)
- See [Quality of Life After Treatment](#) for additional procedures
- See the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for details on QoL assessment

Gilmore, N. et al. (2019). The intensive cognitive-communication rehabilitation program for young adults with acquired brain injury. *American Journal of Speech-Language Pathology*, 28, 341–358.

Hegde, M. N. (2022). *A coursebook on aphasia and other neurogenic language disorders* (5th ed.). Plural Publishing.

Le, K. et al. (2022). Systematic review of discourse and social communication interventions in traumatic brain injury. *American Journal of Speech-Language Pathology*, 31, 991–1022.

Meulenbroek, P. et al. (2019). Social communication following traumatic brain injury part 2: Identifying effective treatment ingredients. *International Journal of Speech-Language Pathology*, 21(2), 128–142..

Rietdijk, R. et al. (2022). Improved conversation outcome after social communication training for people with traumatic brain injury and their communication partners: A clinical trial investigating in-person and telehealth delivery. *Journal of Speech, Language, and Hearing Research*, 63, 615–632.

von Steinbüchel, N. et al. (2010). *Quality of Life After Brain Injury* (QOLIBRI): Scale validity and correlates of quality of life. *Journal of Neurotrauma*, 27(7), 1157–1165.

Traumatic Brain Injury (TBI) in Children. Treatment of communication disorders in children with TBI requires a few special considerations because, unlike in adults, still developing speech-language and other skills may be impaired; TBI at young age affects academic and social development; treatment procedures described in the previous entry are applicable with suitable modifications to accommodate a child's speech-language skill levels, academic and social skills, and the demands the child faces; because the child's language acquisition may be impaired,

the procedures described under Language Disorders in Children also are generally applicable; see the sources cited at the end of this main entry to make a thorough assessment of communication and related disorders before developing an intervention plan; consider the following.

Intervention of Children With TBI: General Guidelines

- **Design a treatment program suitable to the child**
 - Consider the child's social, family, and academic communication needs
 - Develop a family-centered management plan; ask the family members about the most functional and effective communication targets for the child
 - Offer counseling to family members who may be overwhelmed with the multiple problems associated with TBI in children, including aggressive and antisocial behaviors
 - Discuss the child's needs with the child's teachers, educational psychologists, and special educators, including literacy specialists
 - Analyze the child's curricula to select treatment targets that will meet the academic needs; include academic terms in language training
 - Integrate literacy skills into communication training; see Literacy and Literacy Skills
- **Work with the child's classroom teacher**
 - Work with the student's teacher to optimize the classroom for the child's communication needs
 - Discuss with the teacher communication intervention targets and strategies
 - Suggest simplification of academic tasks for the child
 - Shape difficult tasks by moving from simpler to more complex
 - Use simpler language, speak at a slower rate
 - Limit distractions in the classroom
 - Keep the classroom situation organized with minimum variation

- Use gestures and signs along with verbal expressions
- Repeat instructions, give written instructions
- Ask the child to repeat the instructions
- Teach the child to take adequate notes
- Encourage the child to request help and promptly reinforce such attempts
- Accept any mode of expression initially, but shape progressively more refined verbal communication
- Inform the teacher and other educators serving the child of communication treatment targets, general procedures, and expected outcome
- **Implement a comprehensive communication intervention program**
 - Retrain to restore lost skills
 - Treat Memory Impairments
 - Treat Dysarthria associated with TBI
 - Treat language disorders; see Language Disorders in Children
 - Train Social Communication Skills involving the child's peers or siblings in the treatment sessions (e.g., topic initiation, topic maintenance, turn taking, narrative skills)
- **Implement a habilitative program**
 - Assess speech and language skills not yet mastered before the injury
 - Teach grammatical morphemes, variety of sentence types, and social communication skills the child has not mastered
- **Consider augmentative and alternative communication (AAC)**
 - Assess the need for AAC
 - Consider aided communication (Picture Exchange Communication System, drawing, tablets, smart-phones)
 - Consider unaided forms (gestures, formal or informal signs) for the child
 - See Augmentative and Alternative Communication (AAC) for procedural details
- **Reduce undesirable behaviors in treatment sessions**

- Use Differential Reinforcement techniques to replace desirable behaviors
- Use Extinction, Time-Out, and Response Cost to decrease undesirable behaviors
- **Train parents in a home intervention program**
 - Train family members to prompt and reinforce the child's newly acquired skills
 - Use telepractice to view and modify the parents' procedures
 - Select relevant recommendations from "Promote community reentry" under the previous main entry

Coreno, A., & Ciccio, A. H. (2020). Supporting students with TBI: A clinically focused tutorial for speech-language pathologists. *Seminars in Speech and Language, 41*(2), 161–169. doi: 10.1055/s-0040-1701684.

Hegde, M. N. (2022). *A coursebook on aphasia and other neurogenic language disorders* (5th ed.). Plural Publishing.

Laane, S. A., & Cook, L. G. (2020). Cognitive-communication interventions for youth with traumatic brain injury. *Seminars in Speech and Language, 41*(2), 183–194. doi: 10.1055/s-0040-1701686.

Treatment. Application of an independent variable (treatment) to induce changes in dependent variables (speech-language skills); use of effective procedure in teaching new communicative skills; behaviorally, management of contingent relations between antecedents, responses, and consequences; conceptually, a rearrangement of communicative relationships between speakers and their listener; it is what the clinician does to teach target behaviors; never to be confused with target behaviors, although such a confusion is common in treatment reports.

Treatment of Communication Disorders: Procedures That Apply Across Disorders. Common procedures used in treating most if not all disorders of communication; modified to suit the individual client's specific problems, needs, and the performance data. To implement a treatment program:

- **Assess the client**
 - Determine the diagnosis

- Describe the strengths and limitations of the client
- Describe the client's current level of communicative performance
- **Evaluate the client's family constellation**
 - Assess the family communication patterns
 - Describe the family support and resources
 - Describe the client's ethnocultural background and the language status (mono- or bilingual)
 - Describe the social, educational, or occupational demands made on the client
- **Select functional, client-specific target behaviors**
 - Consider the client's and the family members' preference for target behaviors
 - Select behaviors that, when treated, will have the greatest effects on the client's communication in social situations
 - Select both short- and long-term targets
 - Define the dismissal criterion
- **Establish the baselines of target behaviors**
 - Select stimuli for evoking the target behaviors; consider the ethnocultural appropriateness of treatment stimuli
 - Repeat the measures to establish reliability
 - Use the procedures described under Baselines
 - Obtain a home sample if possible
- **Design flexible treatment sessions**
 - Use the degree of control and structure that is necessary; initially use more structured sessions
 - Gradually, loosen the structure to make the treatment environment more like the client's everyday environment
- **Write a treatment program**
 - List the target behaviors
 - Describe the treatment procedures
 - Specify the reinforcers and verbal correction procedures
 - Describe the Criteria for Making Clinical Decisions (moving from one level of treatment to another)
 - Describe the Probe, Maintenance Strategy, Follow-Up, and Booster Treatment

- **Implement the treatment program**
 - Initially use the Discrete Trials procedure for teaching most speech and language targets
 - Use objects, pictures, demonstrated actions, and so forth to evoke the target behaviors
 - Give instructions, demonstrations, and explanations
 - Model the target responses
 - Prompt the target responses
 - Use manual guidance to assist the client in producing the target responses
 - Shape the responses
 - Fade the special stimuli, including pictures, objects, modeling, prompts, and manual guidance
 - Reinforce correct responses with verbal praise; use natural reinforcers; if you used tangible reinforcers, fade them; decrease the reinforcement density
 - Verbally correct wrong responses by saying “No” or “Wrong”; use other procedures as found appropriate (time-out, response cost)
 - Do not always start treatment at the lowest level (e.g., simplest phonemes, isolated phonemes, words), unless shaping is needed; start treatment at a level of complexity the individual can handle (more complex phonemes, words instead of isolated phonemes; more complex sentences instead of the simplest forms); drop down a level only if the performance is not sustained at a higher level
 - Probe for generalized production as often as necessary
 - Shift treatment, in progressive steps, to more complex levels as the client meets the probe criterion at each level
 - Train the target behaviors in conversational speech with natural consequences in the final stages of treatment
- **Implement a maintenance program**
 - Train family members, teachers, friends, and professional caregivers in supporting the client’s communication skills
 - Teach the family members to evoke the target behaviors and reinforce the client naturally

- Invite other persons into the treatment sessions
- Shift training to nonclinical settings
- Train family members to conduct informal treatment sessions at home
- Have teachers reinforce the target skills you teach in conversation with the child and integrate those skills into their literacy and other academic work
- Teach the client to self-monitor errors and correct productions
- Teach the client to count relevant behaviors
- Teach the client to self-correct mistakes
- Teach the client to self-cue
- Teach the client to pause soon after an error response is produced
- Dismiss the client when responses are reliably produced in natural settings
- **Follow up the client**
 - Set up a schedule for follow-up
 - Follow up with a client for a duration necessary to document maintenance
 - Take conversational speech samples during follow-up sessions
 - Measure the production of relevant communication skills
 - Recommend Booster Treatment if the skills have deteriorated
- **Arrange for booster treatment**
 - Give the same or better treatment
 - Probe the response rates
 - Schedule the next follow-up if necessary

Hegde, M. N. (2023). *Clinical methods and practicum in speech-language pathology* (7th ed.) Plural Publishing.

Hegde, M. N. (2024). Elemental treatment for all disorders of communication. *Journal of Indian Speech Language and Hearing Association*, 38(1), 1–12.

Treatment of Communication Disorders: A General Sequence That Applies Across Disorders.

Treatment programs may be designed to facilitate step-wise progression in mastering topographic levels of target

skills; response topography, response modes, multiplicity of targets, training and maintenance, and response consequences dictate treatment sequence.

- **Sequence topographic levels of target behaviors**

- Simplify and sequence the target behaviors topographically (shape or form of responses)
- Do not start treatment at a lower level if the client can perform at the higher level (e.g., when the correct production of sounds in words is imitated, do not drop down to the syllable level)
- Experiment to see if a client can manage at a higher level; always start treatment at the highest level possible
- Syllables or words are the simplest
- Phrases are the next higher level
- Sentences that are imitated or otherwise controlled may be the next higher level
- Sentences that are more spontaneously produced (may be prompted) are next
- Sentences that are fully spontaneously produced (not prompted) are higher
- Conversational speech, narratives, and discourse are at the highest level

- **Sequence response modes**

- Begin with the modeling-imitation sequence
- Fade modeling and begin evoked trials

- **Sequence multiple targets**

- Teach the most useful (functional) skills earlier than the more advanced skills
- Teach relatively more complex behaviors before you teach the most complex behaviors; experiment with different levels of complexity
- Teach first those behaviors that are building blocks for other behaviors (e.g., syllables for words and words for phrases and sentences)
- Select a new target or shift training to a more complex level of the same skills under training when one target reaches the probe criterion

- **Sequence clinical training, generalization, and maintenance strategies**

- Initially establish the behavior under structured clinical situations
- Loosen the structure gradually and make treatment conditions more similar to natural conditions
- Shift treatment to more natural conditions in and around the clinic
- Shift training to natural environments away from the clinic
- Train family members to conduct informal training at home
- Monitor the accuracy of home intervention
- **Sequence patterns of reinforcement and verbal corrections**
 - Reinforce every correct response in the beginning, a fixed ratio 1 (FR1)
 - Offer verbal correction for every incorrect response
 - Reduce reinforcer and verbal correction density as the learning stabilizes (shift the ratio gradually to FR2, FR4, FR6, and so forth)
 - Give tangible reinforcers if necessary and only in the beginning; shift to social and more natural reinforcers (verbal praise, agreement, simile)
 - Train others to give typical feedback in naturalistic settings

Treatment of Communication Disorders: Procedural Modifications. Make prompt changes in treatment procedures if the performance data suggest that the method is ineffective or the effect is less than optimum; antecedents (stimuli), responses, and consequences may need to be modified; treatment procedures, not principles, are modified.

- **Modification of treatment stimuli**
 - Change stimuli that are ineffective in evoking the target responses
 - Shift from pictures to objects
 - Shift from line drawings to photographs
 - Shift from abstract to concrete stimuli
 - Shift from pictorial representation to enacted stimuli

- Discard clinical stimuli in favor of stimuli from the client's home
- Model if evoking is not effective
- Prompt if evoking is not effective
- Provide manual guidance (physical assistance to execute a response) if the evoking techniques are not effective
- Give instructions and repeat them
- Ask effective, common questions to evoke the responses
- Rephrase ineffective questions
- **Modification of target responses**
 - Simplify the response if a more complex response proves difficult for the client
 - Shift downward in progressive steps if the target is not produced in sentences
 - Abandon training on a behavior that is too difficult for the client in favor of one that is easier; use the baseline data for guidance
 - Move quickly to a higher level of complexity if the level being trained seems too easy to the client
 - Abandon training on a behavior that is not imitated in favor of the one that is
 - Return to abandoned behaviors at a later date; shape them in small incremental steps
- **Modification of response consequences**
 - Use the operational definition of consequences; to be called reinforcers, events should increase the behaviors on which they are made contingent; corrective procedures should decrease the behaviors on which they are made contingent
 - Change reinforcers that do not increase behaviors
 - Change corrective procedures that do not decrease behaviors
 - Use primary reinforcers if verbal praise and other social reinforcers do not reinforce
 - Switch back to social reinforcers after the behaviors are established
 - Use tokens backed up by a variety of reinforcers; use time-out if other procedures fail

- Use one of the Differential Reinforcement procedures to simultaneously increase a desirable behavior and decrease an undesirable behavior
- Use biofeedback if other forms fail

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Treatment Discrete Trials. These are temporally separated opportunities for target response production; efficient in initially teaching target responses and to measure their frequency for calculating the percent correct responses, but they are not efficient in promoting generalization and maintenance; there are two kinds of treatment trials described here; there are also two kinds of Baseline Discrete Trials and one kind of Probe Discrete Trials.

Treatment Modeled Trials. The sequence of modeling-imitation-response contingent consequences makes it a modeled treatment trial.

- Present a physical stimulus to evoke a response (e.g., a picture, an object)
- Ask a question that evokes a response (e.g., “What is this?”)
- Model the correct response immediately and ask the client to imitate
- Reinforce the correctly imitated responses and verbally correct incorrect responses
- Move the stimulus away from the individual
- Record the response as correct, incorrect, absent, and so forth
- Wait for a few seconds to mark the end of a trial
- Re-present the stimulus and start a new trial
- Stop modeling when the client gives five consecutively correct imitated responses
- Begin evoked trials

Treatment Evoked Trials. Evoking (eliciting) responses with typical means (e.g., questions) and without modeling but with response-contingent consequences makes it a treatment evoked trial.

- Present a physical stimulus to evoke a response (e.g., a picture, an object)
- Ask a question that evokes a response (e.g., “What is this?”)
- Reinforce the correct responses and verbally correct incorrect responses
- Move the stimulus away from the individual
- Record the response as correct, incorrect, absent, and so forth; use a recording sheet
- Wait for a few seconds to mark the end of a trial
- Re-present the stimulus to start a new trial
- Re-introduce modeled trials if the client gives incorrect responses on three to four evoked trials
- Fade modeling after five correctly imitated responses and reintroduce evoked trials
- Consider an exemplar trained when the client gives 10 consecutively correct evoked responses (training criterion)
- Train five to seven exemplars to the training criterion
- Calculate the percentage of correct evoked responses on the training trials
- Administer probe trials

Treatment Efficacy Research. See Evidence-Based Clinical Practice.

Treatment Targets. Skills or behaviors selected for teaching; they are what the clients do in therapy, as against treatment, which is what the clinicians do; successful treatment depends on careful selection of target behaviors; appropriately selected target skills will ensure improvement in clients’ social communication and promote generalization and maintenance:

- Select treatment targets after a thorough assessment
- Select operationally defined functional targets that are useful to the client; consult with the significant others in the client’s life (teachers, family members, colleagues) to select target skills that will be effective in family, social, educational, or employment contexts

- Select targets that are verbally and culturally appropriate to the client; select targets only after understanding the client's and the family's ethnocultural background
- Select skills that can make an immediate and socially significant difference in the communication skills of the client; integrate verbal and written academic or occupational communication skills with treatment at appropriate stages
- Select behaviors that serve as building blocks for more complex functional skills
- Select social communication skills produced in natural environments as the final clinical target
- Select skills that when expanded and reinforced in social contexts will improve the client's quality of life

Treatment Versus Stimulation. Whether to actively and directly treat skills or to offer indirect stimulation programs can be a dilemma in some contexts, especially for infants and toddlers; see [Stimulation Versus Treatment](#); see also [Language Disorders in Infants and Toddlers](#); in most cases, direct treatment, if possible, is preferred to stimulation—a generally inefficient approach because of unspecified target behaviors and vague intervention procedures.

Treatment Variables. Technical operations performed by the clinician to create, increase, or decrease behaviors; not to be confused with treatment targets, although such a confusion is common; what the clinician does is treatment, and what the client does is the target; treatment variables the clinician manages include:

- Antecedents or stimuli used in treatment, including instructions, demonstrations, modeling, prompts, cues, manual guidance, pictures, objects, re-created events, storytelling (by the clinician), topics of conversation, and so forth
- Consequences or feedback the clinician gives, including verbal praise, tokens, tangible reinforcers, opportunities to indulge in various activities, privileges offered by parents, verbal correction, time-out, token loss, and so forth

Trials. Measurable and sequentially repeatable opportunities to produce a response; may be more or less structured; see [Discrete Trials](#); [Baseline Modeled Trials](#) and [Baseline Evoked Trials](#) described under [Baselines](#); [Treatment Modeled Trials](#); and [Treatment Evoked Trials](#).

Turn Taking. To teach the alternating roles of speaker and listener during conversational exchanges, see [Language Disorders in Children](#).

U

Unconditioned Reinforcers. Reinforcers whose effects do not depend on past learning or conditioning (e.g., food and drink); the same as Primary Reinforcers; see also Conditioned Reinforcers; essential to teach mands (requests) for food and drink.

Unconditioned Response. A response given to unconditioned stimulus; typically a response without a conditioning or learning history (e.g., salivary response to food in the mouth); see Conditioned Response.

Unconditioned Stimulus. A stimulus that elicits a response without the benefit of conditioning or learning (e.g., food in sight or in the mouth that automatically elicits a salivary response); see Conditioned Stimulus.

Unilateral Upper Motor Neuron Dysarthria. To treat this type of motor speech disorder, see Dysarthria and Treatment of Dysarthria: Specific Types.

v

Validity. Validity of standardized tests is the degree to which the tests measure what they purport to measure; see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for validity of standardized tests; validity of treatment research data is the assurance that treatment procedures, not some extraneous variables, produced the claimed effects; validity of treatment procedures is roughly the same as generality; treatment procedures with multiple kinds of validity are preferred to those with no or limited kinds of validity; experimentally demonstrated treatment generalities include:

- **Inferential (statistical) generality:** Assurance that a treatment researched in a group design study may be applied to individuals in the population with the same disorder
- **Logical generality:** Assurance that a treatment researched in a single-subject design study may be extended to the clients who are similar to the study participants
- **Generality across individuals:** Assurance that a method is effective when applied to other individuals with the same disorder
- **Generality across settings:** Assurance that a method is effective when applied in different clinical or educational settings
- **Generality across clinicians:** Assurance that a method is effective when different clinicians apply it
- **Generality across response classes:** Assurance that a method is effective across different kinds of skills or across disorders

Hegde, M. N., & Salvatore, A. P. (2019). *Clinical research in communication disorders* (4th ed.). Plural Publishing.

Vascular Dementia. Use the procedures described under Dementia to treat communication disorders associated with the following three common types of vascular dementia:

- **Large vessel occlusions.** To treat this type of dementia:
 - Target behavioral impairments, including confusion, in the early stage of the disease

- Treat inconsistent memory impairments that may range from total difficulty to recall an event and sudden recall of the same event
- **Lacunar state.** To treat dementia due to lacunar states:
 - Treat Dysarthria
 - Treat other communication problems characteristic of Dementia, especially in the late stage of the disease
- **Binswanger's disease.** To treat dementia associated with this disease:
 - Target symptoms that are similar to those found in patients with lacunar states
 - Target symptoms of Dementia in the later stage of the disease

Hegde, M. N. (2022). *A coursebook on aphasia and other neurogenic language disorders* (5th ed.). Plural Publishing.

Velopharyngeal Dysfunction (VD). Communication-based treatment is not appropriate for certain kinds of VDs; various terms are used interchangeably and confusingly; velopharyngeal insufficiency, inadequacy, incompetence, dysfunction, and mislearning are all in use; if the dysfunction is due to structural limitations (insufficient muscular mass of the velopharyngeal mechanism), behavioral methods of treating the voice consequences are not recommended; intervention may include prosthetic and surgical treatment for children with clefts, those with dysarthria, and those who present voice and resonance disorders; post-surgery or prosthetic fitting, SLPs may treat hypernasality, hyponasality, denasality, cul-de-sac nasality, nasal air emission, weak consonant productions, and compensatory articulation errors; if functional or mislearning (as in children with cleft palate) is confirmed, behavioral intervention may be successful; see cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for differential diagnosis

Treatment for VD: General Guidelines

- If atypical, inadequate, or insufficient mechanism is the sole cause of poor velopharyngeal closure, speech and resonance treatment is unwarranted, as it is likely

to be ineffective; other kinds of intervention (e.g., surgical or prosthetic) are a priority

- Communication treatment and training in using the new structures or prosthetic devices in place may maximize the benefits of surgical and prosthetic interventions
- Compensatory speech treatment may be offered to individuals with physiologically inadequate mechanism only if no other intervention is possible; in such cases, the clinician may try to teach compensatory skills that might minimize the resonance disorders caused by an untreatable velopharyngeal problem
- Reduction in hypernasality and audible nasal emission as well as correct speech sound production may be targeted in children with velopharyngeal inadequacy associated with palatal clefts
- When hypernasality and nasal air emission are mild, and it is therefore difficult to judge the candidacy for speech treatment, it is acceptable to offer trial speech treatment with a clear understanding given to the clients and the families that the treatment is purely exploratory or diagnostic in nature and that it may have to be discontinued if no progress is evident
- Trial treatment also may be offered to clients who show minimal hypernasality and audible nasal emission, which is evident only when fatigued, and the problems are inconsistent (i.e., sometimes there seems to be good closure); it is likely that in such cases, treatment will be successfully concluded
- Any time trial therapy is offered, the clinician should make frequent evaluations of the effect to stop treatment when continuation appears unethical
- Treatment for articulatory placement may be offered any time to any client, even if there is significant velopharyngeal insufficiency, causing prominent hypernasality and audible nasal emission; clients who are waiting for surgery or have not yet been evaluated for it may still be scheduled for articulatory placement therapy but not for resonance therapy; velopharyngeal

insufficiency may lead to compensatory misarticulations but will not prevent learning of correct articulatory placements

Treatment Strategies for VD

- Offer continued treatment for hypernasality, audible nasal emission, or both when there is expectation of improvement because of success during trial therapy
- Treatment techniques include those described under Voice Disorders; essentially, modeling, shaping, and positive reinforcement using words, phrases, and sentences with only oral speech sounds may be effective provided the velopharyngeal mechanism is capable of closure; encourage greater degree of mouth opening to increase oral resonance
- Be cautious in using many techniques promoted to treat speech problems resulting from velopharyngeal closure deficiencies because they have not been experimentally evaluated
- Do not use nonspeech motor exercises that include blowing, sucking, whistling, cheek puffing, swallowing, and playing wind instruments as they are ineffective
- Use such other tactics as increased mouth opening, light articulatory contacts, and decreased rate of speech only experimentally, as their outcomes have been questionable
- **Other potentially useful approaches include the following:**
 - **Temporary speech prostheses:** Have a speech bulb prepared for the client and evaluate its use as a temporary means of closing the velopharyngeal gap; this requires a nasopharyngoscopy examination to properly construct, modify, and fit the prosthetic bulb; evidence on the effectiveness of speech bulbs in reducing hypernasality is weak, and therefore be cautious about its recommendation
 - **Mechanical feedback techniques:** Biofeedback with the help of a nasometer and similar instruments might help shape a more appropriate resonance

pattern; nasal endoscopic feedback gives the client a chance to observe the closure during continuous speech production

Kummer, A. (2011). Types and causes of velopharyngeal dysfunction. *Seminar in Speech and Language*, 32(2), 150–158.

Smetona, J. T. et al. (2024). What's new in cleft palate and velopharyngeal dysfunction management: An update. *Plastic and Reconstructive Surgery*, 154(2), 378e–390e.

Zajac, D. J., & Vallino, L. D. (2026). *Evaluation and management of cleft lip and palate* (2nd ed.). Plural Publishing.

Ventricular Dysphonia. In treating this type of voice disorder characterized by low pitch, monotone, decreased loudness, harshness, and arrhythmic voicing, teach the client to:

- Take a prolonged inhalation through the open mouth and sustained exhalation without phonation
- Practice inhalation phonation (this is usually true vocal fold phonation) and exhalation phonation on the same breath
- Practice exhalation phonation
- Vary the pitch
- Stabilize phonation in conversational speech without inhalation-exhalation

Verbal Correction. Use this essential component of all treatment procedures.

- Verbally correct incorrect responses immediately by saying “No,” “Wrong,” or “That is not correct,” “Try again,” and so forth
- Correct or stop the client when an incorrect response seems imminent
- Give Nonverbal Corrective Feedback when appropriate (gestures that show disapproval of a response)
- Correct in an objective tone
- Give Mechanical Corrective Feedback or Biofeedback whenever possible
- Measure the frequency of incorrect responses to see if the verbal correction and mechanical feedback are effective

- Minimize the use of verbal or other types of correction by modeling and generously reinforcing the correct responses, and by Shaping complex skills

Verbal Praise. A common and effective method of positive reinforcement to strengthen correct target responses.

- Praise the client for imitating or more spontaneously producing correct responses
- Use such phrases as “Good Job!” “I like that!” “That was correct!” “Excellent!” and so forth; select the phrases and words that are appropriate for the client’s age, education, and cultural background
- Deliver verbal praise in a natural manner, with appropriate emotion and facial expression; a common mistake is robotic praise
- Keep an accurate record of response frequency to document that the verbal consequences are indeed functioning as reinforcers (i.e., the response rate is increasing)
- Use other forms of reinforcers if verbal praise does not work (e.g., tokens or primary reinforcement)

Verbal Prompt. Use this effective technique to evoke an unsure response; they are similar to *hints* of everyday life and *verbal cues* in aphasia treatment; commonly used in the treatment of communication disorders; see Prompts.

Visual Alerting Systems. A variety of assistive listening devices for persons with severe hearing loss; generate visual signals (light) that alert the person to incoming phone calls, door bells, and smoke alarms; see Aural Rehabilitation.

Vocally Abusive Behaviors. Several kinds of verbal actions that negatively affect the larynx and result in voice disorders; for intervention, see Voice Disorders; Treatment of Vocally Abusive Behaviors.

Vocal Emphasis. Use this effective treatment technique in which the target feature within a word, phrase, or sentence is highlighted in some manner while modeling it:

- In speech sound production training, emphasize the target phoneme with extra stress, increased loudness, or a slightly slower rate
- In language treatment, emphasize the modeled grammatical morphemes or other targets with increased stress, loudness, or slight prolongation of the vowels

Vocal Fold Paralysis. Treatment of voice disorders associated with vocal fold paralysis requires a determination of whether the paralysis is bilateral or unilateral; respiratory survival and feeding are the main concerns when the paralysis is bilateral; unilateral vocal fold paralysis is more common, often due to trauma or accidental cutting of the recurrent laryngeal nerve, and the resulting aphonia or dysphonia may be treated; Teflon may be injected into the paralyzed fold to make it bulge and help approximate; although appropriate, voice therapy in the case of unilateral vocal fold paralysis may only produce temporary effects, and the goals may be limited to having the client produce functional voice:

- Model and reinforce a higher-pitched voice that might improve the voice
- Use the Half-Swallow Boom
- Use Digital Manipulation of the Larynx
- Try different Head Positioning maneuvers

Vocal Hyperfunction. Treatment of vocally abusive behaviors that cause nodules, polyps, and associated voice disorders is the bulk of routine voice therapy; *hyperfunction* is speaking with excessive muscular effort and force, which must be reduced during treatment.

- Assess the specific kinds of vocally abusive behaviors the client exhibits
- Reduce those behaviors; see Voice Disorders; Treatment of Vocally Abusive Behaviors
- Experiment with different Specific Normal Voice Facilitating Techniques (described under Voice Disorders) to promote normal or vastly improved voice
- Use those techniques to teach the client a more relaxed and normal voice production

Vocal Nodules. Treatment of voice disorders resulting from benign lesions of the vocal folds targets a reduction of hoarseness, harshness, periodic aphonia, frequent throat clearing, hard glottal attacks, vocal tension, and a dry vocal tract; voice therapy designed to reduce vocally abusive behaviors is preferred over surgical intervention; see [Treatment of Vocally Abusive Behaviors](#) under [Voice Disorders](#); essentially:

- Prescribe mandatory vocal rest if surgery has been performed to remove the nodules
- Reduce vocally abusive behaviors; see [Voice Disorders; Treatment of Vocally Abusive Behaviors](#)
- Increase the breath support for speech
- Reduce vocal intensity
- Use [Specific Normal Voice Facilitation Techniques](#) (described under [Voice Disorders](#)) to teach the client to produce a healthy voice

Voice Disorders. Speech-language pathologists often collaborate with laryngologists who treat medical conditions associated with voice disorders; to fully understand voice disorders and make a diagnostic assessment, see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*. Generally, treatment targets include:

- **Disorders of loudness.** Socially inappropriate voice that is too loud or soft.
- **Disorders of pitch.** Voice characterized by inappropriate pitch; speech at the low end of one's pitch range, which requires too much effort and force; or speech at the high end of the range, which causes fatigue.
- **Disorders of resonance.** Voice characterized by inappropriate resonance, including [Hypernasality](#) and [Hyponasality](#).
- **Disorders of vocal quality.** Voice that is characterized by varying degrees of breathiness, hoarseness, harshness, and pitch and loudness deviations.

Treatment of Voice Disorders: General Principles

- Consider the goal of voice therapy as a normal-sounding voice with little or no effort and tension
- Maintain a cooperative working relationship with laryngologists, neurologists, and other medical professionals
- Have a good knowledge of laryngeal anatomy and physiology; understand the surgeries of larynx and related structures, medication to treat various voice disorders or underlying diseases, and the effects of all medical and surgical procedures on, and interactions with, voice production and treatment
- Have medical evaluations done before starting voice therapy and periodically during the course of voice therapy
- Combine techniques designed to reduce vocally abusive behaviors with those that facilitate efficient and normal voice production
- Individualize the facilitating techniques because what works with one client may not work with another
- Reduce vocally abusive behaviors along with teaching healthier (adaptive) vocal behaviors
- Work closely with parents of children and spouses or other family members of adults to help sustain healthy vocal behaviors at home

Treatment of Vocal Loudness Disorders. Loudness problems that need to be modified include socially inadequate soft and loud voice characteristics; treatment for both is Shaping, although the movement to the target is in opposite directions.

- Treatment of Excessively Soft Voice.*** Shape progressively louder voice in a client with too soft a voice:
- Rule out such physical problems as hearing loss and vocal fold pathology as the cause of the excessively soft voice
 - Baseline vocal loudness with an audiorecorded speech sample to give feedback to increase the client's awareness of excessively soft voice

- Instruct, demonstrate, and model voice at different loudness (e.g., role-play whispered voice, a quiet voice, normal conversational voice, loud and louder voice); reinforce acceptable imitative responses and verbally correct off-target responses
- Start with words and phrases and move on to conversational speech; reinforce speech produced at progressively higher intensity until the voice is produced with a desired loudness
- Experiment with pitch changes to see if a slight increase in the pitch level might produce a noticeably louder voice; if so, have the client practice speaking at that pitch level
- Ask the person to take deeper breaths that might increase subglottic air pressure and vocal intensity; if so, reinforce speaking with increased air pressure on repeated trials
- Use the auditory masking on an audiometer to induce the Lombard effect (reflexively increased vocal intensity under auditory masking); fade the noise gradually to maintain louder speech
- Shape a progressively louder voice by reinforcing it until an acceptable level is reached and stabilized
- Use such biofeedback instruments as the Visi-Pitch™ and any available computer programs for voice therapy to shape the desired vocal pitch
- Use the Changing Criterion method to shape up the desired vocal loudness
- Use the Pushing Approach (described later in this section under Specific Normal Voice Facilitating Techniques) only if instruction, modeling, and biofeedback fail

Treatment of Excessively Loud Voice. Shape progressively softer voice in a client who speaks too loudly:

- Use the shaping-down procedure to progressively lower the vocal intensity from the baserate level; alternatively, use the shaping-up procedure by beginning with a very soft voice (well below the baseline) to progressively increase vocal intensity to the desired level

- Use instructions, demonstrations, modeling, positive reinforcement, and verbal correction as treatment methods
- Use the Changing Criterion method to shape down the vocal loudness

Treatment of Vocal Pitch Disorders

Procedure to Raise the Baseline Pitch

- Baseline the client's vocal pitch; increase the client's awareness of pitch by listening to an audiorecorded speech sample
- Instruct the client on pitch, its variations, acceptable range; select the target pitch the client agrees with
- Model different levels of pitch for the client
- Experiment to see if the client can produce a desirable pitch, even if briefly
- Audiorecord the client's desirable pitch and use it as a model for self-imitation
- Provide a live model or a mechanical model on such computerized instruments as Visi-Pitch
- Use instrumental and instantaneous feedback during training trials or durations; supplement mechanical feedback with verbal praise and verbal correction
- Begin treatment with single words, preferably those that begin with vowels; have the client produce them with the desirable pitch
- Shift training to progressively higher levels of words, phrases, sentences, and conversational speech
- Use the Changing Criterion method to shape up the pitch
- Use oral reading to stabilize the desired pitch

Procedure to Lower the Baseline Pitch

- Use the same procedures used for raising the pitch except for shaping down the pitch to a lower targeted level
- Lower the pitch in carefully graded steps, if necessary
- Use the Changing Criterion method to shape down the pitch level

Treatment of Resonance Disorders

General Guidelines

- Modification of hyper- and hyponasality are often the treatment targets
- Do not offer behavioral voice therapy for individuals with cleft palate or congenital palatopharyngeal inadequacy until the organic problems are eliminated or significantly improved by surgical or prosthetic means; see Velopharyngeal Dysfunction for additional information on treating resonance disorders
- Work with a prosthodontist who might fabricate and fit a prosthetic device that may temporarily help close the velopharyngeal port (e.g., a speech bulb)
- Assess speech before and after surgical and prosthetic treatment
- Use biofeedback instruments to monitor and reinforce appropriate oral and nasal resonance

Treatment of Hypernasality

- Assess Hypernasality and treat it only when there is adequate or at least marginal velopharyngeal adequacy and a behavioral management is a good option
- Use treatment procedures described under Velopharyngeal Dysfunction

Treatment of Hyponasality

- Assess Hyponasality and treat it only when an otolaryngologist has ruled out a physical or physiological cause for it; be aware that very few persons exhibit behavioral (functional) hyponasality with no physical basis
- Use treatment procedures described under Hyponasality

Treatment of Vocal Quality Disorders. The typical individual who seeks voice therapy has an abnormal voice quality of breathiness, harshness, hoarseness, and tensed voice because of vocally abusive behaviors; voice therapy in these cases is aimed at reducing such maladaptive behaviors; the therapy is entirely behavioral.

Treatment of Vocally Abusive Behaviors

- Educate the client and the family about the harmful results of vocally abusive behaviors; use charts and diagrams to show vocal nodules and how they affect voice
- Point out the relationship between abusive behaviors, the resulting damage to the vocal folds, and the consequent voice disorders
- Take a detailed case history to list the vocally abusive behaviors and the conditions under which they are most likely to occur (e.g., a child's noisy vocal behaviors while playing with certain toys; an adult's shouting and screaming at the sports stadium); discuss these with the client and the family members
- Make a hierarchical list of vocally abusive behaviors to be reduced
- Ask the client to establish the baselines of vocally abusive behaviors in natural settings; give them a prepared recording sheet; a family member may assist the client
- Ask the parents of young children to count and graph vocally abusive behaviors on a daily basis
- Target for initial elimination the most frequent and/or most damaging vocally abusive behaviors; go down the hierarchy as the more troublesome behaviors are reduced in frequency, using the procedures that follow
- **Reduce vocally abusive behaviors**
 - Use the Changing Criterion method to shape down the frequency of vocally abusive behaviors; set an initial criterion that is just below the baserate number so it is easily met; examples:
 - For an adult with a baserate of 10 shouts/screams a minute inside a sports stadium, set an initial criterion of no more than 7 such abusive episodes; subsequently, set progressively lower criteria of 5, 3, and zero episodes; have an independent observer, a family member, or a friend count the episodes in the stadium

- For a woman whose baserated on-the-phone talking is 50 minutes, reduce the duration in 5-minute decrements; have her or a family member log the number and duration of calls
- For a child who typically plays noisily with toys for 30 minutes on average, have a parent model and reinforce quiet play or less noisy play initially for 5 minutes and increase the duration of quiet play in small increments (8 minutes of quiet play, increased to 12 minutes, 15 minutes, and so forth); encourage the parents to buy quieter toys and reinforce their children for playing with them
- Similarly, decrease the frequency of visits to noisy bars, frequency of throat clearing, excessive conversation with friends, hard glottal attacks, and so forth
- Ask the client, parents, spouse, teacher, or a friend to help establish the reliability of measures of vocally abusive behaviors in the natural environment; give them a prepared recording sheet to document the occurrence of such behaviors before, during, and at the conclusion of treatment
- Eliminate a condition under which a vocally abusive behavior is highly likely (e.g., have the parents buy toys with less aggressive themes; gradually reduce the frequency of play involving toys that promote vocal abuse; in the case of an adult, encourage the client to avoid noisy and smoke-filled bars where the client and conversational partners tend to talk loudly and excessively)
- Implement a token system for reinforcing the child for having met the criterion of reduced frequency, duration, or both (of a specific vocal abuse); in the case of adolescents and adults, have a family member reinforce the vocally healthy behaviors by offering a treat of some kind that is likely a reinforcer
- **Teach healthier vocal behaviors**
 - Initiate speech sounds softly

- Speak with optimum pitch
- Speak at an appropriate loudness
- Reduce the frequency of coughing or throat clearing
- Breathe through the nose
- Use an easy, relaxed breathing pattern when speaking
- Speak with relaxed speech muscles
- Open mouth more widely during talking
- Whole-body relaxation
- Work with parents, teachers, friends, and others to monitor the production of vocally more appropriate behaviors (e.g., gentle vocal onset, soft speech)
- Make a periodic assessment of voice if and when the medical or surgical treatment is repeated
- Measure the effects of the program on targeted voice quality
- Refer the client to a laryngologist for periodic examinations to assess the effects of your behavioral therapy on any structural deviations that may be present (e.g., nodules)
- Follow up with the client and arrange for booster treatment

Sapienza, S., & Ruddy, B. F. (2021). *Voice disorders* (4th ed.). Plural Publishing.

Stemple, J. C. et al. (2024). *Clinical voice pathology* (7th ed.). Plural Publishing.

Specific Voice Treatment Techniques or Programs. *Specific techniques* target a particular vocal behavior and tend to have a narrower focus; *treatment programs* may be broader and may address multiple vocal targets; many voice treatment techniques and programs are based on clinical experience; there is a need for experimental evidence to support their general practice; use them with caution; establish reliable baselines, measure changes in the target vocal behaviors in all treatment sessions, and abandon or modify procedures to obtain desirable outcomes; under each strategy, instruct, demonstrate, and model target behaviors; reinforce correct responses and verbally correct incorrect responses.

Auditory Feedback. Providing feedback of client's own voice to help improve its quality, pitch, or intensity; amplified real-time feedback may be the most effective in modifying speech quality.

- Use an instrument such as the Kay Facilitator (Kay Elemetrics), which provides real-time amplified feedback of the client's voice through headphones; also provides delayed auditory feedback, loop feedback, metronomic pacing, and masking noise
- Ask clients to listen carefully to their own voice as the feedback is provided; teach them to use a different voice facilitation approach (e.g., chanting or glottal fry)
- Ask the client to pay attention to improvement in voice quality when a facilitating approach is used; contrast this with the typical deviant voice of the patient

Chant-Talk Method. A manner of speaking in which spoken words are connected, with even stress on syllables, prolongation of sounds, and soft glottal attack; recommended for clients with hyperfunctional voice including Hard Glottal Attacks; ask the person to:

- Reduce the effort required to speak
- Listen to recorded chant; imitate it
- Imitate your modeled chant
- Read aloud by alternating the chant voice with the regular one
- Read for 20 seconds at a time
- Listen to and differentiate chanted and normal voice on patient's recorded oral reading samples
- Fade the chant and maintain normal conversational voice without the chant

Chewing Technique. Requires individuals to imagine that they are chewing food while voicing; recommended for reducing vocal hyperfunction, improving voice quality, reducing vocal stress, and increasing oral resonance; describe and justify the procedure and let the client face a mirror along with you; teach the client to:

- Simulate the act of chewing some food

- Exaggerate open-mouth chewing motions
- Pretend to move the food from one side of the mouth to the other
- Phonate various sounds softly by constantly moving the tongue around in chewing motions
- Say words while chewing
- Chew and count
- Chew and produce connected speech
- Gradually reduce the chewing movements and talk in a more typical manner (fading)

Digital Manipulation of the Larynx. Physical manipulation of the larynx to promote desirable voice quality; may be used to reduce vocal pitch and decrease laryngeal tension.

- **Use digital pressure to lower the pitch**
 - Ask the patient to prolong a vowel
 - Apply slight finger pressure to the thyroid cartilage as the vowel is prolonged (the pitch will drop)
 - Fade the digital pressure and let the client practice the lower pitch
- **Lower the larynx to reduce tension**
 - Apply a slight downward pressure with the middle finger and the thumb just above the thyroid notch
 - Ask the client to prolong vowels with the larynx in the lowered position
 - Fade the downward pressure on the larynx

Glottal Attack Modifications. This strategy includes two opposite manipulations: (1) reduce hard glottal attacks to treat vocal abuse or (2) teach slightly harder glottal attacks to improve vocal fold approximation in individuals with vocal paralysis.

- **Reduce hard glottal attacks**
 - Educate your clients with symptoms of vocal abuse about harmful effects of hard glottal attacks
 - Model and demonstrate hard and soft glottal attacks; let the clients listen to their own audiorecorded hard glottal attacks to increase awareness of the problem

- Begin training on soft initiation of voice with monosyllabic words that contain the /h/ in the initial position; progress to initiating the /h/ words softly, words containing initial unvoiced consonants, and then with words containing vowels
- Shape gentle onset with a whisper as the initial response; ask the client to whisper the initial vowel and increase the intensity gradually until a soft phonation is heard
- Use such other procedures as Yawn-Sigh and Chant-Talk
- Teach simultaneous chewing and chanting
- Increase the response complexity gradually until the client produces conversational speech with soft glottal contacts
- **Teach slightly harder glottal attacks**
 - Educate your clients with vocal fold paralysis about producing voice with more effort (slightly harder glottal attacks)
 - Demonstrate and model harder glottal attacks by producing such words as *pop*, *peep*, *bob*, and *beet* in a sudden plosive manner; ask the patient to imitate them and reinforce imitated or approximated productions
 - Provide visual feedback of soft and harder glottal attacks on a computer screen
 - Use the Pushing Approach (described in a later entry) to induce harder glottal attacks
 - Increase the response complexity gradually to induce better approximation of vocal folds in conversational speech

Glottal Fry. A normal voice characteristic useful in treating voice disorders associated with vocal nodules, polyps, fold thickening, functional dysphonia, ventricular phonation, and spasmodic dysphonia; produced with relaxed vocal folds, minimal vocal fold tension, and minimal airflow; after letting the client know that relaxed vocal folds are better able to approximate

despite the nodules and other masses on them, teach the client to:

- Say /i/ softly after a slight exhalation
- Produce deep sounds with a slow series of pops by “opening the throat”
- Produce the same tone on inhalation to see whether the fry is better produced on inhalation or exhalation because individuals differ; alternate between inhaled and exhaled phonations
- Say words like *on* and *off* and *in* and *out* while producing the fry; audiorecord the client’s productions that will serve as a model
- Produce phrase or short sentence productions in the fry mode; ask the client to say, “Easy does it,” “Squeeze the peach,” or “See the eagle” and point out the improved voice quality to the client
- Practice speech in the fry mode at home
- Reduce the frequency of glottal fry as it should not be habitual

Head Positioning. Manipulation of head positions to promote better voice quality is recommended especially for clients with neurological disorders, including dysarthria; may be used with clients who have hyperfunctional voice.

- Give instructions, model different head positions, demonstrate their effects on voice, and justify the procedure to the client
- Experiment with different head positions to find the one that promotes better voice (e.g., head rotated toward left or right; neck flexed downward with the face looking down); ask the client to produce some prolonged vowels (e.g., /i/, /I/, /o/, or /u/) as different head positions are tried to identify the best position
- Ask the client to hold the head position that helps produce the best voice and produce words, phrases, and sentences
- Gradually fade the unusual head position into a more typical position

Inhalation Phonation. A technique of voice therapy designed to evoke true vocal fold vibrations in clients who are aphonic or those who exhibit ventricular phonation; teach the client to:

- Raise shoulders, inhale, and phonate a high-pitched hum
- Raise shoulders, inhale, phonate the high-pitched hum, and lower the shoulders; exhale and produce the same sound; repeat
- Produce inhalation phonation
- Produce inhalation and exhalation phonation with corresponding shoulder movements
- Move from the high-pitched voice to the exhaled low-pitched voice
- Fade the shoulder movements
- Have the client practice single words until a normal-sounding voice is stabilized
- Move on to more complex responses (phrases and sentences)

Laryngeal Massage. The laryngeal area is gently massaged and manipulated to improve vocal quality in functional voice disorders with a high-positioned larynx and neck tension during speech; the Yawn-Sigh is another option for these clients.

- Ask the client to prolong vowels as you massage the larynx
- Encircling the hyoid bone with the thumb and the middle finger, make a circular motion with your fingers while applying light pressure
- Repeat the procedure, working the fingers posteriorly from the thyroid notch
- Repeat the procedure at the posterior borders of the thyroid cartilage
- Gently move the larynx in downward and lateral directions by placing the fingers over the superior borders of the thyroid cartilage
- Reinforce improved voice production

Lee Silverman Voice Treatment. Originally developed to treat the voice and speech disorders associated

with hypokinetic dysarthria of Parkinson's disease, it may be used to treat voice disorders in other individuals; because you need specialized training and certification to use, only an overview is offered here; the method involves:

- Intensive treatment (16 one-hour sessions over 4 weeks); may be even three times a week over 8 weeks
- Increased vocal effort and breath support for voice and speech that results in louder, clearer speech and voice
- Instructions to “think loud, think shout”
- Shaping loudness in sustained vowel production and gradually increasing response complexity
- Periodic recalibration of the client's vocal loudness level
- Quantification of loudness improvement over time

Masking. A technique of voice therapy to treat clients with functional aphonia and those with poor voice quality because of inadequate auditory monitoring of one's own voice; masking noise introduced through headphones.

- Use a standard audiometer to introduce masking noise; do so without any explanation
- Ask the client to read orally; turn the masking on and off for brief periods
- Audiorecord the client's reading to document possible changes in voice quality or the emergence of voice in the whispering aphonic patient
- Playback the recorded sample to demonstrate improved voice quality or initiation of voice in an aphonic person; contrast voice with and without masking
- Ask the client to match improved voice or emergence of phonation without masking
- Have the client read aloud under masking and, as the voice improves or phonation emerges, abruptly end masking; repeat this process until the client can sustain the gains
- Positively reinforce even the slightest improvement

Nasal/Glide Stimulation. Use nasal glide consonants to promote better voice quality in clients with functional dysphonia, spasmodic dysphonia, and voice problems associated with vocal fold thickening, nodules, and polyps; words loaded with nasal glides need less effort to produce.

- Prepare a list of words with nasal consonants in them (e.g., *man, moon, many, morning, many men, moon man, morning singing*)
- Prepare also a list of words that are repeated with /a/ inserted between repetitions (e.g., *man a man a man* or *wing a wing a wing*; use this list as an alternative)
- Prepare a list of words with glide consonants /r/ and /l/ combined with nasal sounds (e.g., *marrow, married women, only lonely memory, Laura ran around*)
- Prepare a list of words with /l/ and /r/ that are repeated with /a/ inserted between each repetition (*lee a lee a lee* or *rah a rah a rah*)
- Have the client practice the selected words; reinforce improved voice quality
- Gradually move on to words, phrases, and sentences containing other consonants to improve voice in meaningful, spontaneous speech

Open-Mouth Approach. Oral openness during speech to increase oral resonance, reduce speaking effort, induce more relaxed speech, and promote appropriate loudness, pitch, and quality of voice.

- Give feedback on lack of mouth opening during speech; use a mirror if necessary
- Model greater versus reduced oral openness; use a puppet (greater mouth opening); contrast that with the speech of a ventriloquist (minimum mouth opening)
- Ask the client to imitate the two ways of modeled speaking
- Have the client tilt the head down and speak
- Teach self-monitoring oral openness in natural settings

- Let the client practice speech with oral openness and reinforce for doing so

Pushing Approach. A voice therapy technique to promote better approximation of vocal folds; appropriate for increasing vocal loudness.

- Instruct and demonstrate pushing
- Ask the client to push down on the arm of the chair or push up by trying to lift the chair by gripping the bottom of the seat while seated
- Ask the client to phonate and push simultaneously
- Reinforce the louder voice that typically results
- Increase the length of utterances with the louder voice
- Fade pushing

Relaxation Training. A method to teach deep muscle relaxation with or without the help of biofeedback (e.g., electromyographic feedback); recommended for clients with excessive muscle tension, anxiety, and stress.

- Use biofeedback instruments
- Give instructions to contract and relax muscles if no instruments are used
- Ask the client to alternately tense and relax the targeted muscles (e.g., shoulder, neck, and jaw muscles)
- Select facial, neck, and head muscles for relaxation training; ask the client to alternatively relax and tense target muscles to appreciate the difference
- Manipulate head positions to induce relaxation
- Ask the client to imagine speaking situations that induce greater tension and immediately let the client relax the speech muscles
- Use relaxing head movements (positions) if necessary
- Use other appropriate techniques (e.g., gentle phonatory onset) in combination with relaxation
- Stabilize a relaxed speaking posture and improved voice quality

Respiration Training. Teaching clients to manage inhalation-exhalation cycles optimally for the pur-

pose of phonation and sustained vocalization; recommended for clients with functional voice disorders who do not use their breath stream properly to produce voice.

- Describe the relation between breathing and speaking and between airflow and vocal fold vibrations; model and reinforce all correct responses
- Teach the client to inhale more quickly, more deeply than usual but exhale more slowly and in a controlled manner; to extend exhalation, ask the client to count slowly first to 5 and then to 10
- Ask the client to prolong vowels to teach controlled and prolonged exhalation that would better support speech; in progressive steps, teach the client to prolong a vowel for about 20 seconds
- Teach the client to inhale quickly between utterances
- Teach good posture to promote normal airflow management

Tongue Position Modification. Manipulating tongue position in the oral cavity to affect changes in voice quality and resonance; tongue positioned too far back results in cul-de-sac resonance, whereas carried too far forward in the mouth creates “thin voice,” giving the baby-talk effect.

- Teach clients to carry the tongue in its neutral position
- Modify the excessively backward or forward tongue positions
- Instruct, model, demonstrate, and reinforce correct tongue positions

Vocal Function Exercises. Designed for both hyper- and hypofunctional voice disorders; includes a series of four exercises the clients perform at home, twice a day, to improve voice production in clients of all ages; using their soft voice, teach the clients to:

1. Sustain the vowel /i/ (ee) as long as possible on a musical note (above middle C for females and boys and below middle C for males); sustained duration should be similar to the longest sustained /s/

2. Glide from the lowest note to the highest note on the word “knoll” without voice breaks and with rounded lips to encourage forward placement of the tone and laryngeal opening; considered a stretching exercise
3. Glide from the highest note to the lowest note on the word “knoll” with the same requirements as in the previous exercise; considered a contracting exercise
4. Sustain the musical notes on “ol” (without the *kn* of “knoll”); this is a low-impact adductory power exercise
 - Give a daily recording sheet to the client to record their home practice sessions (see the cited source)
 - Fade the program from two times a day to one time a week

Vocal Rest. A voice therapy technique that requires little or no talking, typically for 4 to 7 days; vocal rest may be complete or partial; in most cases, absolute vocal rest is impractical.

- Recommend vocal rest for clients who have undergone any form of laryngeal surgery to promote normal healing of the surgical wounds
- Recommend vocal rest as initial treatment for clients who have such types of laryngeal lesions as vocal fold hemorrhage and mucosal tear to let the healing process begin
- Recommend partial vocal rest for clients who have a severe cold (and resulting laryngeal inflammation), vocal nodules, and vocal fold edema; note that partial vocal rest means talking only when absolutely essential and with healthy vocal behaviors
- Instruct the client either to totally avoid or markedly reduce shouting, screaming, singing or humming, whispering, coughing or throat clearing, laughing or crying, and lifting or pushing heavy objects
- Have a family member monitor these activities

- Teach the client to keep a record of such activities
- Teach the client to self-monitor

Yawn-Sigh Method. Designed to treat hyperfunctional voice, using the relaxing effects of the inspiratory yawn followed by an expiratory sigh and phonation.

- Instruct and demonstrate the relaxing effects of prolonged inspiration involved in a yawn and the relaxed phonation that results with a sigh
- Ask the client to yawn and then exhale slowly while phonating lightly
- Ask the client to say words that start with /h/ after each yawn
- Teach the client to produce a gentle, voiced sigh while exhaling
- Teach the client to produce an easy, prolonged, open-mouthed exhalation after each yawn
- Ask the client to skip the yawn and teach the client to inhale normally and exhale a prolonged sigh with the open mouth
- Ask the patient to say “hah” after beginning each sigh
- Ask the patient to say additional words, all beginning with the glottal /h/
- Ask the patient to blend in an easy, relaxed phonation during the middle of a sigh
- Fade the sigh and move on to words, phrases, and sentences

Sapienza, C., & Hoffman, B. (2021). *Voice disorders* (4th ed.). Plural Publishing.

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Voice Prosthesis. A small (1.8 to 3.6 cm) silicone device that has a valve at the back end and an opening at the front end; inserted into the tracheoesophageal puncture in patients who have undergone laryngectomy; allows air into the esophagus, which vibrates; the sound is shaped into speech; see Laryngectomy.

Voluntary Stuttering. A treatment target in the fluent stuttering approach of Van Riper; for procedures, see Treatment of Stuttering: Specific Techniques or Programs under the main entry, Stuttering.

w

Wernicke's Aphasia. To treat this type of fluent aphasia, see [Aphasia](#); [Treatment of Aphasia: Specific Types](#).

Whole Word Accuracy (WWA). A criterion measure used in some treatment programs for speech sound disorders in children; instead of judging the accuracy of the target phoneme production, the entire word is judged for accuracy.

Wh-Questions. Questions that begin with *wh*; interrogative statements that begin with *what*, *when*, *where*, and *who*; intervention targets for children with language disorders; see [Language Disorders in Children](#) for teaching strategies.

Wilson's Disease. To treat motor speech disorders associated with this and other neurological disorders, see [Dysarthria](#).

Word Fluency. To teach clients the skill of saying as many words as possible that start with a sound or words that belong to a particular category (e.g., animals or vegetables), see [Aphasia](#).

Word Retrieval Problems. To treat difficulties in recalling names and other kinds of words often associated with recent brain injury, see [Aphasia](#), in which they are a dominant problem; see also [Dementia](#), [Traumatic Brain Injury in Adults](#), and [Right Hemisphere Disorder](#).

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